Classical Algebraic Geometry

— ***EYNTKA*** **Series** —

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Contents

Introduction

For over two millennia, a central domain of study in mathematics has been geometry, which studies shapes, spaces, and their geometric properties. Let us use the term "geometrical object" as a catch-all for what would usually be considered a shape or a space. Examples of geometrical objects studied throughout history include:

- Circles, more generally ellipsoids and cones
- Curves on a plane
- The Platonic solids
- The Cartesian plane
- Spheres and tori
- Möbius bands
- Graphs of functions

and so forth, the list stretching out endlessly. Historically, much of this study concentrated on the *Euclidean approach* or *ruler and compass approach*, taking most of its inspiration from the continuation of Euclid's book *Elements* written around 300 BC. From just 5 axioms, Euclid managed to deduce a large portion of classical geometry. This was a *synthetic* approach, building geometric facts purely from logical deductions about primitive objects like points and lines, without any underlying reference frame. However, there are fundamental limitations to the Euclidean method. Its rigid axiomatic framework (specifically the infamous 5th axiom, known as the *parallel postulate*) eliminates from consideration many important geometric arenas that modern mathematicians and physicists work with, such as the curved geometry on the surface of a sphere.

Today, the standard starting point for many geometric fields involves explicitly defining a space, such as the real vector space $\mathbb{R}^n$, and equipping it with continuous or smooth local structures. This modern transition from synthetic to analytic geometry relies on the idea of an origin and a coordinate system, introduced by *René Descartes* in the 17th century to quantify geometric problems. Descartes' insight created the first formal bridge between the spatial world of geometry and the symbolic world of algebra.

We now recognize that there are many different paradigms in which we can pursue the study of geometry, each focusing on a different area of interest. These paradigms fall under broad, heavily overlapping umbrellas: *Differential Topology*, *Riemann Geometry*, *Discrete Geometry*, *Classical*

Geometry[1], among others. Yet, a major consequence of Descartes' initial bridge lies in the study of *Algebraic Geometry*.

Algebraic geometry is the study of shapes defined by *algebraic equations*, mostly polynomials or ratios of polynomials. But why polynomials? Because they are constructed from the most fundamental arithmetic operations (addition and multiplication), making them purely algebraic and straightforward to manipulate; yet they express a wide range of geometric forms. When we say "shapes given by polynomials," we are referring to the *algebraic sets*, or the *zero-sets* (the vanishing points) of a collection of polynomials. Famously, the polynomial $x^2 + y^2 - 1 \in \mathbb{R}[x, y]$ is the unit circle, read as the set of points where $x^2 + y^2 - 1$ vanishes.

A central theme of classical algebraic geometry is that these shapes can be characterized by the algebraic functions that "live" on them. We will see that there is a symbiotic relationship, a "dictionary", between polynomials and their geometry.[2,3] The properties of quotient polynomial rings, ideals, and dimensions offer information about the geometric zero-sets. Conversely, looking at the geometry of these zero-sets guides our algebraic intuition. Through theorems like Hilbert's *Nullstellensatz* (theorem 2.4.4), we will translate geometric spaces (via the Zariski topology) into algebraic rings, and see how abstract concepts like Krull dimension and regular local rings formalize geometric dimension and smoothness.

Furthermore, classical algebraic geometry provides a framework to resolve the exceptions and anomalies that arise in standard affine geometry, such as the behavior of parallel lines. During the 19th and early 20th centuries, mathematicians (particularly those of the Italian school of algebraic geometry) sought a setting where intersection theory could be studied systematically without the need for special cases. By extending our familiar affine spaces into *Projective Varieties*, we introduce "points at infinity." In projective geometry, parallel lines finally meet, smoothing out exceptions and allowing for uniform intersection theorems. For instance, Bézout's Theorem (theorem 13.1.1), which counts the number of intersection points between curves, was later realized to naturally exist in projective space, where the theorem holds universally.

The goal of this text is to build that dictionary between geometry and algebra in full, so that a problem can be carried across in either direction. Part I sets it up over affine space: algebraic sets and the Zariski topology (chapter 1), the Nullstellensatz that makes the correspondence exact (chapter 2), and then the three invariants that the dictionary translates, namely birational type (chapter 3), dimension (chapter 4), and smoothness (chapter 5), closing with normality and the normalization of a variety (chapter 7). Part III rebuilds the whole picture over projective space, where rational functions and intersections behave as they should. Part IV is the classical payoff: intersection multiplicities, Bézout's theorem, divisors, the group law on a cubic, and the resolution of curve singularities by blowing up.

The book assumes the commutative algebra of *Commutative Algebra* [5]: localization, integral extensions, Noether normalization, and the dimension theory of Noetherian local rings are all taken from there. It is deliberately pre-scheme-theoretic. Everything constructed here is generalized by the scheme theory of *Algebraic Geometry* [3], and chapter 19 says where each construction reappears.

[1] which includes Euclidean, Affine, Conformal, and Analytic geometry, among other fields

[2] While this dictionary is a highly effective tool, it should not be overstated. There are inherently geometric phenomena that transcend pure algebra. For instance, differential geometry routinely utilizes smooth "bump functions," and topological studies may rely on soft sheaves, which possess analytic or infinite properties that lack direct algebraic analogues.

[3] Determining the exact limits of this relationship remains an active area of modern research. For example, the *Hodge Conjecture*, one of the Millennium Prize Problems, explores to what extent the topological spaces of certain complex algebraic varieties are determined entirely by their algebraic subvarieties.

0.0.1 Reading the numbering A star marks *bonus material*: a chapter, section or subsection that belongs in the book but that nothing later depends on. Skipping it costs nothing downstream. The star sits on the number, as in §2.7$\star$; in divisions written before this convention it sits at the front of the title instead, as in **Uniform Spaces*. The two mean the same thing.

Part I

Affine Varieties

A shape cut out by polynomial equations is determined by the polynomial functions on it. This Part makes that statement into a dictionary: to each algebraic set in affine space is attached a ring, its coordinate ring, and the geometry of the set is recoverable from the algebra of the ring. Points become maximal ideals, irreducible subsets become primes, and the maps between varieties become homomorphisms between the rings. The dictionary is global: every entry in it is an invariant of the whole variety, and Part II takes up the invariants attached to a single point.

The ground field is algebraically closed throughout, from chapter 2 onward; the Nullstellensatz, which is what makes the dictionary exact rather than approximate, is false without it. The commutative algebra the Part rests on is cited to *Commutative Algebra* [5] rather than rebuilt.

1

Algebraic Sets and the Zariski Topology

The dictionary this book is about begins with a single construction: to a set of polynomials, associate the points at which they all vanish. This chapter sets up that construction and its converse, which sends a set of points to the polynomials vanishing on it, and finds that the two are almost inverse to each other. The gap between "almost" and "exactly" is what chapter 2 closes.

Along the way the vanishing sets turn out to be the closed sets of a topology, the Zariski topology, which is coarse enough to be unlike any topology from analysis and fine enough to carry the whole theory. The chapter ends with the maps that respect it.

1.1 Affine Space and Regular Functions

In this whole chapter, R is a commutative ring and k be a field.

So far when polynomials were encountered, they were treated as a commutative ring with an indeterminate extension (or equivalently, they can be thought of as a ring over a free monoid). There is a natural way of thinking about any such polynomial as a function (which is probably how most first encounter polynomials): if $f \in k[x_1, x_2, ..., x_n]$, then if $(a_1, a_2, ..., a_n) \in k^n$, let $f(a_1, a_2, ..., a_n) \in k$ be the element which you get when replacing each x_i with a_i. Then we can make f into a function through the following map[1,2]:

$$k[x_1, ..., x_n] \to \mathrm{Hom}_{\mathbf{Set}}(k^n, k) \qquad f \mapsto ((a_1, ...a_n) \mapsto f(a_1, ..., a_n)) \tag{1.1}$$

where we shall consider the constants $c \in k[x_1, ..., x_n]$ to map to the functions that shall return the value of $c \in k$:

$$c \mapsto ((a_1, ..., a_n) \mapsto c)$$

Note that the codomain of the function in (1.1) is be the hom-set over the category **Set**, and not any other category we've extensively worked over so far (**Grp**, **Ring**, **Alg**, etc.); this was done since there are many non-linear functions representing non-linear polynomials (a simple example would be $x^2 \in k[x]$). On the other hand, the image of the above function is not just a collection of all set functions as they must all satisfy polynomials relations (can you find a set-function that is not given/representable by a polynomial?[3] The subset given by the function in (1.1) shall be called the set of *regular functions*: $\mathrm{Hom}_{\mathbf{Reg}}(k^n, k)$. Before properly boxing this definition, we shall take some more time pondering what the domain of the hom-set should be, namely we shall want the domain to represent Cartesian space without the extra algebraic structure that is usually implied by the notation k^n. For example, k^n usually denotes the n-dimensional vector-space over k with an origin, and more generally we think of k^n as having addition and scalar multiplication satisfying the axioms of a vector-space. Vector-spaces are "defined" for linear polynomials, with subspaces that are naturally "linear" (recall that subspaces are given by solutions of linear polynomials, see [11, Classification of Planes])[4]. We shall want a new type of space for which subspaces are solutions to general polynomial equations.

Definition 1.1.1: Affine Space

Let $(k, +, \cdot)$ be any field. Then the collection of the n-tuples of k^n as a set shall be called *affine space*. Its elements shall be called *points* and the space shall be denoted $\mathbb{A}^n_k$.

Hence, a field can be thought of as $(\mathbb{A}^1_k, +, \cdot)$. The space $\mathbb{A}^n_k$ is called *affine* space rather than just "space" as there shall be an equally important (or arguably more important) space called *projective space* that shall be introduced in chapter 8[5]. The term affine can also be thought of as emphasizing the lack of importance of the origin (recall that a linear transformation that doesn't need to map the origin to itself was called affine).

With affine space established, let us now define the image of the function in (1.1):

[1] For those familiar with algebraic geometry, this map can be thought of as selecting the global sections of $\mathbb{A}^n_k$. By the Nullstellensatz, $k^n = \mathrm{Spec}\,(k[x_1, .., x_n])$, and by the representability of affine schemes $\Gamma(\mathrm{Spec}\,(R)) \cong \mathrm{Hom}_{\mathrm{Reg}}(\mathrm{Spec}\,(R), A^1_k)$

[2] advanced comment: if k is not algebraically closed, we are missing points, and hence this technically is *not* a representable functor and is not the sheaf obtains from embedding Var_k into $\mathbf{Sch}_k$

[3] Over finite fields, and only over finite fields, every set function is given by a polynomial: interpolate, using that $a^{|k|} = a$ for every $a \in k$.

[4] In [12], we shall show that the vector-space structure can be recovered using Hopf Algebras, but this adds more structure to the algebraic set

[5] More generally, we shall see that affine space will be the representation space for schemes which generalizes affine space, something which chapter 19 points the reader toward

Definition 1.1.2: Regular Function [on Affine Space]

Let F represent the function in (1.1):

$$F : k[x_1, ..., x_n] \to \mathrm{Hom}_{\mathbf{Set}}(k^n, k) \qquad f \mapsto ((a_1, ...a_n) \mapsto f(a_1, ..., a_n))$$

Then the collection $\mathrm{im}(F)$ is called the *regular functions*, and is usually denoted $\Gamma(\mathbb{A}^n_k)$, $k[\mathbb{A}^n_k]$, or $\mathrm{Hom}_{\mathbf{Reg}}(\mathbb{A}^n_k, \mathbb{A}^1_k)$.

Each of these notations for the collection of regular functions have its use. The $\mathrm{Hom}_{\mathbf{Reg}}(\mathbb{A}^n_k, \mathbb{A}^1_k)$ notion makes it clear what the domain and codomain is, which shall be particularly important once we take different codomain. The $k[\mathbb{A}^n_k]$ notation is particularly useful when we shall take subsets $V \subseteq \mathbb{A}^n_k$ and denote all regular functions on V as $k[V]$; in this case this notation emphasizes that we are working over the particular field k. This shall be useful for there are circumstances where we change which field we are working over[6]. The $\Gamma(\mathbb{A}^n_k)$ notation comes from modern algebraic geometry where gamma is used hint that these are the *global section* over $\mathbb{A}^n_k$ (see [3] for a full explanation). As this chapter is gearing up the reader for modern algebraic geometry, we shall preferentially use the $\Gamma(\mathbb{A}^n_k)$ notation to acclimate the reader. It should be pointed out that another common notation is $\mathcal{O}_{\mathbb{A}^n_k}(\mathbb{A}^n_k)$ which is more useful when it is important to keep track of regular functions on open subsets of $\mathbb{A}^n_k$ (once we define a topology on $\mathbb{A}^n_k$) in which case we shall have the collection of regular function on $U \subseteq \mathbb{A}^n_k$ be written as $\mathcal{O}_{\mathbb{A}^n_k}(U)$. This is not useful to us in this chapter, and so shall not be used.

The set $\Gamma(\mathbb{A}^n_k)$ forms a natural k-algebra structure given by point-wise operations:

$$f + g \equiv f(a) + g(a) \qquad fg \equiv f(a)g(a) \qquad c \cdot f = cf(a) \qquad \forall a \in k \tag{1.2}$$

With this k-algebra structure, it can be shown that $k[x_1, ..., x_n] \cong_k \mathrm{Hom}_{\mathbf{Reg}}(\mathbb{A}^n_k, \mathbb{A}^1_k)$, and so the two sets are often used interchangeably. If the distinction needs to be made clear between the element x_i and the function x_i, we shall write

$$k[X_1, ..., X_n] = \mathrm{Hom}_{\mathbf{Reg}}(\mathbb{A}^n_k, \mathbb{A}^1_k) = k[\mathbb{A}^k_n]$$

where each X_i is a regular function that is the projection onto the ith coordinate, and call an element of $k[X_1, ..., X_n]$ a *polynomial function* or *polynomial map*. This distinction shall become more important once there can be multiple polynomials that will be associated to the same regular function, which shall be the case in section 2.1. When it is clear from context, we may denote a polynomial or a regular function by $f(x)$ and $f(X)$ respectively where x is understood to be the indeterminate variables and X is understood to be the regular functions $X_1, ..., X_n$.

Due to this isomorphism, we can define a function from $k[x_1, ..., x_n]$ to k directly by directly substituting the values for the indeterminates with a point in the affine space. Namely, for each $a \in \mathbb{A}^n_k$, we take:

$$\mathrm{ev}^k_a : k[x_1, ..., x_n] \to k \qquad f(x) \mapsto f(a)$$

where x and a is understood to be n-tuples of indeterminates and values respectively. These are all k-algebra homomorphisms, for

$$\mathrm{ev}^k_a(f(x) + cg(x)) = \mathrm{ev}^k_a(f(x)) + c \cdot \mathrm{ev}^k_a(g(x)) \qquad \mathrm{ev}^k_a(f(x)g(x)) = \mathrm{ev}^k_a(f(x)\mathrm{ev}^k_a(g(x))$$

[6] A good example of the use of group cohomology is measure how much an isomorphism $k[V] \cong k[W]$ is preserved/fails when consider $\overline{k}[V]$ and $\overline{k}[W]$

The subscript of the evaluation function represents the point of evaluation, while the superscript represents the values we choose to plug for every indeterminate, where the values are usually the same as the base field, however they may differ, for example it is common to take $\mathbb{Z}$ or $\mathbb{Q}$[7]. Thus, a regular function can be thought of as the collection of evaluation functions on a polynomial.

1.1.3 Foreshadowing If you know elementary analysis or calculus, then you are probably aware that the types of functions we most commonly use are *continuous*. Continuous functions are a very important tool in analysing the properties of the space we're studying (ex. $\mathbb{R}$, $\mathbb{R}^n$, $L^2(\mathbb{R})$, etc.), continuous functions can be thought of as the natural types of functions that preserve some local properties of the space. In Algebra, we can think of polynomial functions as the natural types of functions. This comparison between continuous functions mapping to $\mathbb{R}$ and polynomial functions mapping to k will in fact be incredibly precise, as we will see in the last section of this chapter.

1.2 Algebraic Sets and Vanishing Ideals

1.2.1 Quick Side-Note A common misconception, born from a strong reliance on graphs in most first-year undergraduate calculus/analysis classes, is to intuitively associate the zero-set of a function with the *graph* of a function.
For example, $f(x) = x$ has a zero-set containing only the point 0, while the function $F(x, y) = x - y$ has a zero-set that forms the graph $\Gamma(f)$. This distinction may seem trivial, but it is surprisingly easy to conflate the two concepts when moving into higher dimensions.

Graphs can be thought of as the collection of the solutions of $f(x_1, \ldots, x_n) = y_0$ as the constant term y_0 varies.

From such polynomials functions, we can create many interesting shapes depending on the zeros', or locus, of polynomial:

[7]These shall be called $\mathbb{Z}$-valued and $\mathbb{Q}$-valued points of the polynomial, or more precisely the variety (as we shall soon see)

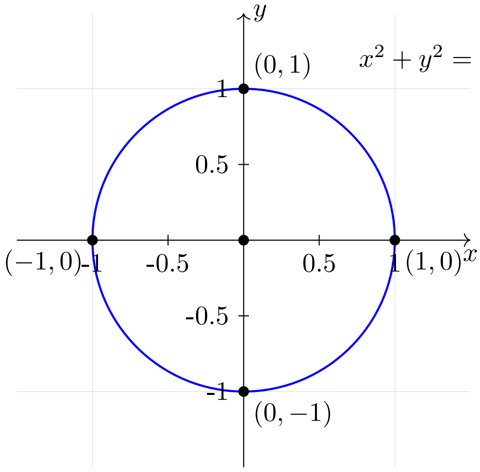
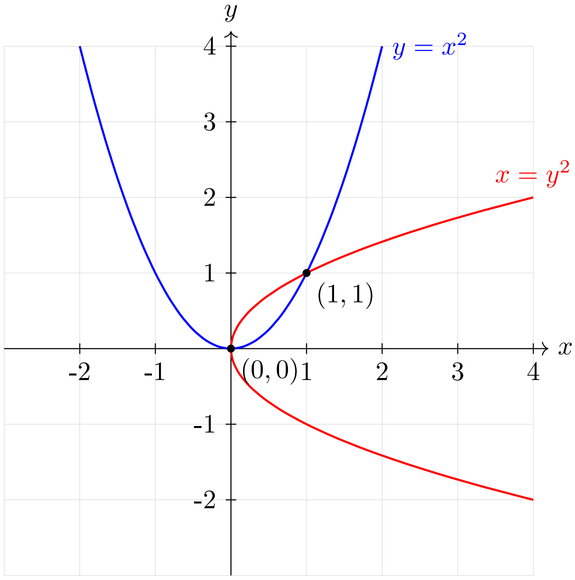


Figure 1.1: Circle represented by the equation $x^2 + y^2 - 1 = 0$

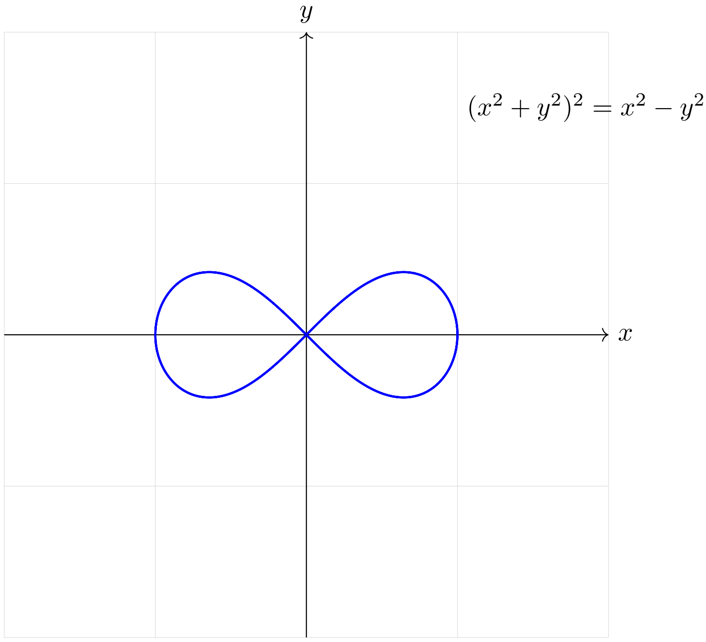
Example 1.2.2: Zeros' of Polynomials

1. The polynomial $x^2+y^2-1 \in \mathbb{R}[x, y]$ gives the circle of fig. 1.1. More precisely, the polynomial shall be associated to an element of $\Gamma(\mathbb{A}^n_k)$, where plugin in any point on from $\mathbb{A}^2_{\mathbb{R}}$ that is on the circle (ex. $(1, 0)$ or $(\sqrt{2}/2, \sqrt{2}/2)$ shall give 0.

2. Take $y - x^2 \in \mathbb{R}[x, y]$. Then this shall give the usual parabola $y = x^2$. If we take $x - y^2 \in \mathbb{R}[x, y]$, we shall get a parabola rotated 90 degree's (or $\pi/2$ radians) clockwise. If we take the polynomial $(y - x^2)(x - y^2) \in \mathbb{R}[x, y]$, then the zero set of this new polynomial shall be the zero-set of the two multiplied polynomials:



3. Take $(x^2+y^2)^2-(x^2-y^2) \in \mathbb{R}[x,y]$. This shape is called the *Bernoulli Lemniscate*:

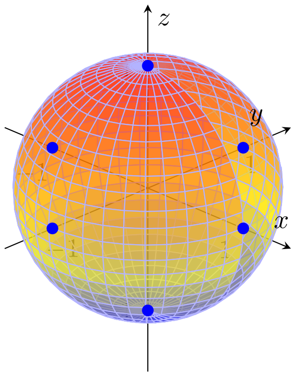


   Notice that the shape crosses over itself. This shows that zero-sets of polynomials differ from graphs, and also differ from the usual smooth manifolds from differential geometry.

4. Take $p(x) \in k[x]$ over any field k. Then the shape associated to $p(x)$ are the *roots* of the polynomials. For example, taking $x^2-2 \in \mathbb{Q}(\sqrt{2})[x]$ shall give $\{\sqrt{2}, -\sqrt{2}\} \subseteq \mathbb{A}^1_{\mathbb{Q}(\sqrt{2})}$ as these are the solution to $x^2-2=0$. More generally, the shape given by a polynomial in one variable is always the collection of roots that is present in the underlying field. This means that $x^2-2 \in \mathbb{Q}[x]$ would give the empty set, as there are no rational roots. By the fundamental theorem of algebra, any polynomial $p(x) \in \mathbb{C}[x]$ of degree d shall have d roots, and hence the shape associated to $p(x)$ shall have d points.

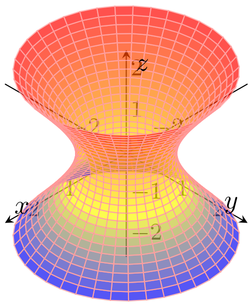
5. Take $x^2+y^2+z^2-1 \in \mathbb{R}[x,y,z]$. Then this is the classical unit sphere centered at the origin:

$$x^2 + y^2 + z^2 = 1$$

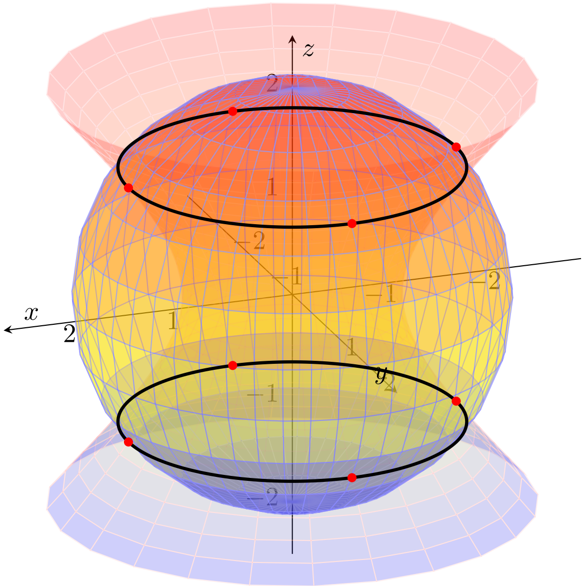


What polynomial would you choose to center the unit sphere at different points of $\mathbb{A}^3_{\mathbb{R}}$? How would you change the radius of the sphere?

6. Let us now consider taking two shapes: the sphere given by $x^2 + y^2 + z^2 - 4$ and the hyperboloid given by $x^2 + y^2 - z^2 - 1$. The hyperboloid looks like so:

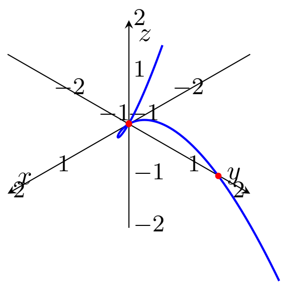


Then the intersection of these two shapes shall produce two circles, visualized like so:



This produce *two disconnected components* (in the usual euclidean topology). We shall show later that no single polynomial can create these two circles, hence we shall often have the case where we get interesting shapes by interesting or taking the union of two zero-sets.

7. The following is an infamous example of a shape that looks like it could be the zero-set of a single polynomial, but can't be: consider the following shape given by $t \mapsto (t, t^2, t^3)$:



It can be proven that this shape is *not* the zero-set of a single polynomial $p(x, y, z) \in k[x, y, z]$. On the other hand, this shape, is the intersection of two zero-sets of the surfaces $y = x^2$ and $z = xy$. This shape is called the *twisted cubic.*

More generally, given any polynomial in $k[x_1, x_2, ..., x_n]$, we can associate with it a geometric shape based on its zero set!

Definition 1.2.3: [Affine] Algebraic Set

Let $\mathbb{A}^n$ be an affine n-space over a field k, and let $S \subseteq \Gamma(\mathbb{A}^n)$. Then:

$$\mathcal{Z}(S) \subseteq \mathbb{A}^n \quad \mathcal{Z}(S) := \{(a_1, a_2, ..., a_n) \mid f(a_1, a_2, ..., a_n) = 0, \text{ for all } f \in S\}$$

If for $V \subseteq \mathbb{A}^n$, there exists an $S \subseteq \Gamma(\mathbb{A}^n)$ such that $V = \mathcal{Z}(S)$, then V is called an *affine algebraic set*, *locus*, or *zero set*. If it is unambiguous, the set may be called an *algebraic set*[a].

[a] In some textbooks, this set is called a *variety*. The term variety shall be reserved for definition 2.3.1

From now on, when taking a subset of $\mathbb{A}^n$, $A \subseteq \mathbb{A}^n$, it will be assumed that A is an algebraic set unless specified otherwise (i.e. it would be stated "for some subset $A \subseteq \mathbb{A}^n$)

Example 1.2.4: Affine Algebraic Sets

All the above are algebraic sets. Let's go over a few more examples:

1. If $n = 1$, then the zeros of a polynomial are always points, that is, the roots of the polynomial. When k is infinite the only polynomial with infinitely many zeros is 0 (more precisely, the regular function associated to 0), which has all of k as its vanishing locus. Infiniteness is the right hypothesis rather than characteristic 0: item (5) below exhibits the failure over $\mathbb{F}_2$, while $\overline{\mathbb{F}_p}$ has characteristic p and the claim still holds there. Furthermore, the polynomial 1 vanishes nowhere, and hence the corresponding algebraic set is $\emptyset$. Therefore, in $\mathbb{A}^1$, the affine sets are

$$\emptyset \qquad \text{finite sets} \qquad k$$

   This also shows that not all sets are algebraic sets (for example, if $k = \mathbb{R}$, then $[0, 1]$ is not an algebraic set in $\mathbb{A}^1_{\mathbb{R}}$)

2. If $n = 2$, Then take $f \in \Gamma(\mathbb{A}^2)$ such that $f(x, y) = x^2 + y^2 - 1$. Then

$$\mathcal{Z}(f) = \left\{(x, y) \in \mathbb{A}^2 \;\middle|\; x^2 + y^2 = 1\right\}$$

   If $k = \mathbb{R}$, then $\mathcal{Z}(f)$ is infinite, so it is not a product of two algebraic subsets of $\mathbb{A}^1$: those are finite or all of $\mathbb{A}^1$, and neither product is a circle.

   Furthermore, this examples shows that there are lot's more zero sets in $\mathbb{A}^2$ than there are in $\mathbb{A}^1 \times \mathbb{A}^1$. This is in stark contrast to how every element in the vector space $\mathbb{R}^2$ can be written as an element in $\mathbb{R}$ and an element in $\mathbb{R}$, i.e. $\mathbb{R}^2 \cong \mathbb{R} \times \mathbb{R}$ as vector spaces.

3. For any $\mathbb{A}^n$, can you always find a set of functions $F \subseteq \Gamma(\mathbb{A}^n)$ such that $|V(F)| = 1$ (i.e. is every singleton an algebraic set for every $\mathbb{A}^n$ over any field?) This shall be answered in section 2.4.

4. What is the vanishing set of linear polynomials?

5. In the first example, we produced the entire space using the zero polynomial. Is it necessary that we use the zero-polynomial to make the zero set the entire space? The answer is no: simply take $k = \mathbb{Z}/2\mathbb{Z}$ and $p(x) = x^2 + x$. However, this is a problem exclusive to finite fields. Can you prove that if k is an infinite field, $p \in k[x_1, x_2, ..., x_n]$ and $p(a_1, a_2, ..., a_n) = 0$ for all $a_i \in k$ then $p \equiv 0$? Freely use that a nonzero one-variable polynomial has finitely many roots, and induct on the number of variables.

6. Let k be a finite field and take any subset $S \subseteq \mathbb{A}_k^n$. Show that S is always an algebraic set. Hence, being algebraic gives no additional information in the finite case.

7. Let $f \in k[x_1, x_2, ..., x_n]$. Then $\mathcal{Z}(f)$ is called a *hyper-surface*. For example $S^1 = \mathcal{Z}(x^2 + y^2 - r^2)$ represents a circle of radius r. Another example would be $\mathcal{Z}(xy - 1)$ which represents the hyperbola $y = 1/x$. More generally, any conic section is a hyper-surface. In higher dimensions, $\mathcal{Z}(x^2 + y^2/4 + z^2/9 - 1)$ represents an ellipsoid in $\mathbb{R}^3$. The twisted cubic is *not* a hyper-surface.

8. For any polynomial function $f : k \to k$, we can define $F : k^2 \to k$ where $F(x, y) = f(x) - y$. Then $\mathcal{Z}(F)$ consists of all points where $f(x) - y = 0$, i.e. $f(x) = y$, which is the pair of points $\{(a, b) \mid f(a) = b\}$, i.e. the graph of f! More generally, if $f : k^n \to k$ is a polynomial, the polynomial $F : k^{2n} \to k$ gives rise to the polynomial $F : (x, y) = f(x) - y$ whose zero set is the graph of f (recall the convention that x, y represent multiple variables when it is clear from context).

Proposition 1.2.5: Properties of Affine Algebraic Sets

Let $S, T \subseteq \Gamma(\mathbb{A}^n)$. Then

1. If $S \subseteq T$ then $\mathcal{Z}(T) \subseteq \mathcal{Z}(S)$ (i.e. $\mathcal{Z}$ is *contravariant* in respect to inclusion)

2. If $(S) = I \subseteq \Gamma(\mathbb{A}^n)$ is the ideal of S, then $\mathcal{Z}(I) = \mathcal{Z}(S)$.

3. For any arbitrary collection of algebraic sets $\{S_i\}_{i \in I}$,
$$\bigcap_{i \in I} \mathcal{Z}(S_i) = \mathcal{Z}\left(\bigcup_{i \in I} S_i\right)$$
that is, the arbitrary intersection of algebraic sets is an algebraic sets, and is equal to the algebraic set of the intersections of the $\{S_i\}_{i \in I}$

4. For any finite collection of algebraic sets $\{S_i\}_{i=1}^n$,
$$\bigcup_{i=1}^n \mathcal{Z}(S_i) = \mathcal{Z}\left(\prod_{i=1}^n S_i\right)$$
that is, the finite union of algebraic sets is an algebraic set, and is equal to the product of the S_i as ideals (from part 2, we can assume $(S_i) = S_i$ without loss of generality).

5. If $0, 1 \in \Gamma(\mathbb{A}^n)$ denote the zero and constant function, then
$$\mathcal{Z}(0) = \mathbb{A}^n \qquad \mathbb{Z}(1) = \emptyset$$

6. If I is an ideal of $\Gamma(\mathbb{A}^n)$, then
$$\mathcal{Z}(I) = \mathcal{Z}(I^m) \ \forall m \in \mathbb{N}_{>0}$$
that is, $\mathcal{Z}$ is invariant to taking higher powers of the ideals

These are all straight-forwards proofs, and so should all be done before reading these proofs for the best learning experience.

***Proof*:**

1. Let $S \subseteq T$, and consider $x \in \mathcal{Z}(T)$. Then $f(x) = 0$ for every $f \in T$. Since $S \subseteq T$, in particular $g(x) = 0$ for every $g \in S$. Thus $x \in \mathcal{Z}(S)$, establishing $\mathcal{Z}(T) \subseteq \mathcal{Z}(S)$.

2. Let $S \subseteq \Gamma(\mathbb{A}^n)$ be any set and let $I = (S)$. We want to show that $\mathcal{Z}(I) = \mathcal{Z}(S)$. Since $S \subseteq I$, the $\mathcal{Z}(I) \subseteq \mathcal{Z}(S)$ conclusion comes from the first part. For the reverse, consider $x\mathcal{Z}(S)$. Then for all $t \in S$, $t(x) = 0$. If we pick any two $f, g \in S$, then $(f+g)(x) = f(x)+g(x) = 0$. If we pick any $c(x) \in \Gamma(\mathbb{A}^n)$ and any $f \in S$, then $(cf)(x) = c(x)f(x) = c(x)0 = 0$. Thus, if we enlarge S to be closed under addition and multiplication by elements of $k^[\mathbb{A}^n]$, those polynomial will also be zero on x, and so $\mathcal{Z}(S) \subseteq \mathcal{Z}(I)$, establishing $\mathcal{Z}(S) = \mathcal{Z}(I)$

3. We want to show that $\bigcap_{i\in I} \mathcal{Z}(S_i) = \mathcal{Z}\left(\bigcup_{i\in I} S_i\right)$ for an arbitrary indexing set I. This is just some set theory: by definition:
$$x \in \mathcal{Z}(\cup_{i\in I} S_i) = \{y \in \mathbb{A}^n \mid f(y) = 0,\ \forall f \in \cup_{i\in I} S_i\} = \mathcal{Z}\left(\sum_\alpha I_\alpha\right)$$
which is true if and only if x is the solution for polynomials in every set S_i. The collection of such elements is:
$$\bigcap_{i\in I} \mathcal{Z}(S_i)$$

4. We want to show that $\bigcup_{i=1}^n \mathcal{Z}(S_i) = \mathcal{Z}\left(\prod_{i=1}^n S_i\right)$. By definition:
$$x \in \mathcal{Z}\left(\prod_{i=1}^n S_i\right) = \left\{ y \in \mathbb{A}^n \;\middle|\; f(y) = 0\ \forall f \in \prod_{i=1}^n S_i \right\}$$
where by definition $f(x) = \sum_{j=1}^m (f_{1_j} f_{2_j} \cdots f_{n_j})(x) = \sum_{j=1}^m f_{1_j}(x) f_{2_j}(x) \cdots f_{n_j}(x) = 0$. Notice that for a particular term in the sum to be zero, we only need on of the f_{i_j} to be zero. Therefore, the sum is zero if and only if one polynomial in every term of the sum is zero. Since this also applies to sums with only one term in it, then if x is the solution to one polynomial in S_i, then it will be part of the zero's of $\prod_{i=1}^n S_i$. But this is just another way of saying:
$$x \in \bigcup_{i=1}^n \mathcal{Z}(S_i)$$

5. Notice that $0(x) = 0$ for all $x \in \mathbb{A}^n$ and $1(x) = 1 \neq 0$ for all $x \in \mathbb{A}^n$, and so
$$\mathcal{Z}(0) = \mathbb{A}^n \qquad\qquad \mathcal{Z}(1) = \emptyset$$

6. Since $I^m \subseteq I$, the first part gives $\mathcal{Z}(I) \subseteq \mathcal{Z}(I^m)$. For the reverse inclusion, let $x \in \mathcal{Z}(I^m)$ and let $f \in I$ be arbitrary. Then f^m is a product of m elements of I, so $f^m \in I^m$, and therefore
$$f(x)^m = f^m(x) = 0$$
As k is a field it has no nonzero nilpotents, so $f(x) = 0$. Since $f \in I$ was arbitrary, $x \in \mathcal{Z}(I)$, giving $\mathcal{Z}(I^m) \subseteq \mathcal{Z}(I)$ and hence $\mathcal{Z}(I) = \mathcal{Z}(I^m)$, as we sought to show.

A couple of remarks are in order. From (2), we can without loss of generality restrict $\mathcal{Z}$ be a function from the ideals of $k[x_1, ..., x_n]$ to the algebraic sets in $\mathbb{A}^n$:

$$\mathcal{Z} : \{\text{ideals in } k[x_1, ..., x_n]\} \to \{\text{affine algebraic sets in } \mathbb{A}^n\}$$

Furthermore, $\mathcal{Z}$ is a surjection from the ideals of $k[x_1, x_2, ..., x_n]$ and the algebraic sets (the ideals I^m for each $m \geq 1$ map to the same algebraic set). Next, the finiteness condition on the product of ideals is a necessary condition for closure of algebraic sets:

Example 1.2.6: Not Closed Under Arbitrary Unions

Take $\mathbb{A}^1_{\mathbb{R}}$ and the sets $V(x-a)$ for $a \in \mathbb{Z}$. Then prove that the union of these sets is *not* an algebraic set (you may freely use the fact that polynomials have at most finite roots, and recall that $\mathbb{R}[x]$ is Noetherian).

Next, we took the product of ideal $\prod_i^n I_i$ instead of the intersection. Since it always the case that $IJ \subseteq I \cap J$ then in the cases where $IJ \subsetneq I \cap J$, $\mathcal{Z}(I \cap J) \subsetneq \mathcal{Z}(I) \cup \mathcal{Z}(J)$[8]. However, since $\mathcal{Z}(I^2) = \mathcal{Z}(I)$, the distinction is invisible to $\mathcal{Z}$, and $\mathcal{Z}(IJ) = \mathcal{Z}(I \cap J)$ always

Perhaps most importantly for this chapter, notice that these properties are the properties of a topology (see [9, basic topology]):

Definition 1.2.7: Zariski Topology

Let $\mathbb{A}^n$ be an n-affine space over k. Then the set

$$\mathcal{T}_{\mathcal{Z}} = \{\mathcal{Z}(I) \mid \text{I is an ideal of } \Gamma(\mathbb{A}^n)\}$$

from the set of closed sets in the *Zariski topology* $(\mathbb{A}^n, \mathcal{T}_{\mathcal{Z}})$

We can therefore try and study this topological space! This is essentially the foundation for studying geometrical concepts in algebra, and vice-versa, it is the foundation for studying algebraic concepts in geometry, and so we will spend a lot of time establishing the link between the ideals of $\Gamma(\mathbb{A}^n)$ and the Zariski topology.

Before such topological considerations, we shall first see if we can find some function that is the inverse of $\mathcal{Z}$, that is we start from an algebraic set in $\mathbb{A}^n$ (or more generally any subset of $\mathbb{A}^n$) and associate it to an ideal in $\Gamma(\mathbb{A}^n)$, letting us strive towards some way of corresponding the two in a manner similar to Galois Theory[9]. Towards the goal of finding an inverse for $\mathcal{Z}$, notice that we already have that $\mathcal{Z}$ is a surjection of ideals of $\Gamma(\mathbb{A}^n)$ to the algebraic sets. Thus, it is meaningful to ask about having a function from the set of algebraic sets to the set of ideals of $k[x_1, x_2, ..., x_n]$. To understand this question, we start off with a slightly more general question (just like we started with subsets of $k[x_1, ..., x_n]$ instead of ideals):

[8]The condition $IJ = I \cap J$ is more delicate than it looks: it holds exactly when $\mathrm{Tor}_1^R(R/I, R/J) = 0$, for which see [7].

[9]In fact,the order reversing nature of this correspondence will end up making this correspondence a *Galois correspondence*, which name is inspired by how the Galois correspondence between intermediate fields and subgroups was one of the first well-known order-reversing correspondences

Definition 1.2.8: Vanishing Ideal

Let $V \subseteq \mathbb{A}^n$ be a subset of the affine n-space. Then

$$\mathcal{I}(V) := \{f \in \Gamma(\mathbb{A}^n) \mid f(v) = 0,\ \forall v \in V\}$$

is the *vanishing ideal* of V.

Notice that $\mathcal{I}(V)$ was not defined as an ideal, however it is indeed an ideal; it is important for the sake of intuition that the reader should prove this at least once. Even better, $\mathcal{I}(V)$ is the *largest* ideal of $\Gamma(\mathbb{A}^n)$ that corresponds to the set V. Just like $\mathcal{Z}$, $\mathcal{I}$ can be interpreted as a function

$$\mathcal{I} : \{V \subseteq \mathbb{A}^n_k\} \to \{\text{ideals of } \Gamma(\mathbb{A}^n)\}$$

Example 1.2.9: Vanishing Ideal

1. Let $\mathbb{A}^2$ be an affine space over the field $\mathbb{R}$ and let $X \subseteq \mathbb{A}^2$ be the x-axis. Then $\mathcal{I}(X) = (y) \subseteq \mathbb{R}[x, y]$
2. Let k be infinite. Then $\mathcal{I}(\mathbb{A}^n) = 0$.
3. Find the vanishing ideals if of subsets $\mathbb{A}^n_k$ if k is finite.

Just like $\mathcal{Z}$, $\mathcal{I}$ shares similar properties:

Proposition 1.2.10: Properties of Vanishing Ideals

let $A, B \subseteq \mathbb{A}^n$ be two algebraic sets. Then

1. If $A \subseteq B$, then $\mathcal{I}(B) \subseteq \mathcal{I}(A)$ (i.e. $\mathcal{I}$ is *contravariant* with respect to inclusion)
2. $\mathcal{I}(A \cup B) = \mathcal{I}(A) \cap \mathcal{I}(B)$
3. $\mathcal{I}(\emptyset) = \Gamma(\mathbb{A}^n)$. If k is infinite, $\mathcal{I}(\mathbb{A}^n) = 0$

***Proof*:**

1. Let $A \subseteq B$ and take $f \in \mathcal{I}(B)$, so $f(b) = 0$ for every $b \in B$. Every $a \in A$ lies in B, so $f(a) = 0$, giving $f \in \mathcal{I}(A)$.
2. A polynomial vanishes on $A \cup B$ exactly when it vanishes on A and vanishes on B, which is the assertion $\mathcal{I}(A \cup B) = \mathcal{I}(A) \cap \mathcal{I}(B)$.
3. The condition defining $\mathcal{I}(\emptyset)$ is vacuous, so every polynomial belongs to it and $\mathcal{I}(\emptyset) = \Gamma(\mathbb{A}^n)$. For the second claim, suppose k is infinite and f vanishes at every point of $\mathbb{A}^n$. Induct on n. For $n = 1$ a nonzero polynomial has at most $\deg f$ roots, while k is infinite, so $f = 0$. For $n > 1$ write $f = \sum_i g_i(x_1, \ldots, x_{n-1})x_n^i$. Fixing any $a \in k^{n-1}$ makes $f(a, x_n)$ a one-variable polynomial vanishing on all of k, so each $g_i(a) = 0$ by the base case. As a was arbitrary, each g_i vanishes on $\mathbb{A}^{n-1}$, and by induction $g_i = 0$. Hence $f = 0$, so $\mathcal{I}(\mathbb{A}^n) = 0$, as we sought to show.

The hypothesis that k be infinite in (3) is not removable: over $\mathbb{F}_q$ the polynomial $x^q - x$ vanishes at every point of $\mathbb{A}^1$, so $\mathcal{I}(\mathbb{A}^1_{\mathbb{F}_q}) \neq 0$.

Notice that we have omitted showing what $\mathcal{I}(A \cap B)$ is. One problem is that $\mathcal{I}(A) \cup \mathcal{I}(B)$ is not necessarily an ideal (since the union of two ideals is not necessarily an ideal). It might be tempting to say that since

$$\langle \mathcal{I}(A) \cup \mathcal{I}(B) \rangle = \mathcal{I}(A) + \mathcal{I}(B)$$

we can conclude that:

$$\mathcal{I}(A \cap B) = \mathcal{I}(A) + \mathcal{I}(B)$$

However, this is not generally the case (consider the algebraic sets given by the ideals $(x^2 - y)$ and (x)). Proving the reverse inclusion needs the Nullstellensatz, but it can already be shown from the definitions that $\mathcal{I}(A \cap B) \supseteq \mathcal{I}(A) + \mathcal{I}(B)$, since a function vanishing on A or on B vanishes on $A \cap B$. With the properties of $\mathcal{I}$ established, we can start getting an idea of what ideals of $\Gamma(\mathbb{A}^n)$ are necessary to restrict $\mathcal{Z}$ to make it a bijection:

Proposition 1.2.11: Zero Sets and Coordinate Ideals

Let $\Gamma(\mathbb{A}^n)$ be the set of polynomial functions from $\mathbb{A}^n$ to k. Then

1. If $V \subseteq \mathbb{A}^n$, then $V \subseteq \mathcal{Z}(\mathcal{I}(V))$. If $I \subseteq \Gamma(\mathbb{A}^n)$ is an ideal of $\Gamma(\mathbb{A}^n)$, then $I \subseteq \mathcal{I}(\mathcal{Z}(I))$.

2. If $V = \mathcal{Z}(I)$ is an algebraic set, then $V = \mathcal{Z}(\mathcal{I}(V))$. Similarly, if $I = \mathcal{I}(V)$ is a vanishing ideal, then $I = \mathcal{I}(\mathcal{Z}(I))$. In other words, for every subset $V \subseteq \mathbb{A}^n$ and every ideal $I \subseteq \Gamma(\mathbb{A}^n)$,
$$\mathcal{I}(\mathcal{Z}(\mathcal{I}(V))) = \mathcal{I}(V) \qquad \mathcal{Z}(\mathcal{I}(\mathcal{Z}(I))) = \mathcal{Z}(I)$$

***Proof*:**

1. For $V \subseteq \mathcal{Z}(\mathcal{I}(V))$, since $\mathcal{I}(V)$ is the set of all functions such that $f(V) \equiv 0$ (including at least the zero polynomial), then V is certainly part of its zero set.

   Conversely, for $I \subseteq \mathcal{I}(\mathcal{Z}(I))$, the coordinate ideal of $\mathcal{Z}(I)$ certainly contains I since for any $f \in I$ and $x \in \mathcal{Z}(I)$, $f(x) = 0$. Thus, $I \subseteq \mathcal{I}(\mathcal{Z}(I))$

2. Let $V = \mathcal{Z}(I)$. By part 1 it suffices to prove $V \supseteq \mathcal{Z}(\mathcal{I}(V))$. Take $x \in \mathcal{Z}(\mathcal{I}(V))$, so that $f(x) = 0$ for every $f \in \mathcal{I}(V)$. Every $f \in I$ vanishes on $\mathcal{Z}(I) = V$ and so lies in $\mathcal{I}(V)$, giving $I \subseteq \mathcal{I}(V)$; hence $f(x) = 0$ for every $f \in I$, which says exactly that $x \in \mathcal{Z}(I) = V$. Therefore $V = \mathcal{Z}(\mathcal{I}(V))$.

   The second claim is the same argument with the roles exchanged. If $I = \mathcal{I}(V)$ then part 1 gives $I \subseteq \mathcal{I}(\mathcal{Z}(I))$; conversely if $g \in \mathcal{I}(\mathcal{Z}(I))$ then g vanishes on $\mathcal{Z}(\mathcal{I}(V))$, which we have just shown equals V, so $g \in \mathcal{I}(V) = I$. Applying $\mathcal{I}$ and $\mathcal{Z}$ to these two identities gives the displayed equations, completing the proof.

What the 2nd part of proposition 1.2.11 shows is that if we restrict to just the algebraic sets and vanishing ideals of $\mathbb{A}^n$ and $\Gamma(\mathbb{A}^n)$ respectively, then $\mathcal{Z}$ and $\mathcal{I}$ are indeed inverses of each other. The goal then would be to be able to characterize vanishing ideals, that is ideals that are of the form $I = \mathcal{I}(V)$ for some algebraic set V. It is in fact a rather involved process to find the nature of the ideal $\mathcal{I}(V)$; a lot of section 2.2 will be building up to get to the theorem which will tell us the answer: *Hilbert's Nullstellensatz*. In the process of building up the theory to answer this question, there are a few other loose ends that we will need to tie up. For example, what properties must the Zariski topology have, and what does that imply for the corresponding ring $\Gamma(\mathbb{A}^n)$? Another important point to address is that as $k[x_1, ..., x_n]$ is Noetherian, hence every ideal is finitely generated: $I = (f_1, ..., f_m) = (f_1) \cup (f_2) \cup \cdots \cup (f_m)$. By the properties of zero sets:

$$\mathcal{Z}(I) = \mathcal{Z}(f_1) \cap \mathcal{Z}(f_2) \cap \cdots \cap (f_m)$$

In other words, there seem to be a natural theory of decomposition of algebraic sets. Furthermore, we have been shown theoretically what properties algebraic sets have, but how is one computed in practice? These questions will receive answers in the following section.

Exercise 1.2.1

1. Show that $z \mapsto \overline{z}$ is not a complex polynomial.
2. Show that $V = \left\{(z, w) \in \mathbb{A}^2_{\mathbb{C}} \;\middle|\; |z|^2 + |w|^2 = 1\right\}$ is not an algebraic subset of $\mathbb{A}^2_{\mathbb{C}}$, but that the same set, viewed in $\mathbb{A}^4_{\mathbb{R}}$, is algebraic.
3. Let k be a perfect field with algebraic closure $\overline{k}$ and Galois group $\mathrm{Gal}_k(\overline{k})$, acting on $\mathbb{A}^n_{\overline{k}}$ coordinatewise. Show that

   $$\mathbb{A}^n_k \cong \left\{p \in \mathbb{A}^n_{\overline{k}} \;\middle|\; p^\sigma = p \text{ for all } \sigma \in \mathrm{Gal}_k(\overline{k})\right\}$$

   and say where perfectness is used.
4. Let $V = \mathcal{Z}(x^2 + y^2) \subseteq \mathbb{A}^2$. Show that V is irreducible over $\mathbb{Q}$ but reducible over $\mathbb{Q}(i)$, so that irreducibility depends on the field of definition. Why does this phenomenon disappear once the field is algebraically closed?
5. Here is a way of recovering $A = k[x_1, \ldots, x_n]$ from $V = k^n$. Write $V^* = \mathrm{Hom}_k(V, k)$ and recall the symmetric algebra ([11, Symmetric Algebra]):

   $$\mathrm{Sym}^\bullet(V^*) = k \oplus V^* \oplus \mathrm{Sym}^2(V^*) \oplus \mathrm{Sym}^3(V^*) \oplus \cdots$$

   Show that $\mathrm{Sym}^\bullet(V^*) \cong A$. Then let $\mathfrak{m} \subseteq A$ be the maximal ideal of a point and show $\mathfrak{m}/\mathfrak{m}^2 \cong V^*$, so that the linear structure is recovered at each point. This is the *Zariski tangent space* of chapter 5.

1.3 The Zariski Topology

In this section, we will be assuming basic topological knowledge from the reader (see [9, basic topology])

As the reader (ought to have) showed in in example 1.2.4, if $k = \mathbb{F}_q$ is finite, then every subset of $\mathbb{A}^n_{\mathbb{F}_q}$ is algebraic, hence the Zariski topology is isomorphic to the discrete topology, showing us that there is no "interesting" behaviour in the finite case. Thus, we shall usually focus on the case where k is not-finite. This doesn't mean that finite fields don't play an important role in algebraic geometry, however their importance shall not be visible until the development of schemes[10] or Class Field Theory[11] and so they shall have negligible importance this chapter.

As the topology was defined using closed sets (sets for which a collection of polynomials is zero on that set), a moment should be given to give a geometric interpretation to open sets. Open sets are those for which a collection of polynomials are *not all zero* at once on those points. If $S = \{f\} \in \Gamma(\mathbb{A}^n_k)$ is a singleton, then it will be precisely the set of points where the function f does not vanish. This makes all open sets of $\mathbb{A}^1_k$ (where k is not finite) the *cofinite sets* and hence it has the cofinite topology. If $n > 1$, the topology is certainly no longer cofinite. Open sets shall become importance when we look to define *rational polynomials* and *rational functions* for they give us the domain on which a function is non-zero. (see definition 1.3.5).

Let us now look at the separatedness of the Zariski topology. Notice that $\mathcal{T}_\mathcal{Z}$ is rarely Hausdorff (also known as T_2): if $\mathbb{A}^1 = k$ is any infinite field, as we've established that the only algebraic sets (and hence closed sets) are the empty set, finite sets, and $\mathbb{A}^1_k$ and so the only non-trivial open sets in $\mathcal{T}_\mathcal{Z}$ would be co-finite sets (i.e. $\mathcal{T}_\mathcal{Z}$ is the co-finite topology). Then as we can see, if we pick any two points $a, b \in \mathbb{A}^1$, we cannot separate them by two open disjoint sets. However, over any $\mathbb{A}^1 = k$, the singleton's are always algebraic sets (giving a hint to those who were working on example 1.2.4), and so it is always a Fréchet space (or T_1)[12].

This result generalizes for any k^n

Proposition 1.3.1: Zariski Topology Always T_1

Let k be any field and give $\mathbb{A}^n_k$ the Zariski topology. Then:

1. $\mathbb{A}^n_k$ is T_1, that is, every singleton is closed.
2. If k is infinite, $\mathbb{A}^n_k$ is not Hausdorff.

The two halves need different hypotheses, and the second genuinely fails without its one: over a finite field every subset is algebraic (example 1.2.4), so the topology is discrete and therefore Hausdorff.

***Proof*:**

1. For any $a = (a_1, \ldots, a_n) \in k^n$ let $I = (x_1 - a_1, \ldots, x_n - a_n)$. A point b lies in $\mathcal{Z}(I)$ exactly when $b_i - a_i = 0$ for every i, so $\mathcal{Z}(I) = \{a\}$ and singletons are closed. No hypothesis on k is used.
2. Let k be infinite and suppose U, V are nonempty open sets. Write $U = \mathbb{A}^n \setminus \mathcal{Z}(I)$ and $V = \mathbb{A}^n \setminus \mathcal{Z}(J)$ with $\mathcal{Z}(I), \mathcal{Z}(J) \neq \mathbb{A}^n$, so neither I nor J lies in $\mathcal{I}(\mathbb{A}^n) = 0$ (proposition 1.2.10); choose $f \in I$ and $g \in J$ nonzero. Then fg is nonzero, since $\Gamma(\mathbb{A}^n) \cong k[x_1, \ldots, x_n]$ is an integral domain, so again by proposition 1.2.10 the product fg does not vanish identically on $\mathbb{A}^n$. Any point where fg is nonzero has both f and g nonzero, hence lies in $U \cap V$. So any two nonempty open sets meet, and $\mathbb{A}^n_k$ is not Hausdorff, as we sought to show.

[10] In particular, for any Scheme X, there is always a scheme morphism $X \to \mathrm{Spec}\,(\mathbb{Z})$ a there is always a ring homomorphism $\mathbb{Z} \to \mathcal{O}_X(X)$; the fibers of this scheme X_p are $\mathbb{F}_p$-schemes. Furthermore, $\mathbb{F}_q$-schemes shall be important in number theory

[11] In particular, the behaviour of unramified field extensions between local fields will be captured via the automorphisms of finite fields

[12] For those familiar with algebraic topology, schemes are almost never T_1, as prime ideals that are not maximal are not closed points

Two topological concepts which should be immediately introduced to be explored as the chapter develops are the following:

Definition 1.3.2: Zariski Dense And Closure

Let $A \subseteq \mathbb{A}^n$ be any subset of the affine space $\mathbb{A}^n$. Then the *zariski closure* of A is the smallest algebraic set containing A, denoted $\overline{A}$. Geometrically, it is the smallest set $\mathcal{A}$ for which every point in A is the zero set of a collection of polynomials.

If $\overline{A} = \mathbb{A}^n$, then A is said to be *zariski dense*. More generally if $A \subseteq V \subseteq \mathbb{A}^n$ such that $\overline{A} = V$, then A is said to be zariski dense in V.

As an example, if we give $\mathbb{R}$ the Zariski topology, the closure of any finite set is itself, while the closure of any infinite set is all of $\mathbb{R}$. The Zariski closure of a set is in fact quite easy to characterize using $\mathcal{Z}$ and $\mathcal{I}$ and follows quickly using proposition 1.2.11:

Corollary 1.3.3: Zariski Closure

Let $A \subseteq \mathbb{A}^n$. Then:

$$\overline{A} = \mathcal{Z}(\mathcal{I}(A))$$

***Proof*:**
The set $\mathcal{Z}(\mathcal{I}(A))$ is algebraic and contains A by proposition 1.2.11(1), so it is one of the algebraic sets whose intersection defines $\overline{A}$, giving $\overline{A} \subseteq \mathcal{Z}(\mathcal{I}(A))$.

Conversely, let W be any algebraic set with $A \subseteq W$. Applying $\mathcal{I}$ reverses the inclusion, so $\mathcal{I}(W) \subseteq \mathcal{I}(A)$, and applying $\mathcal{Z}$ reverses it again:

$$\mathcal{Z}(\mathcal{I}(A)) \subseteq \mathcal{Z}(\mathcal{I}(W)) = W$$

the last equality because W is algebraic (proposition 1.2.11(2)). So $\mathcal{Z}(\mathcal{I}(A))$ is contained in every algebraic set containing A, hence in $\overline{A}$, as we sought to show.

Sticking to $k = \mathbb{R}$ or $k = \mathbb{C}$ for a moment, we can compare the euclidean topology on $\mathbb{R}^n$ with the Zariski topology on $\mathbb{A}^n_{\mathbb{R}}$. (Coarser, and I think is the corsest topology on which all polynomial functions are continuous)

Another important topological consideration would be to find a topological basis for the Zariski topology. Recall a subset of the topology on a space X is a basis B if any open set of X is the union of a family of sets from B.

Proposition 1.3.4: Basis For Zariski Topology

For $f \in \Gamma(\mathbb{A}^n)$ define

$$D(f) = \mathbb{A}^n_k \setminus \mathcal{Z}(f) = \{a \in \mathbb{A}^n_k \mid f(a) \neq 0\}$$

Then the collection $\{D(f) \mid f \in \Gamma(\mathbb{A}^n)\}$ forms a [topological] basis for the Zariski topology on $\mathbb{A}^n$.

The letter D can be thought of as a mnemonic for *doesn't* vanish.

***Proof*:**
It suffices to show that for any open U and any $a \in U$ there is an f with $a \in D(f) \subseteq U$, since then U is the union of those $D(f)$. As U is open, $U = \mathbb{A}^n_k \setminus \mathcal{Z}(I)$ for some ideal $I \subseteq \Gamma(\mathbb{A}^n)$. Now $a \notin \mathcal{Z}(I)$ says precisely that $f(a) \neq 0$ for *some* $f \in I$; fix such an f, so that $a \in D(f)$. Since $(f) \subseteq I$,

$$\mathcal{Z}(I) \subseteq \mathcal{Z}(f)$$

and taking complements gives:

$$a \in D(f) \subseteq U$$

and since a was arbitrary, we see that U is the union of sets $D(f)$, as we sought to show.

The collection of $D(f)$ for each $f \in k[x_1, ..., x_n]$ forms the coarsest basis:

Definition 1.3.5: Distinguished Open Sets

Let $f \in k[x_1, ..., x_n]$. Then the set

$$D(f) = \mathbb{A}^n_k \setminus \mathcal{Z}(f) = \{a \in \mathbb{A}^n_k \mid f(a) \neq 0\}$$

is called he distinguished open set.

Example 1.3.6: Distinguished Open Sets

1. In $\mathbb{A}^1_k$ take $f = x$. Then $D(x) = \mathbb{A}^1_k \setminus \{0\}$, the punctured line. It is not closed, but it is itself an algebraic set in disguise: the map

$$D(x) \to \mathcal{Z}(xy - 1) \subseteq \mathbb{A}^2_k \qquad a \mapsto (a, 1/a)$$

   is a bijection with inverse $(a, b) \mapsto a$, and both are given by polynomials on their domains. So the punctured line is the hyperbola.

2. The same works in general. For $f \in k[x_1, \ldots, x_n]$,

$$D(f) \;\cong\; \mathcal{Z}\big(y\, f(x_1, \ldots, x_n) - 1\big) \;\subseteq\; \mathbb{A}^{n+1}_k$$

   by $a \mapsto (a, 1/f(a))$. Every distinguished open set is therefore an algebraic set in one more variable, which is why they are the right basic open sets: the topology has a basis of sets that are themselves affine.

3. The distinguished opens do form a basis. Any open set is $\mathbb{A}^n_k \setminus \mathcal{Z}(I)$, and

$$\mathbb{A}^n_k \setminus \mathcal{Z}(I) = \mathbb{A}^n_k \setminus \bigcap_{f \in I} \mathcal{Z}(f) = \bigcup_{f \in I} D(f)$$

4. Not every open set is distinguished. In $\mathbb{A}^2_k$ the set $\mathbb{A}^2_k \setminus \{(0,0)\}$ is open, but $D(f)$ misses the whole curve $\mathcal{Z}(f)$, which is infinite for nonconstant f, and equals $\mathbb{A}^2_k$ for constant $f \neq 0$. Neither omits exactly one point.

The next property we get about the Zariski topology is that it's always Noetherian, that is, if there is an ascending chain of open sets, it must stabilize:

Proposition 1.3.7: Zariski Topology Is Noetherian

Let $\mathbb{A}_k^n$ have the Zariski topology. Then $\mathbb{A}_k^n$ is Noetherian.

***Proof*:**
Let $U_1 \subsetneq U_2 \subsetneq \cdots$ be an ascending chain of open sets, so there is a corresponding

$$X_1 \supsetneq X_2 \supsetneq \cdots$$

descending chain of closed sets. Then there is a corresponding

$$I(X_1) \subsetneq I(X_2) \subsetneq \cdots$$

ascending chain of ideals. Then we know that $\Gamma(\mathbb{A}^n)$ is *noetherian*, and so this chain must stabilize at some n:

$$I(X_n) = I(X_{n+1}) = \cdots$$

meaning the respective closed chain, and hence open chain, must stabilize, as we sought to show.

Noetherianity already forces compactness, and by the same argument every subspace of $\mathbb{A}_k^n$ is compact too: given an open cover of a subspace, the closed sets complementary to finite subcovers form a descending chain, which must stabilize. Proposition 2.5.1 nonetheless gives a second proof, via the Nullstellensatz, because that argument produces the explicit partition of unity $1 = \sum g_i f_i$ rather than the bare existence of a finite subcover, and that identity is what later arguments use[13].

1.4 Regular Maps and Continuity

The next thing we should consider the continuous functions. As the topology is the weakest in which the zero-sets is closed, it may not surprise the reader that it is the coarsest topology in which polynomials are continuous

Definition 1.4.1: Regular Map on Affine Space

A map $\varphi : \mathbb{A}_k^n \to \mathbb{A}_k^m$ is called a *regular map* if there exists polynomials $\varphi_1, \varphi_2, ..., \varphi_m \subseteq k[x_1, x_2, ..., x_n]$ such that

$$\varphi(a_1, a_2, ..., a_n) = (\varphi_1(a_1, a_2, ..., a_n), \varphi_2(a_1, a_2, ..., a_n)..., \varphi_m(a_1, a_2, ...a_n))$$

If $m = 1$, a regular map is often called a *regular function*.

[13] Compactness is cheap here and correspondingly weak: almost every scheme is compact, so the useful strengthening is *properness*, which chapter 10 develops for projective varieties.

Lemma 1.4.2: Coordinate Ring and Regular Map

The map $\varphi : \mathbb{A}_k^n \to \mathbb{A}_k^m$ is regular if and only if the induced map $\varphi^\# : \Gamma(\mathbb{A}_k^m) \to \Gamma(\mathbb{A}_k^n)$ given by $\varphi^\#(f) = f \circ \varphi$ is a k-algebra homomorphism

***Proof*:**
Let $\varphi : \mathbb{A}_k^n \to \mathbb{A}_k^m$ be a regular map so that each component is given by a polynomial. Then $\varphi^\#$ maps a function $f \in \Gamma(\mathbb{A}_k^m)$ to $f \circ \varphi \in \Gamma(\mathbb{A}_k^n)$, namely $f \circ \varphi$ is a polynomial map. If $f, g \in \Gamma(\mathbb{A}_k^m)$, then:

$$(f+g) \circ \varphi = f \circ \varphi + g \circ \varphi$$
$$(f \cdot g) \circ \varphi = f \circ \varphi \cdot g \circ \varphi$$
$$(k \cdot f) \circ \varphi = k \cdot (f \circ \varphi)$$

thus it is a k-algebra homomorphism.

Conversely, let $\varphi^\# : \Gamma(\mathbb{A}_k^m) \to \Gamma(\mathbb{A}_k^n)$ is a k-algebra homomorphism that maps:

$$f \mapsto \varphi^\#(f) = f \circ \varphi \in \Gamma(\mathbb{A}_k^n)$$

Let $y_1, ..., y_n : \mathbb{A}_k^m \to k$ be the coordinate functions in $\Gamma(\mathbb{A}_k^m)$. If one of the components of φ where not a polynomial, then take $f = x_i$, so that the composition $x_i \circ \varphi = \varphi_i$ is not a polynomial, giving a contradiction. Hence φ must be a regular map, completing the proof.

Lemma 1.4.3: Ring homomorphism Induces Regular map

Let $f : \Gamma(\mathbb{A}_k^m) \to \Gamma(\mathbb{A}_k^n)$ be a k-algebra homomorphism. Then it induces a map $\varphi : \mathbb{A}_k^n \to \mathbb{A}_k^m$ such that $\varphi^\# = f$.

***Proof*:**
Let $k[y_1, ..., y_m] \cong \Gamma(\mathbb{A}_k^m)$ and $k[x_1, ..., x_n] \cong \Gamma(\mathbb{A}_k^n)$. Then as f is a k-algebra homomorphism, it is enough to map each y_i to some polynomial $f(y_i) = p(x_1, ..., x_m)$ and extend to form a map on all of $\Gamma(\mathbb{A}_k^m)$. Let φ_i be the function determine by $f(y_i)$. Then it is a matter of course to check that $\varphi^\# = f$, completing the proof.

Note that the maps must be k-algebra homomorphism, not just ring homomorphisms. For example, if $n = m = 1$, then $f : \mathbb{C}[\mathbb{A}_k^1] \to \mathbb{C}[\mathbb{A}_k^1]$ sending $p(x) \mapsto \overline{p(x)}$ (i.e. the complex conjugate) is not a $\mathbb{C}$-algebra homomorphism, and the reader could check there can't be a corresponding φ such that $\varphi^\# = f$.

Lemma 1.4.4: Regular Map is Continuous

A regular map $\varphi : \mathbb{A}_k^n \to \mathbb{A}_k^m$ is continuous.

***Proof*:**
It suffices to check that the pre-image of every closed set is closed. If $V = \mathcal{Z}(I) \subseteq \mathbb{A}_k^m$ is an algebraic set, we need to show there exists an ideal J such that

$$\mathcal{Z}(J) = \varphi^{-1}(V)$$

To that end:

$$\begin{aligned}\varphi^{-1}(V) &= \{a \in \mathbb{A}_k^n \mid \varphi(a) \in V\} \\ &= \{a \in \mathbb{A}_k^n \mid f(\varphi(a)) = 0 \ \forall f \in I\} \\ &= \left\{a \in \mathbb{A}_k^n \,\middle|\, \varphi^{\#}(f)(a) = 0 \ \forall f \in I\right\}\end{aligned}$$

Consider the set $\varphi^{\#}(I) = \{g \in \Gamma(\mathbb{A}_k^n) \mid g = f \circ \varphi\}$. This set would be the desired pre-image ideal, except it is in general not an ideal as it need not be closed under multiplication (recall the image of an ideal need not be an ideal). Thus, take $J = \langle \varphi^{\#}(I) \rangle$. Then it is clear that

$$\mathcal{Z}(J) = \varphi^{-1}(V)$$

and as V was an arbitrary closed set, φ is continuous, as we sought to show.

Note that *not all* continuous maps arise from a ring homomorphism:

Example 1.4.5: Continuous, No Associated Ring Homomorphism
Take $f : \mathbb{A}_{\mathbb{R}}^1 \to \mathbb{A}_{\mathbb{R}}^1$ given by:

$$f(a) = \begin{cases} 2 & a = 1 \\ 1 & a = 2 \\ a & \text{otherwise} \end{cases}$$

The closed sets of $\mathbb{A}_{\mathbb{R}}^1$ are the finite sets and the whole space, and f carries each of these to a set of the same size, so the pre-image of every closed set is closed and f is continuous. But f is not regular. If it were given by a polynomial p, then $p(a) = a$ for every $a \notin \{1, 2\}$, so the polynomial $p(x) - x$ would have infinitely many roots and hence be zero. That forces $p(x) = x$, which does not transpose 1 and 2.

Another such example is $k = \mathbb{C}$ with the conjugation map presented earlier.

There needs to be more information for the full and faithful correspondence between regular maps and ring homomorphisms. This extra information is a *sheaf* and a *locally ringed morphism*. We shall not define these in this book, instead the reader may refer to [3].

Example 1.4.6: Regular Maps

1. Take $V = \mathcal{Z}(y - x^2, z - x^3)$. Show that the map $f : \mathbb{A}_k^1 \to V$ given by $t \mapsto (t, t^2, t^3)$ is a regular map. Show it is not an isomorphism[a].
2. Any straight line $\mathbb{A}_k^n$ is isomorphic to any other straight-line. Similarly for any k-dimensional linear subspace.
3. The map $\mathbb{A}_k^1 \to \mathcal{Z}(y - x^2)$ given by $t \mapsto (t, t^2)$ is a regular map with inverse $(x, y) \mapsto x$, and hence is an isomorphism. Note that it was important which coordinate we project to.

4. Here is another version of the above map. Consider the global sections of $\mathbb{A}^1_k \to \mathbb{A}^1_k$ given by

$$k[y] \to k[x] \qquad y \mapsto x^2$$

(note that the k-homomorphism is the contravariant direction of the regular morphism). As a map between varieties, this gives the map $a \mapsto a^2$ (the reader should verify this!).

5. Take $f : \mathbb{A}^2_k \to \mathbb{A}^2_k$ given by $(x, y) \mapsto (x, xy)$. Show that $\mathrm{im}(f)$ is *not* an algebraic set. The image of a regular map is always a *constructible set*, a finite union of locally closed sets, which is theorem 11.3.4.

6. The reader should be careful with their geometric intuitions coming from the real numbers. Consider the map:

$$\mathcal{Z}(x^2 + y^2 - 1) \to \mathbb{A}^1_{\mathbb{C}} \qquad (x, y) \mapsto x$$

Show that this map is *surjective* (in contrast to the map certainly being *injective* when over $\mathbb{R}$). In fact, by taking different cross sessions of the resulting shape interpreted as a 2-real dimensional shape living in 4-real dimensional space, you will see that $\mathcal{Z}(x^2 + y^2 - 1)$ will have both the parabola and the hyperbolic curve as sections. The hyperbolic sections demonstrate why the map is surjective and 2-to-1.

[a] It shall later be shown to be a *birational equivalence*

Let us now take a moment to look at isomorphism of algebraic sets:

Definition 1.4.7: Isomorphism of Varieties

let X, Y be two algebraic sets. Then if there exists a regular map $f : X \to Y$ with regular inverse, then the two varieties are said to be *isomorphic*

Note that this was not called a homeomorphism between two varieties. All isomorphisms are homeomorphisms, but not all homeomorphisms are isomorphisms

Example 1.4.8: Isomorphism of Varieties

1. Consider $X = \mathcal{Z}(y - x^n) \subseteq \mathbb{A}^2_k$. Then $X \cong \mathbb{A}^1_k$ with $\varphi(x, y) = x$, which has inverse $\psi(t) = (t, t^n)$

2. Take $X = \mathcal{Z}(x^3 - y^2) \subseteq \mathbb{A}^2_k$. Then X can be parameterized by the regular map $f : \mathbb{A}^1_k \to X$ by $f(t) = (t^2, t^3)$. However, this is not an isomorphism: the inverse map is:

$$g(x, y) = \frac{y}{x}$$

however, there is no polynomial associated to g (can the reader see/prove why?), and hence it is not an isomorphism.

3. The reader may have recalled that $\mathcal{Z}(x^3 - y^2)$ has a cusp at 0, and believe this is the obstruction to the inverse. It is in fact a bit more subtle: a map can be bijective, with no singularity anywhere, and still fail to be an isomorphism. Let $\mathrm{char} k = p > 0$ and consider the Frobenius $\varphi : \mathbb{A}^1_k \to \mathbb{A}^1_k$ given by $\varphi(t) = t^p$. Over $k = \overline{k}$ this is a bijection, since $t \mapsto t^p$ is injective ($t^p = s^p$ gives $(t - s)^p = 0$) and surjective (k is algebraically closed), and it is

regular. Yet its inverse $t \mapsto t^{1/p}$ is not regular: the induced map on coordinate rings is $k[x] \to k[x]$, $x \mapsto x^p$, whose image is $k[x^p]$, and x does not lie in it. So φ is a bijective regular map that is not an isomorphism.

The hypothesis on the characteristic matters. In characteristic 0, or more generally whenever $p \nmid n$, the map $t \mapsto t^n$ is n-to-1 away from the origin over an algebraically closed field, so it is not even bijective.

On the other hand, the map given in the previous example had a rational polynomial inverse. Rational functions appear as localization of the coordinate ring, and so they offer a different class of functions and a different classification problem that turns out to be much easier, as we shall demonstrate in chapter 3.

It is in general difficult to ascertain isomorphic classes of algebraic sets or varieties. Some isomorphism classes that we shall state here but not prove are:

1. Smooth genus 0 curves are isomorphic to conic curves in $\mathbb{P}^2_k$
2. classes of elliptic curves can be classified by their j-invariant
3. Algebraic surfaces are birationally equivalent (defined in chapter 3), but have different isomorphism classes given their Kodaira dimension.

Exercise 1.4.1

1. Show that the Zariski topology on $\mathbb{A}^1_k$ is the cofinite topology, hence the coarsest topology in which points are closed. Then show that for $n \geq 2$ the Zariski topology on $\mathbb{A}^n_k$ is strictly finer than the cofinite topology, so the characterization fails in higher dimension.
2. Prove that $(I \cap J)^2 \subseteq IJ \subseteq I \cap J$, and conclude that $\mathcal{Z}(I \cap J) = \mathcal{Z}(IJ)$.
3. Let $k = \mathbb{F}_q$ be a finite field. Show that every set function $k^n \to k$ is given by a polynomial. Hint: build a characteristic function $\chi_{\overline{a}}$ with $\chi_{\overline{a}}(\overline{a}) = 1$ and 0 elsewhere, and show these span. Conclude that the Zariski topology on $\mathbb{A}^n_{\mathbb{F}_q}$ is discrete.
4. Show that the Zariski topology on $\mathbb{A}^n_k$ is Hausdorff if and only if k is finite.
5. Let $V \subseteq \mathbb{A}^n_k$ and $W \subseteq \mathbb{A}^m_k$ be algebraic sets. Show that $V \times W$ is an algebraic set in $\mathbb{A}^{n+m}_k$, and that $\mathbb{A}^n_k \times \mathbb{A}^m_k = \mathbb{A}^{n+m}_k$ as algebraic sets[14].
6. Let $f, g \in k[x, y]$ have no common *factor*. Show that $\mathcal{Z}(f, g)$ is finite. Hint: they remain coprime in $k(x)[y]$, a principal ideal domain, by a Gauss-lemma argument.
7. Show that if $V \subseteq \mathbb{A}^n_k$ and $W \subseteq \mathbb{A}^m_k$ are varieties, not merely algebraic sets, then $V \times W \subseteq \mathbb{A}^{n+m}_k$ is a variety.

[14]This easy statement becomes markedly harder for projective algebraic sets; see section 10.1.

2

The Algebra-Geometry Dictionary

Chapter 1 left $\mathcal{Z}$ and $\mathcal{I}$ almost inverse to one another: an algebraic set is recovered from its vanishing ideal, but an ideal need not be recovered from its zero set. This chapter identifies exactly which ideals are recovered. The answer, Hilbert's Nullstellensatz, is that they are the radical ideals, and it holds precisely when the ground field is algebraically closed.

From here to the end of the book, k is algebraically closed unless a result says otherwise. The hypothesis is not decorative: over $\mathbb{R}$ the ideal (x^2+1) is its own radical and has empty zero set, so the correspondence fails at once. With the Nullstellensatz in hand, the geometric side and the algebraic side can be tabulated against each other, and the rest of the book consists of extending that table.

2.1 Coordinate Rings

In this section, we explore more properties of affine sets. As affine sets correspond to ideals, we shall see that there is a correspondence between the two objects. In particular it shall will show that properties of algebraic sets will often have corresponding properties in their coordinate rings, and vice-versa. This theme shall play out often in this chapter, where a property proven in a geometric or algebraic setting shall have a corresponding result in the respective dual.

By lemma 1.4.2, the collection of regular function $\mathrm{Hom}_{\mathbf{Reg}}(\mathbb{A}^n_k, \mathbb{A}^1_k)$ can be thought of as the generalization of the dual vector-space, where polynomial functions are taken instead of linear functions. We saw that for each vector subspace $W \subseteq V$, there is a dual space $W^\vee = \mathrm{Hom}_k(W, k)$ which it is isomorphic to. It is thus natural to ask what is the appropriate dual space for an algebraic set $V \subseteq \mathbb{A}^n_k$ corresponding to the *quotient* of $\Gamma(\mathbb{A}^n_k)$[1].

[1] For the categorically inclined, since the arrows in the dual space are reversed, sub-objects correspond to quotient-objects

In order to find the dual, we must consider more the subspace topology put on $V \subseteq \mathbb{A}_k^n$. Note that all of the properties of the Zariski topology apply to A since $\mathbb{A})k^n$ is T1 (proposition 1.3.1), from which we can conclude that closed sets are preserved in the subspace topology. We will always limit ourselves to taking A to be an algebraic set, since algebraic set are the object of interest. The dual space we wish to establish is a corresponding "polynomial ring" which will generate all of the closed sets of A, just like how $k[x_1, x_2, ..., x_n]$ is needed to generate the closed sets of $\mathbb{A}^n$. Without looking at the next definition, can you see what the definition of such a ring may be?

To elucidate the coming definition, let us come back to the algebraic set V defined given by the vanishing set of $x^2 + y^2 - 1$. Then on V, the function $X^2 + Y^2 - 1$ on V will evaluate to 0, hence these two polynomials are equivalent as functions:

$$X^2 + Y^2 - 1 \equiv 0 \qquad \text{on } V$$

If g is any other polynomial corresponding to the function G, then $(X^2 + Y^2 - 1)G$ evaluates to 0 everywhere as well, showing the set of regular functions taking inputs on V is closed under multiplication by regular functions $\Gamma(\mathbb{A}_k^n)$. Furthermore, if two regular functions f, g which evaluate to zero everywhere are added, the function $f + g$ will also evaluate to zero, showing it is closed under addition. In other words, the collection of functions which evaluates to 0 forms an ideal of $\Gamma(\mathbb{A}_k^n)$. What is this ideal? If we let F be the function

$$F : k[x, y] \to \mathrm{Hom}_{\mathbf{Reg}}(V, k) \qquad f \mapsto ((a, b) \to f(a, b)) \tag{2.1}$$

where $\mathrm{Hom}_{\mathbf{Reg}}(V, k)$ is the set of regular functions which evaluate only on the points V, then we get that this map is surjective and that:

$$\frac{k[x, y]}{\ker(F)} \cong_k \mathrm{Hom}_{\mathbf{Reg}}(V, k)$$

The reader should clue in that $\ker(F)$ is by definition $\mathcal{I}(V)$ as defined in definition 1.2.8! This leads to the following definition:

Definition 2.1.1: [Affine] Coordinate Ring

Let $k[x_1, ..., x_n]$ be a polynomial ring over n variables, V an algebraic set, and $\mathcal{I}(V)$ the corresponding vanishing ideal. Then:

$$\Gamma(V) = k[V] = \frac{k[x_1, ..., x_n]}{\mathcal{I}(V)}$$

is called the *coordinate ring* of V. The corresponding regular functions shall be denoted $k[V]$ or $\Gamma(V)$.

Notice how this notation is the same as for regular functions: the coordinate ring of $V = \mathbb{A}_k^n$ is isomorphic to $k[X_1, ..., X_n]$ since $I(\mathbb{A}^n) = (0)$. In $\Gamma(V)$, we have that $\mathcal{I}(V) = \overline{(0)}$, showing the notation is consistent.

We can naturally bring the notion of regular function to algebraic sets:

Definition 2.1.2: Regular Maps on Algebraic Sets

Let $V \subseteq \mathbb{A}_k^n$, $W \subseteq \mathbb{A}_k^m$ be algebraic sets. Then a map $\varphi : V \to W$ is called a *regular map*, or *regular morphism*, if there exists polynomials $\varphi_1, \varphi_2, ..., \varphi_m \subseteq k[x_1, x_2, ..., x_m]$ such that

$$\varphi(a_1, a_2, ..., a_n) = (\varphi_1(a_1, a_2, ..., a_n), \varphi_2(a_1, a_2, ..., a_n) ..., \varphi_m(a_1, a_2, ...a_n))$$

If $m = 1$, then φ is often called a *regular function*.

This definition also allows us to upgrade definition 1.1.2 and definition 1.4.1. To check that $\varphi : V \to W$ is well-defined, it suffices to check that $\varphi_1, ..., \varphi_m$ as elements of $k[V]$ satisfy the equations of W.

Proposition 2.1.3: Regular Maps Are Continuous

Let $\varphi : V \to W$ be a regular map where V, W are equipped with the Zariski topology. Then φ is continuous.

***Proof*:**
Let $S \subseteq W$ be a closed subset. It suffices to show that $\varphi^{-1}(S)$ is closed. if $\varphi^{-1}(S) = \emptyset$, it is closed by definition, so assume $\varphi^{-1}(S) \neq \emptyset$. As S is closed, there exists polynomials $f_1, ..., f_n$ such that for each $x \in S$, $g_1(x) = \cdots = g_m(x) = 0$. If $a \in \varphi^{-1}(S)$, then $\varphi(a) \in S$, so

$$g_1(\varphi(a)) = \cdots = g_m(\varphi(a)) = 0 \tag{2.2}$$

As $\varphi = (\varphi_1, ..., \varphi_n)$ is a tuple of polynomials, notice that (2.2) is a collection of polynomials in $k[V]$ where a vanishes. As this is true for all points in $\varphi^{-1}(S)$, we see that:

$$\varphi^{-1}(S) = \mathcal{Z}(g_1 \circ \varphi, ..., g_m \circ \varphi)$$

showing it is closed, as we sought to show.

As before, this is not an iff as not all continuous functions are regular functions (simply take $V = W = \mathbb{A}_{\mathbb{C}}^1$ and the conjugation map). That is because continuity in the Zariski topology is not enough to preserve the algebraic relations. In particular, any point $a \in V \subseteq \mathbb{A}_k^n$ satisfying

$$f_1(a) = \cdots f_n(a) = 0$$

must have an image that satisfies:

$$g_1(w) = \cdots = g_m(w) = 0 \qquad \text{i.e.} \qquad g_1(\varphi(a)) = \cdots g_1(\varphi(a)) = 0$$

Example 2.1.4: Coordinate Ring and Regular Mapping

1. Let $V = \mathcal{Z}(xy - 1)$, which represents the hyperbola $y = 1/x$ in $\mathbb{R}^2$. Then the coordinate ring $\mathbb{R}[V] = \mathbb{R}[x, y]/(xy - 1)$. In $\mathbb{R}[V]$, $f(x, y) = x$ and $g(x, y) = x + xy - 1$ are the same function since $f - g \in \mathcal{I}(V)$. In $\mathbb{R}[V]$, $\overline{xy} = \overline{1}$, giving us that $\mathbb{R}[V] \cong \mathbb{R}[x, 1/x]$. Thus, for any $f \in \mathbb{R}[x, y]$ and $(a, b) \in V$, $\overline{f}(a, b) = f(a, 1/a)$ for any f that maps to $\overline{f}$.

   Take the regular map $\varphi : V \to \mathbb{A}_k^1$ mapping $(x, y) \mapsto x$. The image of this map is $\mathbb{A}_k^1 \setminus \{0\}$, which is certainly *not* a closed set (can the reader see why). It is an open set, being the complement of the closed set $\{0\}$. However, if we take the map to be $\varphi' : V \to \mathbb{A}^2_k$ mapping $(x, y) \mapsto (x, 0)$, then the image is no longer an open set, and hence it is possible for the images of regular maps to be neither open or closed. Further exploration the properties of the image of regular maps shall be done section 9.4.

2. Take $k = \mathbb{F}_p$ and consider $\mathbb{A}^n_{\mathbb{F}_p}$. Then the map:

$$\varphi(a_1, ..., a_n) = (a_1^p, ..., a_n^p)$$

is a regular function. If $V \subseteq \mathbb{A}^n_{\mathbb{F}_p}$ is an algebraic set, notice that φ will always take V to itself, since if f is one of the polynomials that define V so that $f(a) \equiv 0 \bmod p$, then:

$$f(a_1^p, ..., a_n^p) = (f(a_1, ..., a_n))^p \equiv (0)^p \equiv 0 \bmod p$$

The regular function φ is called the *Frobenius mapping*

3. Take $t \mapsto (t^3, t^2)$. This is a regular function and maps $\mathbb{A}^1_{\mathbb{R}}$ to the curve given by $x^3 = y^2$ in $\mathbb{A}^2_{\mathbb{R}}$.

4. Take $V = \mathcal{Z}(xy - 1)$ and $W = \mathcal{Z}(v - u^2)$. We can define a map between global sections:

$$\varphi : \frac{k[u, v]}{(v - u^2)} \to \frac{k[x, y]}{(xy - 1)} \qquad u \mapsto \frac{x + y}{2}$$

and extend k-linearly (note that $v \mapsto \left(\frac{x+y}{2}\right)^2$). As the left hand side is a free k-algebra up to the quotient relation, let's insure the relation is preserved, and indeed:

$$\varphi(0) = \varphi(v - u^2) = \frac{(x + y)^2}{2} - \frac{(x + y)^2}{2} = 0$$

showing that the relation is preserved, and hence φ is well-defined. The associated map on the varieties is given by:

$$(a, 1/a) \mapsto \left(\frac{a + 1/a}{2}, \left(\frac{a + 1/a}{2}\right)^2\right)$$

Note the image is 2-to-1. If $k = \mathbb{R}$, the image is *not* surjective (ex. nothing maps to 0). However if $i \in k$ (where $i^2 = -1$), then $i + 1/i = i - i = 0$, which shows the map is surjective and 2-to-1. This demonstrates part of the importance of working over an algebraically closed field.

Theorem 2.1.5: Regular Map Algebraic Correspondence

Let $V \subseteq \mathbb{A}^n$ and $W \subseteq \mathbb{A}^m$. then there is a bijective correpspondence between the morphisms between V and W and the associated k-algebra homomorphisms, more precisely

$$\{\text{regular functions from } V \text{ to } W\} \leftrightarrow \{k\text{-algebra homomorphisms from } k[W] \text{ to } k[V]\}$$

$$\begin{Bmatrix} \text{regular functions} \\ \varphi : V \to W \end{Bmatrix} \underset{\psi^a}{\overset{f\circ\varphi}{\rightleftarrows}} \begin{Bmatrix} k\text{-algebra homomorphism} \\ \varphi^\# : k[W] \to k[V] \end{Bmatrix}$$

More generally, if k-**Aff** is the category whose objects are algebraic sets and morphism are regular functions and k-$\mathbf{Alg}_0$ is the category whose objects are coordinate rings (reduced, commutative, finite-type k-algebras) and morphisms are k-homomorphisms, then $k\text{-}\mathbf{Aff}^{\mathrm{op}} \cong k\text{-}\mathbf{Alg}_0$ (the equivalence of categories requires k algebraically closed, so that every such algebra is realized as a coordinate ring by proposition 2.5.5).

***Proof*:**
The two constructions are already in hand: lemma 1.4.2 sends a regular map φ to the k-algebra homomorphism $\varphi^\#(f) = f \circ \varphi$, and lemma 1.4.3 sends a k-algebra homomorphism back to a regular map. It remains to see they are mutually inverse.

Given $\varphi\colon V \to W$, write ψ for the regular map induced by $\varphi^\#$. Let $y_1, \dots, y_m$ be the coordinate functions on W. By construction the ith component of ψ is $\varphi^\#(y_i) = y_i \circ \varphi = \varphi_i$, the ith component of φ; as the components agree, $\psi = \varphi$.

Conversely let $\theta\colon k[W] \to k[V]$ be a k-algebra homomorphism, φ the regular map it induces, and $\varphi^\#$ the homomorphism φ induces in turn. Both θ and $\varphi^\#$ are k-algebra maps out of $k[W]$, which is generated over k by the $\overline{y_i}$, and both send $\overline{y_i}$ to $\theta(\overline{y_i})$; a k-algebra homomorphism is determined by its values on generators, so $\varphi^\# = \theta$.

For the categorical statement, $\varphi \mapsto \varphi^\#$ is contravariant and functorial, since $(\varphi \circ \psi)^\# = \psi^\# \circ \varphi^\#$ and $\mathrm{id}^\# = \mathrm{id}$. The bijection just established makes it fully faithful, and proposition 2.5.5 makes it essentially surjective onto k-$\mathbf{Alg}_0$ when k is algebraically closed, completing the proof.

Thus, if $f : X \to Y$, then $(f^\#)^a = f$ and if $\varphi : k[Y] \to k[X]$, then $(\varphi^a)^\# = \varphi$. Similarly to the dual theory provided in [11, Dual Spaces and Forms], the dual map gives information about our original map:

Proposition 2.1.6: Regular Map Properties

Let $\varphi : X \to Y$ be a regular map with $\varphi^{\#} : k[Y] \to k[X]$ the associated k-algebra homomorphism. Then:

1. The kernel of $\tilde{\varphi}$ is $\mathcal{I}(\varphi(X))$: $\ker \tilde{\varphi} = \mathcal{I}(\varphi(X))$. Furthermore, the Zariski closure of $\varphi(X)$ is $\mathcal{Z}(\ker(\tilde{\varphi}))$: $\overline{\varphi(X)} = \mathcal{Z}(\ker(\tilde{\varphi}))$.

2. As a consequence, the homomorphism $\tilde{\varphi}$ is injective if and only if $\varphi(X)$ is Zariski dense in Y, or equivalently if $\overline{f(X)} = Y$

3. $\varphi(v) = w$ if and only if $(\varphi^{\#})^{-1}(\mathcal{I}(v)) = \mathcal{I}(w)$, where $\mathcal{I}(v) \subseteq k[X]$ and $\mathcal{I}(w) \subseteq k[Y]$ are the maximal ideals of the two points

***Proof*:**

1. A function $g \in k[Y]$ satisfies $\varphi^{\#}(g) = g \circ \varphi = 0$ exactly when g vanishes at every point of $\varphi(X)$, which is to say $g \in \mathcal{I}(\varphi(X))$. Hence $\ker \varphi^{\#} = \mathcal{I}(\varphi(X))$, and applying $\mathcal{Z}$ gives $\mathcal{Z}(\ker \varphi^{\#}) = \mathcal{Z}(\mathcal{I}(\varphi(X))) = \overline{\varphi(X)}$ by corollary 1.3.3.

2. By part 1, $\varphi^{\#}$ is injective if and only if $\mathcal{I}(\varphi(X)) = 0$ in $k[Y]$, and $\mathcal{Z}(0) = Y$, so this says $\overline{\varphi(X)} = Y$: the image is dense.

3. Suppose $\varphi(v) = w$. For $g \in k[Y]$, $\varphi^{\#}(g)(v) = g(\varphi(v)) = g(w)$, so $\varphi^{\#}(g) \in \mathcal{I}(v)$ if and only if $g \in \mathcal{I}(w)$, which is exactly $(\varphi^{\#})^{-1}(\mathcal{I}(v)) = \mathcal{I}(w)$. Conversely, suppose that identity holds. Both $\mathcal{I}(v)$ and $\mathcal{I}(w)$ are maximal (theorem 2.4.3), and writing $w' = \varphi(v)$ the forward direction gives $(\varphi^{\#})^{-1}(\mathcal{I}(v)) = \mathcal{I}(w')$. Hence $\mathcal{I}(w) = \mathcal{I}(w')$, and since $\mathcal{I}$ is injective on points, $w = w' = \varphi(v)$, completing the proof.

Exercise 2.1.1

1. Show that $k[X \times Y] \cong k[X] \otimes_k k[Y]$ for algebraic sets X, Y, and deduce another proof that $\mathbb{A}^n_k \times \mathbb{A}^m_k \cong \mathbb{A}^{n+m}_k$ as objects of **Reg**.

2. Show that the product is unique up to isomorphism: if $\varphi\colon X \to X'$ and $\psi\colon Y \to Y'$ are isomorphisms, there is a unique isomorphism $X \times Y \to X' \times Y'$ commuting with the projections.

3. Let $T\colon \mathbb{A}^n_k \to \mathbb{A}^n_k$ be an affine isomorphism with $T(p) = q$. Show that $T^{\#}\colon \mathcal{O}_q(\mathbb{A}^n_k) \to \mathcal{O}_p(\mathbb{A}^n_k)$ is an isomorphism, and that for a variety V it induces $\mathcal{O}_q(V) \to \mathcal{O}_p(V^T)$. This is the precise sense in which affine space has no preferred origin.

4. Sharpen theorem 2.1.5: show that a regular map $\varphi\colon X \to Y$ is dominant if and only if $\varphi^{\#}$ is injective, and that φ is a closed immersion, meaning an isomorphism onto a closed subset, if and only if $\varphi^{\#}$ is surjective.

2.2 Radical Ideals

As we've seen in proposition 1.2.5, given some polynomial f or collection of polynomials $S \subseteq k[x_1, ..., x_n]$, the set of zero's of these polynomials does not change if we take higher powers of the polynomials (ex. $f, f^2, f^3, ...$ all produce the same zero set). Thus, studying the set $\mathcal{Z}(I^n)$ is equivalent to studying $\mathcal{Z}(I)$. This property inspires the following definition:

Definition 2.2.1: Radical Ideal

Let $I \subseteq R$ be an ideal of a commutative ring R. Then:

1. The *radical of* I in R is the collection elements for which some power of them lie in I, that is:
$$\sqrt{I} = \text{rad}(I) := \{a \in R \mid a^k \in I \text{ for some } k \in \mathbb{N}\}$$
here $\text{rad}(I)$ stands for the radical of I. It is often written as $\sqrt{I}$.
2. The radical of the zero ideal ($\text{rad}(0) = \sqrt{(0)}$) is called the *nilradical* of R
3. An ideal I is called a *radical ideal* if $I = \text{rad}(I)$ ($I = \sqrt{I}$)

Note that the nilradical by definition contains all the nilpotent elements of R, and that $\text{rad}(\text{rad}(I)) = \text{rad}(I)$.

Proposition 2.2.2: $\text{rad}(I)$ Is An Ideal

Let $I \subseteq R$ be an ideal of a commutative ring. Then $\text{rad}(I)$ is indeed an ideal (containing I). Furthermore, $\text{rad}(I)/I$ is the nilradical of R/I.

The last statement shows us that R/I has no nilpotent elements if and only if $I = \text{rad}(I)$.

***Proof*:**
See [5].

A type of ideal in which we've seen a similar property is the prime ideal, in which $ab \in I$ if and only if $a \in I$ or $b \in I$. One important way of looking at radical is in how they correspond to prime ideals. The prime ideals in a sense "generate" the radical ideals:

Proposition 2.2.3: Radical Through Prime Ideals

Let $I \subsetneq R$ be a proper ideal of a commutative ring and $\text{rad}(I)$ the radical ideal of I. Then $\text{rad}(I)$ is the intersection of all prime ideals containing I. In particular, the nilradical is the intersection of all prime ideals of R.

***Proof*:**
This is the description of the radical as an intersection of primes; see [5].

Corollary 2.2.4: Prime Or Maximal Then Radical

Let $I \subseteq R$ be a prime ideal. Then I is a radical ideal

***Proof*:**
A prime ideal P satisfies $P = \sqrt{P}$; see [5].

Example 2.2.5: Radical Ideals

1. Let R be a UFD. Given an ideal, finds its radical.
2. The ideal $(x^3 - y^2)$ is prime in $k[x, y]$, and hence radical
3. If $l_1, l_2, ..., l_m$ is a collection of linear polynomials in $k[x_1, ..., x_n]$, then $I = (l_1, l_2, ..., l_m)$ is either $k[x_1, x_2, ..., x_n]$ or a prime ideal, and hence radical.

In the context of this chapter, since we are working over polynomials rings, we are working over Noetherian rings. Due to this, the elements of a radical ideal $\text{rad}(I)$ can be related back to elements of I quite naturally:

Proposition 2.2.6: Radical Ideals In Noetherian Rings

Let R be a noetherian ring and $I \subseteq R$ an ideal of R. Then there exists a $k \in \mathbb{N}$ such that $I^k \subseteq \text{rad}(I)$. In particular, if N is the nilradical ideal of R, then $N^k = 0$ for some $k \in \mathbb{N}$ (i.e. N is a nilpotent ideal)

***Proof*:**
See [5].

Now, returning to the fact that $\mathcal{Z}(I) = \mathcal{Z}(I^k)$, we now have that $\mathcal{Z}(I) = \mathcal{Z}(\sqrt{I})$. Remember how we were trying to characterize the ideals for which $\mathcal{Z}$ and $\mathcal{I}$ are inverses, i.e. what are the ideals $\mathcal{I}(\mathcal{Z}(I))$? We can now state one direction of this:

Proposition 2.2.7: Radical and Algebraic Sets

Let $I \subseteq R$ be an ideal where R is a polynomial ring. Then

$$\sqrt{I} \subseteq \mathcal{I}(\mathcal{Z}(I))$$

***Proof*:**
Let $f \in \sqrt{I}$, so $f^m \in I$ for some $m \geq 1$, and let $x \in \mathcal{Z}(I)$ be arbitrary. Every element of I vanishes at x, so in particular

$$f(x)^m = f^m(x) = 0$$

Since k is a field it has no nonzero nilpotents, so $f(x) = 0$. As $x \in \mathcal{Z}(I)$ was arbitrary, f vanishes on $\mathcal{Z}(I)$, which is to say $f \in \mathcal{I}(\mathcal{Z}(I))$, as we sought to show.

It would be great if we could get $\sqrt{I} = \mathcal{I}(\mathcal{Z}(I))$, however this is not true in general: consider $I = (x^2+1) \subseteq \mathbb{R}[x]$. Since $x^2 = 1$ is irreducible, it is a maximal ideal, and so $I = \sqrt{I}$. Then as we are over the reals, $\mathcal{Z}(I) = \emptyset$, so $\mathcal{I}(\mathcal{Z}(I)) = \mathbb{R}[x]$ But $I \not\supseteq \mathbb{R}[x]$, showing equality doesn't hold. The blocker here was that x^2+1 doesn't factor in $\mathbb{R}[x]$: if we were working over $\mathbb{C}$, then we would get that $(x^2+1) = ((x+i)(x-i))$, telling us that $\mathcal{Z}(I) = \{i, -i\}$, which by the basic properties of $\mathcal{I}$ we have that $\mathcal{I}(\mathcal{Z}(I)) = ((x-i)(x+i)) = (x^2+1) = I$, showing us in this case that $I = \sqrt{I}$. In fact, the underlying field k being algebraically closed is the *only* other condition we need for equality to hold; this will be theorem 2.4.4 (Hilbert's Nullstellensatz). It is tricky to show this, since it is in a sense a very simple condition: All we need to characterize the zero sets of the polynomials (over a closed field) $\mathcal{Z}(I)$ in terms of an ideal is to take the radical of I.

2.3 Irreducibility and the Decomposition Theorem

We now return to the earlier fact that $\mathcal{Z}(I) = \mathcal{Z}(f_1, ..., f_n) = \mathcal{Z}(f_1) \cap \cdots \cap \mathcal{Z}(f_n)$. This shows that we can decompose algebraic sets into smaller components. We will therefore take a moment to see if we can find "decomposition" properties of algebraic sets, and see if there is a corresponding property the coordinate ideals will poses

Definition 2.3.1: [Affine] Variety

let $V \subseteq \mathbb{A}^n$ be an algebraic set. The set V is reducible if for $V_1, V_2 \subseteq V$ where $V_1, V_2 \neq V$

$$V = V_1 \cup V_2$$

If there does not exist such sets, we say that V is *irreducible*, or more commonly an *algebraic variety*

2.3.2 Remark Note that this notion of irreducibility is in fact a topological notion, we simply choose the Zariski topology. In particular, X is irreducible if $X = X_1 \cup X_2$ for some closed subsets X_1 and X_2, then either $X = X_1$ or $X = X_2$. For those who are curious, it may be shown that irreducibility implies connected. Furthermore, the condition translates to open sets to say that if

$$\emptyset = U_1 \cap U_2$$

where U_1, U_2 open, then either $U_1 = \emptyset$ or $U_2 = \emptyset$. It follows that it is equivalent to say that every open set is *dense* in an irreducible space. This contrasts heavily with any Hausdorff space, since by definition a Hausdorff space *must* have disjoint open subsets if the space is not empty or a singleton.

The name *algebraic variety* comes from HERE. Note that this condition automatically implies that V_1 and V_2 are non-empty (do you see why?). If V is an irreducible algebraic set, then the corresponding ideal has a very nice property:

Proposition 2.3.3: Coordinate Ideal of Algebraic Variety

A set $V \subseteq \mathbb{A}^n$ is an algebraic variety if and only if $I(V)$ is a prime ideal

***Proof*:**

($\Rightarrow$) Taking the contrapositive, let's assume $I(V)$ is not prime. Then there exists $a, b \in k[x_1, .., x_n]$ such that $ab \in I(V)$ but $a \notin I(V)$ and $b \notin I(V)$. But then:

$$V = (V \cap \mathcal{Z}(a)) \cup (V \cap \mathcal{Z}(b))$$

with $V \cap V(a) \subsetneq V$, $V \cap V(b) \subsetneq V$, showing that V is in fact reducible

($\Leftarrow$) Doing the contrapositive again, let's assume V is reducible, so $V = V_1 \cup V_2$, where both $V_i \subsetneq V$. Then by the contra-positive correspondence of inclusion between algebraic sets and ideals, $I(V) \subsetneq I(V_i)$, $i \in \{1, 2\}$. Since the inclusion is proper, choose $a \in I(V_1)$ and $b \in I(V_b)$ such that $a, b \notin I(V)$. Then since a and b are polynomials which are zero on their respective algebraic sets and $V = V_1 \cup V_2$, $ab \in I(V)$. However, $a, b \notin I(V)$, showing that $I(V)$ is not prime

Corollary 2.3.4: Equivalent characterization

Let $V \subseteq \mathbb{A}^n$ be an algebraic set. Then V is an algebraic variety if and only if $k[V]$ is an integral domain

***Proof*:**
By definition 2.1.1, $k[V] = \Gamma(\mathbb{A}^n)/\mathcal{I}(V)$, and a quotient of a commutative ring is an integral domain exactly when the ideal quotiented out is prime. So $k[V]$ is a domain if and only if $\mathcal{I}(V)$ is prime, which by proposition 2.3.3 holds if and only if V is irreducible, as we sought to show.

This means there is a certain correspondence between being a prime and being irreducible as a variety. To deepen the conection, let's prove that every algebraic set can be decomposed into algebraic varieties:

Theorem 2.3.5: Irreducible Decomposition

Let $V \subseteq \mathbb{A}^n$ be an algebraic set. Then there exist algebraic varieties $V_1, ..., V_r$, unique up to reordering, such that

$$V = \bigcup_{i=1}^{r} V_i \qquad V_i \not\subseteq V_j \quad i \neq j$$

***Proof*:**
Existence. Since $\Gamma(\mathbb{A}^n)$ is Noetherian, every nonempty collection of ideals has a maximal element, and $\mathcal{I}$ reverses inclusion, so every nonempty collection of algebraic sets has a *minimal* element. Let

$$S = \{W \subseteq \mathbb{A}^n \text{ algebraic} \mid W \text{ is not a finite union of algebraic varieties}\}$$

and suppose for contradiction that $S \neq \emptyset$. Choose $W \in S$ minimal. Then W is not itself a variety, so $W = W_a \cup W_b$ with both $W_a, W_b \subsetneq W$. By minimality neither lies in S, so each is a finite union of varieties, say

$$W_a = V_{1,1} \cup \cdots \cup V_{1,n_1} \qquad W_b = V_{2,1} \cup \cdots \cup V_{2,n_2}$$

and then $W = \bigcup_{i,j} V_{i,j}$ is one too, contradicting $W \in S$. Hence $S = \emptyset$.

Given any finite decomposition, discard every $V_{i,j}$ contained in another; what remains still has union V and now satisfies $V_i \not\subseteq V_j$ for $i \neq j$.

Uniqueness. Suppose $V = V_1 \cup \cdots \cup V_r = W_1 \cup \cdots \cup W_m$ are two such decompositions. Fix i. Then

$$V_i = V_i \cap V = \bigcup_j (W_j \cap V_i)$$

expresses the irreducible set V_i as a finite union of closed subsets, so by irreducibility $V_i = W_j \cap V_i$ for some j, that is $V_i \subseteq W_j$. Running the same argument on W_j against the first decomposition gives $W_j \subseteq V_k$ for some k, whence $V_i \subseteq V_k$. The decomposition has no containments among distinct members, so $i = k$, and therefore $V_i = W_j$. This assigns to each i a j with $V_i = W_j$, and symmetrically to each j an i; the two assignments are mutually inverse, so $r = m$ and the two families agree up to reordering, completing the proof.

Definition 2.3.6: Irreducible Decomposition

Let V be an algebraic set. Then the *irreducible decomposition of* V is

$$V = V_1 \cup \cdots V_n \quad V_i \not\subseteq V_j i \neq j$$

for appropriate algebraic varieties V_i, $1 \leq i \leq n$

2.3.7 Remark In a more general topological setting, it can be shown that if X is a Noetherian space (i.e. satisfies A.C.C. on open sets, so D.C.C. on closed sets), X can be decomposed into irreducible sets just liked was done in theorem 2.3.5. It then follows easily that any closed subset of X has this property too.

Example 2.3.8: Affine Varieties

1. All linear polynomials produce affine varieties. Thus, the study of linear algebra can be thought as the study of varieties with multiplicity 1.
2. $\mathcal{Z}(xy - 1) \subseteq \mathbb{A}^2_{\mathbb{R}}$ is an affine variety, as $\mathbb{R}[x, y]/(xy - 1) \cong \mathbb{R}[x, 1/x]$. Note that $xy - 1$ has two branches, however the variety is still irreducible and connected in the Zariski topology. Show that there is no variety that is only one branch of $xy - 1$ (show that if two affine varieties agree on an open subset, they agree everywhere).

The decomposition of a Noetherian variety into irreducible components has an algebraic mirror: the *primary decomposition* of ideals. The minimal primes over $I(X)$ are precisely the ideals of the irreducible components of X, while embedded primes record finer, scheme-theoretic information carried by the coordinate ring but invisible to the underlying set. The full theory (primary ideals, the Lasker-Noether existence and uniqueness theorems, and the associated primes of a module) is developed in [5]; we use it freely when passing between a variety and its coordinate ring.

Exercise 2.3.1

1. Show that an algebraic set is irreducible if and only if any two nonempty open subsets of it meet, and use this to give a second proof that $\mathbb{A}_k^n$ is irreducible.
2. Show that the intersection of two irreducible algebraic sets need not be irreducible. Hint: take $\mathcal{Z}(z)$ and $\mathcal{Z}(xy - z)$ in $\mathbb{A}_k^3$.
3. Show that the closure of an irreducible subset is irreducible, and that a nonempty open subset of an irreducible set is irreducible and dense.
4. Show that a Noetherian topological space is quasicompact, and that every subspace of one is Noetherian.
5. Let X be an algebraic set with irreducible decomposition $X = X_1 \cup \cdots \cup X_r$ as in theorem 2.3.5. Show that the X_i are exactly the maximal irreducible closed subsets of X, which is why the decomposition is unique.

2.4 Hilbert's Nullstellensatz

In this section, we shall make precise the correspondence between which ideals of the coordinate ring of an affine set correspond to closed subsets of the affine set. In this exploration, we shall also produce a new tower condition on ring extensions via Noether's Normalization theorem which builds upon the theme of separating an extension to "property" and "not property" (ex. separable and inseparable, algebraic and transcendental[2]) Recall if $a_1, a_2, ..., a_n \in S$ for a k-algebra S, these elements are called *algebraically independent* if

$$k[x_1, x_2, ..., x_n] \to k[a_1, a_2, ..., a_n] \qquad x_i \mapsto a_i$$

is an isomorphism. Noether's Normalization theorem is the generalization of the idea that any field extension can be split into a transcendental extension, followed by an algebraic extension:

Theorem 2.4.1: Noether's Normalization Theorem

Let k be a field and suppose that $A = k[r_1, ..., r_n]$ is a finitely generated k-algebra. Then for some q, $0 \leq q \leq n$, there are algebraically independent elements $y_1, y_2, ..., y_q \in A$ such that A is integral over $k[y_1, y_2, ..., y_q]$

***Proof*:**
Induct on the number n of generators. If $n = 0$ then $A = k$ and there is nothing to prove, so let $n \geq 1$ and assume the theorem for k-algebras on fewer than n generators.

Step 1: the independent case. If $r_1, \ldots, r_n$ are algebraically independent then A is integral over itself and we may take $q = n$ and $y_i = r_i$. So assume they are algebraically dependent: there is a nonzero $f \in k[x_1, \ldots, x_n]$ with $f(r_1, \ldots, r_n) = 0$. Write d for the total degree of f, where a monomial $ax_1^{m_1} \cdots x_n^{m_n}$ has degree $m_1 + \cdots + m_n$ and $\deg f$ is the largest degree occurring.

Step 2: a change of variables making f monic in x_n. Set $\alpha_i = (1 + d)^i$ for $1 \leq i \leq n - 1$, introduce new variables $X_i = x_i - x_n^{\alpha_i}$, and define

$$g(X_1, \ldots, X_{n-1}, x_n) = f(X_1 + x_n^{\alpha_1}, \ldots, X_{n-1} + x_n^{\alpha_{n-1}}, x_n)$$

[2] Another such example will appear in the next chapter, and shall be ramified and unramified

so $g \in k[X_1, \ldots, X_{n-1}, x_n]$. A monomial $ax_1^{m_1} \cdots x_n^{m_n}$ of f contributes to g a single term in which the X_i do not appear, namely $a\, x_n^e$ with

$$e = m_n + m_1\alpha_1 + \cdots + m_{n-1}\alpha_{n-1} = m_n + m_1(1+d) + \cdots + m_{n-1}(1+d)^{n-1}$$

every other term carries at least one X_i. Since each $m_i \leq d < 1+d$, the tuple $(m_n, m_1, \ldots, m_{n-1})$ is the base-$(1+d)$ expansion of e, so distinct monomials of f give distinct e. Hence the largest such e, call it N, is attained exactly once, and

$$g = cx_n^N + \sum_{i=0}^{N-1} h_i(X_1, \ldots, X_{n-1})x_n^i$$

for some nonzero $c \in k$: after dividing by c, g is monic in x_n over $k[X_1, \ldots, X_{n-1}]$.

Step 3: descending one generator. Put $s_i = r_i - r_n^{\alpha_i}$ and $B = k[s_1, \ldots, s_{n-1}] \subseteq A$. Substituting $X_i \mapsto s_i$ and $x_n \mapsto r_n$ in g recovers $f(r_1, \ldots, r_n) = 0$, so

$$\frac{1}{c} g(s_1, \ldots, s_{n-1}, r_n) = 0$$

exhibits r_n as a root of a monic polynomial over B; thus r_n is integral over B. Each remaining $r_i = s_i + r_n^{\alpha_i}$ lies in $B[r_n]$, so $A = B[r_n]$, and A is integral over B because it is generated over B by the single integral element r_n.

Step 4: induction. B is a k-algebra generated by the $n-1$ elements $s_1, \ldots, s_{n-1}$, so by the inductive hypothesis there are algebraically independent $y_1, \ldots, y_q \in B$ with B integral over $k[y_1, \ldots, y_q]$. Integrality is transitive, so A is integral over $k[y_1, \ldots, y_q]$, and $q \leq n-1 \leq n$, as we sought to show.

Noether's Normalization can be thought of as a very geometric result: we shall see that any finitely generated k-algebra shall be an n-dimensional affine set along with some extra important information given by the integral extension, more on this in chapter 4. One very important is that if we have an extension F/k that is a finite-type k-algebra, is finitely generated if we allow for both addition and multiplication, it is finitely gendered by *just* allowing addition!!

Theorem 2.4.2: Zariski's Lemma

Let F/k be a field extension, and assume F is a finite-type k-algebra. Then F/k is a finite (and hence algebraic) extension.

***Proof*:**
Since F is a finite-type k-algebra, $F = k[r_1, r_2, ..., r_n]$. By Noether's Normalization Theorem, there exists a $q \in \mathbb{N}$ (possibly 0) enumerating variables $y_1, ..., y_q \in F$ such that F is integral over $k[y_1, y_2, ..., y_q]$, i.e. F is a finitely generated $k[y_1, ..., y_q]$-module. We can re-phrase this by saying we have a finite integral extension $F/k[y_1, y_2, ..., y_q]$. By [5, Integral Extension over Fields], since F is an integral domain, F is a field if and only if $k[y_1, y_2, ..., y_q]$ is a field. But then, $k[y_1, y_2, ..., y_q]$ is a field if and only if there are no indeterminate variables, i.e. $q = 0$. Thus, we get the finite integral extension F/k, which is equivalent to saying we get a finite field extension, as we sought to show.

The last two theorem's are true no matter what field we are working over. The following theorem requires k to be algebraically closed.

Theorem 2.4.3: Hilbert's Nullstellensatz - Weak Version

Let k be an algebraically closed field. Then M is a maximal ideal of the polynomial ring $k[x_1, ..., x_n]$ if and only if

$$M = (x_1 - a_1, x_2 - a_2, ..., x_n - a_n)$$

for any $a_1, a_2, ..., a_n \in k$. As a result, the maps:

$$\{\text{Points in } \mathbb{A}^n\} \underset{\mathcal{Z}}{\overset{\mathcal{I}}{\rightleftarrows}} \{\text{Maximal ideals in } \Gamma(\mathbb{A}^n)\}$$

are in bijective correspondence, and if $I \subseteq \Gamma(\mathbb{A}^n)$ is a proper ideal, then $\mathcal{Z}(I) \neq \emptyset$

***Proof*:**
Certainly, $(x_1 - a_1, ..., x_n - a_n)$ is a maximal ideal in $k[x_1, x_2, ..., x_n]$ since its quotient is k. Conversely, suppose M is a maximal ideal in $k[x_1, x_2, ..., x_n]$. Consider $F = k[x_1, x_2, ..., x_n]/M$. Then since $k \subseteq F$, F/k is a field extension that is also a finitely generated k-algebra (generated by $\overline{x_1}, ...\overline{x_n}$). By Zariski's lemma, F/k is a finite field extension, and hence is algebraic over k.

Now, we take advantage of the fact that k is algebraically closed, meaning there does not exist a non-trivial algebraic extension of k ([11, Algebraic Closure Equivalences]), implying that $F \cong_k k$, in particular we $\overline{x_i} \mapsto a_i$ for some $a_i \in k$. Thus, this is a k-algebra homomorphism, and $\overline{a_i} \in F$, $\overline{x_i - a_i} \mapsto a_i - a_i = 0$, which since the to fields are isomorphic, they are in bijective correspondence, implying $\overline{x_i - a_i} = \overline{0}$, i.e. $x_i - a_i \in M$. Thus, we must have $x_i - a_i \in M$ for every x_i, implying that $(x_1 - a_1, ..., x_n - a_n) \subseteq M$, which by maximality shows that $(x_1 - a_1, ..., x_n - a_n) = M$.

As a consequence, if I is any ideal of $k[x_1, x_2, ..., x_n]$, then $I \subseteq M$ for some maximal ideal M. By what we've just shown, for some $a_1, ..., a_n$, $M = (x_1 - a_1, ..., x_n - a_n)$. Thus, $(a_1, ..., a_n) \in \mathcal{Z}(I)$, showing that $\mathcal{Z}(I)$ is non-empty.

Theorem 2.4.4: Hilbert's Nullstellensatz

Let k be an algebraically closed field. Then for any ideal $I \subseteq k[x_1, x_2, ..., x_n]$,

$$\mathcal{I}(\mathcal{Z}(I)) = \sqrt{I} = \text{rad}(I)$$

Furthermore, the maps $\mathcal{I}$ and $\mathcal{Z}$ are order-reversing bijections:

$$\left\{\begin{matrix}\text{Algebraic Sets} \\ (= \text{closed sets})\end{matrix}\right\} \underset{\mathcal{Z}}{\overset{\mathcal{I}}{\rightleftarrows}} \{\text{Radical Ideals}\}$$

Under this bijection the *irreducible* algebraic sets, that is the affine varieties, correspond to the *prime* ideals, and the points to the maximal ideals.

Example 2.4.5: The Nullstellensatz in Practice

1. *Radicals are unavoidable.* In $k[x]$ take $I = (x^2)$. Then $\mathcal{Z}(I) = \{0\}$ and $\mathcal{I}(\{0\}) = (x)$, so

$$\mathcal{I}(\mathcal{Z}(I)) = (x) = \sqrt{(x^2)} \neq (x^2)$$

The theorem recovers $\sqrt{I}$ and nothing finer: passing to zero sets forgets multiplicity, which is the observation chapter 12 has to work around.

2. *A two-variable case.* Let $I = (x^2, xy) \subseteq k[x, y]$. Then $\mathcal{Z}(I) = \mathcal{Z}(x)$, the y-axis, so $\mathcal{I}(\mathcal{Z}(I)) = (x)$. Directly, $\sqrt{I} = (x)$ as well: $x^2 \in I$ gives $x \in \sqrt{I}$, and conversely every element of I is divisible by x.

3. *Algebraic closure is essential.* Over $\mathbb{R}$ take $I = (x^2 + 1) \subseteq \mathbb{R}[x]$. Then $\mathcal{Z}(I) = \emptyset$, so $\mathcal{I}(\mathcal{Z}(I)) = \mathbb{R}[x] = (1)$, while $\sqrt{I} = (x^2 + 1)$ because $(x^2 + 1)$ is maximal. The two differ, and the theorem fails. Over $\mathbb{C}$ the zero set is $\{\pm i\}$ and $\mathcal{I}(\{\pm i\}) = (x^2 + 1) = \sqrt{I}$ as promised.

4. *The bijection at work.* Under theorem 2.4.4 the maximal ideal $(x - a, y - b)$ corresponds to the point (a, b), the prime $(y - x^2)$ to the parabola, and the ideal $(y^2 - x^4) = \big((y - x^2)(y + x^2)\big)$ to the union of two parabolas. That ideal is already radical, being an intersection of the two primes $(y - x^2)$ and $(y + x^2)$, and reducibility of the zero set corresponds to the ideal failing to be prime rather than failing to be radical.

The refinement matters, and mixing the three levels up is the standard way to misread the theorem. An algebraic set is closed and corresponds to a radical ideal; it is a variety exactly when that radical ideal is prime (proposition 2.3.3); and it is a point exactly when the ideal is maximal (theorem 2.4.3). Every prime is radical, so the varieties sit inside the algebraic sets on one side just as the primes sit inside the radicals on the other, and the bijection restricts.

***Proof*:**
First, we've already shown that $\text{rad}(I) \subseteq \mathcal{I}(\mathcal{Z}(I))$ (proposition 2.2.7), so it remains to show that $\text{rad}(I) \supseteq \mathcal{I}(\mathcal{Z}(I))$. By Hilbert's basis theorem, we have that $I = (f_1, f_2, ..., f_n)$. Let $g \in \mathcal{I}(\mathcal{Z}(I))$. If we show that $g^N \in I$ for some $N \in \mathbb{N}_{>0}$, this would imply $g \in \text{rad}(I)$. To that end, we employ the following trick known as the Rabinowitsch trick: introduce a new variable x_{n+1} and consider the ideal:

$$(f_1, ..., f_n, x_{n+1}g - 1) = I' \subseteq k[x_1, ..., x_n, x_{n+1}]$$

By definition of g, at any point where each $f_1, f_2, ..., f_n$ vanish, g also vanishes (since $g \in \mathcal{I}(\mathcal{Z}(I))$). But then, $x_{n+1}g - 1$ is nonzero! Thus, $\mathcal{Z}(I') = \emptyset$ in $\mathbb{A}^{n+1}$, and by Hilbert's weak Nullstellensatz, I' cannot be a proper ideal, and so $1 \in I'$. Since $1 \in I'$, we can write it as:

$$1 = a_1 f_1 + \cdots + a_n f_n + a_{n+1}(x_{n+1}g - 1) \qquad a_i \in k[x_1, ..., x_{n+1}]$$

For the following steps, it is insightful to expand the equation:

$$1 = \sum_i^n a_i(x_1, ..., x_{n+1}) f_1(x_1, ..., x_n) + a_{n+1}(x_1, ..., x_{n+1})(x_{n+1}g(x_1, ..., x_n) - 1)$$

Now, since x_{n+1} is a free variable, this equation still holds true if we substitute x_{n+1} for another expression (it will simply limit the equation to solution which also satisfy the substituted expression). We wil choose $x_{n+1} = 1/g(x_1, ..., x_n)$ so we get the following expression in

$k(x_1, ..., x_n)$:

$$1 = \sum_i^n a_i \left(x_1, ..., x_n, \frac{1}{g(x_1, ... x_n)} \right) f_i(x_1, ..., x_n)$$

Importantly for us, the only terms in the denominator would be some powers of $g(x_1, ..., x_n)$. Thus, for suitably large $N \in \mathbb{N}$, we may re-write the right hand side as:

$$1 = \sum_i^n \frac{b_i\,(x_1, ..., x_n)\, f_i(x_1, ..., x_n)}{g(x_1, ..., x_n)^N}$$

for suitable $b_1, ..., b_n \in k[x_1, ..., x_n]$. Multiplying both sides by the denominator gives

$$g(x_1, ..., x_n)^N = \sum_i^n b_i(x_1, ..., x_n) f_i(x_1, ..., x_n)$$

an identity between polynomials, so it holds in $k[x_1, \ldots, x_n]$ and not just in the fraction field. The right hand side lies in I, so $g^N \in I$ and therefore $g \in \mathrm{rad}(I)$. As g was arbitrary, $\mathcal{I}(\mathcal{Z}(I)) \subseteq \mathrm{rad}(I)$, and with the reverse inclusion already in hand, $\mathcal{I}(\mathcal{Z}(I)) = \mathrm{rad}(I)$.

It remains to see that the two maps are mutually inverse bijections between algebraic sets and radical ideals. If I is radical then $\mathcal{I}(\mathcal{Z}(I)) = \mathrm{rad}(I) = I$, which is one composite. For the other, let A be an algebraic set, say $A = \mathcal{Z}(J)$. Then $\mathcal{Z}(\mathcal{I}(A)) = A$ by proposition 1.2.11(2). Finally $\mathcal{I}(A)$ really is radical: if $f^m \in \mathcal{I}(A)$ then $f(a)^m = 0$ for every $a \in A$, and k is a field, so $f(a) = 0$ and $f \in \mathcal{I}(A)$. Hence the maps are inverse to each other, and both reverse inclusion by proposition 1.2.5 and proposition 1.2.10.

For the refinement, $\mathcal{I}(A)$ is prime if and only if A is irreducible (proposition 2.3.3), and maximal if and only if A is a point (theorem 2.4.3), so the bijection restricts to varieties against primes and to points against maximal ideals, as we sought to show.

Corollary 2.4.6: Nullstellensatz in Algebraic Closure

Let k be any field with algebraic closure $\overline{k}$, and $I \subseteq k[x_1, ..., x_n]$ an ideal. Then:

$$\mathcal{I}_k(\mathcal{Z}_{\overline{k}}(I)) = \sqrt{I}$$

As a particular result, $I = (1)$ if and only if the polynomials in I have no common zero in $\overline{k}^n$.

***Proof*:**
Let $I^e = I\overline{k}[x_1, \ldots, x_n]$ be the extension of I to the algebraic closure. Since $\overline{k}$ is algebraically closed, theorem 2.4.4 applies there and gives $\mathcal{I}_{\overline{k}}(\mathcal{Z}_{\overline{k}}(I)) = \sqrt{I^e}$. Intersecting both sides with $k[x_1, \ldots, x_n]$, the left hand side becomes $\mathcal{I}_k(\mathcal{Z}_{\overline{k}}(I))$ by definition. For the right hand side, $\sqrt{I^e} \cap k[x_1, \ldots, x_n] = \sqrt{I}$: the inclusion $\supseteq$ is immediate, and for $\subseteq$, if $f \in k[x_1, \ldots, x_n]$ has $f^m \in I^e$ then writing f^m as an $\overline{k}$-combination of generators of I and comparing coefficients against a k-basis of $\overline{k}$ exhibits f^m as a k-combination of them, so $f^m \in I$.

For the last claim, $\mathcal{Z}_{\overline{k}}(I) = \emptyset$ gives $\mathcal{I}_k(\emptyset) = k[x_1, \ldots, x_n]$, so $\sqrt{I} = (1)$ and hence $I = (1)$; the converse is immediate, as we sought to show.

2.5 The Dictionary

With these results established, let us recap all the algebraic/geometric correspondences and prove some more. There are some immediate consequence of these result. An immediate easy result is that it can finally be concluded that:

$$\mathcal{I}(A \cap B) = \sqrt{\mathcal{I}(A) + \mathcal{I}(B)}$$

Next, we now have made precise the relation between radical ideals and prime ideals, as we have fully geometrically classified radical and prime ideals. Next, we get the following dictionary between the language of geometry and the language of algebra:

Geometry	Algebra
Affine sets V	coordinate rings $k[V]$
points of V	maximal ideals of $k[V]$
closed subsets of V	radical ideals of $k[V]$
sub-varieties of V	prime ideals of $k[V]$
regular function: $\varphi : V \to W$	k-algebra homomorphism $\tilde{\varphi} : k[W] \to k[V]$

This dictionary is expanded throughout the rest of the book. As a teaser, we shall show that all PID's correspond to smooth curves (see [4, Dedekind domains as smooth curves]).

Now that we have established this definition between algebraic sets and it's algebraic counter-part, we can expand some more on the Zariski Topology:

Proposition 2.5.1: Zariski Topology is Compact

Let $\mathbb{A}^n_k$ have the Zariski topology. Then $\mathbb{A}^n_k$ is compact

***Proof*:**
Let $\mathcal{O}$ index an open cover for $\mathbb{A}^n_k$

$$\mathbb{A}^n_k = \bigcup_{\alpha \in \mathcal{O}} U_\alpha$$

By proposition 1.3.4, any open set is the union of sets of the form $D(f)$, so we may re-write this as:

$$\mathbb{A}^n_k = \bigcup_{\alpha \in \mathcal{O}'} D(f_\alpha)$$

appropriately re-indexing. Then taking the complement, we get that it's equivalent that:

$$\emptyset = \bigcap_{\alpha \in \mathcal{O}'} \mathcal{Z}(f_\alpha) = \mathcal{Z}\left(\sum_{\alpha \in \mathcal{O}'} (f_\alpha)\right)$$

By the Nullstellensatz, we get that it's equivalent that

$$1 \in \sum_{\alpha \in \mathcal{O}'} (f_\alpha)$$

Thus, for appropriate coefficients $g_i \in \Gamma(\mathbb{A}^n)$, not necessarily constants, it is equivalent that:

$$1 = g_1 f_1 + \cdots + g_n f_n \in (f_1, ..., f_n) = \sum_{i=1}^{n} (f_i)$$

But then

$$\emptyset = \bigcap_{i=1}^{n} \mathcal{Z}(f_i) \quad \Leftrightarrow \quad \mathbb{A}_k^n = \bigcup_{i=1}^{n} D(f_i)$$

so finitely many of the $D(f_\alpha)$ already cover $\mathbb{A}_k^n$, as we sought to show.

Recall that for any algebraic subset $V \subseteq \mathbb{A}_k^n$, the corresponding coordinate ring $k[V]$ is used to define the subspace topology. Using the Nullstellensatz, we may deduce that it cannot have nilpotent elements:

Corollary 2.5.2: Coordinate Ring Always Reduced

Let $V \subseteq \mathbb{A}_k^n$. be an algebraic subset and $k[V]$ the associated coordinate ring. Then $k[V]$ is has no nilpotent elements

***Proof*:**
Since $k[V] = k[x_1, x_2, ..., x_n]/I(V)$, and $I(V)$ is a radical ideal by the Nullstellensatz, the quotient cannot contain any nilpotent elements, as we sought to show.

Corollary 2.5.3: Nullstellensatz For Subspaces

Let $V \subseteq \mathbb{A}_k^n$ be an algebraic subset (i.e. a closed subset) with corresponding coordinate ring $k[V]$. Then we can define $\mathcal{Z}$ and $\mathcal{I}$ as:

$$\left\{\begin{matrix}\text{affine algebraic subsets} \\ \text{of } V\end{matrix}\right\} \underset{\mathcal{Z}}{\overset{\mathcal{I}}{\rightleftarrows}} \left\{\begin{matrix}\text{Radical Ideals} \\ \text{of } k[V]\end{matrix}\right\}$$

***Proof*:**
Write $\pi \colon \Gamma(\mathbb{A}_k^n) \to k[V]$ for the quotient by $\mathcal{I}(V)$. The correspondence theorem for rings makes $J \mapsto \pi^{-1}(J)$ a bijection between the ideals of $k[V]$ and the ideals of $\Gamma(\mathbb{A}_k^n)$ containing $\mathcal{I}(V)$, and it preserves radicals in both directions: $\pi^{-1}(\sqrt{J}) = \sqrt{\pi^{-1}(J)}$, since $f^m \in \pi^{-1}(J)$ says exactly $\pi(f)^m \in J$. So the radical ideals of $k[V]$ correspond to the radical ideals of $\Gamma(\mathbb{A}_k^n)$ containing $\mathcal{I}(V)$.

On the geometric side, the algebraic subsets of V are precisely the algebraic subsets of $\mathbb{A}_k^n$ contained in V, and $\mathcal{Z}$ reverses inclusion, so those correspond to the ideals containing $\mathcal{I}(V)$. Composing with theorem 2.4.4, which matches algebraic subsets of $\mathbb{A}_k^n$ against radical ideals of

$\Gamma(\mathbb{A}^n_k)$, gives the stated bijection, as we sought to show.

Next, sticking to the case where $k = \overline{k}$ is algebraically closed, from this section we can deduce that any coordinate ring is finitely generated, nilpotent-free (i.e. reduced) and is a k-algebra over an algebraically closed field. We can show that all such rings appear as coordinate rings for an affine set.

We first have the following:

Definition 2.5.4: Reduced Ring

Let R be a ring. Then R is said to be *reduced* if $x^n = 0$ implies $x = 0$

(finish here)

Proposition 2.5.5: Coordinate Ring Correspondence

Let R be a ring. Then if R is:

1. Finitely generated
2. Reduced
3. over an algebraically closed field

then R is isomorphic to a coordinate ring for an appropriate dimension n of $\mathbb{A}^n_{\overline{k}}$ and $\mathcal{Z}(S) \subseteq \mathbb{A}^n_{\overline{k}}$.

***Proof*:**
Write $R = \overline{k}[t_1, \dots, t_n]$ for a choice of finitely many generators, and let

$$\pi \colon \overline{k}[x_1, \dots, x_n] \to R \qquad x_i \mapsto t_i$$

be the resulting surjection of $\overline{k}$-algebras, with $I = \ker \pi$, so that $R \cong \overline{k}[x_1, \dots, x_n]/I$. Since R is reduced, I is a radical ideal: if $f^m \in I$ then $\pi(f)^m = 0$ in R, so $\pi(f) = 0$ and $f \in I$.

Set $V = \mathcal{Z}(I) \subseteq \mathbb{A}^n_{\overline{k}}$. By theorem 2.4.4 the ground field being algebraically closed gives $\mathcal{I}(V) = \mathcal{I}(\mathcal{Z}(I)) = \sqrt{I} = I$. Therefore

$$\Gamma(V) = \frac{\overline{k}[x_1, \dots, x_n]}{\mathcal{I}(V)} = \frac{\overline{k}[x_1, \dots, x_n]}{I} \cong R$$

so R is the coordinate ring of the algebraic set V, as we sought to show.

Notice that through this result, we can define an affine set without the reference of an ambient space, similar to how we may define a manifold without the need an ambient embedding space. This idea is greatly generalized by scheme theory, which chapter 19 points the reader toward.

Exercise 2.5.1

1. Deduce the strong Nullstellensatz from the weak one by the Rabinowitsch trick: given f vanishing on $\mathcal{Z}(I)$, apply the weak form to the ideal generated by I and $1 - yf$ in one extra variable y.

2. Show that every maximal ideal of $k[x_1, \ldots, x_n]$ is $(x_1 - a_1, \ldots, x_n - a_n)$ for a unique point $(a_1, \ldots, a_n)$, and explain why this fails over $\mathbb{R}$.

3. Show that the Nullstellensatz fails over a field that is not algebraically closed, by computing $\mathcal{I}(\mathcal{Z}(x^2+1))$ in $\mathbb{R}[x]$ and comparing with $\sqrt{(x^2+1)}$.

4. Show that $\mathcal{Z}$ and $\mathcal{I}$ are mutually inverse inclusion-reversing bijections between the radical ideals of $k[x_1, \ldots, x_n]$ and the algebraic subsets of $\mathbb{A}^n_k$, and that under this correspondence prime ideals match irreducible sets.

5. Use corollary 2.4.6 to show that a system of polynomials over $\mathbb{Q}$ with no common zero in $\overline{\mathbb{Q}}$ generates the unit ideal, and contrast this with $x^2 + 1$ over $\mathbb{R}$, which has no real zero yet generates a proper ideal. What exactly does the closure give?

3

Rational Maps and Birational Classification

Isomorphism is a demanding relation between varieties, and the classification problem it poses is largely intractable. Weakening it produces a relation that can be classified: two varieties are birationally equivalent when they agree on a dense open set. Algebraically this replaces the coordinate ring by its field of fractions, and fields of transcendence degree d are far simpler objects than finitely generated reduced algebras.

The price is that the maps involved are no longer defined everywhere. This chapter pays that price carefully: it locates the domain of definition of a rational function, shows the domain is always dense, and then proves that every variety is birationally equivalent to a hypersurface.

3.1 Rational Functions and the Function Field

Let V be a variety and $k = \overline{k}$ be an algebraically closed field. In this section, we take advantage of the fact that we may localize $k[x_1, ..., x_n]/\mathcal{I}(V)$. By Noether's Normalization Theorem (theorem 2.4.1) $k[V]/k$ is a finite transcendental extension followed by an integral extension. As $k[V]$ is an integral domain we may take its field of fractions $k(V)$, and passing to fraction fields in Noether normalization exhibits $k(V)$ as a *finite* extension of the purely transcendental field $k(y_1, \ldots, y_q)$. The extension is genuinely finite and not an equality: for the cuspidal cubic $V = \mathcal{Z}(y^2 - x^3)$ we have $k[V] \cong k[t^2, t^3]$, integral over $k[x]$, while $k(V) = k(t)$ is a degree-two extension of $k(x) = k(t^2)$. Were $k(V)$ always purely transcendental every variety would be rational, which item (4) of example 3.3.8 disproves. What *is* intrinsic is q, the transcendence degree, which chapter 4 identifies with $\dim V$. Geometrically $k(V)$ shall represent all *rational polynomial functions*, or all rational polynomials functions defined on V. Note that we must be a bit more careful in interpreting elements of $k(V)$ as functions: if

$f/g \in k(V)$, then if $g(p) = 0$ for some $p \in V$ then we get the undefined $f(p)/g(p) = f(p)/0$. We shall address this soon.

Rational functions supply a weaker relation than isomorphism for classifying varieties. We shall see that being *birationally equivalent* is weaker than being a (regular) isomorphism, giving coarser isomorphism classes of varieties that shall give us more information about algebraic sets. As an example, elliptic curves[1] shall require three curve invariants to be classified (the genus, the j-invariant, and the Mordell-Weil group of the curve). However, up to birational map it shall only require the genus and we can without loss of generality work with smooth elliptic curves. Though coarser, it is nonetheless a powerful tool in gathering information about varieties.

Definition 3.1.1: Rational Functions

Let V be an affine variety. Then the fraction field of $k[V]$ is called the field of *rational function on* V, and is denoted $k(V)$. An element of $k(V)$ is called a *rational function*.

The *evaluation* of a rational function at p is defined whenever it admits a representative f/g with $g(p) \neq 0$. Two rational functions are equal if they agree at every point where both are defined. The points at which *no* representative has nonvanishing denominator form the *pole set*; being a pole is a property of the function, not of a chosen representative.

If $\varphi \in k(V)$, then $\varphi = \frac{f}{g}$ where $g \notin I(V)$ (or else the denominator would be the zero function, contradicting the construction of the field of fractions). We shall say that two rational polynomial f/g and f'/g' represent the same rational function if and only if $fg' - f'g \in I(V)$. Note that if $k[V]$ is not a UFD, then there can be representatives of $k(V)$ which are not equivalent up to a unit:

Example 3.1.2: Not UFD, Different Polynomial Representatives

Let $V = \mathcal{Z}(xw - yz) \subseteq \mathbb{A}^4_k$. Show that:

$$\frac{\overline{x}}{\overline{y}} = \frac{\overline{z}}{\overline{w}} = f \in k(V)$$

if $y \neq 0$ or $w \neq 0$.

Another way of constructing $k(V)$ is the following: take the collection

$$\mathcal{O}_V = \{\varphi = f/g \in k(x_1, ..., x_n) \mid g \notin I(V)\}$$

then take the subset $\mathfrak{m}_V = \{\varphi = f/g \in \mathcal{O}_V \mid f \in I(V)\}$. This is the unique maximal ideal of the local ring $\mathcal{O}_V$, and

$$k(V) \cong \mathcal{O}_V/\mathfrak{m}_V$$

since a rational function on V is a ratio f/g of polynomials with $g \notin I(V)$, taken up to the relation that identifies two such ratios agreeing on V, and that relation is exactly congruence modulo $\mathfrak{m}_V$. This shall become the dominant way of interpreting the set of rational functions when working with quasi-projective varieties (see section 9.4)[2].

[1] for those familiar with complex geometry, genus 1 curves

[2] In Scheme theory, this shall be the rational functions at the point (closed or non-closed) associated to X

3.2 Domains of Definition and Local Rings

A key difference between regular and rational functions is that a rational function need not be defined at every point. In fact, it *has* to be the case that a rational function is not defined on a finite set of points for it to not be equivalent to a regular function (when working over an algebraically closed field). To see this, consider the following definition:

Definition 3.2.1: Regular Point

Let $\varphi \in k(X)$ be a rational function. Then a *regular point* at $x \in X$ is a point where we can represent φ as f/g, $\varphi(x) = f(x)/g(x)$, and $g(x) \neq 0$. In this case, the element $f(x)/g(x)$ in k is called the *value* of φ at x and is denoted $\varphi(x)$

For example, take $\varphi = 1/x$ on $\mathbb{A}^1_k$. Then all points except 0 are regular points. If $\varphi = f/g$ is regular at a point x, then it is regular in an open neighborhood of X, namely the neighborhood $D(g) \cap X$. Some authors put that a rational function is regular at a point if it is regular in an open neighborhood.

Proposition 3.2.2: Regular at All Points, then Regular

Let $\varphi \in k(X)$ be a rational function that is regular at all points. Then φ is a regular function

***Proof*:**
Let $\varphi \in k(X)$ be a rational function that is regular at every point. Then for each $x \in X$, we can represent φ as f_x/g_x. Take $\mathfrak{a} = (g_x)_{x\in X}$. Then as $k[X]$ is Noetherian, this is a finitely generated ideal:

$$\mathfrak{a} = (g_{x_1}, \ldots, g_{x_n})$$

Note that the g_{x_i} have no common zero in X: at any $x \in X$ the chosen g_x satisfies $g_x(x) \neq 0$, and $g_x \in \mathfrak{a}$. So $\mathcal{Z}(\mathfrak{a}) = \emptyset$ in X, and the weak Nullstellensatz (theorem 2.4.3) applied in $k[X]$ forces

$$\mathfrak{a} = (1)$$

In particular, there exist $u_i \in k[X]$ such that

$$\sum_{i=1}^{n} u_i g_i = 1$$

The key point is that $\varphi = f_i/g_i$ is an identity in the field $k(X)$, so $\varphi g_i = f_i$ holds globally, not just on the neighbourhood where g_i is nonzero. Multiplying the displayed identity by φ therefore gives

$$\varphi = \varphi \cdot 1 = \sum_{i=1}^{n} u_i(\varphi g_i) = \sum_{i=1}^{n} u_i f_i$$

whose right hand side lies in $k[X]$, so φ is regular, completing the proof.

The Nullstellensatz is essential for the proof, i.e. that k is an algebraically closed field, or else this result is false!

Example 3.2.3: Rational Function That is Regular

the function

$$\frac{x}{x+100}$$

is regular on $\mathcal{Z}(x^2+y^2-1) \subseteq \mathbb{A}^2_{\mathbb{R}}$ even though it is not a polynomial: there does not exist a point (x, y) satisfying x^2+y^2-1 such that $x+100=0$ namely since all points x where $x^2+y^2-1=0$ must be within $[-1,1]$[a]

[a]This failure is *not* present when working with the scheme $\operatorname{Spec}\mathbb{R}[x,y]/(x^2+y^2-1)$ as $x+100$ has a zero at the prime ideal $(x-100, y^2+9999)$, and so this "anomaly" is present if we don't work in the right context

Hence we always have points on which a rational function is not defined. This set of points will always be dense:

Lemma 3.2.4: Domain of Definition

Let $\varphi \in k(V)$ be a rational function. Then set of regular points is non-empty, open and dense.

***Proof*:**

Let $\varphi = f/g$ and take $U = \{p \in V \mid g(p) \neq 0\} = D(g) \cap V$. Then in V, this set is certainly open, and as g is not the zero function there is certainly at least one point $p \in V$ such that $g(p) \neq 0$.

To show U is dense, let $O \subseteq V$ be any nonempty open set; it suffices to show $U \cap O \neq \emptyset$. Suppose instead $U \cap O = \emptyset$. Then

$$V = (U \cap O)^c = U^c \cup O^c$$

writes V as a union of two closed sets, and both are *proper*: $U^c \neq V$ because $U \neq \emptyset$, and $O^c \neq V$ because $O \neq \emptyset$. That contradicts the irreducibility of V. Hence U meets every nonempty open set, which is density, completing the proof.

Definition 3.2.5: Domain of Definition

Let $\varphi \in k(V)$ a rational function. Then the set of regular point is called the *domain of definition.*

Corollary 3.2.6: Collection of Rational Points

Let $\varphi_1, \ldots, \varphi_n \in k(X)$ be rational functions on a variety X. Then their common domain of definition is open, non-empty, and dense.

***Proof*:**

Let U_i be the domain of definition of φ_i, which by lemma 3.2.4 is open, nonempty and dense. A finite intersection of open sets is open, so $\bigcap_i U_i$ is open.

For nonemptiness and density it is enough to show that $\bigcap_i U_i$ meets every nonempty open $O \subseteq X$, and by induction it suffices to treat $n=2$. Given nonempty open O, density of U_1 gives $U_1 \cap O \neq \emptyset$, and $U_1 \cap O$ is open, so density of U_2 gives $U_2 \cap (U_1 \cap O) \neq \emptyset$. Hence $U_1 \cap U_2$ meets O. Taking $O = X$ gives nonemptiness, as we sought to show.

In fact, as the open sets in the Zariski topology are "huge", it suffices to define a rational mapping on any open subset:

Proposition 3.2.7: Suffice To Define On Open Set

Let $\varphi \in k(X)$ be a rational function. Then if φ is specified on some non-empty open subset $U \subseteq X$, then there is a unique extension of φ to all of X.

***Proof*:**
Take $\varphi \in k(X)$. If $\varphi = 0$, there is nothing to prove, so assume $\varphi \neq 0$. Say we specify what φ is on an open non-empty subset $U \subseteq X$, call this function $\varphi|_U$. Then let's say that φ' is a rational function where φ' on U is equal to $\varphi|_U$ (where a-priori it is unknown where there is a unique extension, and hence it may be another function). Consider $\varphi - \varphi' = \varphi''$. On U, this function is identically 0. Take the representative $\varphi'' = f/g$ on U so that $f = 0$. Then take $X_1 \subseteq X$ given by $f = 0$. This is not all of X as $\varphi \neq 0$, however we then have $X_1 \cup (X \setminus U) = X$ where both are proper subsets, contradicting irreducibility of X, completing the proof.

We are often interested in all the functions that are defined at a particular point, namely since geometrically much of the information of an object is given by collections of functions at every point (the reader familiar with differential geometry should be thinking of germs). To that end, we box the following

Definition 3.2.8: Stalk At A Point

Let V be an affine variety and $p \in V$. Then $\mathcal{O}_p(V) \subseteq k(V)$ is the collection of rational functions:

$$\mathcal{O}_p(V) = \{f/g \in k(V) \mid g(p) \neq 0\}$$

Proposition 3.2.9: Regular Functions and Stalk

$$k[V] = \bigcap_{p \in V} \mathcal{O}_p(V)$$

***Proof*:**
Certainly $k[V] \subseteq \mathcal{O}_p(V)$ for all $p \in V$ so $k[V] \subseteq \bigcap_{p\in V} \mathcal{O}_p(V)$.

For the converse, fix $\varphi \in \bigcap_{p\in V} \mathcal{O}_p(V)$ and consider its *ideal of denominators*

$$\mathfrak{d}_\varphi = \{h \in k[V] \mid h\varphi \in k[V]\}$$

This is an ideal: it is closed under addition, and if $h\varphi \in k[V]$ then $(ah)\varphi = a(h\varphi) \in k[V]$ for any $a \in k[V]$. Its zero set is exactly the pole set of φ. Indeed, if $p \notin \mathcal{Z}(\mathfrak{d}_\varphi)$ then some $h \in \mathfrak{d}_\varphi$ has $h(p) \neq 0$, and writing $\varphi = (h\varphi)/h$ exhibits a representative with nonvanishing denominator at p, so φ is regular at p; conversely a representative u/v with $v(p) \neq 0$ puts v in $\mathfrak{d}_\varphi$ with $v(p) \neq 0$, so $p \notin \mathcal{Z}(\mathfrak{d}_\varphi)$. In particular the pole set is closed.

Now φ lies in every $\mathcal{O}_p(V)$, so it is regular at every point and its pole set is empty, that is $\mathcal{Z}(\mathfrak{d}_\varphi) = \emptyset$. Lifting $\mathfrak{d}_\varphi$ to $k[x_1, \ldots, x_n]$ and applying the Nullstellensatz (theorem 2.4.3) gives

$\mathfrak{d}_\varphi = k[V]$, so $1 \in \mathfrak{d}_\varphi$ and therefore

$$\varphi = 1 \cdot \varphi \in k[V]$$

as we sought to show.

Proposition 3.2.10: Stalks Are Local Noetherian Domains

let V be a variety and $p \in V$. Then $\mathcal{O}_p(V)$ is a local Noetherian domain

***Proof*:**
Domain. $\mathcal{O}_p(V)$ is by construction a subring of the field $k(V)$, and a subring of a field has no zero divisors.

Local. Put

$$\mathfrak{m}_p = \{f/g \in \mathcal{O}_p(V) \mid f(p) = 0\}$$

This is the kernel of the evaluation map $\mathrm{ev}_p\colon \mathcal{O}_p(V) \to k$, $f/g \mapsto f(p)/g(p)$, which is well defined precisely because $g(p) \neq 0$ and is a surjective ring homomorphism; so $\mathfrak{m}_p$ is an ideal with $\mathcal{O}_p(V)/\mathfrak{m}_p \cong k$, hence maximal. Every element outside $\mathfrak{m}_p$ is a unit: if $f/g \in \mathcal{O}_p(V)$ has $f(p) \neq 0$ then g/f again has nonvanishing denominator at p, so it lies in $\mathcal{O}_p(V)$ and inverts f/g. A proper ideal contains no unit, so it is contained in $\mathfrak{m}_p$; thus $\mathfrak{m}_p$ is the unique maximal ideal and $\mathcal{O}_p(V)$ is local.

Noetherian. $\mathcal{O}_p(V)$ is the localization of $k[V]$ at the prime $\mathcal{I}(p)$, and $k[V]$ is Noetherian as a quotient of the Noetherian ring $\Gamma(\mathbb{A}^n)$. A localization of a Noetherian ring is Noetherian, since every ideal of $S^{-1}R$ is the extension of its contraction to R, and extension carries a finite generating set to a finite generating set. Hence $\mathcal{O}_p(V)$ is Noetherian, completing the proof.

In section 5.1, we shall see that when V is a curve and $\mathcal{O}_p(V)$ is a *discrete valuation ring*, then V is *smooth* at p. This requires a lot of build-up, and so we shall leave stalks alone for now.

3.3 Rational and Birational Maps

Let us now generalize regular maps:

Definition 3.3.1: Rational Map

Let X be a variety and $Y \subseteq \mathbb{A}^m_k$ a variety. A *rational mapping* $\varphi\colon X \dashrightarrow Y$ is a tuple

$$\varphi = (\varphi_1, \ldots, \varphi_m) \qquad \varphi_i \in k(X)$$

of rational functions such that $(\varphi_1(x), \ldots, \varphi_m(x)) \in Y$ for every x in the common domain of definition of the φ_i, which by corollary 3.2.6 is a dense open subset of X. A rational mapping all of whose components happen to be regular is exactly a regular map.

Example 3.3.2: Rational Maps

1. *The circle, parameterized.* Let $C = \mathcal{Z}(x^2 + y^2 - 1) \subseteq \mathbb{A}^2_k$ and consider
$$\varphi\colon C \dashrightarrow \mathbb{A}^1_k \qquad \varphi(x,y) = \frac{y}{1-x}$$
This is a rational map, its single component lying in $k(C)$. It is undefined where $1-x$ vanishes on C, that is at the point $(1,0)$, and defined on the dense open complement. Its inverse is the regular map
$$t \mapsto \left(\frac{t^2-1}{t^2+1}, \frac{2t}{t^2+1}\right)$$
so C and $\mathbb{A}^1_k$ are birational, though not isomorphic: the map misses $(1,0)$ in one direction and the two points with $t^2 = -1$ in the other.

2. *A rational map with no regular representative.* On $\mathbb{A}^2_k$ the assignment $(x,y) \mapsto x/y$ is a rational map $\mathbb{A}^2_k \dashrightarrow \mathbb{A}^1_k$ whose domain of definition is $D(y)$. It cannot be extended to the origin: along the line $y = cx$ it is constantly $1/c$, so every value of k is taken arbitrarily close to the origin and no single value can be assigned there.

3. *Regular maps are the special case.* The map $(x,y) \mapsto (x^2, xy)$ has polynomial components, so its domain of definition is all of $\mathbb{A}^2_k$ and it is regular. Its image, computed in example 11.3.5, is not closed, which is a separate matter from being defined everywhere.

The image of the rational map shall be denoted $\varphi(X)$, even though not every point may be regular. Note that a rational map shall necessarily map an open subset $U \subseteq X$ into Y, and hence it is well-defined by proposition 3.2.7. By proposition 3.2.7, an equivalent definition of a rational map is that it is the collection of equivalence relations $[(\varphi, U)]$ with $U \subseteq X$ open and $\varphi : U \to Y$ where $[(\varphi, U)] \sim [(\psi, V)]$ if and only if
$$\varphi|_{U\cap V} = \psi|_{U\cap V}$$
This equivalent to definition 3.3.1, and is more prominent in the context of algebraic geometry. Its upside is that each representative is a well-defined map and hence can be seen as the appropriate quotient of some hom-sets, while its downside is that a rational mapping is now an equivalence relationship.

The following is the equivalent of being surjective:

Definition 3.3.3: Dominating Rational Map

Let $f : X \dashrightarrow Y$ be a rational map. Then f is said to be *dominating* if $f(U)$ is dense in Y.

Recall that if A, B are local rings where $A \subseteq B$ is a subring, then B dominates A if $\mathfrak{m}_A \subseteq \mathfrak{m}_B$. This vocabulary shall be made clear from the following

Proposition 3.3.4: Properties of Rational Maps

Let $F : X \dashrightarrow Y$ be a dominating rational map, $U \subseteq X$ and $V \subseteq Y$ affine open subsets where $f : U \to V$ represents F. Then:

1. If $\varphi : X \dashrightarrow Y$ and $\psi : Y \dashrightarrow Z$ are dominating rational maps, then $\psi \circ \varphi$ is dominating and $(\psi \circ \varphi)^\# = \varphi^\# \circ \psi^\#$.
2. $f^\# : k[V] \to k[U]$ is injective, hence extends to an injective map $k(V) \to k(U)$, and so gives $F^\# : k(Y) \hookrightarrow k(X)$, i.e. a field extension $k(X)/k(Y)$. It is an *embedding*, not an isomorphism; $F^\#$ is an isomorphism exactly when F is birational (proposition 3.3.6).
3. If p is in the domain of F and $F(p) = q$, then $\mathcal{O}_p(X)$ dominates $F^\#(\mathcal{O}_q(Y))$. Conversely, if $p \in X$ and $q \in Y$ are such that $\mathcal{O}_p(X)$ dominates $F^\#(\mathcal{O}_q(Y))$, then p belongs to the domain of F and $F(p) = q$.
4. Any k-homomorphism $k(Y) \to k(X)$ induces a unique dominating rational map $X \dashrightarrow Y$.

***Proof*:**

1. Let $U' \subseteq X$ be a dense open set on which φ is defined. Then $\psi\circ\varphi$ is defined on $U' \cap \varphi^{-1}(V')$ for V' a dense open set on which ψ is defined, and this is nonempty because $\varphi(U')$ is dense in Y and so meets V'. Its image contains $\psi(\varphi(U') \cap V')$, which is dense in Z since ψ is dominating and $\varphi(U') \cap V'$ is a nonempty open subset of V'. So $\psi \circ \varphi$ is dominating. Functoriality is then the computation $(\psi \circ \varphi)^\#(h) = h \circ \psi \circ \varphi = \varphi^\#(\psi^\#(h))$, so $(\psi \circ \varphi)^\# = \varphi^\# \circ \psi^\#$: pullback reverses composition.
2. Recall $f^\#(\varphi) = \varphi \circ f$. Suppose $f^\#(\varphi) = f^\#(\psi)$, that is $\varphi \circ f = \psi \circ f$. Then φ and ψ agree on $f(U)$, which is dense in V because F is dominating; two regular functions agreeing on a dense subset of a variety are equal (their difference vanishes on a dense set, hence on its closure). So $\varphi = \psi$ and $f^\#$ is injective. An injective map of integral domains extends to their fraction fields, giving $k(V) \hookrightarrow k(U)$, and since $U \subseteq X$ and $V \subseteq Y$ are nonempty open subsets of varieties we have $k(U) = k(X)$ and $k(V) = k(Y)$ by proposition 3.2.7. Hence $F^\#\colon k(Y) \hookrightarrow k(X)$ is a field embedding. It need not be surjective: $F^\#$ is onto exactly when F admits a rational inverse, which is proposition 3.3.6.
3. If $f(p) = q$, then it is a matter of pushing around the definition to see that $\mathcal{O}_p(X)$ dominates $F^\#(\mathcal{O}_q(Y))$. Conversely, suppose $\mathcal{O}_p(X)$ dominates $F^\#(\mathcal{O}_q(Y))$. Take an affine neighbourhood V of p and W of q where
$$k[W] = k[y_1, ..., y_n]$$
and $F^\#(y_i) = \frac{a_i}{b_i}$ for $a_i, b_i \in k[V]$ with $b_i(p) \neq 0$. Take $b = b_1 \cdots b_n$, and take the localization V_b. Then
$$F^\#(k[W]) \subseteq k[V_b]$$
Hence the map $F^\# : k[W] \to k[V_b]$ induces a unique morphism $f : V_b \to W$. Then if $g \in k[W]$ vanishes at q, $F^\#(g)$ vanishes at p, but then it must be that $f(p) = q$
4. If $\varphi : k(Y) \to k(X)$ wit h$\varphi(k[Y]) \subseteq k[X_b]$, for some $b \in k[X]$, φ induces a morphism $f : X_b \to Y$. Hence $f(X_b)$ is dense in Y as $f^\#$ is injective, giving us the desired result.

Definition 3.3.5: Birational Map

Let $\varphi : X \dashrightarrow Y$ be a rational map. Then φ is said to be *birational* if the image of φ is dense and there exists a rational map $\psi : Y \dashrightarrow X$ with dense image such that $\varphi \circ \psi = \psi \circ \varphi = \mathrm{id}$.

Proposition 3.3.6: Birational Maps and Isomorphisms

Let $\varphi\colon X \dashrightarrow Y$ be a dominating rational map of varieties. Then φ is birational if and only if $\varphi^{\#}\colon k(Y) \to k(X)$ is an isomorphism of fields over k.

***Proof*:**
Suppose φ is birational, with rational inverse ψ. Both composites are the identity, so applying proposition 3.3.4(1) gives $\varphi^{\#} \circ \psi^{\#} = (\psi \circ \varphi)^{\#} = \mathrm{id}$ and likewise $\psi^{\#} \circ \varphi^{\#} = \mathrm{id}$. Hence $\varphi^{\#}$ is an isomorphism with inverse $\psi^{\#}$.

Conversely, suppose $\varphi^{\#}$ is an isomorphism. Then $(\varphi^{\#})^{-1}\colon k(X) \to k(Y)$ is a k-homomorphism of function fields, so by proposition 3.3.4(4) it induces a dominating rational map $\psi\colon Y \dashrightarrow X$ with $\psi^{\#} = (\varphi^{\#})^{-1}$. Then $(\psi \circ \varphi)^{\#} = \varphi^{\#} \circ \psi^{\#} = \mathrm{id}$, and the uniqueness clause of proposition 3.3.4(4) forces $\psi \circ \varphi = \mathrm{id}$ as rational maps; symmetrically $\varphi \circ \psi = \mathrm{id}$. So φ is birational, as we sought to show.

The next definition uses dimension, which chapter 4 develops formally; for now read $\dim V$ as the number of independent parameters a general point of V carries.

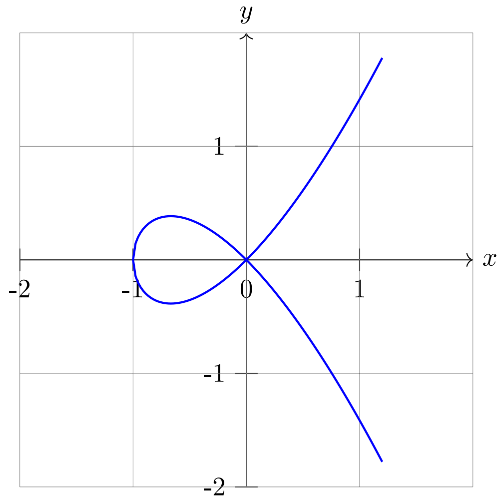
Definition 3.3.7: Rational Closed Set

Let V be a variety. Then if V is birational to $\mathbb{A}^n_k$ or $\mathbb{P}^n_k$, then V is called *affine rational* or *projective rational* respectively.

If V satisfies one of the above two conditions and has dimension $1, 2$ or codimension 1, then it is a *rational curve*, *rational surface*, or *rational hypersurface* respectively.

Example 3.3.8: Rational Mapping and Birational Maps

1. Naturally, all curves of order 1 (i.e., straight lines) are tautologically rational. More generally any linear subspace is rational.
2. Consider the curve given by
$$y^2 = x^2 + x^3$$
which looks like:



Label this curve C. Then this is a rational curve. To see this, consider that the line given by $y = tx$ passing through the origin intersects the curve at a single point (not counting the origin). Then substituting $y = tx$ into our equation, we get:

$$x^2(t^2 - x - 1) = 0$$

The root $x = 0$ corresponds to the origin, and the other roots $x = t^2 - 1$ corresponds to non-trivial solution. Hence, we get $x = t^2 - 1, y = t(t^2 - 1)$, and the rational map:

$$\varphi(t) = (t^2 - 1, t(t^2 - 1))$$

The inverse recovers the slope from the point: given $p = (x_0, y_0) \in C$ other than the origin, the line through p and the origin has slope y_0/x_0, so

$$\psi \colon C \dashrightarrow \mathbb{A}^1_k \qquad \psi(x, y) = \frac{y}{x}$$

is a rational map, regular away from $x = 0$. It inverts φ: substituting gives $\psi(\varphi(t)) = t(t^2 - 1)/(t^2 - 1) = t$ whenever $t^2 \neq 1$, and the two composites agree with the identity on dense open sets, which is equality of rational maps. So φ is birational. It is not an isomorphism: φ sends both $t = 1$ and $t = -1$ to the origin, the node of C, so it is not even injective.

If we take $k = \mathbb{Q}$, then this gives all rational solutions to the equation $y^2 = x^3 + x^2$. The case is a little more subtle if we were looking for integer solutions to the equation, which is explored in [4, Relating Algebra to Number Theory].

3. Let us show that all irreducible curves C of order 2 are rational (note that the above example is an order 3 curve, the maximal degree being 3). Let C be given by the equation $f(x, y)$. Take any point $(x_0, y_0) \in C$, and equation of the line through (x_0, y_0) with slope t

$$y - y_0 = t(x - x_0)$$

Label this line L_t. Let us find the points of intersection of the curve C with line L_t. To do so, we substitute:

$$f(x, y_0 + t(x - x_0)) = 0$$

which is equivalent to the mapping $\mathbb{A}_k^1$ to the curve. This mapping is well-defined (notice it is still of degree 2). Then one of the roots is certainly x_0. Then let a denote the coefficient of x after dividing the equation $f(x, y_0 + t(x - x_0))$ by the coefficient of x^2. Then we can see that the remaining roots must be:

$$x + x_0 = -a \qquad \Rightarrow \qquad x = -x_0 - a$$

Then x is a rational function of t, and substituting the expression for x into L_t, we obtain an expression for y as a rational function of t.

Note that if the coordinates x_0, y_0 of $(x_0, y_0) \in C$ and the coefficients of $f(x, y)$ belong to some subfield $k_0 \subseteq k$, then so do the coefficients of the function that gives the parameterization. In this case, we can find the general form of the solution in rational numbers of a Diophantine equation of degree 2 by knowing *at least* one solution! However, it turns out that knowing there is at least one solution can sometimes be tricky[a]

4. More generally, an irreducible hypersurface $X \subseteq \mathbb{A}_k^n$ given by a polynomial of degree 2 $f(x_1, ..., x_n) = 0$ is rational. The proof generalizes the above and is left as an exercise. The geometric picture is projection the hypersurface to a hyper-plane via a projection that does not pass through some point x. The point x can be any point that is not a "vertex" on X, namely that for at least one variable:

$$\frac{\partial f}{\partial x_i}(x) \neq 0$$

5. The curve order 3 given by $x^3 + y^3 = 1$ is *not* rational. More generally, $x^n + y^n = 1$ is not rational for $n \geq 3$, provided $\mathrm{char} k \nmid n$. To see this, let C_n be the curve given by $x^n + y^n = 1$ and suppose it is rational so that there exists rational functions φ, ψ such that $x = \varphi(t), y = \psi(t)$. Then we may find polynomials p, q, r relatively prime such that

$$\varphi(t) = \frac{p(t)}{r(t)} \qquad \psi(t) = \frac{q(t)}{r(t)}$$

where the denominator is the same for simplicity (the same argument follows with more steps by taking p/r and q/s). Then plugging into our equation we get:

$$p(t)^n + q(t)^n - r(t)^n = 0$$

for all $t \in k$. Differentiating is legitimate and does not annihilate the exponent, since $\mathrm{char} k \nmid n$, and so we have:

$$p(t)^{n-1}p'(t) + q(t)^{n-1}q'(t) - r(t)^{n-1}r'(t) = 0$$

We now have two systems equations. Treating $p^{n-1}, q^{n-1}, -r^{n-1}$ as the unknowns, we can consider the above two equations as being a system of linear equations which has has the matrix:

$$\begin{pmatrix} p & q & r \\ p' & q' & r' \end{pmatrix}$$

Solving this system of linear equations, we get that $p^{n-1}, q^{n-1}, -r^{n-1}$ are proportional to

$$qr' - rq' \qquad rp' - pr' \qquad pq' - qp'$$

As p, q, r are relatively prime, we get that:

$$p^{n-1} | (qr' - rq') \qquad q^{n-1} | (rp' - pr') \qquad r^{n-1} | (pq' - qp')$$

Finally, if a, b, c are the degrees of p, q, r . If $a \geq b \geq c$, then

$$(n-1)a \leq b + c - 1$$

But $a \geq b \geq c$ gives $b + c - 1 \leq 2a - 1$, so $(n-1)a \leq 2a - 1$, forcing $(n-3)a \leq -1$, which is impossible for $n \geq 3$ and $a \geq 0$. The other orderings of a, b, c give the same contradiction by symmetry.

[a] see Legendre theorem in cite:HERE

3.4 Classification up to Birational Equivalence

Birational mappings are weaker than isomorphisms (of varieties). This problem can be translated algebraically to the difference between having $k[X]$ and $k[Y]$ be isomorphic and $k(X)$ and $k(Y)$ be isomorphic. The reader may recall from section 2.4 that $k[X]$ and $k[Y]$ must be finitely generated reduced k-algebras, $k(X)$ and $k(Y)$ finite field extensions. As the structure of field extensions is much simpler, we expect birational mappings to be much simpler. In fact, we may even classify all affine variety up to birational isomorphism:

Theorem 3.4.1: Classification of Varieties Up To Birational Map

Let X be a variety of dimension d over an algebraically closed field k. Then X is birationally isomorphic to a hypersurface in $\mathbb{A}_k^{d+1}$.

Thus, birationality of varieties reduces to the classification of hypersurfaces. Note that being birationally isomorphic to a hypersurface does not imply a curve or plane is rational. For example, the curve given by $x^3 + y^3 = 1$ is already a hypersurface in $\mathbb{A}^2$ as it is given by $\mathcal{Z}(x^3 + y^3 - 1)$, however it is not a rational curve. Its projective closure is the Fermat cubic $\mathcal{Z}(x^3 + y^3 - z^3)$, which is nonsingular, and corollary 14.4.3 shows no nonsingular plane cubic is birational to $\mathbb{A}_k^1$; the invariant responsible is the genus, which chapter 14 constructs.

That claim deserves more than a forward reference, because the curve looks as though it ought to yield. Every technique available at this point can be tried on it, and watching each one fail is the most economical argument for building the machinery of part IV.

Example 3.4.2: Why $x^3 + y^3 = 1$ Resists Parameterization

Assume $\operatorname{char} k \neq 3$, so that the projective closure $\mathcal{Z}(x^3 + y^3 - z^3)$ is nonsingular; the excluded case is degenerate and treated at the end. Write $C = \mathcal{Z}(x^3 + y^3 - 1) \subseteq \mathbb{A}_k^2$.

Attempt 1: the chord construction. A conic is parameterized by fixing a point on it and sweeping the lines through that point, as in example 9.4.4(3). Run the identical computation on three curves, substituting the line through the chosen point and factoring out the known root.

- The conic $x^2 + y^2 = 1$ at $(1, 0)$, with $y = t(x - 1)$:

$$(x-1)\big[\,x + 1 + t^2(x-1)\,\big] = 0$$

  leaving a factor *linear* in x.

- The nodal cubic $y^2 = x^2 + x^3$ at the node $(0, 0)$, with $y = tx$:

$$x^2\big[\,t^2 - 1 - x\,\big] = 0$$

  again leaving a factor *linear* in x.

- The Fermat cubic $x^3 + y^3 = 1$ at $(1, 0)$, with $y = t(x - 1)$:

$$(x-1)\big[\,x^2 + x + 1 + t^3(x-1)^2\,\big] = 0$$

  leaving a factor *quadratic* in x.

In the first two cases the leftover factor is linear, so x is a rational function of t and the construction succeeds: the conic gives $x = (t^2 - 1)/(t^2 + 1)$, and the nodal cubic gives $x = t^2 - 1$, recovering the parameterization of example 3.3.8.

In the third it is quadratic, and solving it introduces a square root. The reason is visible in the factorizations: the method needs the marked point to absorb all but one of the intersections. A line meets a conic twice and the marked point takes one; a line through the node of $y^2 = x^2 + x^3$ meets the cubic three times and the node, being a double point, takes two. A smooth cubic has no double point, so the marked point takes only one of three and two are left over. *The construction does not fail by accident; it fails because the curve is smooth*, which is exactly the hypothesis under which the conclusion is true.

A warning about what this does and does not show. Given *two* points of a cubic the chord still produces a third, and over $\mathbb{Q}$ that is the classical method for generating new solutions from old. That is a statement about individual points, not about parameterizing the whole curve by one variable, and the two are easy to conflate.

Attempt 2: symmetric functions. The equation is symmetric in x and y, so set $u = x + y$ and $v = xy$. Then $x^3 + y^3 = u^3 - 3uv$, and the curve becomes

$$u^3 - 3uv = 1 \qquad \text{that is} \qquad v = \frac{u^3 - 1}{3u}$$

which parameterizes the pair $\{x, y\}$ rationally by u alone. But recovering x and y individually means solving $T^2 - uT + v = 0$, whose discriminant is

$$u^2 - 4v = u^2 - \frac{4(u^3 - 1)}{3u} = \frac{4 - u^3}{3u}$$

So a square root of $(4 - u^3)/3u$ is needed. Setting $W = 3uw$ where w is that square root gives

$$W^2 = 3u\,(4 - u^3) = 12u - 3u^4$$

a square root of a quartic with four distinct roots, namely $u = 0$ and the three cube roots of 4. That is another curve of the same kind: section 14.4 will assign it genus 1, exactly as for C. The attempt is circular, and it is circular in a way one can see before having any theory: it trades the problem for an equal one.

Attempt 3: look for a parameterization directly. Suppose C were rational, so that $x = p(t)/r(t)$ and $y = q(t)/r(t)$ for polynomials p, q, r with no common factor, not all constant. Substituting and clearing denominators,

$$p(t)^3 + q(t)^3 = r(t)^3$$

This is Fermat's Last Theorem for polynomials in the exponent 3, and unlike its arithmetic namesake it has a short proof. The Mason–Stothers theorem states that for coprime polynomials $a + b = c$ in $k[t]$, not all constant and with $\operatorname{char} k = 0$,

$$\max(\deg a, \deg b, \deg c) \;\leq\; n_0(abc) - 1$$

where n_0 counts *distinct* roots. Apply it to $a = p^3$, $b = q^3$, $c = r^3$ and set $d = \max(\deg p, \deg q, \deg r)$. The left side is $3d$. On the right, $p^3q^3r^3$ has the same distinct roots as pqr, so $n_0 \leq \deg(pqr) \leq 3d$, giving

$$3d \;\leq\; 3d - 1$$

which is absurd. So no such p, q, r exist and C is not rational.

What the third attempt costs. It is a complete proof, but only in characteristic zero: the Mason–Stothers argument differentiates, and in characteristic $\rho > 0$ all three of p, q, r may be polynomials in t^ρ, with vanishing derivatives, and the argument collapses. And in the excluded characteristic 3 the curve is not what it appears: $x^3 + y^3 = (x + y)^3$, so $C = \mathcal{Z}((x + y)^3 - 1)$ is a union of lines and is rational after all.

So one attempt fails structurally, one is circular, and one succeeds in a single characteristic by an argument owing nothing to geometry. What is missing is an invariant of the curve that decides the question uniformly, is computable, and explains why the conic and the nodal cubic behave differently from this one. That invariant is the genus. Theorem 14.4.2 shows a plane curve is rational exactly when its genus vanishes, definition 14.4.1 computes the genus of a smooth plane curve of degree d as $\binom{d-1}{2}$, giving 1 here and 0 for a conic, and section 14.4 shows a node drops the genus by exactly 1, which is why the nodal cubic is rational and this one is not. The argument is characteristic-free.

***Proof*:**
Write $K = k(X)$, the fraction field of $k[X] = \Gamma(\mathbb{A}^n)/\mathcal{I}(X)$. It is generated over k by the images $t_1, \ldots, t_n$ of the coordinate functions, a *finite* set, so from it we may extract a maximal algebraically independent subset; after renumbering call it $t_1, \ldots, t_d$, so that $d = \operatorname{trdeg}_k K$ and every remaining t_j is algebraic over $k(t_1, \ldots, t_d)$.

Step 1: each algebraic generator can be taken separable. Fix $j > d$. Then $t_1, \ldots, t_d, t_j$ are algebraically dependent, so some irreducible $f \in k[x_1, \ldots, x_{d+1}]$ has $f(t_1, \ldots, t_d, t_j) = 0$. We claim $\partial f/\partial x_i \neq 0$ for some i. In characteristic 0 this is immediate, as f is nonconstant. In characteristic p, suppose every $\partial f/\partial x_i$ vanished. Then each variable occurs in f only in exponents divisible by p, so

$$f = \sum a_{i_1,\ldots,i_{d+1}}\, x_1^{pi_1} \cdots x_{d+1}^{pi_{d+1}}$$

Since k is algebraically closed it is perfect, so each coefficient has a p-th root: pick $b_{i_1,\ldots,i_{d+1}}$ with

$b^p_{i_1,\ldots,i_{d+1}} = a_{i_1,\ldots,i_{d+1}}$. Then $g = \sum b_{i_1,\ldots,i_{d+1}} x_1^{i_1} \cdots x_{d+1}^{i_{d+1}}$ satisfies $f = g^p$, since the Frobenius is a ring homomorphism in characteristic p. That contradicts irreducibility of f.

Step 2: putting the separable variable last. Say $\partial f/\partial x_i \neq 0$, so x_i genuinely occurs in f and hence t_i is algebraic over the field generated by the other d elements of $\{t_1, \ldots, t_d, t_j\}$. Those d elements are still algebraically independent: if they were not, the transcendence degree of $k(t_1, \ldots, t_d, t_j)$ would fall below d, contradicting the independence of $t_1, \ldots, t_d$. Renaming so that the independent ones are $t_1, \ldots, t_d$ and the algebraic one is t_{d+1}, we have $\partial f/\partial x_{d+1} \neq 0$, which says precisely that t_{d+1} is separable over $k(t_1, \ldots, t_d)$.

Step 3: collapsing to one generator. K is generated over $k(t_1, \ldots, t_d)$ by the finitely many elements $t_{d+1}, \ldots, t_n$, each separable by Step 1 applied in turn, so $K/k(t_1, \ldots, t_d)$ is a finite separable extension. The primitive element theorem then supplies a single z_{d+1} with

$$K = k(t_1, \ldots, t_d)(z_{d+1})$$

Writing $z_i = t_i$ for $i \leq d$, let f be an irreducible polynomial with $f(z_1, \ldots, z_{d+1}) = 0$, and set $Y = \mathcal{Z}(f) \subseteq \mathbb{A}_k^{d+1}$, a hypersurface. By construction $k(Y) \cong K = k(X)$.

Step 4: an isomorphism of function fields is a birational map. By proposition 3.3.6, a dominating rational map is birational exactly when the induced map of function fields is an isomorphism, and proposition 3.3.4(4) produces such a dominating rational map $X \dashrightarrow Y$ from the isomorphism $k(Y) \to k(X)$. Hence X and Y are birationally isomorphic, as we sought to show.

Note that $k(X)/k(z_1, ..., z_d)$ is a finite separable extension, and that the primitive elements $z_1, ..., z_{d+1}$ can be chosen as linear combinations of the $x_1, ..., x_n$ for appropriate c_{ij}:

$$z_i = \sum_j^n c_{ij} x_j$$

Then the mapping $(x_1, ..., x_n) \to (z_1, ..., z_{d+1})$ is a projection from $\mathbb{A}_k^n$ that is parallel to the linear subspace given by:

$$\sum_j^n c_{ij} x_j = 0 \qquad 1 \leq i \leq d+1$$

Thus, the birational mapping theorem says that there is an appropriate hypersurface (not necessarily a hyperplane) on which we may project in a rational manner the variety! From this perspective, linear algebra results which are birationally invariant can be translated over to varieties, which we shall soon see gives us some interesting results!

One direction of the classification is easy to get wrong. A variety dominated by $\mathbb{A}_k^n$, meaning one admitting a dominant rational map from affine space, is called *unirational*; a variety birational to $\mathbb{A}_k^n$ is *rational*. Rational implies unirational trivially. In dimension one the converse also holds, and that is Lüroth's theorem. In higher dimensions it fails, which is one of the deepest facts in the subject and well outside this book.

Theorem 3.4.3: Lüroth's Theorem

Let L be a field with $k \subsetneq L \subseteq k(t)$, where t is transcendental over k. Then $L = k(u)$ for some $u \in L$.

***Proof*:**
Choose $u_0 \in L \setminus k$. Then t is algebraic over $k(u_0)$: writing $u_0 = a(t)/b(t)$ with $a, b \in k[t]$ not both constant, t is a root of $a(X) - u_0 b(X) \in k(u_0)[X]$, which is nonzero because a and b are not proportional. So t is algebraic over L as well; put $n = [k(t) : L]$ and let

$$f(X) = X^n + c_{n-1}X^{n-1} + \cdots + c_0 \ \in L[X]$$

be the minimal polynomial of t over L.

Step 1: some coefficient is not in k. If every c_j lay in k then t would be algebraic over k, contradicting transcendence. Fix j with $u = c_j \notin k$ and write $u = a(t)/b(t)$ in lowest terms, with $m = \max(\deg a, \deg b) \geq 1$.

Step 2: $[k(t) : k(u)] = m$. The polynomial $P(X) = a(X) - u\,b(X) \in k(u)[X]$ has degree m and kills t. It is irreducible over $k(u)$: regard it in $k[X][u]$, where it is linear in u and primitive, since a and b are coprime; Gauss's lemma then makes it irreducible in $k(u)[X]$. Hence it is the minimal polynomial of t over $k(u)$ and $[k(t) : k(u)] = m$.

Step 3: $n \leq m$. We have $k(u) \subseteq L \subseteq k(t)$ because $u = c_j \in L$, so $n = [k(t) : L]$ divides $[k(t) : k(u)] = m$; in particular $n \leq m$.

Step 4: $m \leq n$. Clear denominators: let $d(t) \in k[t]$ be a common denominator for $c_0, \ldots, c_{n-1}$, chosen so that $F(t, X) = d(t)f(X)$ lies in $k[t][X]$ and is primitive as a polynomial in X over $k[t]$. Its degree in X is n. Since f is the minimal polynomial of t over L and $P(X) = a(X) - ub(X)$ also kills t, we get $f \big| P$ in $L[X]$; clearing denominators and applying Gauss's lemma in $k[t][X]$,

$$F(t, X) \ \Big| \ a(X)b(t) - a(t)b(X)$$

The right side has degree at most m in t. On the other hand $u = a(t)/b(t)$ appears among the coefficients of f, so after clearing denominators F has degree at least m in t. Hence $\deg_t F = m$ and the quotient is a constant, so

$$F(t, X) = c\,\big(a(X)b(t) - a(t)b(X)\big) \qquad c \in k^*$$

Comparing degrees in X gives $n = \deg_X F = m$.

Conclusion. Steps 3 and 4 give $n = m$, so $[k(t) : L] = [k(t) : k(u)]$; since $k(u) \subseteq L$, the two fields coincide and $L = k(u)$, as we sought to show.

Corollary 3.4.4: Unirational Curves Are Rational

Let C be a curve admitting a dominant rational map $\mathbb{P}^1_k \dashrightarrow C$. Then C is rational, that is birational to $\mathbb{P}^1_k$.

***Proof*:**
A dominant rational map induces an inclusion of function fields $k(C) \hookrightarrow k(\mathbb{P}^1_k) = k(t)$ by proposition 3.3.4. Its image is a subfield of $k(t)$ containing k, and it properly contains k because C is a curve, so $k(C)$ has transcendence degree 1. Theorem 3.4.3 makes that image $k(u)$ for some u, hence $k(C) \cong k(u) \cong k(t)$, and isomorphic function fields mean birational varieties by theorem 3.4.1 and its proof.

The failure in higher dimension is worth naming, since it is easy to assume the pattern continues. There are unirational surfaces over fields of positive characteristic that are not rational, and unirational threefolds over $\mathbb{C}$ that are not rational; the first examples over $\mathbb{C}$ are due to Clemens and Griffiths, Iskovskikh and Manin, and Artin and Mumford, all in 1971–72. Deciding rationality is a live problem, and nothing in this book bears on it.

Exercise 3.4.1

1. In the proof of proposition 3.2.2, verify the identity $\varphi g_i = f_i$ in $k(X)$, and explain why it holds on all of X even though the representation $\varphi = f_i/g_i$ was chosen only near x_i.
2. Let X be irreducible. Show that any finite collection of rational functions on X has a non-empty common domain of definition
3. Show that the collection of poles of a rational function $f \in k(V)$ is an algebraic set
4. Show that for each $a \in k$, the stalk $\mathcal{O}_a(\mathbb{A}^1_k)$ at a is a DVR with uniforming parameter $x - a$.
5. Define $\mathcal{O}_\infty = \left\{ \frac{f}{g} \;\middle|\; \deg(f) \leq \deg(g) \right\}$. Show that $\mathcal{O}_\infty$ is a DVR with uniforming parameter $1/x$. (Note: This is foreshadowing that it is natural consider curves in *projective space* rather than affine space; see chapter 8).

Part II

Local Properties

Everything in Part I is an invariant of a whole variety. This Part changes the question: fix a point p of a variety X and ask what can be said about X near p. The answer throughout is that the local ring $\mathcal{O}_{X,p}$ carries it. The dimension of X at p is the Krull dimension of that ring, smoothness at p is the equality of that dimension with the dimension of the cotangent space $\mathfrak{m}_p/\mathfrak{m}_p^2$, the multiplicity of X at p is read from the associated graded ring, and normality is the condition that every $\mathcal{O}_{X,p}$ is integrally closed.

The passage runs both ways: A global statement is repeatedly proved by checking a condition at every point. The regular functions on a variety are the intersection of its local rings, a rational function regular everywhere is regular, and a variety is normal exactly when each of its local rings is.

Dimension theory, regular local rings, completion and the Cohen structure theorem are the commutative algebra the Part runs on, and they come from *Commutative Algebra* [5].

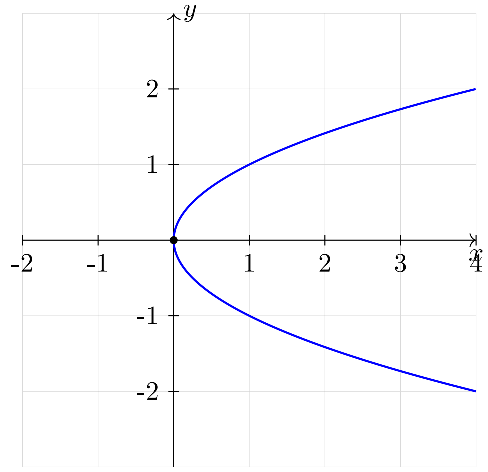
4

Dimension

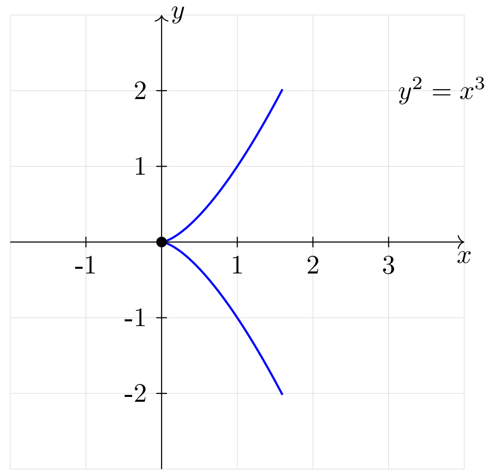
Every notion of dimension that works in analysis or topology fails on the Zariski topology, whose open sets are too large for covering arguments and whose only irreducible subsets in the Euclidean sense are points. What survives is a definition built from chains of irreducible closed subsets, and its algebraic mirror, chains of prime ideals. This chapter establishes that the two agree, and that for a variety both agree with the transcendence degree of its function field.

4.1 What Dimension Should Mean

When looking at $\mathbb{A}_k^n$, we may naturally think of this space as being n-dimensional, and in this special case this matches with our notion of a 3-dimensional vector-space. We may think that varieties also have a notion of dimension, for example the curve with coordinate ring $\frac{k[x,y]}{(y^2-x)}$ gives the impression that it should be thought of as a 1-dimensional variety:



This variety should even be thought of as being smooth, as compared to the variety with coordinate ring $\frac{k[x,y]}{(y^2-x^3)}$ which has a cusp at $(x, y) = (0, 0)$.



We may hope that the notion of dimension is as simple as it was for vector spaces, however this turns out to be a more subtle problem. For example, continuous maps do not preserve dimension (there is a continuous map from $\mathbb{R}$ onto $\mathbb{R}^2$ given by the Peano space-filling curve). Homeomorphisms between open sets of $\mathbb{R}^n$ and $\mathbb{R}^m$ do force the condition that $n = m$[1], however we still need to define the notion of dimension so that it can be preserved.

One of the most general definition of dimension can be defined in the topological level and is called the *Lebesgue covering dimension* where a space X is dimension at most n if every open cover can be refined to an open cover in which each point lies in at most $n + 1$ sets. However, this definition will fail spectacularly in the Zariski topology as most open sets are huge (in fact, most Zariski topologies will have infinite Lebesgue covering dimension).

[1] see [16]

There are a few other notion of dimension (to name some notable example: the Brouwer-Menger-Urysohn boundary dimension, the Hausdorff Dimension, Deviation of Poset). One that the Nullstellensatz has already put within reach is the transcendence degree of a field extension. For example, if we take $k[V]$ to be some coordinate ring for a variety (so that $k[V]$ is an integral domain), we may take the field of fractions $k(V)$. Then taking the transcendental degree of $k(V)/k$, we may define this to be the *transcendental dimension* of $k[V]$. This works well on smooth shapes, and theorem 10.4.4 and theorem 10.4.3 will show that every affine or projective variety admits a finite surjection onto $\mathbb{A}_k^n$ or $\mathbb{P}_k^m$ respectively, so the transcendence degree is always attained. It is also invariant under isomorphism, being defined from the function field. However this definition fails when there the coordinate ring is not an integral domain[2]. Furthermore, it does not generalize to the spectrum of an arbitrary ring: the geometric object attached to $\mathbb{Z}$ has transcendence degree 0 over no field at all, while the definition settled on below gives it dimension 1.

One limitation is that these notions of dimension are global; the entire space has some dimension. This turns out to be a misnomer, as the example of the curve $y^2 = x^3$ mentioned earlier demonstrates (it *drops* a dimension at $(x, y) = (0, 0)$). To anyone who has done differential geometry, they may already be comfortable with the fact that the dimension of a smooth manifold is given by the vector space dimension of the tangent space at a point (which for a smooth-manifold should be invariant to the choice of point, and hence a well-defined global notion). Then the reader well-versed in the equivalent definition's of tangent spaces of a manifold may recall that the tangent space can be defined as the dual of $\mathfrak{m}/\mathfrak{m}^2$ where $\mathfrak{m}$ is the maximal ideal of C_a^∞ (the set of germs of C^∞ on a manifold at the point a). This definition is agnostic of the differential structure, and so can be used on varieties. It shall also capture the local subtleties; to reiterate the same example the tangent space at $(x, y) = (0, 0)$ of $y^2 = x^3$ will be dimension 2, representing the fact that it should be treated differently than the other points. Section 5.1 takes up this notion.

4.2 Krull Dimension and Transcendence Degree

What has been settled on as the most useful notion of global dimension that translates well between algebra and geometry is the following:[3]

Definition 4.2.1: Topological Krull Dimension

Let X be a topological space. Then the *Krull dimension* of X is the supremum of the length of all chains:

$$Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_n$$

of distinct irreducible closed subsets of X. The Krull dimension of a topological space will be denoted kdim

This definition is not interesting in euclidean topology, for example for $\mathbb{R}^n$ as only points are irreducible we have $\text{kdim}(\mathbb{R}^n) = 0$. If the dimension is finite, then it is equal to the length of the longest chain of irreducible closed subsets. The Krull dimension of $\mathbb{A}_k^1$ is always 1, $\text{kdim}(\mathbb{A}_k^1) = 1$, as the only irreducible closed subsets of $\mathbb{A}_k^1$ are the single point and whole space. As irreducible closed subsets

[2]It also fails as an adequate definition when localizing a ring or taking its completion, which are two important tools that we shall often use on shapes

[3]The Krull dimension turns out to be equivalent to the Brouwer-Menger-Urysohn definition of dimension, which is useful for cohomological intuitions

correspond to prime ideals, we have the equivalent notion for primes:

Definition 4.2.2: Height of Prime

Let $\mathfrak{p} \subseteq R$ be a prime ideal. Then the *height* of a prime ideal is the supremum of all integers n for which there is a chain of distinct prime ideals:

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n = \mathfrak{p}$$

The *dimension* of R is the supremum of the heights of all prime ideals.

A field has exactly one prime, (0), of height 0, so its Krull dimension is 0. A principal ideal domain that is not a field has Krull dimension 1: its nonzero primes are maximal, so the longest chain is $(0) \subsetneq \mathfrak{p}$.

Proposition 4.2.3: Dimension of Affine Algebraic Set

Let $Y \subseteq \mathbb{A}^n_k$ be an affine algebraic set. Then $\operatorname{kdim}(Y)$ is equal to the dimension of the affine coordinate ring $k[Y]$

***Proof*:**
By proposition 2.3.3 the irreducible closed subsets of Y correspond to the prime ideals of $\Gamma(\mathbb{A}^n)$ containing $\mathcal{I}(Y)$, which by the correspondence theorem are the prime ideals of $k[Y] = \Gamma(\mathbb{A}^n)/\mathcal{I}(Y)$. The correspondence *reverses* inclusion, so a strictly ascending chain of length d of irreducible closed subsets of Y is the same thing as a strictly descending chain of length d of primes of $k[Y]$, and conversely. Taking suprema, $\operatorname{kdim}(Y) = \operatorname{kdim}(k[Y])$, as we sought to show.

When covering Noether's Normalization (theorem 2.4.1), it was observed that this can be used to define the dimension of a ring. The following definition captures this idea:

Definition 4.2.4: Transcendental Dimension

Let X be a variety. Then the *transcendence dimension* of X over k is the transcendence degree of its function field $k(X)$, and is denoted $\dim X$. (Proposition 4.2.5 shows this agrees with $\operatorname{kdim} X$, after which the two notations are used interchangeably.) The transcendence degree of a field extension, and the fact that it is well-defined, are established in *Core Algebra* [11].

Proposition 4.2.5: Transcendental and Krull Dimension

Let k be a field and R a finitely generated k-algebra which is an integral domain. Then

$$\operatorname{kdim}(R) = \operatorname{trdeg}_k \operatorname{Frac}(R)$$

Furthermore, for every prime $\mathfrak{p} \subseteq R$,

$$\operatorname{height}(\mathfrak{p}) + \operatorname{kdim}(R/\mathfrak{p}) = \operatorname{kdim}(R)$$

Neither hypothesis can be dropped. Transcendence degree is an invariant of a field extension, so R must be a domain for $\operatorname{Frac}(R)$ to exist; and finite generation cannot be dropped, as example 4.2.6

shows. The second identity is the statement that finitely generated domains over a field are catenary and equidimensional, which is what makes codimension behave additively.

***Proof*:**
This is the equality $\mathrm{kdim} R = \mathrm{trdeg}_k R$ for a finitely generated k-algebra, together with the dimension formula $\mathrm{height}(\mathfrak{p}) + \mathrm{kdim}(R/\mathfrak{p}) = \mathrm{kdim} R$; both are proved via Noether normalization in [5].

The conditions are necessary for the equality of both equations:

Example 4.2.6: Different Krull and Transcendental Dimension
Take $R = k(x)$, the field of rational functions in one variable. As a field it has exactly one prime, so $\mathrm{kdim} R = 0$, while $\mathrm{trdeg}_k R = 1$. There is no contradiction with proposition 4.2.5: $k(x)$ is *not* a finitely generated k-algebra. It is finitely generated as a *field* extension, which is a strictly weaker condition, and the proposition needs the algebra version.

Geometrically the collapse is familiar to a reader who has met schemes: $k(x)$ is the local ring at the generic point of $\mathbb{A}^1_k$, a single non-closed point, and a one-point space has Krull dimension 0 however large the field attached to it.

Corollary 4.2.7: Dimension of Affine Space

The Krull dimension of affine space $\mathbb{A}^n_k$ is n.

***Proof*:**
$\Gamma(\mathbb{A}^n) \cong k[x_1, \ldots, x_n]$ is a finitely generated k-algebra and an integral domain, so proposition 4.2.5 applies and gives $\mathrm{kdim} \mathbb{A}^n = \mathrm{trdeg}_k k(x_1, \ldots, x_n)$. The x_i are algebraically independent over k and generate the field, so they form a transcendence basis and that degree is n, as we sought to show.

4.3 Codimension, Curves, and Surfaces

Definition 4.3.1: Codimension

Let $Y \subseteq X$ be a closed subvariety of a variety X. The *codimension* of Y in X, written $\mathrm{codim}_X Y$, is the supremum of the lengths of chains

$$Y = Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_c \subseteq X$$

of irreducible closed subsets of X starting at Y. The ambient is part of the data and is dropped from the notation only when it is fixed by context.

Proposition 4.3.2: Codimension Is a Difference of Dimensions

Let $Y \subseteq X$ be a closed subvariety of a variety X over a field. Then

$$\operatorname{codim}_X Y = \dim X - \dim Y$$

***Proof*:**
Passing to an affine open subset changes neither side, so assume X affine with coordinate ring $k[X]$, a finitely generated domain over k. Chains of irreducible closed subsets of X starting at Y correspond, reversing inclusion, to chains of primes of $k[X]$ descending from $\mathcal{I}(Y)$, so $\operatorname{codim}_X Y = \operatorname{height}(\mathcal{I}(Y))$. Likewise $\dim Y = \operatorname{kdim}(k[X]/\mathcal{I}(Y))$ by proposition 4.2.3. The dimension formula of proposition 4.2.5 applied to $\mathfrak{p} = \mathcal{I}(Y)$ then reads $\operatorname{codim}_X Y + \dim Y = \dim X$, as we sought to show.

The identity fails without the hypotheses of proposition 4.2.5: it is exactly the statement that the ring is catenary and equidimensional, which is why codimension is *defined* by chains and only *computed* as a difference.

One consequence is used constantly and deserves a name: dimension cannot be detected locally, in the sense that shrinking to a smaller open set does not shrink the dimension.

Corollary 4.3.3: Dimension Is Determined by Any Open Subset

Let X be an irreducible variety and $U \subseteq X$ a nonempty open subset. Then $\dim U = \dim X$.

***Proof*:**
U is dense in X by irreducibility, so restriction identifies the two function fields, $k(U) = k(X)$. Both dimensions equal the transcendence degree of that common field by proposition 4.2.5, completing the proof.

Verify as a check on the definitions that planar curves have dimension 1, that finite sets have dimension 0, and that $\mathbb{A}^n_k$ and $\mathbb{P}^n_k$ have dimension n.

Definition 4.3.4: Curves and Surfaces

Let X be an algebraic variety. Then:

1. If X is 1-dimensional, it is called a *curve*
2. If X is 2-dimensional, it is called a *surface*
3. If X is 1-codimensional, it is called a *hypersurface*.

Proposition 4.3.5: Dimension Between Irreducible Varieties

Let $Y \subseteq X$ be a closed subvariety of a variety X. Then $\dim Y \leq \dim X$, with equality if and only if $Y = X$.

***Proof*:**
Every chain of irreducible closed subsets of Y is a chain of irreducible closed subsets of X, since Y is closed in X; taking suprema gives $\dim Y \leq \dim X$.

Suppose $\dim Y = \dim X = d$ and $Y \subsetneq X$. Take a chain $Z_0 \subsetneq \cdots \subsetneq Z_d = Y$ of irreducible closed subsets of Y realizing its dimension. As Y is irreducible, closed and properly contained in X, appending X extends this to

$$Z_0 \subsetneq \cdots \subsetneq Z_d = Y \subsetneq X$$

a chain of length $d+1$ in X, so $\dim X \geq d+1$, contradicting $\dim X = d$. Hence $Y = X$; the converse is trivial, completing the proof.

4.4 Hypersurface Sections and Codimension One

Cutting a variety by one equation ought to drop its dimension by exactly one. That is the geometric content of Krull's principal ideal theorem, and it is the fact that makes dimension usable: it lets a variety be built up, or cut down, one equation at a time. This section records it, and then proves the converse in affine space, where codimension one turns out to be the same thing as being cut out by a single polynomial.

Theorem 4.4.1: Hypersurface Sections Drop Dimension by One

Let X be a variety and $f \in k[X]$ neither zero nor a unit. Then every irreducible component of $\mathcal{Z}_X(f) = \{p \in X \mid f(p) = 0\}$ has dimension $\dim X - 1$.

***Proof*:**
The irreducible components of $\mathcal{Z}_X(f)$ correspond to the primes of $k[X]$ minimal over (f). Let $\mathfrak{p}$ be such a prime. Krull's principal ideal theorem ([5, Krull's Height Theorem]) gives $\operatorname{height}(\mathfrak{p}) \leq 1$.

For the reverse bound, $k[X]$ is a domain because X is a variety (corollary 2.3.4), so (0) is prime; and $f \neq 0$ forces $\mathfrak{p} \neq (0)$. Hence $(0) \subsetneq \mathfrak{p}$ is a chain of length 1 and $\operatorname{height}(\mathfrak{p}) \geq 1$, so $\operatorname{height}(\mathfrak{p}) = 1$.

The component attached to $\mathfrak{p}$ has coordinate ring $k[X]/\mathfrak{p}$, so by proposition 4.2.5,

$$\dim \mathcal{Z}_X(\mathfrak{p}) = \operatorname{kdim}(k[X]/\mathfrak{p}) = \operatorname{kdim} k[X] - \operatorname{height}(\mathfrak{p}) = \dim X - 1$$

as we sought to show.

The theorem says nothing about $\mathcal{Z}_X(f)$ being irreducible, and it need not be: $\mathcal{Z}(xy) \subseteq \mathbb{A}^2$ is the union of the two axes, two components each of dimension 1. What the theorem guarantees is that no component is smaller than expected. The hypothesis that f be a nonunit is what keeps $\mathcal{Z}_X(f)$ nonempty in the first place, and $f \neq 0$ is what stops it from being all of X.

In affine space the converse holds, and it is a genuine dictionary entry: codimension one on the geometric side is principality on the algebraic side.

Theorem 4.4.2: Codimension One in Affine Space Means Hypersurface

Let $Y \subsetneq \mathbb{A}_k^n$ be an irreducible closed subset with $\operatorname{codim}_{\mathbb{A}^n} Y = 1$. Then there is an irreducible polynomial f, unique up to a nonzero scalar, with

$$\mathcal{I}(Y) = (f) \qquad Y = \mathcal{Z}(f)$$

Conversely $\mathcal{Z}(f)$ is irreducible of codimension 1 for every irreducible f.

***Proof*:**
$\mathcal{I}(Y)$ is prime because Y is irreducible (proposition 2.3.3), and $\operatorname{height}(\mathcal{I}(Y)) = \operatorname{codim}_{\mathbb{A}^n} Y = 1$ by proposition 4.3.2. So it suffices to show that a height-one prime $\mathfrak{p}$ in a unique factorization domain is principal, generated by an irreducible element.

The ring $\Gamma(\mathbb{A}^n) \cong k[x_1, \ldots, x_n]$ is a UFD, being a polynomial ring in finitely many variables over the field k ([11, Polynomial Rings in Several Variables over a UFD are UFDs]). Choose $0 \neq g \in \mathfrak{p}$ and factor it into irreducibles, $g = f_1 \cdots f_r$. As $\mathfrak{p}$ is prime, some $f_i \in \mathfrak{p}$; write $f = f_i$. In a UFD an irreducible element is prime ([11, Prime If And Only If Irreducible In UFD]), so (f) is a nonzero prime ideal with

$$(0) \subsetneq (f) \subseteq \mathfrak{p}$$

If the second inclusion were strict, that chain would have length 2 and give $\operatorname{height}(\mathfrak{p}) \geq 2$. Hence $(f) = \mathfrak{p}$. Uniqueness up to a scalar is uniqueness of the generator of a principal ideal in a domain, and $Y = \mathcal{Z}(\mathcal{I}(Y)) = \mathcal{Z}(f)$ by proposition 1.2.11.

Conversely let f be irreducible. Then (f) is prime, so $\mathcal{Z}(f)$ is irreducible, and theorem 4.4.1 applied to $X = \mathbb{A}^n$ gives $\dim \mathcal{Z}(f) = n - 1$, that is codimension 1, as we sought to show.

Example 4.4.3: Hypersurfaces and Their Ideals

1. The cuspidal cubic $\mathcal{Z}(y^2 - x^3) \subseteq \mathbb{A}^2$ has $y^2 - x^3$ irreducible, so its vanishing ideal is $(y^2 - x^3)$ and it is a curve: codimension 1 in $\mathbb{A}^2$.
2. $\mathcal{Z}(xy) \subseteq \mathbb{A}^2$ is not irreducible, and correspondingly (xy) is not prime. Its two components $\mathcal{Z}(x)$ and $\mathcal{Z}(y)$ each have codimension 1, as theorem 4.4.1 predicts.
3. The twisted cubic $\mathcal{Z}(y - x^2, z - x^3) \subseteq \mathbb{A}^3$ has dimension 1, hence codimension 2, and so by theorem 4.4.2 it cannot be cut out by a single equation. This is the promise made when the twisted cubic first appeared in example 1.2.2 that it is not a hypersurface.

Exercise 4.4.1

1. Let X, Y be irreducible affine varieties. Show that $\dim(X \times Y) = \dim(X) + \dim(Y)$ directly from the transcendence-degree definition. Corollary 10.5.4 proves this a different way, by reading the projection as a family of fibres, so the two arguments are worth comparing.
2. Define the *coheight* of a prime $\mathfrak{p} \subseteq R$ to be the supremum of the lengths of chains of primes $\mathfrak{p} = \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_d$. Show that $\operatorname{coht}(\mathfrak{p}) = \operatorname{kdim}(R/\mathfrak{p})$ for any ring, give an example where

it is infinite, and show it is finite whenever R is a finitely generated k-algebra.

3. Let R be a Noetherian ring. Show that $\mathrm{kdim} R[x] = \mathrm{kdim} R + 1$.
4. Compute the dimension of $\mathcal{Z}(xy, xz) \subseteq \mathbb{A}^3_k$, identify its irreducible components, and observe that they do not all have the same dimension. Which hypothesis of proposition 4.3.2 does this violate?
5. Show that $\dim \mathcal{Z}(f) = n - 1$ for any nonconstant $f \in k[x_1, \ldots, x_n]$, and deduce that a proper closed subset of $\mathbb{A}^n_k$ has dimension at most $n - 1$.
6. Let $X \subseteq \mathbb{A}^n_k$ be an irreducible variety of dimension d. Show that X is contained in no linear subspace of dimension less than d, and give an example where it is contained in one of dimension exactly d.
7. Verify from the definitions that a plane curve has dimension 1, that a finite set has dimension 0, and that $\mathbb{A}^n_k$ has dimension n.

5

Smooth Points and Regular Local Rings

The dimension theory the previous chapter used, and which this one uses again, rests on the Hilbert-Samuel function of an $\mathfrak{m}$-primary ideal and the Dimension Theorem $\delta(R) = d(R) = \mathrm{kdim} R$ it yields, together with systems of parameters. Those are in *Commutative Algebra* [5].

This chapter takes the one consequence that carries geometric content, the tangent-space inequality $\mathrm{kdim} R \leq \dim_k(\mathfrak{m}/\mathfrak{m}^2)$, and turns it into a criterion. A point is *smooth* when the inequality is an equality, which says the tangent space has no more directions than the variety has dimensions. The Jacobian gives a computable test for this, regular local rings give an intrinsic one free of any embedding, and the two agree. The chapter closes with the case that matters most for Part IV: in dimension one, regular is the same as being a discrete valuation ring.

5.1 The Tangent Space Bound

For a Noetherian local ring $(R, \mathfrak{m})$ the Dimension Theorem ([5]) identifies the Krull dimension with the degree of the Hilbert-Samuel polynomial and with the least number of generators of an $\mathfrak{m}$-primary ideal. The one consequence we need geometrically bounds the dimension by the embedding dimension.

Corollary 5.1.1: Dimension and Tangent Space

Let $(R, \mathfrak{m})$ be a Noetherian local ring with residue field $k = R/\mathfrak{m}$. Then

$$\mathrm{kdim}(R) \leq \dim_k(\mathfrak{m}/\mathfrak{m}^2)$$

***Proof*:**
Choose $r_1, \ldots, r_s \in \mathfrak{m}$ whose images form a basis of $\mathfrak{m}/\mathfrak{m}^2$; by Nakayama they generate $\mathfrak{m}$, hence an $\mathfrak{m}$-primary ideal on s generators, so the Dimension Theorem ([5]) gives $\mathrm{kdim} R \leq s = \dim_k(\mathfrak{m}/\mathfrak{m}^2)$.

5.2 The Jacobian Criterion and Regular Local Rings

One of the properties most geometric objects carry is *smoothness*, or nonsingularity. In this section, we shall introduce an intuitive way of defining smoothness given an embedding into an affine/projective space. We shall then show a more intrinsic definition that will not require any embedding.

Definition 5.2.1: Nonsingular Variety - Embedding Variant

Let $X \subseteq \mathbb{A}^n_k$ be an affine variety of dimension r, and let $f_1, \ldots, f_t$ generate its vanishing ideal $\mathcal{I}(X) \subseteq \Gamma(\mathbb{A}^n)$. Then X is *nonsingular* at $p \in X$ if the Jacobian matrix evaluated at p,

$$J(p) = \left(\frac{\partial f_i}{\partial x_j}(p) \right)_{\substack{1 \leq i \leq t \\ 1 \leq j \leq n}}$$

has rank $J(p) = n - r$. We say X is *nonsingular* if it is nonsingular at every point.

If the reader knows differential geometry, this is saying that the tangent space must be "full rank" at every point. The matrix is called the *Jacobian matrix*. Its rank at p does not depend on the chosen generators, even though the matrix itself does: the proof of proposition 5.2.2 below identifies rank $J(p)$ with $n - \dim_k \mathfrak{m}_p/\mathfrak{m}_p^2$, an invariant of the local ring alone. This definition is geometrically well-motivated, but is not intrinsic to the variety: it is dependent on the "global" information of the embedding space and generators. We shall immediately work towards the intrinsic definition of nonsingular variety. The key is that at each point p the local ring $\mathcal{O}_{X,p}$ has a single maximal ideal $\mathfrak{m}_p$ representing where functions evaluate to 0, and $\mathfrak{m}_p/\mathfrak{m}_p^2$ "remembers" only linear terms, making it the best analogy to the cotangent space at a point. Recall that $\dim_k \mathfrak{m}_p/\mathfrak{m}_p^2 \geq \mathrm{kdim} k[X]$ by corollary 5.1.1. The inequality is strict exactly at a singular point, where the tangent space picks up extra dimensions. We thus seek conditions where the local ring at each point shall have the same dimension everywhere:

A local Noetherian ring of dimension d with maximal ideal $\mathfrak{m}$ and residue field k is *regular* when $\dim_k(\mathfrak{m}/\mathfrak{m}^2) = d$, equivalently when $\mathfrak{m}$ can be generated by d elements, equivalently when the associated graded ring $G_{\mathfrak{m}}(R)$ is the polynomial ring $k[t_1, \ldots, t_d]$ on independent degree-one indeterminates. The definition, the equivalence of the three conditions, and the consequences used below are commutative algebra and are developed in *Commutative Algebra*: see [5, Regular Local Ring] and [5, Characterizations of a Regular Local Ring].

Proposition 5.2.2: Jacobian Criterion and Regular Local Rings

Let $X \subseteq \mathbb{A}^n_k$ be a variety. Then X is nonsingular at p iff the local ring at p is regular.

***Proof*:**
Let $p = (a_1, \ldots, a_n) \in \mathbb{A}^n$ and let $\mathfrak{a}_p = (x_1 - a_1, \ldots, x_n - a_n)$ be the corresponding maximal ideal of $A = \Gamma(\mathbb{A}^n) \cong k[x_1, \ldots, x_n]$. Define a k-linear map $\theta_p \colon A \to k^n$ by

$$\theta_p(f) = \left\langle \frac{\partial f}{\partial x_1}(p), \ldots, \frac{\partial f}{\partial x_n}(p) \right\rangle$$

The images $\theta_p(x_i - a_i)$ are the standard basis vectors, so θ_p is surjective and its restriction to $\mathfrak{a}_p$ is still surjective. That restriction has kernel exactly $\mathfrak{a}_p^2$: the product rule gives $\theta_p(fg) = f(p)\theta_p(g) + g(p)\theta_p(f)$, which vanishes when $f, g \in \mathfrak{a}_p$, and conversely a Taylor expansion of $f \in \mathfrak{a}_p$ at p has linear part determined by $\theta_p(f)$. Note the restriction is essential: θ_p kills every constant, so $\ker \theta_p \neq \mathfrak{a}_p^2$ on all of A. Thus θ_p induces an isomorphism $\theta_p' \colon \mathfrak{a}_p/\mathfrak{a}_p^2 \to k^n$.

Now let $\mathfrak{b} = \mathcal{I}(X)$ be the ideal of X in A, and let $f_1, \ldots, f_t$ be a set of generators of $\mathfrak{b}$. Then the rank of the Jacobian matrix $J = (\partial f_i / \partial x_j)(p)$ is just the dimension of $\theta_p(\mathfrak{b})$ as a subspace of k^n. Using the isomorphism θ_p', this is the same as the dimension of the subspace $(\mathfrak{b} + \mathfrak{a}_p^2)/\mathfrak{a}_p^2$ of $\mathfrak{a}_p/\mathfrak{a}_p^2$. On the other hand, the local ring $\mathcal{O}_p$ of p on X is obtained from A by dividing by $\mathfrak{b}$ and localizing at the maximal ideal $\mathfrak{a}_p$. Thus if $\mathfrak{m}$ is the maximal ideal of $\mathcal{O}_p$, we have

$$\mathfrak{m}/\mathfrak{m}^2 \cong \mathfrak{a}_p/(\mathfrak{b} + \mathfrak{a}_p^2)$$

Counting dimensions of vector spaces, we have $\dim \mathfrak{m}/\mathfrak{m}^2 + \operatorname{rank}(J) = n$.

But this gives what we want. If $\dim X = r$ then $\mathcal{O}_p$ has Krull dimension r, since localizing $k[X]$ at the maximal ideal of p leaves the dimension of the irreducible X unchanged by proposition 4.3.2. So $\mathcal{O}_p$ is regular if and only if $\dim_k \mathfrak{m}/\mathfrak{m}^2 = r$. But this is equivalent to $\operatorname{rank}(J) = n - r$, which says that p is a nonsingular point of X, completing the proof.

Example 5.2.3: The Jacobian Criterion on Three Curves
Each of the following is a curve in $\mathbb{A}^2_k$, and the criterion is applied at the origin, which lies on all three. Assume $\operatorname{char} k \neq 2, 3$.

1. *The parabola* $f = y - x^2$. The Jacobian is $(-2x,\ 1)$, of rank 1 everywhere. So every point is nonsingular, and $\dim_k \mathfrak{m}_p/\mathfrak{m}_p^2 = 1 = \dim C$ at each.
2. *The node* $f = y^2 - x^2 - x^3$. The Jacobian is $\left(-2x - 3x^2,\ 2y\right)$, which vanishes at $(0,0)$ and at $(-2/3, 0)$; only the first is on the curve. So the origin is the unique singular point, and there $\dim_k \mathfrak{m}/\mathfrak{m}^2 = 2 > 1$.
3. *The cusp* $f = y^2 - x^3$. The Jacobian is $\left(-3x^2,\ 2y\right)$, vanishing only at the origin, which is on the curve. Again the origin is the unique singular point with $\dim_k \mathfrak{m}/\mathfrak{m}^2 = 2$.

The criterion separates the parabola from the other two but says nothing distinguishing the node from the cusp: both are singular points of multiplicity 2. Telling them apart needs the tangent cone of definition 6.1.2, taken up in the next chapter, which sees two tangent lines at the node and one doubled at the cusp.

From this, we can extend the definition naturally to any quasiprojective variety.

Definition 5.2.4: Nonsingular Variety

Let X be a quasi-projective variety. Then X is *nonsingular* or *regular* if the local ring at every point is regular.

5.3 The Singular Locus

The set of points that are singular are usually "measure 0". We shall not go so far to prove this statement, but the following simpler result that a variety cannot *only* consist of singular points:

Proposition 5.3.1: Singular Points of Variety

Let X be a variety. Then the set of singular points of X forms a proper closed subset of X

The fact that it is closed should be unsurprising as it is essentially the pre-image of the closed set 0 for the right continuous function, the key take-away is that it must be proper.

***Proof*:**
Let $\mathrm{Sing}(X)$ denote the set of singular points of X.

Closed. Being closed is local, so covering X by affine open sets reduces us to X affine, say of dimension r in $\mathbb{A}^n$ with $\mathcal{I}(X) = (f_1, \dots, f_t)$. Combining corollary 5.1.1 with the dimension count in the proof of proposition 5.2.2 gives rank $J(p) \leq n - r$ at every p, with equality exactly at the nonsingular points. So the singular points are those where the rank is *strictly* less than $n - r$. Thus $\mathrm{Sing}(X)$ is the algebraic set defined by the ideal generated by $I(X)$ together with all determinants of $(n-r) \times (n-r)$ sub-matrices of the matrix $\|\partial f_i / \partial x_j\|$. From these, we can see that $\mathrm{Sing}\, X$ is closed.

Proper. By theorem 3.4.1, X is birationally isomorphic to a hypersurface $Y \subseteq \mathbb{A}^{d+1}$, where $d = \dim X$; note the theorem produces an *affine* hypersurface, not a projective one. By proposition 9.5.2 a birational isomorphism identifies a dense open subset of X with one of Y, and nonsingularity is a local condition, so it is enough to exhibit a single nonsingular point on Y. Thus we may assume X is a hypersurface in $\mathbb{A}^n$ cut out by one irreducible polynomial $f(x_1, \dots, x_n)$.

Now, $\mathrm{Sing}(X)$ is the set of points $p \in X$ such that $(\partial f / \partial x_i)(p) = 0$ for $i = 1, \dots, n$. If $\mathrm{Sing}\, X = X$, then the functions $\partial f / \partial x_i$ are zero on X, and hence $\partial f / \partial x_i \in I(X)$ for each i. But $I(X)$ is the principal ideal generated by f, and $\deg(\partial f / \partial x_i) \leq \deg f - 1$ for each i, so we must have $\partial f / \partial x_i = 0$ for each i.

In characteristic 0 this is already impossible, because if x_i occurs in f, then $\partial f / \partial x_i \neq 0$. So we must have char $k = p > 0$, and then the fact that $\partial f / \partial x_i = 0$ implies that f is a polynomial in x_i^p. This is true for each i, so by taking pth roots of the coefficients (which is possible since k is algebraically closed), we get a polynomial $g(x_1, \dots, x_n)$ such that $f = g^p$. But this contradicts the hypothesis that f was irreducible, and so $\mathrm{Sing}\, X \subsetneq X$, as we sought to show.

5.4 Homological Characterization: Serre's Theorem

Before stating the main theorem on regular local rings, we need a homological invariant[1]:

Definition 5.4.1: Projective and Global Dimension

Let R be a ring and M an R-module. The *projective dimension* of M, denoted $\mathrm{pd}_R(M)$, is the minimal length n of a projective resolution of M:

$$0 \to P_n \to \cdots \to P_0 \to M \to 0$$

If no finite resolution exists, $\mathrm{pd}_R(M) = \infty$.

The *global dimension* of R, denoted $\mathrm{gldim}(R)$, is the supremum of the projective dimensions of all R-modules:

$$\mathrm{gldim}(R) = \sup\{\mathrm{pd}_R(M) \big| M \in \mathrm{Mod}_R\}$$

Example 5.4.2: Infinite Global Dimension When There is A Singularity

Take $R = k[x, y]/(x^2 - y^3)$. This corresponds to the cusp singularity at the origin. Let us compute the global dimension by finding the projective dimension of the residue field $k \cong R/(x, y)$ (where $\mathfrak{m} = (\overline{x}, \overline{y})$). We begin a free resolution of k by killing $\mathfrak{m} = (\overline{x}, \overline{y})$:

$$R^2 \xrightarrow{d_1} R \to k \to 0$$

where $d_1 = \begin{pmatrix} x & y \end{pmatrix}$. The kernel of this map consists of relations between x and y. In the polynomial ring, the only relation is trivial ($yx - xy = 0$). However, in our quotient ring R, we have the relation defining the ring: $x^2 - y^3 = 0$. Thus, the kernel is generated by two elements:

1. The "Koszul" relation: $y \cdot (x) - x \cdot (y) = 0$
2. The "Curve" relation: $x \cdot (x) - y^2 \cdot (y) = 0$

So we have a map $R^2 \xrightarrow{d_2} R$ given by the matrix $\begin{pmatrix} y & x \\ -x & -y^2 \end{pmatrix}$. (Check: $\begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} y & x \\ -x & -y^2 \end{pmatrix} = \begin{pmatrix} xy - yx & x^2 - y^3 \end{pmatrix} = \begin{pmatrix} 0 & 0 \end{pmatrix}$).

Now we need the kernel of d_2. Calculating the determinant of the matrix, we find $\det(d_2) = -y^3 + x^2 = 0$. Because the determinant vanishes, the columns are linearly dependent, and the kernel is non-trivial. In fact, the resolution becomes *periodic*. The resolution continues forever with alternating matrices:

$$\cdots \xrightarrow{d_3} R^2 \xrightarrow{d_2} R^2 \xrightarrow{d_1} R \to k \to 0$$

where $d_{\text{odd}} = \begin{pmatrix} y & x \\ -x & -y^2 \end{pmatrix}$ and $d_{\text{even}} = \begin{pmatrix} y^2 & x \\ -x & -y \end{pmatrix}$. Since the resolution never terminates (it never reaches a 0 module), the projective dimension of k is infinite. Therefore, the global dimension of R is infinite.

[1]These terms will be central in [7, Homological Dimension] and will be reintroduced there

For a local ring $(R, \mathfrak{m})$ with residue field k, a fundamental result of Auslander and Buchsbaum states that $\mathrm{gldim}(R) = \mathrm{pd}_R(k)$. Thus, the global dimension is determined entirely by the resolution of the residue field.

Theorem 5.4.3: Auslander-Buchsbaum-Serre Theorem

A local Noetherian ring $(R, \mathfrak{m})$ has finite global dimension if and only if it is a regular local ring.

The proof is homological rather than geometric, and belongs with the machinery that produces it: it runs through the Auslander-Buchsbaum identity $\mathrm{gldim}\, R = \mathrm{pd}_R\, k$ and the Koszul resolution of the residue field, neither of which this book develops. *Homological Algebra* [7, Auslander-Buchsbaum-Serre Theorem] gives it in full; the two books are co-requisites for exactly this reason.

***Proof Key ideas*:**
One direction is the Koszul resolution. If R is regular of dimension d, then $\mathfrak{m}$ is generated by a regular sequence $x_1, \ldots, x_d$, and the Koszul complex on that sequence is a free resolution of $k = R/\mathfrak{m}$ of length d. Hence $\mathrm{pd}_R\, k \leq d < \infty$, and by the Auslander-Buchsbaum identity $\mathrm{gldim}\, R = \mathrm{pd}_R\, k$ is finite.

The converse is the substantive half. Finiteness of $\mathrm{pd}_R\, k$ forces the minimal free resolution of k to terminate, and one shows by induction on $\dim_k \mathfrak{m}/\mathfrak{m}^2$ that this can happen only when a minimal generating set of $\mathfrak{m}$ is a regular sequence, which is regularity. See [7, Auslander-Buchsbaum-Serre Theorem] for the details.

5.5 Regular Local Rings in Dimension One

Let us now take a closer look at the properties regular local rings must have, especially in low dimension. In one dimension, this shall produce a very restrictive result that will be very powerful, and in higher dimensions we shall find an algebraic condition that eventually lets the rings of integers of *Algebraic Number Theory* [4] be read as nonsingular $\mathbb{Z}$-curves. For curves this is the decisive case, and section 7.3 uses it to show that each birational class of curves contains a unique nonsingular projective model, so that a field extension K/k of transcendence degree one determines *the* curve attached to it.

One consequence is needed before the dimension-one case, and it is not visible from the numerical definition: a regular local ring has no zero-divisors. The mechanism is that regularity makes the associated graded ring a polynomial ring, hence a domain, and a domain downstairs lifts to a domain upstairs whenever $\bigcap_n \mathfrak{m}^n = 0$, which Krull's theorem supplies in any Noetherian local ring. Both steps are commutative algebra and are carried out in [5, A Regular Local Ring Is a Domain].

Corollary 5.5.1: Regular Local Rings of Dimension 1

Let R be a local Noetherian ring of dimension 1. Then R is a regular local ring if and only if R is a DVR.

***Proof*:**
Suppose $(R, \mathfrak{m})$ is regular of dimension 1, so $\dim_{R/\mathfrak{m}} \mathfrak{m}/\mathfrak{m}^2 = \mathrm{kdim} R = 1$. Choose $m \in \mathfrak{m}$ whose class spans $\mathfrak{m}/\mathfrak{m}^2$. Nakayama's lemma ([5, Projective Module over Local Ring]) then gives $\mathfrak{m} = (m)$: the single element m generates. By the discussion above R is a domain, and a Noetherian local domain of dimension 1 with principal maximal ideal is a discrete valuation ring ([5, Characterizations of a DVR]).

Conversely let R be a discrete valuation ring with uniformizer m. It is local: its unique maximal ideal is (m), the elements of positive valuation. Note it is *not* true that (m) is its only prime, as (0) is prime too; indeed the chain $(0) \subsetneq (m)$ is what makes $\mathrm{kdim} R = 1$, and no longer chain exists because every nonzero ideal is $(m)^n$. Finally $(m)/(m)^2$ is spanned by the class of m over $R/(m)$ and is nonzero, so $\dim_{R/(m)}(m)/(m)^2 = 1 = \mathrm{kdim} R$ and R is regular, as we sought to show.

This gives a nice condition for 1-dimensional regular local rings. It will sometimes be useful to identify regular local rings in different context:

Proposition 5.5.2: Equivalences of 1 Dimensional Regular Local Rings

Let $(R, \mathfrak{m})$ be a Noetherian local domain of dimension 1. Then the following are equivalent:

1. R is a regular local ring
2. R is a DVR
3. R is normal (integrally closed)
4. $\mathfrak{m}$ is principal
5. every nonzero ideal of R is a power of $\mathfrak{m}$

***Proof*:**
This is a statement about Noetherian local domains with no geometric content, and it is proved in full upstream: see [5, Characterizations of a DVR], which lists exactly these five conditions.

The equivalence of (1) and (3) is the one chapter 7 consumes. It says that for a curve, being smooth at a point and being normal at that point are the same condition, which is why normalization resolves the singularities of a curve completely while leaving higher-dimensional singularities behind.

For higher-dimensions, the situation is more subtle, and the full account belongs with divisors on a scheme; this book records only the following:

Corollary 5.5.3: Regular Local Ring, then Normal

Let R be a regular local Noetherian ring. Then R is normal.

***Proof*:**
This is [5, A Regular Local Ring Is Normal], where it is obtained from factoriality: a regular local ring is a unique factorization domain by the theorem of Auslander and Buchsbaum, and a unique factorization domain is integrally closed. In dimension one the statement is already

contained in proposition 5.5.2, which is the case this book uses.

The converse does not hold in general in dimension > 1, a normal domain need not be regular, so normality alone does not detect nonsingular varieties. Chapter 7 takes up exactly this gap.

Proposition 5.5.4: Completion and Regularity

Let $(R, \mathfrak{m})$ be a Noetherian local ring and $\hat{R}$ its $\mathfrak{m}$-adic completion. Then R is regular if and only if $\hat{R}$ is regular.

***Proof*:**
This is [5, Regularity and Completion]. The mechanism is that completion leaves the associated graded ring alone, $G_{\mathfrak{m}}(R) \cong G_{\hat{\mathfrak{m}}}(\hat{R})$, and leaves the dimension alone; the graded criterion for regularity depends on nothing else, so it holds for one exactly when it holds for the other, as we sought to show.

In this case $\hat{R}$ is again an integral domain.

Exercise 5.5.1

1. Compute the singular locus of $\mathcal{Z}(y^2 - x^3)$, of $\mathcal{Z}(y^2 - x^2 - x^3)$, and of $\mathcal{Z}(xy - z^2) \subseteq \mathbb{A}^3_k$, using proposition 5.2.2 in each case.
2. Show that the singular locus of a variety is a proper closed subset, so that the nonsingular points are dense. Where does the argument use irreducibility?
3. Let $X = \mathcal{Z}(f) \subseteq \mathbb{A}^n_k$ be a hypersurface. Show that $p \in X$ is nonsingular if and only if some $\partial f / \partial x_i$ fails to vanish at p, and give an example in characteristic $p > 0$ where every partial derivative vanishes identically on a nonconstant f.
4. Show that $\dim_k \mathfrak{m}_p / \mathfrak{m}_p^2 \geq \dim X$ at every point of a variety, with equality exactly at the nonsingular points, and identify which inequality of corollary 5.1.1 is being used.
5. Show that a regular local ring is a domain, and give an example of a Noetherian local domain that is not regular.
6. Show that the local ring of the node $\mathcal{Z}(y^2 - x^2 - x^3)$ at the origin is not a domain after completion, although it is a domain before. (This is why $\mathcal{O}_p$ alone cannot distinguish a node from a cusp, and why proposition 5.5.4 is stated for completions.)
7. Use theorem 5.4.3 to show that a regular local ring is a unique factorization domain in dimension one, and explain why the same argument does not immediately give the general case.

6

Tangent Cones and Completions

The Jacobian criterion of chapter 5 decides whether a point is smooth and stops there. It cannot tell the node $y^2 = x^2 + x^3$ from the cusp $y^2 = x^3$: both are singular at the origin, both have a one-dimensional variety with a two-dimensional cotangent space, and no invariant built from $\mathfrak{m}_p/\mathfrak{m}_p^2$ alone separates them. Yet they are visibly different curves, the node crossing itself and the cusp not.

This chapter builds the invariants that see the difference. The first keeps more of the filtration than $\mathfrak{m}_p/\mathfrak{m}_p^2$ does: the associated graded ring of $\mathcal{O}_{X,p}$, whose zero set is the cone of tangent directions and whose Hilbert polynomial is the multiplicity. The second is finer still. Passing to the completion $\hat{\mathcal{O}}_{X,p}$ makes available power series that converge in no algebraic sense, and in that larger ring the node's local equation factors while the cusp's does not; the factors are the branches of the curve through the point. The chapter closes by using regularity to show that on a smooth variety every codimension-one subvariety is locally cut out by one equation, which is the fact section 7.5 needs and could not prove.

6.1 The Tangent Cone and Multiplicity

Throughout, $X \subseteq \mathbb{A}^n_k$ is an affine variety and $p \in X$ a point, and after a translation p is the origin. Write $A = \Gamma(\mathbb{A}^n_k) = k[x_1, \ldots, x_n]$ and $\mathfrak{n} = (x_1, \ldots, x_n)$ for the maximal ideal at the origin.

Definition 6.1.1: Initial Form

For $0 \neq f \in A$ write $f = f_m + f_{m+1} + \cdots$ with each f_j homogeneous of degree j and $f_m \neq 0$. The *initial form* of f is $f_* = f_m$, and m is the *order* of f at the origin. For an ideal $I \subseteq A$, the *initial ideal* is $I_* = (f_* : 0 \neq f \in I)$, the ideal generated by all initial forms.

The initial form records the lowest-order behaviour of f at the origin, so it is the first place a curve's shape shows up: for the node $f = y^2 - x^2 - x^3$ it is $y^2 - x^2$, which factors into two distinct lines when $\operatorname{char} k \neq 2$, while for the cusp $f = y^2 - x^3$ it is y^2, a single line doubled. In characteristic 2 the node's initial form is $(y + x)^2$ and the distinction collapses, so that hypothesis is carried explicitly wherever the node is used. That the initial ideal is generated by initial forms and not merely by the initial forms of a generating set is a real distinction, and the next proposition is what makes the definition intrinsic in spite of it.

Definition 6.1.2: Tangent Cone

The *tangent cone algebra* of X at p is the associated graded ring

$$\operatorname{gr}_{\mathfrak{m}_p} \mathcal{O}_{X,p} = \bigoplus_{j \geq 0} \mathfrak{m}_p^j / \mathfrak{m}_p^{j+1}$$

a graded k-algebra. The *tangent cone* $TC_p(X) \subseteq \mathbb{A}_k^n$ is the affine algebraic set $\mathcal{Z}(I_*)$, where $I = \mathcal{I}(X)$.

Proposition 6.1.3: The Tangent Cone Algebra Is Intrinsic

With the notation above, there is an isomorphism of graded k-algebras

$$\operatorname{gr}_{\mathfrak{m}_p} \mathcal{O}_{X,p} \cong A/I_*$$

In particular the tangent cone algebra does not depend on the embedding $X \subseteq \mathbb{A}_k^n$, and $TC_p(X) = \mathcal{Z}(I_*)$ is the zero set of the tangent cone algebra.

***Proof*:**
Step 1: the local ring may be replaced by the coordinate ring. Put $R = \Gamma(X) = A/I$ with maximal ideal $\mathfrak{m} = \mathfrak{n}/I$ at the origin. Each $\mathfrak{m}^j/\mathfrak{m}^{j+1}$ is annihilated by $\mathfrak{m}$, hence is a vector space over $R/\mathfrak{m} = k$, and localizing at $\mathfrak{m}$ acts as the identity on such a module. So $\operatorname{gr}_{\mathfrak{m}_p} \mathcal{O}_{X,p} \cong \operatorname{gr}_{\mathfrak{m}} R$.

Step 2: the surjection and its kernel. The quotient $A \twoheadrightarrow R$ carries $\mathfrak{n}^j$ onto $\mathfrak{m}^j$, hence induces a surjection of graded rings $\operatorname{gr}_{\mathfrak{n}} A \twoheadrightarrow \operatorname{gr}_{\mathfrak{m}} R$, and $\operatorname{gr}_{\mathfrak{n}} A \cong A$ as graded rings since $\mathfrak{n}^j/\mathfrak{n}^{j+1}$ is the space of forms of degree j. In degree j the kernel is

$$\big((I \cap \mathfrak{n}^j) + \mathfrak{n}^{j+1}\big)/\mathfrak{n}^{j+1}$$

and it remains to identify this with the degree-j part of I_*.

Step 3: the kernel is I_*. For one inclusion, let $f \in I$ have initial form f_* of degree m and let g be homogeneous of degree $j - m$. Then $gf \in I$, and since A is a domain the initial form of gf is gf_*, of degree j; so $gf \in I \cap \mathfrak{n}^j$ and $gf \equiv gf_* \pmod{\mathfrak{n}^{j+1}}$. Hence every degree-$j$ element of I_* lies in the kernel. Conversely let $a \in I \cap \mathfrak{n}^j$. Its initial form a_* has degree at least j. If the degree exceeds j then $a \in \mathfrak{n}^{j+1}$ and a contributes 0; if it equals j then $a \equiv a_* \pmod{\mathfrak{n}^{j+1}}$ with $a_* \in I_*$. So the kernel is exactly the degree-j part of I_*, completing the proof.

The proposition is what licenses the name: the left-hand side is built from $\mathcal{O}_{X,p}$ alone and knows nothing of the embedding, so the right-hand side cannot depend on it either, even though I_* is written in the coordinates of $\mathbb{A}_k^n$.

One caution follows immediately, and it is the first place this book runs into the limitation that chapter 19 takes up. The tangent cone algebra A/I_* and the tangent cone $TC_p(X) = \mathcal{Z}(I_*)$ are not equivalent data: passing from the ideal to its zero set discards nilpotents, so the coordinate ring of $TC_p(X)$ is $A/\sqrt{I_*}$ rather than A/I_*. For the cusp $I_* = (y^2)$, so the tangent cone algebra is $k[x,y]/(y^2)$ while the tangent cone is the line $\mathcal{Z}(y)$, and the doubling is visible only in the algebra. The algebra is therefore the finer invariant, and it is the one the multiplicity is defined from.

Definition 6.1.4: Multiplicity

Let $d = \mathrm{kdim}\mathcal{O}_{X,p}$. The function $j \mapsto \ell(\mathcal{O}_{X,p}/\mathfrak{m}_p^j)$ agrees for $j \gg 0$ with a polynomial in j ([5, The Hilbert-Samuel Polynomial]) whose degree is d ([5, Dimension Theorem]), and whose leading coefficient may be written $e/d!$ for a positive integer e. This e is the *multiplicity* of X at p, written $m_p(X)$.

Two things in that definition need saying. First, the leading coefficient is $e/d!$ with e a positive integer because a polynomial taking integer values at all large integers is an integer combination $\sum_i c_i \binom{j}{i}$ of binomial coefficients, so its degree-d coefficient is $c_d/d!$ with $c_d \in \mathbb{Z}$; here the values $\ell(\mathcal{O}_{X,p}/\mathfrak{m}_p^j)$ are lengths, hence non-negative integers, and $c_d > 0$ since the polynomial has degree exactly d and is eventually increasing. Second, the first difference of $j \mapsto \ell(\mathcal{O}_{X,p}/\mathfrak{m}_p^j)$ is $\dim_k \mathfrak{m}_p^j/\mathfrak{m}_p^{j+1}$, the Hilbert function of the tangent cone algebra, so $m_p(X)$ is determined by that algebra, and by proposition 6.1.3 by the point alone.

Proposition 6.1.5: Multiplicity of a Hypersurface

Let $n \geq 2$ and let $X = \mathcal{Z}(f) \subseteq \mathbb{A}_k^n$ be a hypersurface with f irreducible, and let $p \in X$. Then $I_* = (f_*)$ and

$$m_p(X) = \deg f_* = \text{the order of } f \text{ at } p$$

***Proof*:**
Since f is irreducible, $\mathcal{I}(X) = (f)$ by the Nullstellensatz (theorem 2.4.4), (f) being prime hence radical. Every element of (f) is gf, and A being a domain gives $(gf)_* = g_* f_*$, so every initial form of an element of (f) is a multiple of f_*; hence $I_* = (f_*)$. Write $m = \deg f_*$.

By proposition 6.1.3 the tangent cone algebra is $A/(f_*)$, whose degree-j piece has dimension $\binom{j+n-1}{n-1} - \binom{j-m+n-1}{n-1}$ for $j \geq m$, multiplication by the form f_* being injective on a domain. This is a polynomial in j of degree $n-2$ with leading coefficient $m/(n-2)!$; the identity holds for $j \geq m$, and the finitely many earlier terms shift the sum below by a constant without affecting its degree or leading coefficient. Summing over j raises the degree by one, so $\ell(\mathcal{O}_{X,p}/\mathfrak{m}_p^j)$ has degree $n-1$ and leading coefficient $m/(n-1)!$. By the dimension theorem that degree is $\mathrm{kdim}\mathcal{O}_{X,p}$, so $d = n-1$, and definition 6.1.4 gives $m_p(X) = m$, as we sought to show.

Proposition 6.1.6: Smooth Points Have Multiplicity One

If p is a smooth point of X then $m_p(X) = 1$. For a hypersurface the converse holds: an irreducible hypersurface with $m_p(X) = 1$ is smooth at p.

***Proof*:**
If p is smooth then $\mathcal{O}_{X,p}$ is regular of dimension d (proposition 5.2.2), so its tangent cone algebra is the polynomial ring $k[t_1, \ldots, t_d]$ by [5, Characterizations of a Regular Local Ring]. Then $\dim_k \mathfrak{m}_p^j/\mathfrak{m}_p^{j+1} = \binom{j+d-1}{d-1}$, a polynomial of degree $d-1$ with leading coefficient $1/(d-1)!$, and summing gives leading coefficient $1/d!$; so $m_p(X) = 1$.

For the converse on a hypersurface $X = \mathcal{Z}(f)$, proposition 6.1.5 makes $m_p(X) = 1$ say that f_* is a nonzero linear form. That linear form is $\sum_i (\partial f/\partial x_i)(p)\, x_i$ up to translation, so some partial derivative of f is nonzero at p and the Jacobian has rank one there, which is smoothness by definition 5.2.1, completing the proof.

The converse in fact holds for every variety, not only for hypersurfaces, but its proof needs an unmixedness theorem of Nagata that this book does not develop; see [15]. What the general statement does require is irreducibility. On a reducible algebraic set multiplicity one can fail to force regularity: for $\mathcal{Z}(xz, yz) \subseteq \mathbb{A}^3_k$, a plane meeting a line, the local ring at the origin has $\dim_k \mathfrak{m}^j/\mathfrak{m}^{j+1} = j+2$ for $j \geq 1$, so $\ell(R/\mathfrak{m}^j) = j^2/2 + 3j/2 - 1$ and $m = 1$ at a point where the ring is not even a domain.

Example 6.1.7: Tangent Cones of the Node, the Cusp, and the Quadric Cone

1. *The node* $C = \mathcal{Z}(y^2 - x^2 - x^3)$ at the origin. The initial form is $y^2 - x^2 = (y-x)(y+x)$, so the tangent cone is the pair of lines $\mathcal{Z}(y-x) \cup \mathcal{Z}(y+x)$ and $m_0(C) = 2$. The two lines are the two directions in which the curve leaves the origin.

2. *The cusp* $C = \mathcal{Z}(y^2 - x^3)$ at the origin. The initial form is y^2, so $m_0(C) = 2$ as well, but the tangent cone algebra is $k[x,y]/(y^2)$ and the tangent cone is the single line $\mathcal{Z}(y)$, doubled. So the multiplicity does not separate the node from the cusp, while the tangent cone does: two distinct lines against one line with multiplicity two. This is the distinction example 5.2.3 promised.

3. *The quadric cone* $X = \mathcal{Z}(xy - z^2) \subseteq \mathbb{A}^3_k$ at the origin. Here f is already homogeneous, so $f_* = f$ and the tangent cone is X itself: a cone is its own tangent cone at its vertex. By proposition 6.1.5, $m_0(X) = 2$, and by proposition 6.1.6 the vertex is singular, which exercise set 5.5.1 found by computing the Jacobian.

4. *A smooth point.* For the parabola $\mathcal{Z}(y - x^2)$ at the origin the initial form is y, the tangent cone is the tangent line $\mathcal{Z}(y)$, and $m_0 = 1$. At a smooth point the tangent cone is always the tangent space: [5, Characterizations of a Regular Local Ring] makes the tangent cone algebra a polynomial ring on d degree-one generators, so I_* is generated by linear forms.

For a plane curve the tangent cone is a form in two variables, hence a product of linear factors over the algebraically closed k; its factors are the *tangent lines* at the point and their exponents are the tangent multiplicities. That is the description chapter 12 works with, where the multiplicity of a plane curve at a point is defined directly as the order of its defining polynomial. For an *irreducible* f the two definitions agree, by proposition 6.1.5. They part company otherwise, and deliberately so: that chapter takes a plane curve to be a polynomial up to scalar rather than a zero set, so $\mathcal{Z}(y^2)$ carries multiplicity two there while as a variety it is the line $\mathcal{Z}(y)$, of multiplicity one.

6.2 Completion and Local Parameters

The tangent cone keeps more of the local ring than $\mathfrak{m}_p/\mathfrak{m}_p^2$ does, but it is still a linearization: it records the graded pieces $\mathfrak{m}_p^j/\mathfrak{m}_p^{j+1}$ and forgets how they are glued. The completion forgets nothing. It is the inverse limit of the quotients $\mathcal{O}_{X,p}/\mathfrak{m}_p^j$, so an element of it is a compatible system of Taylor approximations to arbitrary order, and the ring it lives in is large enough to contain power series that no polynomial fraction represents.

The construction of the $\mathfrak{m}$-adic completion, its Noetherianity, and the Cohen structure theorem are in *Commutative Algebra* [5, Completion Of A Ring] and are used here without redevelopment. Write $\hat{\mathcal{O}}_{X,p} = \varprojlim_j \mathcal{O}_{X,p}/\mathfrak{m}_p^j$ and $\hat{\mathfrak{m}}_p$ for its maximal ideal. The natural map $\mathcal{O}_{X,p} \to \hat{\mathcal{O}}_{X,p}$ is injective, since its kernel is $\bigcap_j \mathfrak{m}_p^j$, which vanishes in a Noetherian local ring by [5, Krull Theorem, Hausdorff Topology]. So completing loses no functions and adds many.

Theorem 6.2.1: The Completion at a Smooth Point

Let X be a variety over the algebraically closed field k and let $p \in X$ be a smooth point with $\dim_p X = d$. Then

$$\hat{\mathcal{O}}_{X,p} \cong k[[t_1, \ldots, t_d]]$$

as k-algebras, and the t_i may be taken to be the images of any regular system of parameters of $\mathcal{O}_{X,p}$.

***Proof*:**
Write $(R, \mathfrak{m}) = (\mathcal{O}_{X,p}, \mathfrak{m}_p)$, a Noetherian local ring of dimension d. Its residue field is k: it equals $k[X]/\mathcal{I}(p)$ for an affine neighbourhood, and Zariski's lemma (theorem 2.4.2) makes that a finite extension of the algebraically closed k, hence k itself. Since p is smooth, R is regular (proposition 5.2.2).

Step 1: the completion is complete regular local of dimension d. The ring $\hat{R}$ is Noetherian local with maximal ideal $\hat{\mathfrak{m}}$ and the same residue field k, by [5, Noetherian Closed Under Completion], and $\mathrm{kdim}\hat{R} = \mathrm{kdim} R = d$ by [5, Completion Preserves Dimension]. It is regular by proposition 5.5.4.

Step 2: Cohen's theorem gives a presentation. The ring $\hat{R}$ is complete Noetherian local, contains k, and has residue field k, so [5, Cohen Structure Theorem] presents it as

$$\hat{R} \cong k[[x_1, \ldots, x_n]]/I, \qquad n = \dim_k \hat{\mathfrak{m}}/\hat{\mathfrak{m}}^2$$

Regularity of $\hat{R}$ makes $n = \mathrm{kdim}\hat{R} = d$, so the presentation is $k[[x_1, \ldots, x_d]]/I$, and Cohen's theorem takes the x_i to any lift of a basis of $\hat{\mathfrak{m}}/\hat{\mathfrak{m}}^2$, which a regular system of parameters of R provides.

Step 3: the ideal vanishes. This is lemma 6.2.2 below, applied to $\hat{R}$, completing the proof.

The last step is worth isolating, since it is used again in section 6.3 and its argument is independent of the geometry.

Lemma 6.2.2: A Complete Regular Local Ring Is a Power Series Ring

Let $(T, \mathfrak{n})$ be a complete Noetherian regular local ring of dimension d containing a field k with $T/\mathfrak{n} = k$. Then $T \cong k[[x_1, \ldots, x_d]]$, the variables mapping to any regular system of parameters.

***Proof*:**
[5, Cohen Structure Theorem] presents $T \cong S/I$ with $S = k[[x_1, \ldots, x_n]]$ and $n = \dim_k \mathfrak{n}/\mathfrak{n}^2$, which regularity makes d; its proof sends the variables to any lift of a basis of $\mathfrak{n}/\mathfrak{n}^2$, that is to any regular system of parameters. It remains to show $I = 0$.

First, S is a domain. A nonzero element of S has a well-defined initial form, its lowest-degree homogeneous part, and the initial form of a product is the product of the initial forms because the polynomial ring $k[x_1, \ldots, x_d]$ is a domain; so a product of nonzero elements has a nonzero initial form and is nonzero. (Completion does not preserve domains in general, so this needs the argument rather than a citation.)

Second, $\mathrm{kdim} S = d$: it is the completion of $k[x_1, \ldots, x_d]_{(x_1,\ldots,x_d)}$, whose dimension is d by corollary 4.2.7 and proposition 4.3.2, and [5, Completion Preserves Dimension] applies.

Now suppose $I \neq 0$ and pick $0 \neq f \in I$; it is a nonunit, I being proper. Any chain of primes $\mathfrak{q}_0 \subsetneq \cdots \subsetneq \mathfrak{q}_e$ in S/I lifts to a chain of primes of S all containing f, hence all nonzero, so prepending the zero ideal, legitimate as S is a domain, gives a chain of length $e+1$ in S. Therefore $e + 1 \leq \mathrm{kdim} S = d$, so $\mathrm{kdim} S/I \leq d - 1$. But $S/I \cong T$ has dimension d, a contradiction. So $I = 0$, as we sought to show.

Definition 6.2.3: Local Parameters

A *system of local parameters* at a smooth point p of a d-dimensional variety is a regular system of parameters $t_1, \ldots, t_d$ of $\mathcal{O}_{X,p}$, that is, d elements of $\mathfrak{m}_p$ generating it. For a curve, $d = 1$ and a single generator t of $\mathfrak{m}_p$ is a *local parameter* or *uniformizer* at p.

Corollary 6.2.4: Power Series Expansion

Let p be a smooth point of X with local parameters $t_1, \ldots, t_d$. Every $f \in \mathcal{O}_{X,p}$ has a unique expansion as a formal power series in $t_1, \ldots, t_d$ with coefficients in k, and $f \in \mathfrak{m}_p^j$ if and only if that series has no terms of degree less than j.

***Proof*:**
The composite $\mathcal{O}_{X,p} \hookrightarrow \hat{\mathcal{O}}_{X,p} \cong k[[t_1, \ldots, t_d]]$ is injective by the remark before theorem 6.2.1, which gives existence and uniqueness of the expansion. Under the isomorphism $\hat{\mathfrak{m}}_p^j$ corresponds to the series of order at least j, so it remains to see that $\mathcal{O}_{X,p} \cap \hat{\mathfrak{m}}_p^j = \mathfrak{m}_p^j$. For a Noetherian local ring, [5, Quotient in Completion] identifies $\hat{\mathcal{O}}/\hat{\mathfrak{m}}_p^j$ with $\mathcal{O}_{X,p}/\mathfrak{m}_p^j$ compatibly with the map from $\mathcal{O}_{X,p}$, so that map composed with the projection is the projection $\mathcal{O}_{X,p} \to \mathcal{O}_{X,p}/\mathfrak{m}_p^j$, whose kernel is $\mathfrak{m}_p^j$, completing the proof.

For a curve the corollary is the statement that a regular function near a smooth point is a power series in the uniformizer, and the order of that series is the order of vanishing; definition 12.2.4 will take it as the definition. In higher dimension it is the algebraic substitute for a coordinate chart: a smooth point has no neighbourhood isomorphic to an open set of $\mathbb{A}^d_k$ in the Zariski topology, but it always has a formal one.

Example 6.2.5: Completions at Smooth and Singular Points

1. *Affine space.* At the origin of $\mathbb{A}^d_k$ the coordinates $x_1, \dots, x_d$ are local parameters and $\hat{\mathcal{O}} \cong k[[x_1, \dots, x_d]]$, which is [5, Completion of Rings] read geometrically.
2. *A smooth plane curve.* On $C = \mathcal{Z}(y - x^2)$ the function x generates $\mathfrak{m}_0$, since $y = x^2$ in $\Gamma(C)$, so x is a uniformizer and $\hat{\mathcal{O}}_{C,0} \cong k[[x]]$. Every regular function near the origin is a power series in x alone, the variable y contributing nothing new.
3. *The parabola meets its axis.* On the same curve the function y has expansion x^2, so $\mathrm{ord}_0(y) = 2$: the axis $\mathcal{Z}(y)$ meets C doubly at the origin. That doubling is the intersection multiplicity chapter 12 will count.
4. *A singular point is not a power series ring.* At the origin of the cusp $\mathcal{Z}(y^2 - x^3)$ the ring $\hat{\mathcal{O}}$ is $k[[x, y]]/(y^2 - x^3)$, computed in proposition 6.3.1 below. It is not isomorphic to $k[[t]]$, since $\dim_k \hat{\mathfrak{m}}/\hat{\mathfrak{m}}^2 = 2$ there while $k[[t]]$ has $\dim_k (t)/(t^2) = 1$; theorem 6.2.1 of course forbids it, the origin being singular.

6.3 Formal Branches

The completion separates the node from the cusp, and it does so in the most direct way available: at the node the local equation factors, and at the cusp it does not. Neither factorization exists in $\mathcal{O}_{X,p}$, because the factors involve a square root that is a power series and not a rational function. This section makes that precise and attaches to each singular point of a curve a number, the count of branches through it.

Proposition 6.3.1: The Completion of a Plane Curve

Let $C = \mathcal{Z}(f) \subseteq \mathbb{A}^2_k$ be a plane curve with f irreducible and $p = (0, 0) \in C$. Then

$$\hat{\mathcal{O}}_{C,p} \cong k[[x, y]]/(f)$$

***Proof*:**
Write $A = k[x, y]$, $\mathfrak{n} = (x, y)$ and $R = \Gamma(C) = A/(f)$, with $\mathfrak{m}$ the image of $\mathfrak{n}$.

Step 1: the local ring may be replaced by the coordinate ring. For every j the ideal $\mathfrak{m}^j$ is $\mathfrak{m}$-primary, so $R/\mathfrak{m}^j$ is already local with maximal ideal $\mathfrak{m}/\mathfrak{m}^j$ and localizing at $\mathfrak{m}$ does nothing: $R/\mathfrak{m}^j \cong R_\mathfrak{m}/(\mathfrak{m}R_\mathfrak{m})^j$. Taking inverse limits, the $\mathfrak{m}$-adic completion of R agrees with the $\mathfrak{m}_p$-adic completion of $\mathcal{O}_{C,p} = R_\mathfrak{m}$. The same argument identifies the $\mathfrak{n}$-adic completion of A with that of $A_\mathfrak{n}$, and the former is $k[[x, y]]$ by [5, Completion of Rings].

Step 2: complete the presentation. Completion is exact on finitely generated modules over a Noetherian ring ([5, Completion Preserves Exactness]), so applying it to $0 \to (f) \to A \to R \to 0$ gives $\hat{R} \cong k[[x,y]]/(f)k[[x,y]]$, and $(f)k[[x,y]]$ is the ideal generated by f there. Combining with Step 1 gives the claim, as we sought to show.

Definition 6.3.2: Formal Branch

Let p be a point of a variety X. A *formal branch* of X at p is a minimal prime of $\hat{\mathcal{O}}_{X,p}$. The number of formal branches is the number of such primes, which is finite because $\hat{\mathcal{O}}_{X,p}$ is Noetherian.

A smooth point has one branch, since $\hat{\mathcal{O}}_{X,p}$ is then a power series ring and hence a domain, whose unique minimal prime is 0. The content of the definition is at singular points, where a curve may pass through p more than once.

Proposition 6.3.3: The Node Has Two Branches, the Cusp One

At the origin,

1. if $\operatorname{char} k \neq 2$, the node $N = \mathcal{Z}(y^2 - x^2 - x^3)$ has exactly two formal branches, and $\hat{\mathcal{O}}_{N,0} \cong k[[u,v]]/(uv)$;
2. in every characteristic, the cusp $C = \mathcal{Z}(y^2 - x^3)$ has exactly one formal branch, and $\hat{\mathcal{O}}_{C,0}$ is a domain that is not a power series ring.

***Proof*:**
The node. In $k[[x]]$ the series $1 + x$ has a square root when $\operatorname{char} k \neq 2$. Seek $w = 1 + \sum_{n \geq 1} a_n x^n$ with $w^2 = 1 + x$; comparing coefficients gives $2a_1 = 1$ and, for $n \geq 2$,

$$2a_n = -\sum_{0<i<n} a_i a_{n-i}$$

Each equation determines a_n from its predecessors, division by 2 being the only division performed, so w exists, and it is a unit since $w(0) = 1$. Then in $k[[x,y]]$

$$y^2 - x^2 - x^3 = y^2 - x^2(1+x) = (y - xw)(y + xw)$$

Put $u = y - xw$ and $v = y + xw$. Both have order 1 with linear parts $y - x$ and $y + x$, which are linearly independent as $\operatorname{char} k \neq 2$, so their classes span $\mathfrak{n}/\mathfrak{n}^2$ for $\mathfrak{n}$ the maximal ideal of $k[[x,y]]$, and Nakayama's lemma ([5, Nakayama's Lemma I]) makes u, v generate $\mathfrak{n}$. Thus u, v is a regular system of parameters of the complete regular local ring $k[[x,y]]$, and lemma 6.2.2 gives $k[[x,y]] = k[[u,v]]$. By proposition 6.3.1, $\hat{\mathcal{O}}_{N,0} \cong k[[u,v]]/(uv)$, whose minimal primes are (u) and (v): these are prime because the quotients $k[[v]]$ and $k[[u]]$ are domains, they are minimal because $(uv) \subseteq \mathfrak{p}$ forces $u \in \mathfrak{p}$ or $v \in \mathfrak{p}$, and they are distinct. So there are exactly two branches.

The cusp. By proposition 6.3.1, $\hat{\mathcal{O}}_{C,0} \cong k[[x,y]]/(y^2 - x^3)$, so it suffices to show $y^2 - x^3$ is irreducible in $k[[x,y]]$; then $(y^2 - x^3)$ is prime, it is the unique minimal prime of the quotient, and the quotient is a domain. Suppose $y^2 - x^3 = gh$ with g, h nonunits. Setting $x = 0$ gives $y^2 = g(0,y)h(0,y)$ in $k[[y]]$, so the orders of $g(0,y)$ and $h(0,y)$ in y sum to 2; neither can be 0, since a factor with $g(0,y)$ a unit would be a unit in $k[[x,y]]$, so both orders are 1. The

Weierstrass preparation theorem ([5, Weierstrass Preparation Theorem]) then writes each as a unit times a monic degree-one polynomial in y over $k[[x]]$, so

$$y^2 - x^3 = \varepsilon\,(y - a(x))(y - b(x)), \qquad a, b \in (x) \subseteq k[[x]]$$

with ε a unit. Here $(y-a)(y-b)$ is a Weierstrass polynomial of degree two, its lower coefficients lying in (x), and so is $y^2 - x^3$ itself; the uniqueness clause of the same theorem therefore forces $\varepsilon = 1$. Comparing coefficients gives $a + b = 0$ and $ab = -x^3$, so $b = -a$ and $a^2 = x^3$. Taking orders in $k[[x]]$ gives $2\,\mathrm{ord}(a) = 3$, impossible. So $y^2 - x^3$ is irreducible.

Finally $\hat{\mathcal{O}}_{C,0}$ is not a power series ring, since $\dim_k \hat{\mathfrak{m}}/\hat{\mathfrak{m}}^2 = 2$ while $\mathrm{kdim} = 1$, completing the proof.

Definition 6.3.4: Analytically Isomorphic Singularities

Two pointed varieties (X, p) and (Y, q) are *analytically isomorphic* if $\hat{\mathcal{O}}_{X,p} \cong \hat{\mathcal{O}}_{Y,q}$ as k-algebras.

The computation says more than that the node has two branches: it says the node is analytically isomorphic to the crossing of the two coordinate axes $\mathcal{Z}(xy)$ at the origin, since both completions are $k[[u, v]]/(uv)$, while the cusp belongs to a different class. This is a genuinely finer relation than isomorphism of neighbourhoods: the node and the pair of axes are not isomorphic as varieties near the origin, one being irreducible and the other not, and it is exactly the passage to the completion that makes them agree. The same argument applies to any plane curve whose tangent cone at p is two distinct lines, which definition 12.1.6 will call an ordinary double point.

The characteristic hypothesis in the first case is not a technicality. In characteristic two, $y^2 - x^2 - x^3 = (y+x)^2 - x^3$, so substituting $u = y + x$ turns the node's equation into $u^2 - x^3$: the curve is analytically a cusp there, with one branch rather than two. The tangent cone sees the same collapse, its initial form being the doubled line $(y + x)^2$.

The count of branches is a local invariant that a global construction also computes: the normalization of a curve separates its branches, so the points lying over p should correspond to them. Making that precise needs the normalization, which is chapter 7, and the comparison is carried out there as proposition 7.2.4.

6.4 Local Factoriality and Weil Divisors

The last local property of this chapter is the one section 7.5 depends on. A Weil divisor on a normal variety is a formal sum of codimension-one subvarieties, and the class group measures how far such a sum is from being the divisor of a function. That measurement is only meaningful once one knows when a codimension-one subvariety is cut out by a single equation near each point, and the answer is that smoothness suffices.

Corollary 6.4.1: A Smooth Variety Is Locally Factorial

Let X be a variety and $p \in X$ a smooth point. Then $\mathcal{O}_{X,p}$ is a unique factorization domain.

***Proof*:**
By proposition 5.2.2 the ring $\mathcal{O}_{X,p}$ is regular local, and a regular local ring is factorial by [5, Regular Local Rings Are Factorial], as we sought to show.

Theorem 6.4.2: Codimension One Is Locally Principal at a Smooth Point

Let X be a variety, $Y \subseteq X$ an irreducible closed subvariety of codimension one, and $p \in Y$ a point at which X is smooth. Then the ideal of Y in $\mathcal{O}_{X,p}$ is principal: there is $f \in \mathcal{O}_{X,p}$ generating it.

***Proof*:**
Step 1: reduce to the affine case. Choose an affine open $U \subseteq X$ containing p. Passing to U changes neither $\mathcal{O}_{X,p}$ nor the codimension, so $Y \cap U$ is an irreducible closed subvariety of U of codimension one, and its ideal $\mathcal{I}(Y \cap U) \subseteq k[U]$ is prime and contained in the maximal ideal at p, the point p lying on Y.

Step 2: the ideal has height one. By definition 4.3.1 the codimension of $Y \cap U$ in U is the supremum of lengths of chains of irreducible closed subsets ascending from $Y \cap U$, and the correspondence between irreducible closed subsets and prime ideals reverses inclusions (theorem 2.4.4), so such chains correspond to chains of primes descending from $\mathcal{I}(Y \cap U)$. Hence $\operatorname{ht} \mathcal{I}(Y \cap U) = \operatorname{codim}_U(Y \cap U) = 1$. Localizing at p preserves both primality and height, since the primes contained in $\mathcal{I}(Y \cap U)$ are exactly those surviving in the localization.

Step 3: apply factoriality. The ring $R = \mathcal{O}_{X,p}$ is a Noetherian local domain by proposition 3.2.10 and factorial by corollary 6.4.1, p being a smooth point. A height-one prime of a Noetherian factorial domain is principal, by [5, Factoriality by Height-One Primes], which gives the generator f, completing the proof.

The theorem is what makes the class group of section 7.5 a global invariant on a smooth variety: every prime divisor is locally the zero set of one function, so a nonzero class records an obstruction to patching those local equations into a single global one, not a failure at any individual point. On a singular normal variety the obstruction is already local, and the quadric cone is the standard witness.

Proposition 6.4.3: The Quadric Cone Is Not Locally Factorial at Its Vertex

Let $X = \mathcal{Z}(xy - z^2) \subseteq \mathbb{A}^3_k$ with vertex v the origin. Then $\mathcal{O}_{X,v}$ is not a unique factorization domain, and the ideal of the ruling $L = \mathcal{Z}(x, z) \subseteq X$ is a height-one prime of $\mathcal{O}_{X,v}$ that is not principal.

***Proof*:**
Write $R = \mathcal{O}_{X,v}$ with maximal ideal $\mathfrak{m}$, and let $\overline{x}, \overline{y}, \overline{z}$ be the images of the coordinates. Each lies in $\mathfrak{m} \setminus \mathfrak{m}^2$: the images of x, y, z form a basis of $\mathfrak{m}/\mathfrak{m}^2$, because the defining equation $xy - z^2$ lies in $(x, y, z)^2$ and so imposes no linear condition on $\mathfrak{m}/\mathfrak{m}^2$. An element of $\mathfrak{m} \setminus \mathfrak{m}^2$ is irreducible in R: a factorization into two nonunits would place it in $\mathfrak{m}^2$. So $\overline{x}, \overline{y}, \overline{z}$ are irreducible, and no two are associates, their classes in $\mathfrak{m}/\mathfrak{m}^2$ being linearly independent while an associate would differ by a unit and hence have proportional class.

The relation $\overline{x}\,\overline{y} = \overline{z}^{\,2}$ therefore exhibits two factorizations of one element into irreducibles that do not match up to associates and reordering, so R is not factorial. Here R is a Noetherian local domain by proposition 3.2.10, the cone being a variety, so [5, Factoriality by Height-One Primes] applies and some height-one prime of R fails to be principal.

That prime may be named. The ideal $\mathfrak{p} = (\overline{x}, \overline{z})$ is the ideal of L in R, and it has height one, L having codimension one in X. Suppose $\mathfrak{p} = (f)$. Then f divides $\overline{x}$ and is a nonunit, and $\overline{x}$ is irreducible, so $(f) = (\overline{x})$. From $\overline{z} \in (\overline{x})$ write $\overline{z} = \overline{x}h$; then

$$\overline{x}\,\overline{y} = \overline{z}^{\,2} = \overline{x}^{\,2}h^2$$

and cancelling $\overline{x}$, legitimate in a domain, gives $\overline{y} = \overline{x}h^2$. As $\overline{y}$ is irreducible and $\overline{x}$ is not a unit, h^2 must be a unit, making $\overline{y}$ and $\overline{x}$ associates and contradicting what was shown above. So $\mathfrak{p}$ is not principal, as we sought to show.

The failure is concentrated at the vertex alone: away from v the cone is smooth, so theorem 6.4.2 applies at every other point of L and makes the ideal of L principal there. Note that L is nonetheless the zero set of a single function, since $\mathcal{Z}(x) \cap X = \mathcal{Z}(x, z^2) \cap X = L$ as a set; what fails is that no one function generates the ideal near v, and $\operatorname{div}(x) = 2L$ in section 7.5 is the quantitative form of that failure, the function that does vanish exactly on L vanishing to order two.

Exercise 6.4.1

1. Compute the tangent cone and the multiplicity at the origin of $C = \mathcal{Z}(y^2 - x^5) \subseteq \mathbb{A}^2_k$, and show that C is irreducible. How do the answers compare with the cusp $\mathcal{Z}(y^2 - x^3)$, and what does the comparison say about whether the tangent cone alone classifies plane singularities?
2. Let $\operatorname{char} k \neq 3$ and let $F = \mathcal{Z}(x^3 + y^3 - 3xy) \subseteq \mathbb{A}^2_k$ be the folium. Compute its tangent cone and multiplicity at the origin, and decide whether the origin is an ordinary double point.
3. Show that if $f \in k[x_1, \ldots, x_n]$ is homogeneous and irreducible, then the tangent cone of $\mathcal{Z}(f)$ at the origin is $\mathcal{Z}(f)$ itself. Why is this what the word *cone* is meant to record?
4. Let p be a smooth point of a variety X with $\dim_p X = d$. Show that $TC_p(X)$ is a linear subspace of $\mathbb{A}^n_k$ of dimension d, and identify it with the tangent space.
5. The node $N = \mathcal{Z}(y^2 - x^2 - x^3)$ passes through $(-1, 0)$ as well as the origin. Show that N is smooth at $(-1, 0)$ and compute $m_{(-1,0)}(N)$ and the tangent cone there.
6. Let $\operatorname{char} k \neq 2$ and let $C = \mathcal{Z}(x^2 + y^2 - 1)$. Show that y is a local parameter at $(1, 0)$ and that $\hat{\mathcal{O}}_{C,(1,0)} \cong k[[y]]$. Write the expansion of $x - 1$ as a power series in y to order three.
7. Let C be a curve and $p \in C$. Show that if $\hat{\mathcal{O}}_{C,p} \cong k[[t]]$ then p is a smooth point of C, so the converse of theorem 6.2.1 holds in dimension one.
8. Show that $k[[x, y]]/(xy)$ has exactly two minimal primes, and conclude that the union of the two coordinate axes has two formal branches at the origin.
9. Show that analytically isomorphic pointed varieties have the same multiplicity and the same number of formal branches. Deduce that the node and the cusp are not analytically isomorphic when $\operatorname{char} k \neq 2$, and explain what happens to this deduction in characteristic two.
10. Show that every irreducible closed subvariety of $\mathbb{A}^n_k$ of codimension one is the zero set of a

single irreducible polynomial, and that its ideal is principal in every local ring of $\mathbb{A}^n_k$. Which is the stronger statement, and why does theorem 6.4.2 not already give it?

11. Let p be a smooth point of X. Show that $\mathcal{O}_{X,p}$ is integrally closed in its fraction field, so that X is normal at every smooth point.

12. For the quadric cone $X = \mathcal{Z}(xy - z^2)$ and the ruling $L = \mathcal{Z}(x, z)$, show directly that $\mathcal{I}(L)$ is a prime of height one in $k[X]$, and that $\mathcal{Z}(x) \cap X = L$ as sets.

13. Show that the tangent cone algebra of the cusp at the origin is $k[x, y]/(y^2)$, and compute $\dim_k \mathfrak{m}^j/\mathfrak{m}^{j+1}$ for every j. Confirm that the multiplicity read from this Hilbert function is 2.

14. Let R be the local ring of $\mathcal{Z}(xz, yz) \subseteq \mathbb{A}^3_k$ at the origin, a plane meeting a line. Show $\dim_k \mathfrak{m}^j/\mathfrak{m}^{j+1} = j + 2$ for $j \geq 1$, deduce $m = 1$, and explain why this does not contradict proposition 6.1.6.

7

Normal Varieties and Normalization

A variety can fail to be smooth in two quite different ways, and only one of them is repairable by an isomorphism of the function field. Normality is the condition that isolates the repairable case: a variety is normal when its coordinate ring is integrally closed in its function field, and every variety admits a normalization, a finite birational map onto it from a normal one.

For curves normality and smoothness coincide, and the normalization is the smooth model. That gives this chapter's structural result: a field of transcendence degree one over k determines a unique nonsingular projective curve, so a curve and its function field are the same information.

7.1 Integrally Closed Domains and Normal Varieties

The map $\varphi\colon \mathbb{A}^1_k \to \mathcal{Z}(x^3 - y^2)$ given by $t \mapsto (t^2, t^3)$ of example 1.4.8 is a bijective birational morphism that is not an isomorphism. Its inverse is y/x, a rational function that is not regular at the origin. That failure has a purely algebraic signature. The coordinate ring is $k[t^2, t^3]$, its function field is $k(t)$, and the missing element $t = t^3/t^2$ satisfies

$$T^2 - t^2 = 0$$

a monic equation over $k[t^2, t^3]$. So t is integral over the coordinate ring without belonging to it: the ring is not integrally closed in its own fraction field. Normality is the condition that this cannot happen, and it will turn out to be exactly the condition under which the pathology above is absent.

Integral dependence, integral closure, and the term *normal domain* are developed in *Commutative Algebra* [5, Integral Elements] and are used here without redevelopment.

Definition 7.1.1: Normal Variety

Let X be an affine variety, so that $k[X]$ is an integral domain with fraction field $k(X)$. Then X is *normal* if $k[X]$ is integrally closed in $k(X)$.

More locally, X is *normal at* $p \in X$ if the stalk $\mathcal{O}_p(X)$ is integrally closed in $k(X)$.

The two notions agree, which is what makes normality a usable local condition rather than a global accident.

Proposition 7.1.2: Normality Is Local

An affine variety X is normal if and only if it is normal at every point.

***Proof*:**
By proposition 3.2.9, $k[X] = \bigcap_{p \in X} \mathcal{O}_p(X)$, and each $\mathcal{O}_p(X)$ is the localization of $k[X]$ at the maximal ideal $\mathcal{I}(p)$. Integral closure commutes with localization ([5, Normalization Commutes with Localization]), which gives exactly the statement that a domain is normal precisely when all of its localizations at maximal ideals are, as we sought to show.

The first supply of normal varieties comes from a purely algebraic observation.

Proposition 7.1.3: Unique Factorization Implies Normal

Every unique factorization domain is integrally closed in its fraction field.

***Proof*:**
Let R be a UFD with fraction field K and let $x = a/b \in K$ be integral over R, written in lowest terms so that a and b share no irreducible factor. An integral relation

$$\left(\frac{a}{b}\right)^n + r_{n-1}\left(\frac{a}{b}\right)^{n-1} + \cdots + r_0 = 0 \qquad r_i \in R$$

becomes, after multiplying through by b^n,

$$a^n = -b\left(r_{n-1}a^{n-1} + r_{n-2}a^{n-2}b + \cdots + r_0 b^{n-1}\right)$$

So every irreducible factor of b divides a^n, and being prime in a UFD, divides a. As a and b are coprime, b has no irreducible factors at all, so b is a unit and $x \in R$, as we sought to show.

Example 7.1.4: Normal and Non-Normal Varieties

1. Affine space $\mathbb{A}_k^n$ is normal. Its coordinate ring is the polynomial ring $k[x_1, \ldots, x_n]$, a UFD ([11, Polynomial Rings in Several Variables over a UFD are UFDs]), hence normal by proposition 7.1.3.
2. The cuspidal cubic $C = \mathcal{Z}(x^3 - y^2) \subseteq \mathbb{A}_k^2$ is not normal, by the computation opening this section: $t = y/x$ lies in $k(C)$ and is integral over $k[C]$ but not in it. The failure is

concentrated at one point, the cusp: away from the origin x is invertible, so y/x is already regular there and C is normal at every other point.

3. The nodal cubic $N = \mathcal{Z}(y^2 - x^2 - x^3)$ is not normal either. Its parameterization $t \mapsto (t^2 - 1, t(t^2 - 1))$ from example 3.3.8 identifies $k(N)$ with $k(t)$, and $t = y/x$ is again integral over $k[N]$ without lying in it. Here the failure sits at the node, where two branches of the curve cross.

4. The quadric cone $Q = \mathcal{Z}(xy - z^2) \subseteq \mathbb{A}^3_k$ *is* normal, and it is singular at the origin. Section 7.4 returns to this: in dimension one normality and smoothness coincide, but from dimension two onward normality is strictly weaker.

Items (2) and (3) share a shape. In both, the curve is the image of a map from a smooth curve that is injective away from a small set, and the element of the function field that fails to be regular is the one that undoes the map. Item (4) shows the shape is special to curves.

7.2 The Normalization of an Affine Variety

Given a variety that is not normal, one can ask for the closest normal variety above it. The construction is forced: take the integral closure of the coordinate ring, and read it back as a variety. The only thing that needs proof is that the result is again the coordinate ring of a variety, and that is exactly the finiteness theorem for integral closures.

Theorem 7.2.1: Existence of the Normalization

Let X be an affine variety and let $\widetilde{k[X]}$ be the integral closure of $k[X]$ in $k(X)$. Then there is an affine variety $\tilde{X}$, the *normalization* of X, with

$$k[\tilde{X}] = \widetilde{k[X]}$$

together with a morphism $\nu\colon \tilde{X} \to X$ induced by the inclusion $k[X] \hookrightarrow \widetilde{k[X]}$. Moreover:

1. $\tilde{X}$ is normal;
2. $k[\tilde{X}]$ is a finite $k[X]$-module;
3. ν is surjective with finite fibres, and induces an isomorphism $k(\tilde{X}) \cong k(X)$, so ν is birational;
4. ν restricts to an isomorphism over the set of points at which X is already normal.

***Proof*:**
Step 1: the ring is a coordinate ring. $k[X]$ is a domain, finitely generated as a k-algebra, so [5, Finiteness of Integral Closure] applies with $L = \operatorname{Frac}(k[X]) = k(X)$: the integral closure $\widetilde{k[X]}$ is a finite $k[X]$-module, and in particular is itself a finitely generated k-algebra. This gives (2). Being a subring of the field $k(X)$ it is a domain, hence reduced, so proposition 2.5.5 realizes it as the coordinate ring of an affine algebraic set $\tilde{X}$, which is a variety by corollary 2.3.4. The inclusion of k-algebras corresponds under theorem 2.1.5 to a morphism $\nu\colon \tilde{X} \to X$.

Step 2: normality. An element of $k(\tilde{X})$ integral over $\widetilde{k[X]}$ is integral over $k[X]$ by transitivity of integral dependence, hence already lies in $\widetilde{k[X]}$. So $\widetilde{k[X]}$ is integrally closed and $\tilde{X}$ is normal, giving (1).

Step 3: birational and surjective. Every element of $\widetilde{k[X]}$ lies in $k(X)$, so $k(\tilde{X}) = \mathrm{Frac}(\widetilde{k[X]}) = k(X)$ and ν is birational. Since $k[\tilde{X}]$ is integral over $k[X]$, the going-up property gives a prime of $k[\tilde{X}]$ over every prime of $k[X]$, and in particular a point of $\tilde{X}$ over every point of X; finiteness of the fibres follows because $k[\tilde{X}]$ is a finite $k[X]$-module. This is (3), and it is the first instance of what section 10.3 will call a *finite morphism*.

Step 4: isomorphism over the normal locus. Suppose X is normal at p, so $\mathcal{O}_p(X)$ is integrally closed. Localizing the inclusion $k[X] \subseteq \widetilde{k[X]}$ at $\mathcal{I}(p)$ and using that integral closure commutes with localization ([5, Normalization Commutes with Localization]) gives $\mathcal{O}_p(X) = \widetilde{\mathcal{O}_p(X)}$, so ν is an isomorphism of local rings there and hence an isomorphism on a neighbourhood, which is (4), completing the proof.

Part (4) says the normalization only does work where the variety was not already normal, and part (3) says it never changes the function field. Together these make ν the minimal repair: it is birational, so it preserves everything visible to rational functions, and it is an isomorphism wherever no repair was needed. The universal property makes "minimal" precise.

Proposition 7.2.2: Universal Property of Normalization

Let X be an affine variety with normalization $\nu\colon \tilde{X} \to X$. If Y is a normal affine variety and $f\colon Y \to X$ is a dominating morphism, then f factors uniquely through ν:

$$f = \nu \circ g \qquad \text{for a unique morphism } g\colon Y \to \tilde{X}$$

***Proof*:**
Since f is dominating, $f^{\#}\colon k[X] \to k[Y]$ is injective (proposition 3.3.4(2)) and extends to an embedding $k(X) \hookrightarrow k(Y)$. Any $u \in \widetilde{k[X]}$ satisfies a monic equation over $k[X]$; applying $f^{\#}$ to that equation shows $f^{\#}(u)$ is integral over $k[Y]$, and $k[Y]$ is integrally closed because Y is normal, so $f^{\#}(u) \in k[Y]$. Hence $f^{\#}$ extends to $\widetilde{k[X]} = k[\tilde{X}] \to k[Y]$, and this extension is unique because $\widetilde{k[X]}$ sits inside $k(X)$, on which $f^{\#}$ is already determined. Applying theorem 2.1.5 converts this into the asserted unique factorization $f = \nu \circ g$, as we sought to show.

Example 7.2.3: Normalizing the Cuspidal and Nodal Cubics

1. For the cuspidal cubic $C = \mathcal{Z}(x^3 - y^2)$ with $k[C] = k[t^2, t^3]$, the integral closure in $k(t)$ is all of $k[t]$: it contains t by the opening computation, and $k[t]$ is normal by proposition 7.1.3, so nothing larger is needed. Hence $\tilde{C} = \mathbb{A}^1_k$ and ν is the parameterization $t \mapsto (t^2, t^3)$. It is bijective, and an isomorphism away from the cusp.
2. For the nodal cubic N, the same computation gives $\tilde{N} = \mathbb{A}^1_k$ with $\nu(t) = (t^2 - 1, t(t^2 - 1))$. Here ν is *not* injective: both $t = 1$ and $t = -1$ map to the origin. The normalization has pulled the two branches of the node apart, which is what separating a self-crossing requires.

3. Both examples end at $\mathbb{A}^1$: by corollary 7.4.2 below, normalizing a curve always produces a smooth one.

The contrast between (1) and (2) is worth holding on to. A cusp is repaired without changing the underlying set, since ν is a bijection; a node is repaired by separating points that had been glued. Both are invisible to the function field, which is why birational geometry cannot see them and why a normal model is the right thing to ask for.

What the normalization separates is exactly what the completion sees, which is the comparison section 6.3 deferred to here.

Proposition 7.2.4: Branches and the Normalization of a Curve

Let C be an affine curve, $p \in C$ a point, and $\nu\colon \tilde{C} \to C$ the normalization of theorem 7.2.1. Then the number of formal branches of C at p, in the sense of definition 6.3.2, equals the number of points of $\tilde{C}$ lying over p.

***Proof Key ideas*:**

Both counts are computed by the same ring. The points of $\tilde{C}$ over p correspond to the maximal ideals of the integral closure $\tilde{\mathcal{O}}$ of $\mathcal{O}_{C,p}$ in its fraction field, which is finite over $\mathcal{O}_{C,p}$ by theorem 7.2.1 together with [5, Normalization Commutes with Localization], hence semilocal; and the minimal primes of $\hat{\mathcal{O}}_{C,p}$ correspond to the factors in the decomposition of $\hat{\tilde{\mathcal{O}}}$ into complete local rings, one for each maximal ideal of $\tilde{\mathcal{O}}$. Matching the two requires knowing that completion commutes with integral closure for a curve over a field, which is the statement that $\mathcal{O}_{C,p}$ is analytically unramified; that is a theorem about excellent rings and is not developed in this series. See [15].

The two cases this book uses are checked directly instead. Proposition 6.3.3 gives two branches at the node and one at the cusp, and the fifth and first exercises of exercise set 7.5.1 compute the normalizations: two points lie over the node, at $t = \pm 1$, and one over the cusp, at $t = 0$. So the counts agree in both cases, completing the argument in the cases used.

7.3 Curves, Function Fields, and Discrete Valuation Rings

Definition 7.3.1: Local Ring of Field

Let K/k be a field extension. Then A is a *local ring of* K if $\operatorname{Frac}(A) = K$ and $k \subseteq A$.
If A is a DVR, we shall say that A is a *discrete valuation ring of* K.

The prototypical example of local fields over a ring are the stalks $\mathcal{O}_p(V)$ of $\kappa(V)$ for a variety V.

Theorem 7.3.2: Dominating DVR

Let C be a projective curve, $K = \kappa(C)$, and let L/K be a field extension. Let R be a DVR of L containing k. Assume $K \not\subseteq R$. Then there is a unique point $p \in C$ such that R dominates $\mathcal{O}_p(C)$.

***Proof*:**
First, such a point is unique; if R dominates $\mathcal{O}_p(C)$ and $\mathcal{O}_q(C)$, then choose $f \in \mathfrak{m}_p(C)$ so $1/f \in \mathcal{O}_q(C)$. Then $\operatorname{ord}(f) > 0$ and $\operatorname{ord}(1/f) \geq 0$, a contradiction.

Thus, let's show it exists. Let $C \subseteq \mathbb{P}_k^n$ be a projective curve (a closed one-dimensional subvariety. Without loss of generality, we may take C to not be strictly contained in any x_i-hyperplane so that $C \cap U_i \neq \emptyset$. Then in each $k_{\mathbb{P}}[C] = k[x_1, ..., x_n + 1]/I(C) \cong k[u_1, ..., u_{n+1}]$. Let

$$N = \max_{i,j}(x_i/x_j)$$

As the max is achieve, without loss of generality assume $\operatorname{ord}(x_j/x_{n+1}) = N$ for some j. Then for all i:

$$\operatorname{ord}\left(\frac{u_i}{u_{n+1}}\right) = \operatorname{ord}\left(\frac{(u_j/u_{n+1})}{(u_i/u_j)}\right) = N - \operatorname{ord}(u_j/u_i) \geq 0$$

If C_* is the affine curve corresponding to $C \cap U_{n+1}$, then $\Gamma(C_*)$ can be defined with

$$k\left[\frac{u_1}{u_{n+1}}, ..., \frac{u_n}{u_{n+1}}\right]$$

hence, $\Gamma(C_*) \subseteq R$.

Now, if $\mathfrak{m}$ is the maximal ideal of R, let $J = \mathfrak{m} \cap \Gamma(C_*)$. Then J is a prime ideal, so J corresponds to a closed subvariety $W \subseteq C_*$. IF $W = C_*$, $J = 0$, and every nonzero element of $\Gamma(C_*)$ is a unit of R. But then $K \subseteq R$, contradiction our assumption. So $W = \{p\}$ must be a point. Then R certainly dominates $\mathcal{O}_p(C_*)$, and hence dominates $\mathcal{O}_p(C)$, completing the proof.

Corollary 7.3.3: Rational Map of Curves

Let f be a rational map from C' to a projective curve C. Then the domain of f must include every simple point of C'. If C' is nonsingular, f is a morphism.

***Proof*:**
If F is not dominating, it is constant (as shown in corollary 10.2.4), hence its domain must be all of C. Thus, assume $F^{\#}$ embeds $K = \kappa(C)$ as subfield of $L = \kappa(C')$. Let P be a simple point of C', $R = \mathcal{O}_p(C')$. Then it is enough to show that $K \not\subseteq R$. Suppose that $K \subseteq R \subseteq L$ The L is a finite algebraic extension of K, so R is in fact a field. But DVR's are by definition *not* a field.

Corollary 7.3.4: Function Field and Morphism Correspondence

Let C be a projective curve, C' a nonsingular curve. Then there is a natural one-to-one correspondence between dominant morphisms $f : C' \to C$ and homomorphism $f^{\#} : \kappa(C) \to \kappa(C')$.

***Proof*:**
immediate

Corollary 7.3.5: Characterizing Nonsingular Projective Curves

Two nonsingular projective curves are isomorphic if and only if their function fields are isomorphic if and only if they are birationally equivalent.

***Proof*:**
immediate

Corollary 7.3.6: Function Field associated to Curve

Let C be a nonsingular projective curve over k, $K = \kappa(C)$. Then there is a natural one-to-one correspondence between the points of C and the discrete valuation rings of K, namely

$$p \in C \text{ correspond to } k \subseteq \mathcal{O}_p(C) \subseteq K$$

***Proof*:**
Certainly $\mathcal{O}_p(C)$ is a DVR in K. On the other hand if R is any such DVR, then we saw that R dominates a unique $\mathcal{O}_p(C)$, and as these are both DVR's of K, $R = \mathcal{O}_p(C)$, as we sought to show.

From this, notice that we can create the Zariski topology on the collection of DVRs: Let X be the set of all DVRS of $\kappa(C)/k$. Give the cofinite topology on X. Then the correspondence $p \mapsto \mathcal{O}_p(C)$ is a homeomorphism! We shall use this observation when looking at "smooth $\mathbb{Z}$-curves" in [4, Dedekind domains as smooth curves] which shall how how to interpret Dedekind domains as smooth curves!

7.4 Normality and the Serre Conditions

Regularity implies normality (corollary 5.5.3), and in dimension one the implication reverses (proposition 5.5.2). The question is how far the reverse implication reaches. The answer is: exactly one codimension. A normal variety is smooth in codimension one, and no further.

Proposition 7.4.1: Normal Implies Regular in Codimension One

Let X be a normal variety and $p \in X$ a point whose closure has codimension 1 in X. Then $\mathcal{O}_p(X)$ is a discrete valuation ring, and in particular X is nonsingular at p.

***Proof*:**
The ring $\mathcal{O}_p(X)$ is a Noetherian local domain (proposition 3.2.10) of Krull dimension 1, since its dimension is the codimension of p in X by proposition 4.3.2. It is integrally closed, because X is normal and normality is local (proposition 7.1.2). A Noetherian local domain of dimension one that is integrally closed is a discrete valuation ring, and hence regular, by proposition 5.5.2. Regularity of the local ring is nonsingularity of the point by proposition 5.2.2, as we sought to show.

Corollary 7.4.2: A Normal Curve Is Smooth

A normal curve is nonsingular at every point. Consequently the normalization $\nu\colon \tilde{C} \to C$ of any curve is a smooth curve, and ν resolves every singularity of C.

***Proof*:**
Every point of a curve has codimension 1 in it, so proposition 7.4.1 applies at every point. The normalization $\tilde{C}$ is normal and is a curve, being birational to C (theorem 7.2.1(3)) and hence of the same dimension, completing the proof.

This is the structural payoff of the chapter, and it is what makes curves so much better behaved than surfaces. For a curve, "normal" and "smooth" are the same word, so a single algebraic operation, taking an integral closure, removes every singularity at once. Section 7.3 converts this into the statement that each birational class of curves contains exactly one smooth projective model.

In dimension two the two conditions come apart.

Example 7.4.3: A Normal Singular Surface
Let $\operatorname{char} k \neq 2$ and let $Q = \mathcal{Z}(xy - z^2) \subseteq \mathbb{A}^3_k$, the quadric cone. It is singular at the origin: the Jacobian of $xy - z^2$ is $(y,\ x,\ -2z)$, which vanishes at $(0,0,0)$, so by definition 5.2.1 the rank there is 0 rather than the required $3 - 2 = 1$.

Nonetheless Q is normal. Its coordinate ring is isomorphic to the subring $k[u^2, uv, v^2] \subseteq k[u,v]$ under $x \mapsto u^2$, $y \mapsto v^2$, $z \mapsto uv$, which is the ring of invariants of $k[u,v]$ under $(u,v) \mapsto (-u,-v)$. A ring of invariants of a normal domain under a finite group is normal: if w in the fraction field is integral over the invariants, it is integral over $k[u,v]$, hence lies in $k[u,v]$ by proposition 7.1.3, and it is invariant, so it lies in the invariant subring.

So Q is normal and singular, and its singular locus is a single point, of codimension 2. That is the smallest codimension proposition 7.4.1 permits.

The example is sharp, and it explains the shape of the general criterion. Normality controls codimension one and says nothing beyond it, so a full characterization needs a second condition governing the deeper codimensions. That condition is a statement about depth, and it is what Serre's criterion supplies.

A Noetherian domain is integrally closed exactly when it satisfies two conditions, one asking regularity in codimension one and one asking a lower bound on depth. Write (R_1) for "$R_{\mathfrak{p}}$ is regular for every prime $\mathfrak{p}$ of height at most one" and (S_2) for "$\operatorname{depth} R_{\mathfrak{p}} \geq \min(2, \operatorname{height} \mathfrak{p})$ for every prime $\mathfrak{p}$". The statement that normality is their conjunction is *Serre's Criterion for Normality*, [7, Serre's Criterion for Normality]; it runs on the theory of depth and regular sequences, which is developed there and not here.

The geometric content of the two halves is worth separating. Condition (R_1) is proposition 7.4.1 read backwards: it requires smoothness exactly in codimension one. Condition (S_2) says a function defined outside a closed set of codimension two extends across it, which is why a normal variety cannot be repaired by deleting a small set and gluing differently. The quadric cone of example 7.4.3 satisfies both: it is smooth away from a codimension-two point, and it is a hypersurface, which forces (S_2).

Read geometrically, the criterion says normality is precisely "smooth in codimension one, plus no missing codimension-two information". For curves the second condition is vacuous and the first is everything, which recovers corollary 7.4.2. From dimension two onward both are needed, and normality becomes a genuinely weaker and more flexible condition than smoothness, which is why it is normality rather than smoothness that appears as a hypothesis throughout the theory of divisors in chapter 14.

7.5 Weil Divisors and the Class Group

Proposition 7.4.1 says that a normal variety is regular in codimension one, so at every codimension-one subvariety the local ring is a discrete valuation ring. That is exactly the hypothesis needed to make sense of the order of vanishing of a function along a subvariety, and with it the divisor theory of chapter 14 generalizes from curves to normal varieties of any dimension.

Definition 7.5.1: Prime Divisor and Weil Divisor

Let X be a normal variety. A *prime divisor* on X is an irreducible closed subvariety $Y \subseteq X$ of codimension one. A *Weil divisor* is a formal sum

$$D = \sum_{Y} n_Y Y \qquad n_Y \in \mathbb{Z}$$

over the prime divisors, with all but finitely many n_Y zero. They form a group $\mathrm{Div}(X)$.

For a curve a prime divisor is a point, and definition 14.1.1 is the case $\dim X = 1$.

Proposition 7.5.2: The Order Along a Prime Divisor

Let X be a normal variety and $Y \subseteq X$ a prime divisor. Then the local ring $\mathcal{O}_Y(X)$ of X along Y, that is the localization of $k[U]$ at the prime $\mathcal{I}(Y \cap U)$ for any affine open U meeting Y, is a discrete valuation ring with fraction field $k(X)$. Writing ord_Y for its valuation, every nonzero $f \in k(X)$ has $\mathrm{ord}_Y(f) = 0$ for all but finitely many Y.

***Proof*:**
The ring $\mathcal{O}_Y(X)$ is Noetherian and local, and it is a domain with fraction field $k(X)$, being a localization of a coordinate ring of X. Its dimension is $\mathrm{codim}_X Y = 1$ by proposition 4.3.2. It is normal, being a localization of a normal domain, as the sixth exercise of exercise set 7.5.1 shows. So proposition 5.5.2 applies through its equivalence of normality with being a discrete valuation ring.

For the finiteness, fix an affine open U and write $f = g/h$ with $g, h \in k[U]$. Then $\mathrm{ord}_Y(f) \neq 0$ forces $Y \cap U$ to lie in $\mathcal{Z}(g) \cup \mathcal{Z}(h)$, a proper closed subset of U, which has finitely many irreducible components of codimension one. Covering X by finitely many affine opens gives the claim, completing the proof.

Definition 7.5.3: Principal Divisor and the Class Group

For $f \in k(X)^*$ the *principal divisor* is $\operatorname{div}(f) = \sum_Y \operatorname{ord}_Y(f)\, Y$. The principal divisors form a subgroup $\operatorname{Prin}(X) \subseteq \operatorname{Div}(X)$, and the *class group* is

$$\operatorname{Cl}(X) = \operatorname{Div}(X)/\operatorname{Prin}(X)$$

Two divisors with the same class are *linearly equivalent.*

For a nonsingular projective curve this is $\operatorname{Pic}(C)$ of definition 14.3.1; the two names coexist because on a general variety the divisor class group and the group of line bundles are different objects, agreeing exactly when X is nonsingular. The distinction is developed in [3].

Example 7.5.4: Class Groups

1. *Affine space.* $\operatorname{Cl}(\mathbb{A}_k^n) = 0$. A prime divisor is a hypersurface $\mathcal{Z}(f)$ with f irreducible, by theorem 4.4.2, and $\operatorname{div}(f) = \mathcal{Z}(f)$; so every prime divisor is principal, and hence so is every divisor.
2. *Projective space.* $\operatorname{Cl}(\mathbb{P}_k^n) \cong \mathbb{Z}$, by degree. A prime divisor is $\mathcal{Z}(F)$ for an irreducible form F, and assigning it $\deg F$ gives a surjection $\operatorname{Div}(\mathbb{P}_k^n) \to \mathbb{Z}$. Its kernel is exactly the principal divisors: a rational function on $\mathbb{P}_k^n$ is a ratio F/G of forms of equal degree, so $\operatorname{div}(F/G) = \mathcal{Z}(F) - \mathcal{Z}(G)$ has degree zero, and conversely a degree-zero divisor $\sum n_i \mathcal{Z}(F_i)$ is $\operatorname{div}\big(\prod F_i^{n_i}\big)$.
3. *The quadric cone.* Let $X = \mathcal{Z}(xy - z^2) \subseteq \mathbb{A}_k^3$, which is normal by the fourth exercise of exercise set 7.5.1 although singular at the origin. Let $L = \mathcal{Z}(x, z) \subseteq X$ be the line through the vertex. Then L is a prime divisor, and

$$\operatorname{div}(x) = 2L$$

To see this, note $y \notin (x, z)$, so y is a unit in the local ring $\mathcal{O}_L(X)$ along L. The relation $xy = z^2$ then gives $x = z^2/y$, so the maximal ideal $(x, z)\mathcal{O}_L(X)$ is generated by z alone and $\operatorname{ord}_L(z) = 1$; hence $\operatorname{ord}_L(x) = 2\operatorname{ord}_L(z) - \operatorname{ord}_L(y) = 2$. No other prime divisor occurs, since $\mathcal{Z}(x) \cap X = L$ as a set. So $2L$ is principal but L is not, by proposition 6.4.3, and $\operatorname{Cl}(X) \cong \mathbb{Z}/2\mathbb{Z}$ generated by the class of L.

The last example is the point of the section. On a nonsingular variety every codimension-one subvariety is locally cut out by a single equation, which is theorem 6.4.2, so the class group there measures a global obstruction to patching those local equations together. On a singular normal variety the obstruction is already local, and Cl detects the singularity: the line L on the quadric cone is not the zero set of any single function near the vertex, by proposition 6.4.3, which is what $\operatorname{div}(x) = 2L$ records.

Exercise 7.5.1

1. Let $A = k[x, y]/(y^2 - x^3)$ be the coordinate ring of the cuspidal cubic and put $t = y/x$ in $\operatorname{Frac}(A)$. Show that t is integral over A but does not lie in A, so A is not normal, and show that $A[t] = k[t]$. Conclude that the normalization of the cuspidal cubic is $\mathbb{A}_k^1$.

2. Show that $\mathbb{Z}[\sqrt{-3}]$ is not integrally closed in its fraction field by exhibiting an element of $\mathbb{Q}(\sqrt{-3})$ that is integral over it but not in it. Which ring is its integral closure?

3. Let $A \subseteq B$ be an integral extension of domains with $\mathrm{Frac}(A) = \mathrm{Frac}(B)$. Show that if A is normal then $A = B$. Deduce that the normalization map of theorem 7.2.1 is an isomorphism exactly when the variety is already normal.

4. Show that the quadric cone $\mathcal{Z}(xy - z^2) \subseteq \mathbb{A}^3_k$ is normal but singular at the origin, and explain why this does not contradict corollary 7.4.2. Which of Serre's two conditions in [7, Serre's Criterion for Normality] is doing the work?

5. Let $C = \mathcal{Z}(y^2 - x^2 - x^3) \subseteq \mathbb{A}^2_k$ be the nodal cubic. Using the parameterization $t \mapsto (t^2 - 1,\, t(t^2 - 1))$ of example 3.3.8, show that $k[C]$ is not normal and that its normalization is $k[t]$. How many points of the normalization lie over the node, and why is that different from the cuspidal case?

6. Show that a localization of a normal domain is normal, and use this together with proposition 7.1.2 to show that normality of a variety may be checked on any affine open cover.

7. Give an example of a birational morphism of affine varieties that is bijective on points but not an isomorphism. (Chapter 17 will give a systematic supply.)

8. Let C be an irreducible curve with function field $K = k(C)$. Show that a discrete valuation ring R with $\mathrm{Frac}(R) = K$ and $k \subseteq R$ determines at most one point of any nonsingular projective model of C, and identify the valuation ring attached to the point at infinity of $\mathbb{P}^1_k$.

Part III

Projective Varieties

The affine theory of Part I is incomplete in two ways that turn out to be the same way. A rational function need not be defined everywhere, and two curves of degrees m and n need not meet in mn points. Both failures are failures of points to exist, and both are repaired by the same construction: adjoin a point for every direction in which something can run off to infinity.

The resulting space is projective space, and the varieties inside it are the subject of this Part. The affine dictionary carries over almost unchanged once polynomials are replaced by homogeneous ones, but the geometry is genuinely better behaved: projective varieties are complete, so the image of a regular map out of one is closed, and the only regular functions on one are the constants. Those two facts drive everything in Part IV.

8

Projective Space and Homogeneous Ideals

This chapter builds projective space and transplants the dictionary of chapter 2 onto it. The transplant is not free: a polynomial is not a function on projective space, and the zero set of one is well defined only when the polynomial is homogeneous. Working out what replaces the coordinate ring, and which ideals the Nullstellensatz now describes, occupies the chapter.

8.1 Points at Infinity

So far, all of the algebraic sets were called *affine* algebraic sets, as they were all a subset of $\mathbb{A}_k^n$. These offer great geometrical intuition, however they are lacking in certain algebraic qualities, namely rational functions need not be defined at all points, and not all algebraic curves intersect. As an example of the former point, it is simplest to resort to an analogy from calculus. The reader may recall that the function $f : \mathbb{R} \to \mathbb{R}$ defined as $f(x) = 1/x$ is famously undefined at $x = 0$, as we cannot divide by 0.

However, we can enlarge $\mathbb{R}$ to make this function well-defined by adding a *point at infinity*. Namely, we may take $\mathbb{R}_\infty = \mathbb{R} \cup \{\infty\}$ where (for the sake of limits)[1] we extend the arithmetic of $\mathbb{R}$ to $\mathbb{R}_\infty$ by doing:

$$a + \infty = \infty + a = \infty \qquad b \cdot \infty = \infty \cdot b = \infty$$

where $a \in \mathbb{R}$ and $0 \neq b \in \mathbb{R}_\infty$. We then say that if (x_n) diverges, then

$$\lim_{n\to\infty} x_n = \infty$$

[1]The arithmetic defined here certainly does not satisfy the axioms of a ring; instead it is a structure called a *wheel*; we shall not study these in this textbook

In this case we now have that the function $1/x$ has a well-defined value of ∞ at 0, and even better under the right frame-work (provided in differential geometry), the function $f(x) = 1/x$ can be shown to be smooth on $\mathbb{R}_\infty$. In this way, we have ensured that we cannot have an undefined rational polynomials on $\mathbb{R}_\infty$! The space $\mathbb{R}_\infty$ is called the *projective line*. This "closure" on the set of definable rational polynomials makes the projective line an algebraically nicer space to work with, in a similar way that $\mathbb{C}$ is the better space to work in as all polynomials have roots.

For the latter interpretation (the one about there always being an intersection), consider a polynomial $f(z)$ defined over $\mathbb{C}$ so that there is no worry about roots not existing. Let us regard $\mathbb{C}$ as $\mathbb{R}^2$ with coordinates x, y so that $f(z) = f(x, y)$. Consider the line $L_{(a,b)}$ which has angle α given by (a, b):

$$L_{(a,b)} : t \mapsto (at, bt)$$

Then either $f = L$, in which case their intersection is an entire line, or f and $L_{(a,b)}$ intersect where these two equations agree:

$$f(at, bt) = f_n(a, b)t^n + \cdots + f_1(a, b)t + f_0(a, b)$$

where $f_n(a, b)$ represents the constant in front of the t-term. if $f_n(a, b) \neq 0$ then there are n roots, and hence n points of intersections. However, if $f_n(a, b) = 0$, then there must be a smallest m where

$$f_n(a, b) = 0,\ f_{n-1}(a, b) = 0,\ \cdots f_{m+1}(a, b) = 0,\ f_m(a, b) \neq 0$$

or else $f = L_{(a,b)}$, which we are taking to not be the case. But then we would have a number of intersections lower than the number of roots. There is a remedy to this, which is to consider the remaining $(n-m)$ points to be the intersection of L with f at a point at infinity. To see this, consider the substitution $s = 1/t$. Then:

$$f_n(a, b) + f_{n-1}(a, b)s + \cdots + f_0(a, b)s^n = 0$$

Notice that if $f_n(a, b) \neq 0$, then at $s = 0$ (conceptually can be thought of as $1/t = 0$) there is no root. But if there are $n - m$ leading terms that are zero, then this polynomial has a root of order $n - m$ at $s = 0$. Thus, we recover the intersection of L with f as an invariance! In particular, for each line $L_{(a,b)}$ we can think of there being an extra point at infinity. Notice that by combining the coordinates s, t we get a function:

$$f_n(a, b)t^n + f_{n-1}t^{n-1}s + \cdots + f_0(a, b)s^n = 0$$

which allows us to find all points of intersections at once. This motivates the use of homogeneous functions and the linear equivalence construction that shall be at the start of this section.

Why is the space called projective? This is understood from the following construction of projective space that extends to higher dimensions. Let us start in one dimension. We may define $\mathbb{R}_\infty$ as the

$$\mathbb{R}\mathrm{P}^1 := \frac{\mathbb{R}^2 \setminus \{(0,0)\}}{\{(x,y) \sim (\lambda x, \lambda y), \forall \lambda \in \mathbb{R} \setminus \{0\}\}}$$

by taking the representative of each equivalence class to have norm 1, notice that we may associate all the points of $\mathbb{R}\mathrm{P}^1$ to S^1, and we can associate points of $\mathbb{R}$ with the equivalence class where $y \neq 0$. The equivalence class $[(\lambda x, 0)]$ can be thought of as the "point at infinity" that we've added, hence we may think of $\mathbb{R}\mathrm{P}^1$ as

$$\mathbb{R}\mathrm{P}^1 = \mathbb{R}^1 \sqcup \mathbb{R}^0 \tag{8.1}$$

The term "projective" comes from the idea that parallel lines will meet at the "point at infinity". In terms of "closure" that we talked about earlier, the reader may recall that in classical geometry, two non-parallel lines will always have a solution. In $\mathbb{R}\mathrm{P}^1$, we have added the extra property that there is one point at which parallel lines meet; they also have a solution. In terms of linear algebra, every system of linear equations has a solution.

We may extend this further, and take $\mathbb{R}^2$ and add "points at infinity". In this case, there are many possible pairs of parallel lines. We may consider that up to scaling, we have many new points that we may add. Formally, let:

$$\mathbb{R}\mathrm{P}^2 := \frac{\mathbb{R}^3 \setminus \{(0,0,0)\}}{\{(x,y,z) \sim (\lambda x, \lambda y, \lambda z), \forall \lambda \in \mathbb{R} \setminus \{0\}\}}$$

We may imagine that this insures that any rational function in two variables always has a solution. Adding multiple points at infinity insures that the function $1/y$, $1/x$, $1/xy$ and so forth are all considered distinct when evaluating ($1/y$ is zero when evaluated at one infinity but not another). In terms of parallel lines, we may think that parallel lines in any angle will have a unique solution (note that the distance between the lines does not matter, they still "meet" at the same point at infinity). The shape $\mathbb{R}\mathrm{P}^2$ can be thought of as $\mathbb{R}^2$ with a circle of points at infinity added where opposites are glued. Can the reader use this intuition to generalize (8.1) to:

$$\mathbb{R}\mathrm{P}^2 = \mathbb{R}^2 \sqcup \mathbb{R}^1 \sqcup \mathbb{R}^0$$

The reader familiar with differential geometry will know that $\mathbb{R}\mathrm{P}^2$ is a two dimensional smooth manifold, and is one of the simplest example of a shape that is 2-dimensional but cannot be embedded in 3-dimensions without self-intersection. The following image from Wikipedia may help in trying to visualize (note that this is an *immersion*, meaning it is an approximation of the shape by embedding it in a space in which there is not enough dimensions to visualize it without intersection)

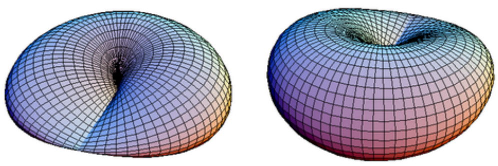


Figure 8.1: Wikipedia: Immersion of $\mathbb{R}\mathrm{P}^2$

We may continue this process and define $\mathbb{R}\mathrm{P}^n$, or more generally $\mathbb{P}^n_k$ for some field k. These shall have a larger class of functions that we may define on them (we shall eventually say that there are an "ample" amount of function[2]). Thus, we shall treat $\mathbb{P}^n_k$ as an extension of $\mathbb{A}^n_k$, and consider the equivalence classes of $\mathbb{P}^n_k$ as *points*.

An important technicality that we may encounter is that we must convert a function that is defined on $\mathbb{A}^n_k$ to a function defined on $\mathbb{P}^n_k$, namely the points of $\mathbb{P}^n_k$ are now equivalence relations and hence a function whose domain is $\mathbb{P}^n_k$ must be independent of representative. This shall form an important class of rings that are "good representatives" to extend polynomials to $\mathbb{P}^n_k$. As we are extending results from $\mathbb{A}^n_k$, this class of rings shall have a similar dictionary as that provided between affine varieties and radical ideals of coordinate rings, namely we shall show that polynomials solutions over $\mathbb{P}^n_k$ shall correspond to *graded ideals* of *graded rings*.

[2]More technically speaking, we shall say that $\mathbb{P}^n_k$ has *ample line bundles*, a notion belonging to scheme theory

8.2 Projective Space and Homogeneous Polynomials

Let us first formally define projective space

Definition 8.2.1: Projective Space

Let k be a field. Then *projective space* over k of dimension n is:

$$\mathbb{P}_k^n = \frac{k^{n+1} \setminus \{0\}}{x \sim \lambda x \text{ for all } \lambda \in k^\times}$$

The points of $\mathbb{P}_k^n$ are often denoted $[x_1 : x_2 : \cdots : x_{n+1}]$.

The first thing we must do is define the topology on this space which parallels the Zariski topology on $\mathbb{A}_k^n$. Polynomial functions need not give well-defined zero sets on projective space, for example $f(x, y) = x^2 + y$ does not map two elements in the same equivalence class to zero. We must limit to polynomials that maps the representatives of an equivalence class to the same point. These are precisely the homogeneous polynomials:

Definition 8.2.2: Homogeneous Polynomial (Algebraic Form)

Let $f \in k[x_1, \ldots, x_n]$ be a polynomial. Then f is a *homogeneous polynomial* of degree d, or an *algebraic d-form*, if every monomial occurring in f has total degree d. Equivalently, as a formal identity in $k[x_1, \ldots, x_n, \lambda]$,

$$f(\lambda x_1, \ldots, \lambda x_n) = \lambda^d f(x_1, \ldots, x_n)$$

The identity is the useful form, and it is the reason homogeneous polynomials have well defined zero sets in projective space. It characterizes homogeneity only when read formally, or when k is infinite: over $\mathbb{F}_2$ the only scalar available is $\lambda = 1$, so the identity holds vacuously for every f.

The term *algebraic form* is a bit out-dated, but is often still used for the degree two case, where such algebraic forms are called *quadratic forms*. This condition extends to a more general class of functions. The reader comfortable with some analysis will see in exercise set 8.5.1 that any (at least) d-times differential degree d homogeneous function is in fact a homogeneous polynomial[3].

Example 8.2.3: Homogeneous Polynomials

For example:

1. $f(x, y, z) = x^3 + xy^2 - 2y^3$ is homogeneous
2. $f(x, y) = x^2 + y^3$ is not homogeneous

Notice that if $f_1, ..., f_n$ are all homogeneous polynomial of the same degree k, then $(f_1 + ... + f_n)^m$ is a homogeneous polynomial of degree mk. Hence, homogeneous polynomials behave well under addition and multiplication.

[3] homogeneity does not imply differentiability or continuity, can you think of a counter-example?

If $(a_1, ..., a_n) \in k^n$ is a solution to a homogeneous polynomials, then all points on the line passing through this point and the origin is also a solution, hence homogeneous polynomials are the "natural" polynomials for the zero set of projective space, as projective space is an equivalence relation on lines. It is tempting conclude the collection of homogeneous polynomials form the set $k[\mathbb{P}^n_k]$ where the set is either equal to $\mathrm{Hom}_{\mathbf{Reg}}(\mathbb{P}^n_k, \mathbb{A}^1_k)$ or $\mathrm{Hom}_{\mathbf{Reg}}(\mathbb{P}^n_k, \mathbb{P}^1_k)$ and we get the same theory as with affine spaces. However, neither of these sets is the collection of homogeneous polynomials. If we consider $\mathrm{Hom}_{\mathbf{Reg}}(\mathbb{P}^n_k, \mathbb{P}^1_k)$, we have the problem that $[0 : 0] \notin \mathbb{P}^1_k$ and so we lose the fundamental property that the topology is defined by the zero-sets. The reader can further check that if $f : \mathbb{P}^n_k \to \mathbb{A}^1_k$ is homogeneous, then extending to $\mathbb{P}^1_k$ by mapping to $[f(x_1, ..., x_n) : 1]$ is not well-defined[4] (we shall see that $\mathbb{A}^1_k$ is dense in $\mathbb{P}^1_k$ and the given extension is the *only* continuous extension, and it does not work).

For the first option; even when $n = 1$ there is a problem of well-definedness. Consider $f : \mathbb{P}^1_k \to \mathbb{A}^1_k$ given by $f([x : y]) = f\left(\left[\frac{x}{y} : 1\right]\right) = \frac{x}{y}$, i.e. the closest we can do to define the identity. The reader may immediately see the problem: when $y = 0$, x/y is not defined (the codomain is $\mathbb{A}^1_k$). Even if we "ignore" this problem by simply not considering where $y = 0$, we have that f is no well-defined:

$$\frac{x}{y} = f\left(\left[\frac{x}{y} : 1\right]\right) = f([x : y]) = f\left(\left[1 : \frac{y}{x}\right]\right) = 1$$

But then f is not well-defined as it is not the case that for all $x, y \in k$, $x/y = 1$. If we instead try $f([x : y]) = xy$ then

$$xy = f([x : y]) = f\left(\left[\frac{x}{y} : 1\right]\right) = \left(\frac{x}{y}\right) \cdot 1 = \frac{x}{y}$$

but it is not always the case that $xy = \frac{x}{y}$ (ex. let $x = y = 2$), and hence it is once again not well-defined. The reader shall find that any such attempt to define a (non-constant) polynomial shall not be well-defined, namely since $f(\lambda x, \lambda y) \neq f(x, y)$.

This may unsettle the reader, as a "regular function" can't be well-defined unless the polynomial is constant. As it turns out, we can't extend (non-constant) polynomials from affine space to projective in a well-defined manner (see section 9.1) unless it is a constant, which the above exploration is a first demonstration on why this is the case.

However, we will be able to treat homogeneous polynomials as regular functions whenever we limit to one of the variables x_i to be nonzero. In particular, if $f \in k[x_1, ..., x_{n+1}]$ is homogeneous, and $x_i \neq 0$, then we can interpret f to be a regular function on $U_i = \{x \in \mathbb{P}^n_k \mid x_i \neq 0\}$:

$$f([x_1 : ... : x_i : ... : x_{n+1}]) = f\left(\left[\frac{x_1}{x_i} : ... : 1 : ... : \frac{x_{n+1}}{x_i}\right]\right) = f(x'_1, ..., x'_{n+1})$$

where we write $x'_j = \frac{x_j}{x_i}$ as $x_i \neq 0$. Notice that since the ith position can be fixed to 1 by dividing by x_i, we have chosen a representative of this equivalence relation, which is why we dropped the square brackets in the above equation. For example, for the homogeneous polynomial $x \in k[x, y]$, when $y \neq 0$, we can think of it as defining the function:

$$f(x) = x$$

and when $x \neq 0$, we can think of it as defining the function:

$$f(y) = 1$$

[4]There shall be well-defined regular maps from $\mathbb{P}^n_k$ to $\mathbb{P}^1_k$, see section 9.4, but these are not extensions of affine regular maps

This mirrors how $f([x : y]) = f([x/y : 1]) = f([x' : 1]) = x'$ and $f([x : y]) = f([1 : y/x]) = f([1 : y']) = 1$. Notice that it was important to define two functions and how each behaves when $y \neq 0$ and $x \neq 0$ respectively. If we want to recover the homogeneous polynomial that defined these functions but only defined it on $x \neq 0$, we see that the $y \neq 0$ case can be λx or simply λ for any nonzero λ, showing we require the function to be defined in both situations (we shall see that having the function defined for each $x_i \neq 0$ and is compatible via "transition mappings" is enough to recover the homogeneous polynomial).

This also means that homogeneous polynomials on $\mathbb{P}^n_k$ act less like functions and more like vector-fields which "locally" are given by functions (i.e. are "local trivializations"). For the reader familiar with Differential Geometry, they may recognize that this vocabulary is used to define *vector bundles*, and that homogeneous polynomials can be seen as the sections of this vector bundle. This is in fact exactly what is going on, but it is beyond the scope of this book to get into the details of this realization (see [3] for a full exploration).

For now, we shall use homogeneous polynomial to define the closed subsets of $\mathbb{P}^n_k$. To that end, let's examine ideals generate by homogeneous elements. For each $d \in \mathbb{N}$, we can take the collection of homogeneous polynomials $R_d \subseteq k[x_1, ..., x_{n+1}]$. Then these form a k-vector space (exercise set 8.5.1(1)). Furthermore, $R_d R_e \subseteq R_{d+e}$, and so these form an $\mathbb{N}$-graded algebra over $R_0 = k$ (recall [5, More on Graded Algebras]). By [11, Characterizing Graded Ideals] and a bit of work, we may consider $k[x_1, ..., x_{n+1}]$ as:

$$k[x_1, ..., x_{n+1}] = R_0 \oplus R_1 \oplus \cdots \oplus R_d \oplus \cdots$$

with the $\mathbb{N}$-graded structure given by homogeneity.

Definition 8.2.4: Projective Algebraic Set

Take $R \subseteq k[x_1, x_2 ..., x_{n+1}]$ with the homogeneous $\mathbb{N}$-graded structure. Let $I \subseteq R$ be a homogeneous ideal. Then a *projective algebraic set* is:

$$\mathcal{Z}_{\mathbb{P}}(I) = \{p \in \mathbb{P}^n_k \mid f(p) = 0, \ \forall f \in I\}$$

If it is clear from context, the $\mathbb{P}$ subscript will be omitted and it shall be written as $\mathcal{Z}(I)$.

Note that we could have taken an arbitrary homogeneous set $S \subseteq R$, and then taken the homogeneous closure ${}^h\langle S \rangle = I$. By [11, Arithmetic of Homogeneous Ideals], we get that homogeneous ideals are closed under ideal addition, multiplication, intersection, and taking the radical:

$$I + J \qquad IJ \qquad I \cap J \qquad \sqrt{I}$$

Furthermore, to check that a homogeneous ideal I is prime, it suffices to check the condition $fg \in I$ iff $f \in I$ or $g \in I$ with homogeneous polynomials (see [11, Prime Graded Ideals]). The addition can naturally be extended to arbitrary sums, allowing us to define the Zariski topology on $\mathbb{P}^n_k$ when focusing on the homogeneous ideals of a $k[x_1, ..., x_{n+1}]$:

Definition 8.2.5: Zariski Topology on Projective Space

Letting the projective algebraic sets to be the closed subsets of $\mathbb{P}^n_k$, we get the [projective] Zariski topology on $\mathbb{P}^n_k$.

The vanishing ideal transplants as well, once restricted to homogeneous elements.

Definition 8.2.6: Projective Vanishing Ideal

Let $V \subseteq \mathbb{P}^n_k$. Then $\mathcal{I}_{\mathbb{P}}(V)$ is the ideal generated by

$$\{f \in R \mid f \text{ homogeneous and } f(p) = 0 \text{ for all } p \in V\}$$

It is a homogeneous ideal by construction, and it is radical by the same argument as in the affine case.

Definition 8.2.7: Projective Coordinate Ring

Let $V \subseteq \mathbb{P}^n_k$ be a projective algebraic set. Then a *projective coordinate ring* is defined as:

$$\Gamma_{\mathbb{P}}(V) = k_{\mathbb{P}}[V] = \frac{R}{(\mathcal{I}_{\mathbb{P}}(V))}$$

where we drop the $\mathbb{P}$ subscript when unambiguous.

Calling these the coordinate ring is a little misleading as the elements are not functions on V, but the similar algebraic properties between affine and projective coordinate rings shall warrant this nomenclature.

With a topology, we may define all of the topological and geometric notion such as compactness, connectedness, dimension, irreducibility, and so forth. The irreducibility condition in projective space becomes a condition on homogeneous prime ideals, from which we define

Definition 8.2.8: Projective Variety

A *projective variety* is an irreducible projective algebraic set.

As mentioned in the introduction of this section, projective space and projective algebraic sets extend affine space. Let us now make this precise. Let us first consider an algebraic subset $V = \mathcal{Z}_{\mathbb{P}}(I) \subseteq \mathbb{P}^n_k$. Then as $\mathbb{P}^n_k = \mathbb{A}^{n+1}_k / \sim$, we see that we can consider $\mathcal{Z}_{\mathbb{A}}(I) \subseteq \mathbb{A}^{n+1}_k$:

Before turning to the cone, one family of projective varieties is worth classifying completely, because it can be done with linear algebra alone and because its members supply most of the small examples in this book.

Definition 8.2.9: Quadric and Its Rank

A *quadric* in $\mathbb{P}^n_k$ is $\mathcal{Z}(Q)$ for a nonzero quadratic form $Q \in k[x_0, \dots, x_n]$. Assume $\operatorname{char} k \neq 2$ and write $Q(x) = x^\top A x$ for the unique symmetric matrix A. The *rank* of the quadric is the rank of A.

Proposition 8.2.10: Classification of Quadrics by Rank

Let $\operatorname{char} k \neq 2$. Every quadric of rank $r+1$ in $\mathbb{P}^n_k$ is carried by a linear change of coordinates to

$$\mathcal{Z}\big(x_0^2 + x_1^2 + \cdots + x_r^2\big)$$

and two quadrics are projectively equivalent if and only if they have the same rank. The singular locus of the quadric above is the linear subspace $\mathcal{Z}(x_0, \ldots, x_r)$, of dimension $n-r-1$, so the quadric is nonsingular exactly when $r = n$, that is when A is invertible.

***Proof*:**
A symmetric bilinear form over a field of characteristic $\neq 2$ admits an orthogonal basis, by the standard diagonalization; in that basis $Q = \sum_{i=0}^{r} \lambda_i x_i^2$ with all $\lambda_i \neq 0$ and $r+1$ the rank. Since k is algebraically closed each λ_i has a square root, and rescaling $x_i \mapsto x_i/\sqrt{\lambda_i}$ brings Q to $\sum_{i \le r} x_i^2$. Rank is unchanged by an invertible linear substitution, since $A \mapsto T^\top A T$, so it is a projective invariant and two quadrics of equal rank are equivalent to the same normal form, hence to each other.

For the singular locus, $\partial Q/\partial x_i = 2x_i$ for $i \le r$ and 0 otherwise, so by proposition 5.2.2 the singular points of $\mathcal{Z}(Q)$ are those with $x_0 = \cdots = x_r = 0$, which all lie on the quadric. That is the linear subspace $\mathcal{Z}(x_0, \ldots, x_r) \cong \mathbb{P}^{n-r-1}_k$, empty exactly when $r = n$, as we sought to show.

Example 8.2.11: Quadrics of Small Rank
Take $\operatorname{char} k \neq 2$ throughout.

1. *Rank* 1. $\mathcal{Z}(x_0^2)$ is a hyperplane doubled: the same point set as $\mathcal{Z}(x_0)$, but a different curve in the sense of definition 12.1.1, which is one reason that definition keeps the polynomial rather than its zero set.
2. *Rank* 2. $\mathcal{Z}(x_0^2 + x_1^2) = \mathcal{Z}\big((x_0 + ix_1)(x_0 - ix_1)\big)$ is a pair of distinct hyperplanes meeting in a $\mathbb{P}^{n-2}$, which is its singular locus.
3. *Rank* 3 *in* $\mathbb{P}^2_k$. A nonsingular conic. It is isomorphic to $\mathbb{P}^1_k$, by the projection computed in example 9.4.4(3), and this is the genus-zero case of theorem 14.4.2.
4. *Rank* 3 *in* $\mathbb{P}^3_k$. Singular along a point, namely $\mathcal{Z}(x_0, x_1, x_2) = [0 : 0 : 0 : 1]$. It is the cone over a nonsingular conic with that vertex, and it is the quadric cone of example 7.4.3 in projective form: normal, and singular at one point.
5. *Rank* 4 *in* $\mathbb{P}^3_k$. A nonsingular quadric surface. Example 10.1.2 identifies it with $\mathbb{P}^1_k \times \mathbb{P}^1_k$ under the Segre embedding, and its two rulings are the two families of lines on it.

So in $\mathbb{P}^3_k$ the quadrics come in exactly four projective equivalence classes, distinguished by a single integer.

8.3 The Affine Cone and Homogenization

Definition 8.3.1: Affine Cone

Let f be a homogeneous polynomial. Then its zero set in $\mathbb{A}^{n+1}$ is called an *affine cone*.

More generally, if $V \subseteq \mathbb{P}^n_k$ is a projective algebraic subset, then $C(V) \subseteq \mathbb{A}^{n+1}_k$ is the affine cone of V.

The cone translates the projective dictionary into the affine one, and the two directions are worth recording, since section 8.4 rests on them.

Proposition 8.3.2: Cone Identities

Let $\emptyset \neq V \subseteq \mathbb{P}^n_k$ be a projective algebraic set and let I be a homogeneous ideal with $\mathcal{Z}_{\mathbb{P}}(I) \neq \emptyset$. Then

$$\mathcal{I}_{\mathbb{A}}(C(V)) = \mathcal{I}_{\mathbb{P}}(V) \qquad C(\mathcal{Z}_{\mathbb{P}}(I)) = \mathcal{Z}_{\mathbb{A}}(I)$$

***Proof*:**

For the first, a point of $C(V) \subseteq \mathbb{A}^{n+1}$ is either the origin or a nonzero vector whose class lies in V. A homogeneous f vanishes on all of $C(V)$ exactly when it vanishes at every such vector, which is exactly when it vanishes at every point of V; and f automatically vanishes at the origin, since a homogeneous polynomial of positive degree has no constant term, while a nonzero constant vanishes nowhere on the nonempty V. Since both ideals are homogeneous, agreement on homogeneous elements gives equality.

For the second, $\mathcal{Z}_{\mathbb{A}}(I)$ is stable under scaling because I is homogeneous, so it is a union of lines through the origin together with the origin itself; the set of classes of its nonzero vectors is precisely $\mathcal{Z}_{\mathbb{P}}(I)$, and taking the cone recovers $\mathcal{Z}_{\mathbb{A}}(I)$. The hypothesis $\mathcal{Z}_{\mathbb{P}}(I) \neq \emptyset$ is what rules out the degenerate case where $\mathcal{Z}_{\mathbb{A}}(I)$ is the origin alone, as we sought to show.

Let us consider a concrete example. The affine cone of the (homogeneous) polynomial $x^2 + y^2 - z^2$ is the zero-set in $\mathbb{A}^{2+1}_k$ that looks like:

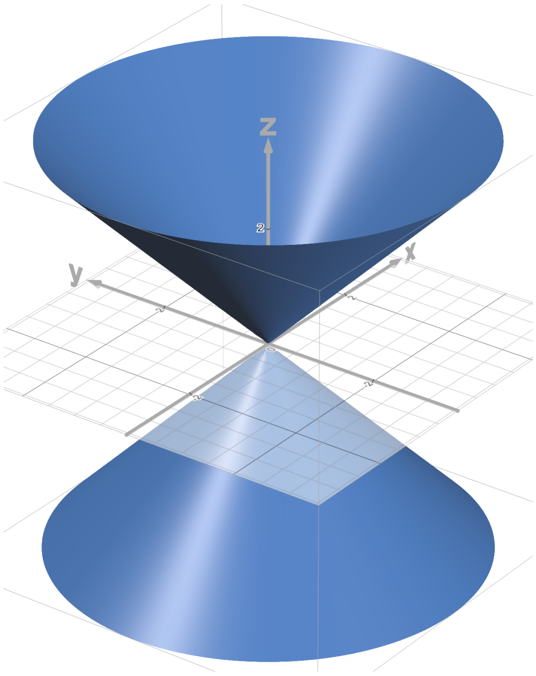


Figure 8.2: zero set of $x^2 + y^2 - z^2$

When $z \neq 0$, This equation in $\mathbb{P}^2_{\mathbb{R}}$ would resemble a circle, which we can see by intersecting the above zero-set with the plane $z = 1$, and hence the usual solution to $x^2 + y^2 = 1$ shall be in this zero set. By setting $z = 0$, we see that $x^2 + y^2 = 0$ if and only if $x = y = 0$, but $[0 : 0 : 0] \notin \mathbb{P}^2_k$, and so all the points of the zero-set $x^2 + y^2 - z^2 = 0$ are precisely the points given by lines passing through $x^2 + y^2 - 1 = 0$

Hence, we see that $x^2 + y^2 - z^2$ becomes the equation of a circle in projective space $\mathbb{P}^2_{\mathbb{R}}$. This matches how the circle has no points going "off to infinity", and so we should expect recovering the same set! However, if we instead consider $xy - z^2$, we see that when $z \neq 0$, we can take the plane where $z = 1$ and get the set $xy = 1$, which is the graph of $y = 1/x$. This has a point at infinity, and indeed if we let $z = 0$, we get the equation $xy = 0$ which is true if either x or y is zero (not that both x and y can both be zero as that would imply $[0 : 0 : 0] \in \mathbb{P}^2_{\mathbb{R}}$). Hence, we gained two additional points corresponding to the two direction the points go off to infinity (see example 8.3.4 for a visual).

The process by which we extend transform a polynomial to a homogeneous polynomial to extend the zero-set is given a name:

Definition 8.3.3: Homogenization

Let $f \in k[x_1, \dots, x_n]$ be a polynomial of degree d. Its *homogenization* is

$$f^*(x_1, \dots, x_{n+1}) = x_{n+1}^d \, f\left(\frac{x_1}{x_{n+1}}, \dots, \frac{x_n}{x_{n+1}}\right)$$

Clearing denominators shows f^* lies in $k[x_1, \dots, x_{n+1}]$ and is homogeneous of degree d. The exponent must be at least $\deg f$ for f^* to be a polynomial at all, and taking it equal to $\deg f$ is what makes f^* divisible by no power of x_{n+1}.

Conversely, a homogeneous $F \in k[x_1, \dots, x_{n+1}]$ is *dehomogenized* with respect to x_i by setting that variable to 1:

$$F_*(x_1, \dots, \widehat{x_i}, \dots, x_{n+1}) = F(x_1, \dots, x_{i-1}, 1, x_{i+1}, \dots, x_{n+1})$$

the hat marking the omitted variable. Unless said otherwise $i = n+1$. These two operations are written $(-)^*$ and $(-)_*$ throughout; the alternative f^* for f^* is not used.

Example 8.3.4: Homogenizing Polynomials

1. Take $x^2 + y^2 - 1$. Homogenizing (with degree 2) and labelling $x_3 = z$, we get $x^2 + y^2 - z^2$. If we homogenizing in degree n, we would get:

   $$z^{n-2}x^2 + z^{n-2}y^2 = z^n$$

   Homogenizing in a degree larger than $\deg f$ multiplies the result by a power of z, which adds the whole line $z = 0$ to the zero set. Those points are an artifact of the padding, so the convention is always to homogenize in degree exactly $\deg f$.

   An affine solution (x_0, y_0) with $x_0^2 + y_0^2 = 1$ becomes the projective point $[x_0 : y_0 : 1]$. For points at infinity, set $z = 0$ and solve $x^2 + y^2 = 0$. Over $\mathbb{R}$ the only solution is $x = y = 0$, which is not a point of $\mathbb{P}^2_{\mathbb{R}}$, so the real circle acquires nothing at infinity. *Over an algebraically closed field of characteristic* $\neq 2$ *this fails*, and that is the standing hypothesis of this book. Choosing i with $i^2 = -1$,

   $$x^2 + y^2 = (x + iy)(x - iy)$$

   so the circle meets the line at infinity in the two points

   $$[1 : i : 0] \qquad [1 : -i : 0]$$

   These are the *circular points at infinity*, and every circle in $\mathbb{P}^2_{\bar{k}}$ passes through both of them, whatever its centre or radius. The intuition that a circle is bounded and so gains nothing at infinity is an artifact of working over $\mathbb{R}$; the projective conic is a conic like any other, and chapter 13 counts its intersections accordingly.

2. Take $xy - 1$. Its shape looks like:

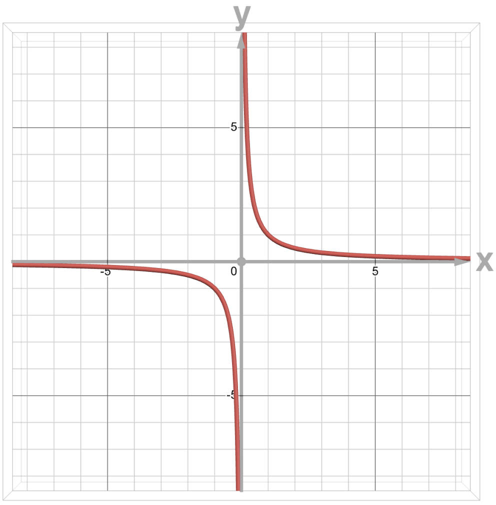


Figure 8.3: zero-set of $xy = 1$

Homogenizing, we get $xy - z^2$. A solution $x_0 y_0 - 1 = 0$ in projective coordinates is of the form:

$$[x_0 : y_0 : 1]$$

setting $z = 0$, we can ask if there are any points where:

$$[x_0 : y_0 : 0]$$

is a solution. And indeed, if $x_0 = 0$ or $y_0 = 0$, we get a solution, since

$$0y_0 - 0 = 0 \qquad x_0 0 - 0 = 0$$

We can see by visualizing the zero set that there seems to be two extra point added at infinite, one for the y-axis and one for the x-axis (remember that opposite points are considered the same).

If we de-homogenize at x, we get $y - z^2 = 0$, equivalently $y = z^2$, and de-homogenizing at y we get $x - z^2 = 0$, equivalently $x = z^2$. In both of these cases, we see that $x = z = 0$ and $y = z = 0$ is a solution. Hence, by de-homogenizing at each variable, we can find the vanishing set of a homogeneous polynomial over projective space. This can be seen more clearly when looking at the affine cone of the homogenized polynomial:

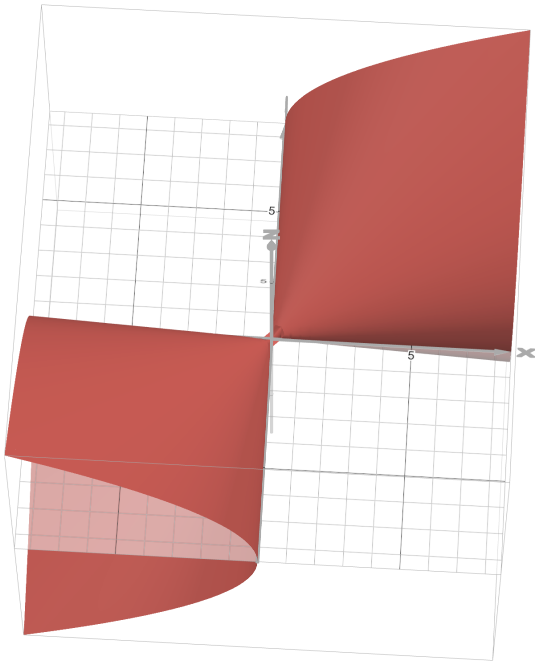


Figure 8.4: affine cone: $xy - z^2$

3. Take $x^3 + xy^2 - 2y^3 - 1$. Its shape looks like:

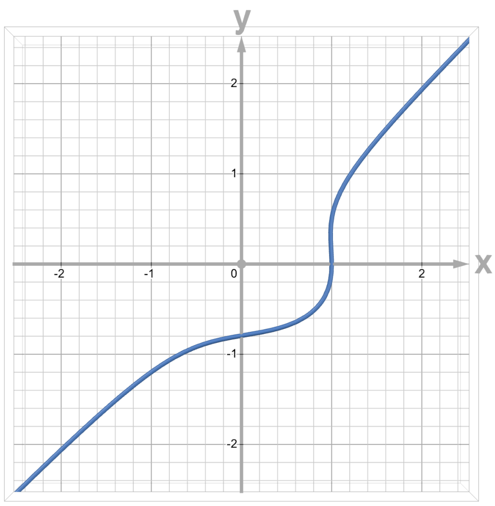


Figure 8.5: $x^3 + xy^2 - 2y^3 = 1$

When homogenizing, the affine cone gives:

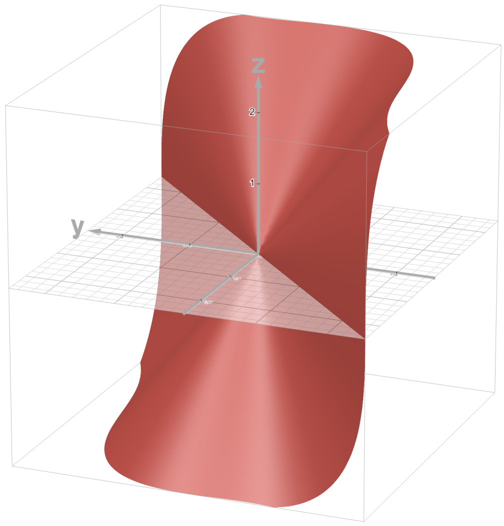


Figure 8.6: affine cone of $x^3 = xy^2 - 2y^3 = z^3$

are there any added points, and if so what are they?

4. Take $f(x) = x^2$. Its vanishing set is the single point 0. Then $f^*(x, x_\infty) = x^2$ is the degree 2 homogenization. Then this has a single zero, namely $[0 : 1]$, which correspond to the single root of x^2 in $\mathbb{A}^1_k$

5. Take $f(x, y) = x^2 - y$, whose zero set is the usual parabola. Its degree-2 homogenization in the variable z is $x^2 - zy$. Setting $z = 1$ recovers the affine points, which are

$$[x_0 : x_0^2 : 1] \qquad x_0 \in k$$

and setting $z = 0$ leaves $x^2 = 0$, so $x = 0$ and the only point at infinity is

$$[0 : 1 : 0]$$

in particular, we added a single point at infinity. This is the point where the parabola "meets". Notice that it wasn't that there was two points at infinite but just 1. The idea is that the "angle" with which the function approaches infinity tends towards $\pi/2$, and hence the edges of the graph shall meet at infinity. This shows that parabola's can be thought of as infinitely large ellipsoids[a].

Can the reader see different sorts of connections between circles, parabolas, ellipses, and hyperbolas through projective space?

[a] for those comfortable with some Riemann geometry, can the reader see why projective space can be thought of as having positive curvature?

Proposition 8.3.5: Properties of Homogenization

Let $F, G \in k[x_1, \ldots, x_{n+1}]$ be homogeneous and $f, g \in k[x_1, \ldots, x_n]$ arbitrary, with $(-)^*$ and $(-)_*$ taken at x_{n+1} as in definition 8.3.3. Then:

1. $(FG)_* = F_*G_*$ and $(fg)^* = f^*g^*$;
2. $(F+G)_* = F_* + G_*$; and writing $r = \deg f$, $s = \deg g$ and $u = \deg(f+g)$,
$$x_{n+1}^{r+s-u}(f+g)^* = x_{n+1}^s f^* + x_{n+1}^r g^*$$
3. if $F \neq 0$ and r is the largest power with $x_{n+1}^r \big| F$, then $x_{n+1}^r (F_*)^* = F$; in particular $(F_*)^* = F$ when $x_{n+1} \nmid F$.

Proof:
Throughout write $z = x_{n+1}$ and use that dehomogenizing is the ring homomorphism $z \mapsto 1$.

1. Dehomogenization is a ring homomorphism, so $(FG)_* = F_*G_*$ is immediate. For the second, $\deg(fg) = \deg f + \deg g$ because k is a domain, and
$$(fg)^* = z^{\deg f + \deg g}(fg)\big(\tfrac{x_1}{z}, \ldots\big) = z^{\deg f} f\big(\tfrac{x_1}{z}, \ldots\big) \cdot z^{\deg g} g\big(\tfrac{x_1}{z}, \ldots\big) = f^*g^*$$
2. Additivity of $(-)_*$ is again the homomorphism property. For the second identity, all three of $f^*, g^*, (f+g)^*$ arise from the same substitution $x_i \mapsto x_i/z$ scaled by different powers of z, namely z^r, z^s and z^u. Multiplying each by the power of z that brings it to the common exponent $r+s$ gives
$$z^s f^* + z^r g^* = z^{r+s}(f+g)\big(\tfrac{x_1}{z}, \ldots\big) = z^{r+s-u}(f+g)^*$$
The correction z^{r+s-u} is needed exactly when the top-degree terms of f and g cancel, so that $u < \max(r, s)$.
3. Write $F = z^r H$ with $z \nmid H$, so H is homogeneous of degree $\deg F - r$. Setting $z = 1$ gives $F_* = H_*$, and since $z \nmid H$ the polynomial H_* has degree exactly $\deg H$; homogenizing therefore returns $(H_*)^* = H$, whence $z^r(F_*)^* = z^r H = F$, as we sought to show.

Corollary 8.3.6: Factorization of Homogeneous Polynomials

Let $F \in k[x_1, \ldots, x_{n+1}]$ be a homogeneous polynomial. Then up to a power of x_{n+1}, factorizing F is the same as factoring F_*.

In particular, if $F \in k[x, y]$, then F factors into a product of linear factors.

Proof:
The first result is immediate from proposition 8.3.5. For the second, as we are asuming k is

algebraically closed in this section we get that F_* must factor into a product of lienar terms,

$$F_* = c\prod(x-\lambda_i)$$

Then by homogenizing we get:

$$F = cy^r\prod(x-\lambda_i y)$$

as we sought to show.

From this, we are justified in decomposing projective algebraic sets by their behaviour when dehomogenizing elements of the generating ideal:

Example 8.3.7: Zero Sets of Projective Space

Example 8.3.4 contained multiple examples of zero sets. All of those examples were from a single homogeneous polynomial.

1. Every point $[a_1 : \cdots : a_{n+1}] \in \mathbb{P}^n_k$ is a projective algebraic set. A point $[x_1 : \cdots : x_{n+1}]$ equals it exactly when

$$(x_1, \ldots, x_{n+1}) = \lambda(a_1, \ldots, a_{n+1}) \qquad \text{for some } \lambda \neq 0$$

that is, exactly when the two vectors are proportional, which holds exactly when each 2×2 minor of the matrix

$$\begin{pmatrix} x_1 & x_2 & \cdots & x_{n+1} \\ a_1 & a_2 & \cdots & a_{n+1} \end{pmatrix}$$

vanish, namely that

$$\begin{vmatrix} x_i & x_j \\ a_i & a_j \end{vmatrix} = a_j x_i - a_i x_j = 0$$

These are $\binom{n+1}{2}$ equations, of which n already suffice (one for each variable beyond a fixed nonzero coordinate). In other words, we get a point via linear homogeneous polynomials with the appropriate coefficients. This shows how "linear" projective space is. These equations can be thought of as the intersection of all the hyper-planes passing through $[a_0 : \cdots : a_n]$. Equivalently, if you know some differential geometry, it shows that we can view $\mathbb{P}^n_k$ as the Grassmannian $G(1, n+1)$.

The coordinate ring associated to a point would be:

$$\frac{k[x_1, \ldots, x_{n+1}]}{\langle a_j x_i - a_i x_j \big| 1 \leq i, j \leq n+1\rangle} \cong k[x_{n+1}]$$

That is, the coordinate ring is *not* just k at a single point! The reader may think of this as the grading condition for the coordinate rings.

2. In $\mathbb{P}^1_k$, the only closed sets are points, the empty set, and the entire space. We shall see why when we show that any closed subset of projective space can be restricted to a closed subset of affine space, in which these are the only possibilities

3. $\mathbb{P}^2_k$ has many more closed sets just like $\mathbb{A}^2_k$. Any line is given by $a_1x_1 + a_2x_2 + a_3x_3$ and any conic by $ax_1^2 + bx_1x_2 + cx_2^2 + dx_1x_3 + ex_2x_3 + fx_3^2$. A famous set of curves is the Fermat curves given by $x_1^n + x_2^n + x_3^n$. There are many more equations.

To give one example of a coordinate ring, the coordinate ring for a line is isomorphic to $k[s,t]$ with $\deg(s) = \deg(t) = 1$.

4. In $\mathbb{P}^3_k$ the twisted cubic is cut out by three equations,

$$x_1x_3 - x_2^2, \quad x_1x_4 - x_2x_3, \quad x_2x_4 - x_3^2$$

and its homogeneous coordinate ring is

$$\frac{k[x_1, x_2, x_3, x_4]}{(x_1x_3 - x_2^2,\ x_1x_4 - x_2x_3,\ x_2x_4 - x_3^2)} \cong k[s^3, s^2t, st^2, t^3] \subseteq k[s,t]$$

the subring of $k[s,t]$ generated by the degree-three monomials. That isomorphism exhibits the parameterization $[s : t] \mapsto [s^3 : s^2t : st^2 : t^3]$, so the twisted cubic is a *rational* curve: it is the image of $\mathbb{P}^1$. What is notable about it is different: it needs three equations in $\mathbb{P}^3$ although its codimension is two, so it is not a complete intersection.

Another example in $\mathbb{P}^3_k$ are the quadratic surfaces given by $\sum_{i,j} a_{ij}x_ix_j$.

8.4 The Projective Nullstellensatz and Irrelevant Ideals

As we are considering the case where k is algebraically closed, we may naturally want the Nullstellensatz to apply between projective space and homogeneous polynomials. In particular, if $I \subseteq k[x_1, ..., x_n]$ then $Z(I) = \emptyset$ if and only if $I = (1)$. However, this is not the case:

Example 8.4.1: Nullstellensatz False in Projective Space

Let $\mathfrak{m} = (x_1, \ldots, x_{n+1})$, the ideal of all polynomials with zero constant term. Its affine zero set is the origin, and the origin is not a point of $\mathbb{P}^n_k$, so $\mathcal{Z}_{\mathbb{P}}(\mathfrak{m}) = \emptyset$ while $\mathfrak{m} \neq (1)$.

The same happens for every power: writing $\mathfrak{m}^s$ for the ideal generated by all monomials of degree s, its affine zero set is again just the origin, so $\mathcal{Z}_{\mathbb{P}}(\mathfrak{m}^s) = \emptyset$. So the projective Nullstellensatz cannot say "empty zero set implies the unit ideal"; the best it can say is "empty zero set implies the ideal contains some $\mathfrak{m}^s$", which is lemma 8.4.2.

Fortunately, the above ideals are the only ideals for which this could happen, namely because these ideals correspond to ideals that contain the origin in affine space:

Lemma 8.4.2: Characterizing Irrelevant Ideals

Let k be algebraically closed and let $I \subseteq k[x_1, \ldots, x_{n+1}]$ be a homogeneous ideal. Then $\mathcal{Z}_{\mathbb{P}}(I) = \emptyset$ if and only if $\mathfrak{m}^s \subseteq I$ for some $s > 0$, where $\mathfrak{m} = (x_1, \ldots, x_{n+1})$.

***Proof*:**

If $\mathfrak{m}^s \subseteq I$ then every x_i^s lies in I, so any point of $\mathcal{Z}_{\mathbb{P}}(I)$ has all coordinates zero, and $[0 : \cdots : 0]$ is not a point of $\mathbb{P}^n_k$. Hence $\mathcal{Z}_{\mathbb{P}}(I) = \emptyset$.

Conversely assume $\mathcal{Z}_{\mathbb{P}}(I) = \emptyset$. As $k[x_1, \ldots, x_{n+1}]$ is Noetherian, write $I = (f_1, \ldots, f_r)$ with each f_i homogeneous. Fix an index j and dehomogenize at x_j: the polynomials

$$(f_i)_* = f_i(x_1, \ldots, 1, \ldots, x_{n+1}) \qquad i = 1, \ldots, r$$

have no common root in $\mathbb{A}_k^n$, since a common root α would lift to the projective point obtained by inserting 1 in the jth slot, which would lie in $\mathcal{Z}_{\mathbb{P}}(I)$. By the weak Nullstellensatz (theorem 2.4.3) there are g_i with

$$\sum_{i=1}^{r} (f_i)_* \, g_i = 1$$

Homogenizing this identity and clearing denominators multiplies the right hand side by a power of x_j, giving $x_j^{m_j} \in I$ for some $m_j \geq 1$. Running the argument for each j produces exponents $m_1, \ldots, m_{n+1}$.

Put $m = \max_j m_j$ and $s = (n+1)(m-1)+1$. Any monomial $x_1^{a_1} \cdots x_{n+1}^{a_{n+1}}$ of total degree at least s must have some $a_j \geq m \geq m_j$, by the pigeonhole principle: were every $a_j \leq m-1$, the total degree would be at most $(n+1)(m-1) < s$. Such a monomial is therefore divisible by $x_j^{m_j} \in I$ and so lies in I. As $\mathfrak{m}^s$ is generated by the monomials of degree s, this gives $\mathfrak{m}^s \subseteq I$, as we sought to show.

Definition 8.4.3: Irrelevant Ideal

The ideal $\mathfrak{m} = (x_1, \ldots, x_{n+1}) \subseteq k[x_1, \ldots, x_{n+1}]$, consisting of the polynomials with zero constant term, is the *irrelevant ideal*. Its powers $\mathfrak{m}^s$ are the ideals generated by the monomials of degree s.

The name records the point of lemma 8.4.2: $\mathfrak{m}$ and its powers are precisely the homogeneous ideals invisible to projective space, so the projective correspondence must quotient them out. Note that $\mathfrak{m}^s$ and $(x_1^s, \ldots, x_{n+1}^s)$ are different ideals, though each contains a power of the other, which is all the lemma needs.

8.5 The Hilbert Function of a Projective Variety

An affine variety carries its coordinate ring, and dimension is read off from it as a Krull dimension. A projective variety carries a *graded* ring, and the grading holds more information than the dimension alone: counting how fast the graded pieces grow recovers the dimension, and the constant in front of that growth is a genuinely new invariant, the degree. This section extracts both, and the arithmetic genus that chapter 14 needs.

Throughout let $S = k[x_0, \ldots, x_n]$ with its grading by degree, $X \subseteq \mathbb{P}_k^n$ a projective variety, and $S(X) = S/\mathcal{I}_{\mathbb{P}}(X)$ its homogeneous coordinate ring, which inherits a grading because $\mathcal{I}_{\mathbb{P}}(X)$ is homogeneous.

Definition 8.5.1: Hilbert Function of a Projective Variety

The *Hilbert function* of X is

$$H_X : \mathbb{N} \to \mathbb{N} \qquad H_X(m) = \dim_k S(X)_m$$

the dimension over k of the degree-m graded piece of $S(X)$.

The definition is a count: $H_X(m)$ is the number of linearly independent forms of degree m on $\mathbb{P}^n$, modulo those vanishing on X. Equivalently it is the number of independent degree-m conditions X can impose. That it eventually behaves polynomially is Hilbert's theorem on graded modules.

Theorem 8.5.2: Hilbert Polynomial

There is a polynomial $h_X \in \mathbb{Q}[T]$, the *Hilbert polynomial* of X, and an integer m_0, such that

$$H_X(m) = h_X(m) \qquad \text{for all } m \geq m_0$$

Moreover $\deg h_X = \dim X$.

***Proof Key ideas*:**
Polynomiality is a statement about finitely generated graded modules over a polynomial ring and carries no geometric content, so it is cited rather than rebuilt: see *Commutative Algebra* [5], where it appears as the graded companion of the Hilbert-Samuel theory this book already cites in section 5.1. The argument is an induction on n, using the exact sequence obtained by multiplying by a variable.

That $\deg h_X = \dim X$ is the geometric half. Intersecting X with a general hyperplane drops the dimension by one (theorem 4.4.1) and drops the degree of the Hilbert polynomial by one, since $H_{X\cap H}(m) = H_X(m) - H_X(m-1)$ for large m and taking that difference lowers the degree of a polynomial by exactly one. Inducting until the dimension reaches zero, where the Hilbert polynomial is the constant counting points, matches the two numbers.

With polynomiality in hand, the leading coefficient is available to be named.

Definition 8.5.3: Degree and Arithmetic Genus

Let $X \subseteq \mathbb{P}^n_k$ be a projective variety of dimension d, so that h_X has degree d and its leading coefficient is c. The *degree* of X is

$$\deg X = d!\, c$$

and the *arithmetic genus* of X is

$$p_a(X) = (-1)^d\,(h_X(0) - 1)$$

The normalization by $d!$ is what makes $\deg X$ an integer and makes $\deg \mathbb{P}^d = 1$, as the first example below checks. Both invariants depend on the embedding $X \subseteq \mathbb{P}^n$ through the grading, not on X alone; a variety embedded two different ways can have two different degrees, which is why the degree of a curve is always quoted together with the projective space it sits in.

Example 8.5.4: Hilbert Polynomials

1. *Projective space.* $S(\mathbb{P}^n) = S$, and the monomials of degree m in $n+1$ variables number $\binom{m+n}{n}$, so

$$H_{\mathbb{P}^n}(m) = \binom{m+n}{n} = \frac{(m+n)(m+n-1)\cdots(m+1)}{n!}$$

This is already a polynomial in m of degree n with leading coefficient $1/n!$. So $\dim \mathbb{P}^n = n$

and $\deg \mathbb{P}^n = n! \cdot \frac{1}{n!} = 1$, and $h_{\mathbb{P}^n}(0) = \binom{n}{n} = 1$ gives $p_a(\mathbb{P}^n) = 0$.

2. *A hypersurface.* Let $X = \mathcal{Z}(F) \subseteq \mathbb{P}^n$ with F irreducible of degree e. Multiplying by F is injective on S and raises degree by e, so

$$H_X(m) = \binom{m+n}{n} - \binom{m-e+n}{n}$$

The two leading terms $m^n/n!$ cancel, leaving degree $n-1$ with leading coefficient $e/(n-1)!$. So $\dim X = n-1$ and $\deg X = e$: *the degree of a hypersurface is the degree of its equation.* This is the fact Bézout's theorem counts with, and it is the reason chapter 13 can speak of the degree of a plane curve without ambiguity.

3. *A plane curve.* Taking $n = 2$ in (2), a smooth plane curve of degree e has

$$h_X(T) = \binom{T+2}{2} - \binom{T-e+2}{2} = eT - \frac{e(e-3)}{2}$$

so $h_X(0) = -e(e-3)/2$ and

$$p_a(X) = (-1)^1\left(-\tfrac{e(e-3)}{2} - 1\right) = \frac{(e-1)(e-2)}{2}$$

This is the *genus-degree formula.* Section 14.4 takes it as the definition of the genus and puts it to work, showing that it vanishes exactly for the curves isomorphic to $\mathbb{P}^1_k$. A line and a conic have genus 0, a smooth cubic has genus 1, which is why smooth plane cubics are exactly the curves chapter 15 finds a group law on.

4. *The twisted cubic.* From example 8.3.7, $S(X) \cong k[s^3, s^2t, st^2, t^3]$, whose degree-$m$ piece is spanned by the monomials of degree $3m$ in s, t, of which there are $3m+1$. So $h_X(T) = 3T+1$: dimension 1, degree 3, and $p_a = -(1-1) = 0$. It is a rational curve of degree 3 in $\mathbb{P}^3$, matching example 8.3.7(4).

Items (1) and (4) both describe a curve of genus 0, but with different degrees, 1 and 3. That is the embedding-dependence made concrete: the twisted cubic is abstractly isomorphic to $\mathbb{P}^1$, and the two are told apart by their Hilbert polynomials only because they sit in projective space differently.

Exercise 8.5.1

1. Let $R_d \subseteq k[x_1, \dots, x_n]$ be the space of homogeneous polynomials of degree d. Show that $\dim_k R_d = \binom{d+n-1}{n-1}$.

2. Let $T\colon \mathbb{A}^{n+1}_k \to \mathbb{A}^{n+1}_k$ be a linear isomorphism, inducing a map on $\mathbb{P}^n_k$, and let $V \subseteq \mathbb{P}^n_k$ be a subset with $V^T = T^{-1}(V)$.
   a) Show that V is a projective variety if and only if V^T is.
   b) Show that $V \cong V^T$, and that T induces an isomorphism of homogeneous coordinate rings $\Gamma_h(V) \cong \Gamma_h(V^T)$.

3. Let $V = \mathbb{P}^1_k$ with $\Gamma_h(V) = k[x, y]$ and set $t = x/y$. Show that $k(V) \cong k(t)$, and that the points of $\mathbb{P}^1_k$ correspond to the discrete valuation rings R with $\mathrm{Frac}(R) = k(V)$ and $k \subseteq R$. Which valuation ring belongs to the point at infinity?

4. Show that any two hyperplanes in $\mathbb{P}^n_k$ meet, for $n \geq 2$: if L_1, L_2 are nonzero linear forms then $\mathcal{Z}(L_1) \cap \mathcal{Z}(L_2) \neq \emptyset$. Then show the corresponding statement for two *lines* is true in $\mathbb{P}^2_k$ and false in $\mathbb{P}^3_k$, by exhibiting two disjoint lines there.

5. (Euler's relation) Let $\frac{\partial f}{\partial x_i}$ denote the formal partial derivative. Show that if $f \in k[x_0, \dots, x_n]$ is homogeneous of degree d, then
$$df = \sum_{i=0}^{n} x_i \frac{\partial f}{\partial x_i}$$
It suffices to check both sides on a monomial, since both are k-linear.

6. Let $R = \bigoplus_{j \geq 0} R_j$ be a graded ring and write $1 = \sum_j r_j$ with $r_j \in R_j$. Show that r_0 is a multiplicative identity for R_0, and in fact that $1 = r_0$.

7. Show that $\mathcal{I}(\mathcal{Z}(I)) = \sqrt{I}$ fails for the irrelevant ideal of definition 8.4.3, and state the projective Nullstellensatz that repairs it.

8. Compute the Hilbert polynomial of the twisted cubic in $\mathbb{P}^3_k$ directly from definition 8.5.1, and check the degree and arithmetic genus against example 8.5.4(4).

9. Show that a homogeneous polynomial in two variables over an algebraically closed field factors into linear forms, and deduce corollary 8.3.6 for $n = 1$.

10. (Analysis aside, over $\mathbb{R}$.) Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be d times continuously differentiable and homogeneous of degree d, meaning $f(\lambda v) = \lambda^d f(v)$ for all $\lambda > 0$. Show that f is a homogeneous polynomial of degree d. So for smooth enough functions, homogeneity in the scaling sense already forces homogeneity in the algebraic sense.

9

Quasi-Projective Varieties and Regular Maps

Projective space is covered by $n+1$ copies of affine space, and that single observation converts every local question about a projective variety into an affine one. This chapter develops the consequence: the class of varieties closed under both taking closed subsets and taking open subsets, the quasi-projective varieties, which contains the affine and the projective ones and is the setting in which the rest of the book works.

9.1 The Affine Cover of Projective Space

A common characterization of $\mathbb{P}_k^n$ is by showing that it can be constructed by gluing together $n+1$ copies of $\mathbb{A}_k^n$. Let $R = k[x_1, x_2, ..., x_{n+1}]$, and let H_i be the zero set of the (homogeneous) polynomial x_i for each i, giving us $H_1, H_2, ..., H_{n+1}$. These sets certainly cover $\mathbb{P}_k^n$ (their union is $\mathbb{P}_k^n$) and are closed as

$$H_i = \mathcal{Z}(x_i)$$

If we consider the map $\mathbb{A}_k^n \to \mathbb{P}_k^n$ mapping $(a_1, ..., a_n) \mapsto [1 : a_1 : \cdots : a_n]$, then the hyperplane H_{n+1} is called the *hyperplane at infinity*, being the hyperplane where $x_{n+1} = 0$ (it contains the points at infinity). For any hyperplane H_i, we have a map

$$H_i \to \mathbb{P}_k^{n-1} \qquad [a_1, ..., a_{i-1}, 0, a_{i+1}, ..., a_n] \to [a_1, ..., a_{i-1}, a_{i+1}, ..., a_n]$$

Once we define regular maps, the reader can verify that $H_i \cong \mathbb{P}_k^{n-1}$. Let

$$D(x_i) = U_i = \mathbb{P}_k^n - H_i$$

Then these sets are open and cover $\mathbb{P}_k^n$, as H_i is closed, and they contain all points where $x_i \neq 0$. Define a map:

$$\varphi_i \colon U_i \to \mathbb{A}_k^n \qquad [a_1 : \cdots : a_{n+1}] \mapsto \left(\frac{a_1}{a_i}, \ldots, \frac{a_{n+1}}{a_i}\right)$$

where we omit $\frac{a_i}{a_i}$. Notice how this is similar to de-homogenization. Verify that the map is well-defined, namely that it is independent of representative of points $U_i \subseteq \mathbb{P}_k^n$. Then:

Proposition 9.1.1: Projective Space Affine Cover

Let φ_i be defined as above. Then φ_i is a homeomorphism:

$$U_i \cong \mathbb{A}_k^n$$

***Proof*:**
First, φ_i is certainly a bijection: if

$$[x_1 : \cdots : 1_i : \cdots : x_n] = \varphi_i(x_1, ..., x_n) = \varphi(y_1, ..., y_n) = [y_1 : \cdots : 1_i : \cdots : y_n]$$

these two are equal if each $x_j = \lambda y_j$ for some $\lambda \in k$. Indeed, since $1_i = \lambda 1_i$, we must have $\lambda = 1$, which shows that $x_j = y_j$ for each coefficient, giving injectivity. For surjectivity, given any $[x_1, ..., x_i, ..., x_n]$ where $x_i \neq 0$, take:

$$[x_1 : \cdots : x_i : \cdots x_n] = \left[\frac{x_1}{x_i} : \cdots : 1_i : \cdots : \frac{x_n}{x_i}\right]$$

We now see that $(\frac{x_1}{x_i}, ..., \frac{x_n}{x_i})$ map to this point, giving surjectivity, and hence bijectivity.

For continuity, notice that $\varphi_i = \pi \circ \psi_i$ where $\psi_i : \mathbb{A}_k^n \to \mathbb{A}_k^{n+1} \setminus \{0\}$ is the natural embedding with ith-coordinate shifted by 1, and π is the projection. The map π is continuous as $\mathbb{P}_k^n$ is given the natural quotient from $\mathbb{A}_k^n$, and ψ_i is continuous as any pre-image of closed subsets of $\mathbb{A}_k^{n+1} \setminus \{0\}$ is given by the dehomogenization of the polynomials at the ith coordinate.

Furthermore, the map is open, since ψ_i is open (its image is a shifted-by-1 copy of $\mathbb{A}_k^n$, so any open set in $\mathbb{A}_k^n$ is open in $\mathcal{Z}(y_i - 1) \cap \mathbb{A}_k^{n+1} \setminus \{0\}$) and π is open since for any open subset $U \subseteq \mathbb{A}_k^{n+1} \setminus \{0\}$ the set

$$\bigcup_{\lambda \in k^*} \lambda U$$

is open, since the map $x \mapsto \lambda x$ is a homeomorphism (as is quickly verifiable). This is the pre-image of $\pi(U)$ and so since π is a quotient map we have that $\pi(U)$ is also open. Thus, the composition is open, making φ open.

Combining all the results, we get that φ_i is a homeomorphism, as we sought to show.

Given $H_i \cong \mathbb{P}_k^{n-1}$, this implies that we have a decomposition:

$$\mathbb{P}_k^n = \mathbb{A}_k^n \sqcup \mathbb{P}_k^{n-1}$$

For example, if $n = 2$, then:

$$\mathbb{P}_k^2 = \mathbb{A}_k^2 \sqcup \mathbb{P}_k^1$$

For this reason, we often think of $\mathbb{P}_k^2$ to be $\mathbb{A}_k^2$ along with a "line at infinity" (the reader should take the time to internalize this idea). This decomposition can be extended further, breaking down $\mathbb{P}_k^{n-1}$ into an affine and projective space. Repeating inductively, we get:

$$\mathbb{P}_k^n = \mathbb{A}_k^n \sqcup \mathbb{A}_k^{n-1} \sqcup \cdots \sqcup \mathbb{A}_k^1 \sqcup \mathbb{A}_k^0$$

Another "immediate" result from our above discussion is that projective space $\mathbb{P}_k^n$ can be covered with $n+1$ copies of $\mathbb{A}_k^n$. This shall become the dominant way of viewing projective space in modern algebraic geometry (similar to how smooth manifolds are viewed as patches of euclidean space glued together), making this result important enough to be a theorem:

Theorem 9.1.2: Projective Variety Affine Cover

Let $Y \subseteq \mathbb{P}_k^n$ be a projective algebraic set. Then the sets $Y \cap U_i$, $1 \leq i \leq n+1$, form an open cover of Y, and each one is isomorphic to an *affine* algebraic set in $\mathbb{A}_k^n$.

***Proof*:**
The U_i cover $\mathbb{P}_k^n$, since a point has some nonzero coordinate, so the $Y \cap U_i$ cover Y and are open in Y. Fix i and identify U_i with $\mathbb{A}_k^n$ by the homeomorphism φ_i of proposition 9.1.1.

Let $Y = \mathcal{Z}_{\mathbb{P}}(F_1, \ldots, F_r)$ with each F_j homogeneous. Under φ_i a point of U_i is the class of a vector whose ith coordinate is 1, and F_j vanishes there exactly when its dehomogenization $(F_j)_*$ at x_i vanishes at the corresponding point of $\mathbb{A}_k^n$. Hence

$$\varphi_i(Y \cap U_i) = \mathcal{Z}_{\mathbb{A}}\big((F_1)_*, \ldots, (F_r)_*\big)$$

an affine algebraic set. Conversely φ_i^{-1} carries that set back, so the two are isomorphic, as we sought to show.

The content is the second clause: it is not just that Y is covered by sets homeomorphic to open subsets of $\mathbb{A}^n$, but that each piece is a genuine affine algebraic set with its own coordinate ring. Every local question about a projective variety can therefore be answered affinely, which is the working method for the rest of the book.

For the reader's awareness, we box the following[1]:

[1] this shall be the transition maps to verify the gluing is well-defined

Corollary 9.1.3: Transition Between Affine Cover

For $i \neq j$ the two charts agree on the overlap: the composite

$$\varphi_{ij} = \varphi_j \circ \varphi_i^{-1} \colon \varphi_i(U_i \cap U_j) \to \varphi_j(U_i \cap U_j)$$

is an isomorphism of affine varieties, given in coordinates by:

$$\begin{aligned}(x_1, ..., \hat{x}_i, ..., x_{n+1}) &\mapsto [x_1, ..., x_{i-1}, 1, x_{i+1}, ..., x_{n+1}] \\ &\mapsto \left[\frac{x_1}{x_j}, ..., \frac{x_{i-1}}{x_j}, \frac{1}{x_j}, \frac{x_{i+1}}{x_j}, ..., \frac{x_j}{x_j} ..., \frac{x_{n+1}}{x_j}\right] \\ &\mapsto \left(\frac{x_0}{x_j}, ..., \frac{x_{i-1}}{x_j}, \frac{1}{x_j}, \frac{x_{i+1}}{x_j}, ..., \hat{1}_j ..., \frac{x_{n+1}}{x_j}\right)\end{aligned}$$

where the hats mark omitted variables, and $i < j$ is assumed without loss of generality.

***Proof*:**
On $U_i \cap U_j$ both x_i and x_j are nonzero, so each displayed coordinate is a ratio with nonvanishing denominator and the composite is regular. Reversing the roles of i and j gives φ_{ji}, and $\varphi_{ji} \circ \varphi_{ij} = \mathrm{id}$ because both sides come from the identity on $U_i \cap U_j$. So φ_{ij} is an isomorphism, completing the proof.

9.2 Projective Closure

Let us now formalize extending closed sets of $\mathbb{A}_k^n$ to $\mathbb{P}_k^n$, and conversely restrict closed sets of $\mathbb{P}_k^n$ to any $U_i \cong \mathbb{A}_k^n$. Let us first take a closed subset $V \subseteq \mathbb{P}_k^n$, and consider its restriction to $U_i \cong \mathbb{A}_k^n$. As $U_i = D(x_i)$ is open, $V \cap U_i = W_i$ is open in V (the subspace topology on U_i and the Zariski topology transported by φ_i agree). Let us show that W_i is closed in $U_i \cong \mathbb{A}_k^n$, that is it is the zero set of the appropriate polynomials. If $f_1, \ldots, f_r$ are the homogeneous polynomials defining V[2] so that

$$f_1 = \cdots = f_n = 0$$

with $\deg f_i = n_i$, then on U_i each f_j is converted to:

$$x_i^{n_j} f_j \left(\frac{x_1}{x_j}, ..., 1_j, ..., \frac{x_{n+1}}{x_j}\right)$$

For example $f(x, y, x_3) = xy - x_3^2$ would be converted to $xy - 1$ in U_3. Then we can see that $V = \cup W_i$, that is each V can be covered by closed subsets from each U_i.

Conversely, if $W \subseteq \mathbb{A}_k^n$ is a closed subset, we may take the closure in $\mathbb{P}_k^n$, $\overline{W}$, which is the intersection of all closed sets containing W where we interpret W as being set in one of the U_i's. Some closed W are also closed in $\mathbb{P}_k^n$. For example $x^2 + y^2 - 1$ over $\mathbb{C}[x, y]$ when embedded becomes the zero set of $x^2 + y^2 - z^2$ whose points are $[x_0 : y_0 : 1]$ with $x_0^2 + y_0^2 = 1$, together with the two circular points at

[2] Recall that we are working over Noetherian rings, and hence each ideal is finitely generated

infinity of example 8.3.4. On the other hand, if we take $xy - 1$ and consider $xy - z^2$, we get that the traditional zero's become:

$$\left[a : \frac{1}{a} : 1\right]$$

and we now get a new 0 when $z = 0$ giving two more points:

$$[1 : 0 : 0] \qquad [0 : 1 : 0]$$

which we can think of as adding the two points at infinity of the graph of $y = 1/x$. Another example would be $y - x^2$. When extending to $\mathbb{P}^2_k$, we get $yz - x^2$. By setting $x = 0$, we get that $y = 0$ is a solution (which already exists in $\mathbb{A}^2_k$) as well as the new solution:

$$[0 : 0 : 1]$$

which is a new point at infinity that makes the parabola shape akin to an infinite ellipse.

Summarizing our discussion and the pertinent consequences:

Proposition 9.2.1: Closure of Affine Variety

Let $I \subseteq k[x_1, ..., x_n]$ be a (not necessarily homogeneous) ideal, $I^* = \{f^* \mid f \in I\}$, $V^* = \mathcal{Z}(I^*) \subseteq \mathbb{P}^n_k$. Take the last coordinate of $\mathbb{P}^n_k$ to be the coordinate at infinity, so that $\mathbb{A}^n_k$ embeds in $\mathbb{P}^n_k$ via $\iota_{n+1} \colon \mathbb{A}^n_k \xrightarrow{\sim} U_{n+1} \subseteq \mathbb{P}^n_k$. Let $(-)_*$ be the dehomogenization operation so that if $V = \mathcal{Z}(J) \subseteq \mathbb{P}^n_k$, then $I_* = \{f_* \mid f \in J\}$ where f_* is the dehomogenization at the coordinate at infinity, and $V_* = \mathcal{Z}(I_*)$. Let H_∞ represent the hyperplane at infinity (i.e. $\mathcal{Z}(x_{n+1})$). Then:

1. $\iota_{n+1}(V) = V^* \cap U_{n+1}$
2. $(V^*)_* = V$
3. If $V \subseteq W \subseteq \mathbb{A}^n_k$, then $V^* \subseteq W^* \subseteq \mathbb{P}^n_k$
4. If $V \subseteq W \subseteq \mathbb{P}^n_k$, then $V_* \subseteq W_* \subseteq \mathbb{A}^n_k$
5. If V is irreducible in $\mathbb{A}^n_k$, then V^* is irreducible in $\mathbb{P}^n_k$
6. If $V = \cup_i V_i$ is the irreducible decomposition of V in $\mathbb{A}^n_k$, then $V^* = \cup_i V_i^*$ is the irreducible decomposition of V^* in $\mathbb{P}^n_k$
7. If $V \subseteq \mathbb{A}^n_k$, then V^* is the closure of $\iota_{n+1}(V) \subseteq \mathbb{P}^n_k$ in the Zariski topology (i.e. it is the smallest algebraic set in $\mathbb{P}^n_k$ containing $\iota_{n+1}(V)$
8. If $\emptyset \neq V \subseteq \mathbb{A}^n_k$, then no irreducible component of V^* is contained in H_∞, though components may meet it
9. If $V \subseteq \mathbb{P}^n_k$ has no irreducible component contained in H_∞, then $(V_*)^* = V$

***Proof*:**
Parts (3) and (4) are immediate from the definitions, since homogenizing and dehomogenizing preserve containment of ideals and $\mathcal{Z}$ reverses it.

(1) and (2). A point of $\mathbb{A}^n_k$ satisfies f exactly when its image in U_{n+1} satisfies f^*, because f^* restricted to $x_{n+1} = 1$ is f by proposition 8.3.5(3). Intersecting with U_{n+1} therefore recovers $\iota_{n+1}(V)$, which is (1), and dehomogenizing recovers V, which is (2).

(7). By (1), V^* is a closed set whose trace on U_{n+1} is $\iota_{n+1}(V)$, so $V^* \supseteq \overline{\iota_{n+1}(V)}$. Conversely let W be any closed set containing $\iota_{n+1}(V)$, say $W = \mathcal{Z}_{\mathbb{P}}(J)$ with J homogeneous. Each $F \in J$ vanishes on $\iota_{n+1}(V)$, so F_* vanishes on V, so $F_* \in \mathcal{I}(V)$ and hence $F = x_{n+1}^r (F_*)^* \in I^*$ for some r, again by proposition 8.3.5(3). Therefore $V^* \subseteq W$, and V^* is the smallest such closed set.

(5) and (6). If V is irreducible then $\mathcal{I}(V)$ is prime, and the homogenization of a prime ideal is prime, because a product of homogeneous elements lies in I^* exactly when the product of their dehomogenizations lies in $\mathcal{I}(V)$. So V^* is irreducible, giving (5). Applying (5) and (7) to each component of $V = \bigcup_i V_i$ gives $V^* = \bigcup_i V_i^*$ with each V_i^* irreducible, and no containments among them since (3) and (2) make $V \mapsto V^*$ inclusion-reflecting; that is (6).

(8) and (9). A component of V^* contained in H_∞ would meet U_{n+1} in the empty set, so by (1) it would contribute nothing to V, contradicting (7)'s minimality. That is (8). For (9), the hypothesis makes V the projective closure of V_*, and (7) identifies that closure with $(V_*)^*$, as we sought to show.

Definition 9.2.2: Projective Closure

Let $V \subseteq \mathbb{A}^n_k$ be an algebraic set. Then $V^* \subseteq \mathbb{P}^n_k$ as defined in proposition 9.2.1 is called the *projective closure*.

Note that the homogenization of an ideal is *not* the homogenization of its generators

Example 9.2.3: Homogenization of Ideal and Generators

Let $I = (y - x^2, z - x^3)$ and $V = \mathcal{Z}(I)$ i.e. the twisted cubic in $\mathbb{A}^3_k$. Show that

1. $zw - xy \in I(V)^*$
2. $zw - xy \notin \big((y - x^2)^*, (z - x^3)^*\big)$

In other words, the ideal $(f_1^*, ..., f_n^*)$ is too small to capture all the new relations (in particular it usually adds a line at infinity).

Next, we define $k_{\mathbb{P}}(V)$ to be the localization of $k_{\mathbb{P}}[V]$. Note that if $f/g \in k(V)$ where $\deg(f) = \deg(g) = d$, then f/g is in fact a well-defined function on V, since:

$$\frac{f(\lambda x)}{g(\lambda x)} = \frac{\lambda^d f(x)}{\lambda^d g(x)} = \frac{f(x)}{g(x)}$$

there are thus many more rational functions defined on V than regular functions. The set of well-defined functions shall be given a name:

Definition 9.2.4: Function Field

Let V be a projective variety. Then:

$$\kappa(V) = \left\{ \frac{f}{g} \;\middle|\; f, g \in k_{\mathbb{P}}[V] \text{ homogeneous of the same degree, } g \notin \mathcal{I}_{\mathbb{P}}(V) \right\}$$

taken inside the fraction field of $k_{\mathbb{P}}[V]$, and modulo the usual identification $f/g = f'/g'$ when $fg' - f'g \in \mathcal{I}_{\mathbb{P}}(V)$. Elements of $\kappa(V)$ are the *rational functions* on V.

Homogeneity of f and g is what makes the ratio well defined on V: degree is not defined on a non-homogeneous element of $k_{\mathbb{P}}[V]$, and it is equality of the two degrees that makes the scaling factors λ^d cancel.

Proposition 9.2.5: Rational Functions over Projective and Affine Variety

Let $V \subseteq \mathbb{A}^n_k$ be an affine variety. Then:

$$k(V) \cong \kappa(V^*)$$

Similarly, if $V \subseteq \mathbb{P}^n_k$ is a projective variety not contained in the hyperplane at infinity. Then:

$$\kappa(V) \cong k(V_*)$$

***Proof*:**
Define a map in the direction of the claim by homogenizing:

$$\alpha\colon k(V) \to \kappa(V^*) \qquad \alpha\left(\frac{f}{g}\right) = \frac{x_{n+1}^{\deg g - \deg f} f^*}{g^*}$$

the power of x_{n+1} inserted so that numerator and denominator have the same degree, as definition 9.2.4 requires; when $\deg f > \deg g$ the correcting factor goes on the denominator instead.

Well defined. If $f/g = f'/g'$ in $k(V)$ then $fg' - f'g \in \mathcal{I}(V)$, and homogenizing that relation, using $(uv)^* = u^*v^*$ from proposition 8.3.5(1), puts the corresponding difference in $\mathcal{I}_{\mathbb{P}}(V^*)$ up to a power of x_{n+1}, which is a unit on U_{n+1} and so does not change the class.

Inverse. Dehomogenizing gives $\beta(F/G) = F_*/G_*$, and $\beta \circ \alpha = \mathrm{id}$ because $(f^*)_* = f$ for every f by proposition 8.3.5(3). For $\alpha \circ \beta = \mathrm{id}$, part (3) of that proposition gives $(F_*)^* = F$ up to a power of x_{n+1}; that power cancels between numerator and denominator since both are corrected to a common degree. Hence α is a bijection, and it is a ring homomorphism because homogenization is multiplicative and additive up to the same powers of x_{n+1}.

The second isomorphism is the first applied to V_*, using $(V_*)^* = V$ from proposition 9.2.1(9), which is exactly the hypothesis that V is not contained in the hyperplane at infinity, as we sought to show.

On the one hand, it may seem counter-intuitive that though we are restricting to degree 0 rational polynomials, we somehow get as many rational functions as when we allowed for any degree $r - s$

rational polynomial. On the other hand, this should not be too strange for the reader: the domain of definition is a dense open subset, and we shall show that V^* is the closure of V when embedded in $\mathbb{P}^n_k$.

We can define the *stalk* at p to be

$$\mathcal{O}_p(V) = \{f/g \in \kappa(V) \mid g(p) \neq 0\}$$

The same argument as proposition 3.2.10 shows $\mathcal{O}_p(V)$ is a local Noetherian domain.

Proposition 9.2.6: Stalk Defined Locally

Let $V \subseteq \mathbb{P}^n_k$ be a projective variety not contained in the hyperplane at infinity, let V_* be its dehomogenization, and let $p \in V_*$. Then:

$$\mathcal{O}_p(V) \cong \mathcal{O}_p(V_*)$$

This shows that the properties of a stalk are inherently "local", depending only on the information around the point p.

***Proof*:**
Restrict the isomorphism α of proposition 9.2.5. A rational function on V lies in $\mathcal{O}_p(V)$ exactly when it has a representative F/G with $G(p) \neq 0$; applying α and its inverse turns such representatives into representatives f/g on V_* with $g(p) \neq 0$, since dehomogenizing at x_{n+1} does not change whether a form vanishes at a point of U_{n+1}. So α carries $\mathcal{O}_p(V)$ onto $\mathcal{O}_p(V_*)$ and is a ring isomorphism between them, as we sought to show.

Next, notice there is more than one natural set of coordinates on projective space. It is natural to consider the coordinates of $\mathbb{P}^n_k$ to be given by degree 1 homogeneous functions, namely each point $x \in \mathbb{P}^n_k$ is of the form $[x_1 : \cdots : x_{n+1}]$ where $[\lambda x_1 : \cdots : \lambda x_{n+1}] = \lambda[x_1 : \cdots : x_{n+1}]$. However, we may also just as naturally choose to represent each point in any degree of homogeneity, namely:

$$[\lambda w_1 : \cdots : \lambda w_{n+1}] = [w_1 : \cdots : w_{n+1}]$$

where we can have $w_i = (x_i)^k$.

Finally, as all homogeneous polynomials must be of the same degree, we can give a rough classification of all hypersurfaces of projective space $\mathbb{P}^n_k$ of degree d. A hypersurface is defined by a single homogeneous equation. As there are $n+1$ variables, there are $\binom{n+d}{d} = N$ possible degree d monomials. Labeling them $M_1, ..., M_N$, any degree d homogeneous polynomial is of the form:

$$\sum_i^N a_i M_i$$

These hypersurface are naturally defined up to multiplication of a constant $c \in k$. But then, the coefficients determine the hypersurface up to a constant, meaning they can be associated to a point in $\mathbb{P}^N_k$, or said differently $\mathbb{P}^N_k$ *parameterizes* the degree d hypersurfaces of $\mathbb{P}^n_k$ or is the *moduli space* for the degree d hypersurfaces of $\mathbb{P}^n_k$.

Note that some points classify the same hypersurface, for example $x^2y = 0$ and $xy^2 = 0$ in $\mathbb{P}^2_k$, both of which cut out the union of the lines $x = 0$ and $y = 0$. The difference is in the multiplicity, information that would be distinguishable when working with *closed subschemes* instead of closed subsets.

9.3 Quasi-Projective Varieties

As we saw in the above, affine and projective varieties are naturally linked to each other. Many results that are proven in classical algebraic geometry apply both to projective and affine varieties. Furthermore, the open subset of projective variety (for example $\mathbb{P}^1_k$ minus the point at infinity and zero) have natural algebraic structure but are not the vanishing set of polynomials (they are the opposite, they are the non-vanishing set of polynomials). The properties of such sets often are strongly tied to varieties, or are even varieties themselves (for example $\mathbb{P}^1_k$ without the point at infinity is the closed set $\mathbb{A}^1_k$ in $\mathbb{A}^1_k$, but is an open set in $\mathbb{P}^1_k$). The definition of quasi-projective varieties aims to unify all of these.

If $X \subseteq \mathbb{P}^n_k$ is closed and $X_i = X \cap U_i$ is open in X, they are in fact *closed* in $U_i \cong \mathbb{A}^n_k$. This can be seen by de-homogenizing: if X is given by equations $f_0, ..., f_n$ where $\deg f_i = n_i$, then U_i is given by:

$$x_i^{-n_j} f_j = f(x_1, .., x_{i-1}, 1, x_{i+1}, ..., x_{n+1}) = 0$$

If we have $X_i \subseteq U_i$ which is closed, then we may consider the closure $\overline{X_i} \subseteq \mathbb{P}^n_k$ by taking the polynomials that generate X_i and homogenizing them, namely if f is one such polynomial where $\deg f = d$, then the equations describing $\overline{X_i}$ are of the form

$$x_i^l f(x_1/x_i, ..., x_{n+1}/x_i) = 0$$

Which implies that:

$$X_i = \overline{X_i} \cap U_i$$

Hence, if $X \subseteq \mathbb{P}^n_k$, then an open subset of X will correspond to a closed subset of $\mathbb{A}^n_k$. As X is an open subset of itself, it corresponds to an open subset of a closed projective algebraic set. Thus, open subsets of closed projective sets encapsulate affine and projective varieties:

Definition 9.3.1: Quasi-projective Variety

A *quasi-projective variety* is an open subset of a closed projective variety.

Example 9.3.2: Quasi-Projective Varieties

1. *Affine varieties are quasi-projective.* $\mathbb{A}^n_k$ is the standard chart $U_0 \subseteq \mathbb{P}^n_k$, an open subset, and a closed $V \subseteq \mathbb{A}^n_k$ is $\overline{V} \cap U_0$ for its projective closure. So every affine variety is open in a projective one.

2. *Projective varieties are quasi-projective*, taking the open set to be everything.

3. *Neither affine nor projective.* $\mathbb{A}^2_k \setminus \{(0,0)\}$ is open in $\mathbb{A}^2_k$, hence quasi-projective. It is not projective, carrying the nonconstant regular function x, and it is not affine: its ring of regular functions is $k[x, y]$, whose maximal ideals include one for the missing origin. The details are the fifth exercise of exercise set 9.5.1.

4. *The complement of a hypersurface.* For a form F of degree d, the set $\mathbb{P}^n_k \setminus \mathcal{Z}(F)$ is quasi-projective and in fact affine, being isomorphic to a closed subvariety of $\mathbb{A}^N$ under the Veronese of definition 11.1.1, which turns $\mathcal{Z}(F)$ into a hyperplane section.

5. *A non-example.* Over $\mathbb{C}$, the subset $\mathbb{Z} \subseteq \mathbb{C} = \mathbb{A}^1_{\mathbb{C}}$ is not quasi-projective. A locally closed

subset of $\mathbb{A}^1_{\mathbb{C}}$ is finite or cofinite, by the sixth exercise of exercise set 11.6.1, and $\mathbb{Z}$ is neither. So it is not even constructible, and by theorem 11.3.4 it is not the image of any regular map.

A closed subset $C \subseteq U$ of a quasi-projective variety is the intersection of C with a closed subset $\mathbb{P}^n_k$. The irreducible decomposition of theorem 2.3.5 carries over verbatim, since it used only that the space is Noetherian. If $Y \subseteq X \subseteq \mathbb{P}^n_k$ with Y quasi-projective, then Y is a *subvariety* of X.

It must be noted that the collection of quasi-projective varieties is larger than the collection affine and projective varieties. For example $\mathbb{A}^2_k \setminus \{(0,0)\}$ is a quasi-projective variety, but is not affine or projective.

Let X be a quasi-projective variety and $U \subseteq X$ an open subset. Define $\Gamma(U, \mathcal{O}_X) \subseteq \kappa(X)$ to be the collection of all rational functions on X that are defined at each $p \in U$. Equivalently:

$$\Gamma(U, \mathcal{O}_X) = \bigcap_{p \in U} \mathcal{O}_p(X)$$

Note that if $U = X$, then by proposition 3.2.9 and proposition 9.2.6,

$$\Gamma(X, \mathcal{O}_X) = k[X] = \Gamma(X)$$

showing this notation is consistent with our earlier definitions.

Next, let's see that checking that a quasi-projective variety is closed is an open-local condition:

Lemma 9.3.3: Closedness of Quasi-projective Variety is Open-local

Let $Y \subseteq X$ be a quasi-projective variety. Then Y is closed if and only if $X = \cup_\alpha U_\alpha$ where U_α is open and $Y \cap U_\alpha$ is closed for every α. Namely, closedness is open-local for quasi-projective varieties

***Proof*:**
If Y is closed then $Y \cap U_\alpha$ is closed in U_α for any open cover, which is the easy direction.

Conversely let $X = \bigcup_\alpha U_\alpha$ with each U_α open and each $Y \cap U_\alpha$ closed in U_α. Write $U_\alpha = X \setminus Z_\alpha$ with Z_α closed, and $Y \cap U_\alpha = T_\alpha \cap U_\alpha$ with T_α closed in X. We claim

$$Y = \bigcap_\alpha (Z_\alpha \cup T_\alpha)$$

which exhibits Y as an intersection of closed sets and so finishes the proof.

For $\subseteq$: let $y \in Y$ and fix α. Either $y \notin U_\alpha$, in which case $y \in Z_\alpha$; or $y \in U_\alpha$, in which case $y \in Y \cap U_\alpha \subseteq T_\alpha$. Either way $y \in Z_\alpha \cup T_\alpha$.

For $\supseteq$: let y lie in every $Z_\alpha \cup T_\alpha$. The U_α cover X, so $y \in U_\beta$ for some β, and then $y \notin Z_\beta$, forcing $y \in T_\beta$. Hence $y \in T_\beta \cap U_\beta = Y \cap U_\beta \subseteq Y$, as we sought to show.

The following shows that any variety is fundamentally locally affine:

Lemma 9.3.4: Varieties Are Locally Affine

Let X be a quasi-projective variety. Then every $x \in X$ has an open neighborhood that is isomorphic to an affine variety

Proof:
Take $X \subseteq \mathbb{P}^n_k$. If $x \in U_{n+1}$, then $x \in X \cap U_{n+1}$ and by definition of a quasi-projective variety:

$$X \cap U_{n+1} = Y - Y_1$$

where Y_1, Y are closed subsets of U_{n+1}. As $x \in Y$, there exists a polynomial f with coordinates in U_{n+1} where $f = 0$ on Y_1 and $f(x) \neq 0$. Let $D(f) = Y \setminus \mathcal{Z}(f)$, the subset of Y on which f does not vanish. It is open in Y and contains x, and it misses Y_1 entirely since f vanishes there, so $D(f) \subseteq Y \setminus Y_1 = X \cap U_{n+1}$ is an open neighbourhood of x inside X.

I claim this neighborhood is isomorphic to an affine variety. If $(f_1, ..., f_m)$ is the ideal associated to $Y \subseteq U_{n+1}$, then define the variety in $\mathbb{A}^{n+1}_k$ via:

$$f_1(x_1, ..., x_n) = \cdots = f_m(x_1, ..., x_n) = 0 \qquad f(x_1, ..., x_n)x_{n+1} = 1$$

i.e.

$$Z = \mathcal{Z}(f_1, ..., f_m, f \cdot x_{n+1} - 1)$$

Then the mapping

$$\varphi(x_1, ..., x_{n+1}) \to (x_1, ..., x_n)$$

gives a regular mapping from Z into $D(f)$, and

$$\psi(x_1, ..., x_n) \to (x_1, ..., x_n, f(x_1, ..., x_n)^{-1})$$

is a regular mapping of $D(f)$ into Z. The two composites are the identity: $\varphi \circ \psi$ drops the last coordinate that ψ appended, and $\psi \circ \varphi$ restores it, since a point of Z satisfies $f \cdot x_{n+1} = 1$ and so has $x_{n+1} = f^{-1}$ determined by the others. Hence $D(f) \cong Z$, an affine variety, as we sought to show.

This result also showed that principal open sets (definition 1.3.5) are naturally associated to an affine algebraic set, and that the regular function on $D(f)$ are:

$$k[D(f)] = k[X][1/f]$$

In particular f is nonzero on $D(f)$, so this is well defined. Overall, when checking if a mapping between quasi-projective is regular, it suffices to check that any affine variety in the codomain has a closed pre-image.

9.4 Regular Maps

Let us now define regular mappings on quasi-projective varieties. This definition should extend that of affine varieties where a polynomial function was given for each coordinate.

Recall that a rational map on $\mathbb{P}^n_k$ is the quotient of two homogeneous polynomials, with degree $r - s$ where r, s is the degree of the numerator and denominator respectively. A regular point x of a rational function f would be a point where f can be represented by p/q with $q(x) \neq 0$. When working over projective varieties, we shall have that most rational polynomials are not rational functions; we shall require that $r = s$, that is it must be of degree 0.

Definition 9.4.1: Regular Point [on Quasi-projective Variety]

Let X be a quasi-projective variety and $f \in \kappa(X)$ a rational function. Then f is *regular at* $x \in X$ if it admits a representative $f = p/q$ with p, q homogeneous of the same degree and $q(x) \neq 0$. Such a q is then nonzero on the whole open set $D(q)$, so being regular at x is a condition on a neighbourhood of x rather than on x alone.

If f is regular at every point of X it is called *regular* on X. The *set* of regular functions on X is written $k[X]$.

When we defined a regular function before, it was an element of the coordinate ring. We have now defined it to be locally regular at all points. When k is algebraically closed, these two are in fact the same. If X is irreducible, then this is the result of proposition 3.2.2. For the general case, say $f = p_x/q_x$ where $q_x \neq 0$ on U_x so that on U_x:

$$q_x f = p_x \tag{9.1}$$

Then this equation holds on all of X. To see this, choose a regular functions that vanishes on $X \setminus U_x$ but not at x (recall that U_x is dense in X), and multiply it by $p_x q_x$. Then we see that (9.1) must hold outside of U_x as both sides of the equality will vanish. Then like the irreducible case, we can now find points $x_1, ..., x_n$ and regular functions $h_1, ..., h_n$ such that

$$\sum_i^n q_x h_i = 1$$

Then for each x_i, multiplying by h_i and adding up we get:

$$f = \sum_i^n p_x h_i$$

and hence f is regular in the same sense given in definition 2.1.2. Recall that it was mentioned that if $X = \mathbb{P}_k^n$ that $k[\mathbb{P}_k^n]$ consists of only constants functions. We shall soon show that if $X = V \subseteq \mathbb{P}_k^n$ is any projective variety that $k[V]$ is the collection of constant functions.

Definition 9.4.2: Regular Map [On Quasi-projective Variety]

Let $f \colon X \to Y$ be a map of quasi-projective varieties with $Y \subseteq \mathbb{P}_k^m$. Then f is *regular* if for every $x \in X$ there is a standard affine chart $U_i \subseteq \mathbb{P}_k^m$ containing $f(x)$ and an open neighbourhood W of x with $f(W) \subseteq U_i$, such that each of the m coordinate functions of $f|_W$ is regular in the sense of definition 9.4.1.

The collection of regular function is denoted $k[X]$, $\mathcal{O}_X(X)$, or $\Gamma(X)$[a], and the collection of regular mapping between two quasi-projective varieties is denoted $\mathrm{Hom}_{\mathbf{Reg}}(X, Y)$.

[a]This notation shall become more prominent in modern algebraic geometry

This definition of regular function is well-defined and is independent of choice of U_i: if $f(x) = (y_0, ..., \hat{1}_i, ..., y_m) \in U_i$ (where the hat indicates omission), it is also contained in U_j when $y_j \neq 0$, and the coordinates of this point in U_i are of the form:

$$(y_0/y_j, ..., 1/y_j, ..., \hat{1}_j, ..., y_m/y_j)$$

Hence, given the mapping $f : U \to U_i$ is given by

$$(f_0, .., \hat{1}_i, .., f_m)$$

then $f : U \to U_j^m$ is given by $(f_0/f_j, ..., 1/f_j, ..., \hat{1}_j, ..., f_m/f_j)$. As $f_j(x) \neq 0$ and the set U' of points of U where $f_j = 0$ is open. Then on U', the functions $f_0/f_j, ..., 1/f_j, ..., \hat{1}_j, ..., f_m/f_j$ are regular, and so $f : U' \to U_j^m$ is regular.

If we have an affine closed set, by theorem 2.1.5 the regular mapping $f\colon X \to Y$ is determined by the pullback $f^{\#}\colon k[Y] \to k[X]$. This is not the case if X, Y are projective, for as mentioned a few times already we shall have $k[X]$ and $k[Y]$ consist of only constant functions. However, it must be noted that for a general quasi-projective variety, $k[X]$ could be quite large[3]. If X is isomorphic to a closed subset of an affine space, it shall be called an *affine variety*, and if it isomorphic to a closed subset of projective space, it shall be called *projective variety*. Not all quasi-projective varieties are affine or projective (for example $\mathbb{A}_k^2 \setminus \{(0,0)\}$).

Let us now characterize regular maps form quasi-projective varieties to projective space[4]. To understand such a map, start with a regular mapping f on X and a point $x \in X$ where $f(x) \in \mathbb{A}_k^m = U_0^m$ so that there is an open neighborhood where $f = (f_1, ..., f_m) : U \to U_i^m$. Then as it is regular we can write $f_i = p_i/q_i$ where p_i, q_i are of the same degree in homogeneous coordinates and $q_i(x) \neq 0$. Taking the l.c.m. of such fractions we get that $f_i = F_i/F_0$ where $F_0, ..., F_m$ are forms of the same degree and $F_0(x) \neq 0$; overall we can think of the point $f(x) \in U_i^m$ as the point:

$$f(x) = [F_0(x) : \cdots : F_m(x)] \in \mathbb{P}_k^m$$

in projective space. Note that there is more than one representative for such a regular function, namely if

$$F(x) = [F_0(x) : \cdots : F_m(x)] \qquad G(x) = [G_0(x) : \cdots : G_m(x)]$$

then they are the same point if and only if $F_iG_j = F_jG_i$ on X for all $0 \leq i, j \leq m$. This gives us an explicit condition for when $Y = \mathbb{P}_k^m$:

Proposition 9.4.3: Equivalent Regular Function

Let $X \subseteq \mathbb{P}_k^n$ be an irreducible quasi-projective variety and $f\colon X \to \mathbb{P}_k^m$ a map of sets. Then f is regular if and only if it can be written as

$$f = [F_0 : \cdots : F_m]$$

for forms F_i of a common degree in the homogeneous coordinates of $\mathbb{P}_k^n$, not all vanishing at any point of X. Two such presentations $[F_0 : \cdots : F_m]$ and $[G_0 : \cdots : G_m]$ define the same map exactly when

$$F_iG_j = F_jG_i \text{ on } X \text{ for all } 0 \leq i, j \leq m$$

***Proof*:**
Suppose f has such a presentation and fix $x \in X$. Some $F_j(x) \neq 0$, and on the open set $D(F_j) \cap X$ the coordinates of f in the chart U_j are the ratios F_i/F_j, each a quotient of forms of equal degree with nonvanishing denominator. By definition 9.4.1 these are regular, so f is

[3]Shafarevich mentions that Rees and Nagata constructed an example where $k[X]$ is not finitely generated
[4]This is in fact a larger exploration that culminates in the theory of line bundles on a projective scheme

regular.

Conversely let f be regular and fix x with $f(x) \in U_j$. In that chart f has coordinates $f_i = p_i/q_i$ with p_i, q_i forms of equal degree and $q_i(x) \neq 0$. Clearing denominators over $Q = q_0 \cdots q_m$ rewrites these as $f_i = F_i/F_j$ with all F_i forms of one degree and $F_j(x) \neq 0$, which is the required presentation near x. Two such local presentations agree on the overlap of their domains, which is a nonempty open subset of the irreducible X and hence dense, so they agree wherever both are defined and glue.

For the last clause, the two presentations give the same point of $\mathbb{P}^m_k$ at x exactly when the vectors $(F_0(x), \dots, F_m(x))$ and $(G_0(x), \dots, G_m(x))$ are proportional, which is the vanishing of every 2×2 minor $F_iG_j - F_jG_i$, as in example 8.3.7(1). Requiring it at every point of X is the stated condition, as we sought to show.

Example 9.4.4: Regular Mappings

1. *Projection*: Let $E \subseteq \mathbb{P}^n_k$ be a d-dimensional subspace given by $n-d$ linearly independent linear equations $(L_1, ..., L_{n-d})$. In homogeneous coordinates, the equations L_i are called *linear forms*. The mapping:
$$\pi(x) = (L_1(x) : \cdots : L_{n-d}(x))$$
is well-defined and is called the projection with center E. Notice this set is regular on $\mathbb{P}^n_k \setminus E$ as the L_i's do not vanish simultaneously. Then for any $X \subseteq \mathbb{P}^n_k$ where $X \cap E = \emptyset$ and X is closed, we have a regular mapping
$$\pi : X \to \mathbb{P}^{n-d-1}_k$$
Geometrically, we can interpret the mapping as follows. Take any $(n-d-1)$-dimensional subspace $H \subseteq \mathbb{P}^n_k$ with $H \cap E = \emptyset$. Through any point $x \in \mathbb{P}^n_k \setminus E$ and the space E, there passes a unique $(d+1)$-dimensional subspace E_x. Then this subspace intersects H at a unique point, namely $\pi(x)$. If X intersects E, but does not contain it, then this is a rational mapping.
We have seen the case where $d = 0$ before (think of the hyperbola to $\mathbb{A}^1_k$)
2. Take $V = \mathcal{Z}(y - x^2, z - x^3)$ (i.e. the twisted cubic). This is a rational curve with parameterization:
$$t \mapsto (t, t^2, t^3)$$
Let us now homogenize this variety. The homogenization of this map is:
$$[s : t] \mapsto \left[1 : \frac{t}{s} : \frac{t^2}{s^2} : \frac{t^3}{s^3}\right] = \left[s^3 : ts^2 : t^2 s : t^3\right]$$
3. Take $V = \mathcal{Z}(x^2 + y^2 - z^2)$; the projectivization of the circle. Define the map:
$$f : \mathbb{P}^1_k \to V \qquad [s : t] \to [s^2 - t^2 : 2st : s^2 + t^2]$$
This is an isomorphism, with inverse $[x : y : z] \mapsto [x + z : y]$. Checking the composite, $(s^2 - t^2)^2 + (2st)^2 = (s^2 + t^2)^2$, so f lands in V; and at $f([s : t])$ we have $x + z = 2s^2$ and $y = 2st$, so $[x + z : y] = [2s^2 : 2st] = [s : t]$ whenever $s \neq 0$, the case $t \neq 0$ being symmetric. So a smooth conic in $\mathbb{P}^2_k$ is isomorphic to $\mathbb{P}^1_k$, which is the genus-zero case of example 8.5.4(3).

4. *The Veronese and Segre embeddings* give further regular maps into larger projective spaces, and are the tools that turn products and degree-d hypersurfaces into linear data. Section 10.1 constructs the Segre embedding and chapter 11 the Veronese.

9.5 Rational Functions on Quasi-Projective Varieties

When working with affine varieties, we had the quotient of two regular functions. However, mentioned a few times, on $\mathbb{P}^n_k$ there are only constant regular functions, and a rational function, being the ratio of two regular functions, would still be constant.

The solution is to recall the definition of $\mathcal{O}_X$, namely:

$$\mathcal{O}_X = \{\varphi = f/g \in k(x_1, \dots, x_{n+1}) \mid f, g \text{ homogeneous of equal degree, } g \notin \mathcal{I}_{\mathbb{P}}(X)\}$$

where f and g are of the same degree. Then take the subset $\mathfrak{m}_X = \{\varphi = f/g \in \mathcal{O}_X \mid f \in I(X)\}$.

Definition 9.5.1: Rational Function On quasi-projective Variety

Let X be a quasi-projective variety. Then the set of rational functions on X is given by:

$$k(X) \cong \mathcal{O}_X/\mathfrak{m}_X$$

By a reasoning similar to proposition 3.2.7, a homogeneous function vanishes on X if and only if it vanishes on an open subset $U \subseteq X$. Hence $k(X) = k(U)$. As a consequence:

$$k(X) = k(\overline{X})$$

Due to this, when studying rational functions on quasi-projective spaces, we may reduce to studying rational functions to affine or projective varieties, which often leads most results to focus on the irreducible case. The set of regular points shall be called the *domain of definition*. If $f\colon X \to \mathbb{P}^m_k$, then it is determined by specifying $m+1$ forms $[F_0 : \cdots : F_m]$ of $n+1$ homogeneous coordinates of projective space $\mathbb{P}^n_k$ containing X (where at least one form must not vanish on X). Furthermore, if f can be given by functions $[f_0 : \cdots f_m]$ where each f_i are regular at all $x \in X$ and not all vanish at this point, then the map is regular at x, and hence determines a regular mapping of some neighborhood x into $\mathbb{P}^m_k$.

We saw before that birational maps are weaker than isomorphisms. However, rational maps are regular on an open dense set (as the reader should verify with quasi-projective varieties). This leads to the following result:

Proposition 9.5.2: Reduction To Open Subset

Let X, Y be irreducible varieties. Then X and Y are birationally isomorphic if and only if they contain isomorphic open subsets $U \subseteq X$ and $V \subseteq Y$.

***Proof*:**
If X and Y contain isomorphic open subsets then those subsets are dense, since X and Y are irreducible, and an isomorphism between dense open sets is a birational map. That is the easy direction.

Conversely let $f\colon X \dashrightarrow Y$ be birational with inverse g. Let $U_1 \subseteq X$ and $V_1 \subseteq Y$ be their domains of definition, both dense open by lemma 3.2.4. Put

$$U = U_1 \cap f^{-1}(V_1) \qquad V = V_1 \cap g^{-1}(U_1)$$

Each is open, and each is nonempty: $f(U_1)$ is dense in Y so it meets the nonempty open V_1, and symmetrically.

Shrinking once more, replace U by $U \cap g(V)$ and V by $V \cap f(U)$, which are again nonempty and open because f and g are dominating. On these sets both composites are defined everywhere and agree with the identity on a dense subset, hence equal it: two regular maps into a variety agreeing on a dense subset are equal, since their equalizer is closed. So $f|_U : U \to V$ and $g|_V : V \to U$ are mutually inverse isomorphisms, as we sought to show.

Exercise 9.5.1

1. Write down the $n+1$ standard charts of $\mathbb{P}^n_k$ explicitly and compute the transition map between two of them for $n = 1$ and $n = 2$. Verify that the transitions are regular where defined, as corollary 9.1.3 asserts.
2. Show that the projective closure of the parabola $\mathcal{Z}(y - x^2) \subseteq \mathbb{A}^2_k$ is $\mathcal{Z}(yz - x^2)$, and identify the point it acquires at infinity. Do the same for the hyperbola $\mathcal{Z}(xy - 1)$.
3. Let $V \subseteq \mathbb{A}^3_k$ be the twisted cubic, with $\mathcal{I}(V) = (y - x^2,\, z - x^3)$. Homogenize those two generators and show that their common projective zero set is strictly larger than the projective closure of V. Conclude that the projective closure is obtained by homogenizing every element of $\mathcal{I}(V)$, not just a set of generators.
4. Show that a regular function on a quasi-projective variety is continuous for the Zariski topology, and deduce that the locus where two regular maps agree is closed.
5. Show that $\mathbb{A}^2_k \setminus \{0\}$ is a quasi-projective variety and that every regular function on it extends to $\mathbb{A}^2_k$. Deduce that it is not isomorphic to any affine variety.
6. Verify the irreducible decomposition theorem for quasi-projective varieties: every quasi-projective variety is a finite union of irreducible closed subsets, none containing another, and the decomposition is unique.
7. Show that $[s : t] \mapsto [s^2 : st : t^2]$ is an isomorphism of $\mathbb{P}^1_k$ onto the conic $\mathcal{Z}(xz - y^2) \subseteq \mathbb{P}^2_k$, and compare with example 9.4.4(3).
8. Show that a rational function on an irreducible quasi-projective variety is determined by its restriction to any nonempty open subset, which is proposition 9.5.2, and give an example showing this fails for reducible varieties.

10

Completeness and Finite Maps

The property that makes projective varieties worth the trouble is completeness: the projection $X \times Y \to Y$ is a closed map whenever X is projective. It is the algebraic counterpart of compactness, and it yields at once that the image of a projective variety under a regular map is closed and that the only regular functions on a projective variety are constant.

Proving it requires knowing what a product of projective varieties is, which is not the naive thing. From there the chapter turns to finite maps, the algebraic mirror of integral extensions, and the theorem that every variety is a finite cover of a space as simple as $\mathbb{A}^m$ or $\mathbb{P}^m$. It closes by measuring a map against its fibres: the dimension of the source is the dimension of the target plus the dimension of the generic fibre, and the special fibres can only be bigger.

10.1 Products and the Segre Embedding

The direct product of algebraic sets $X \subseteq \mathbb{A}_k^n$ and $Y \subseteq \mathbb{A}_k^m$ is again algebraic, and lives in $\mathbb{A}_k^{n+m}$. If $X' \subseteq \mathbb{A}_k^p$ and $Y' \subseteq \mathbb{A}_k^q$ are algebraic sets with $X \cong X'$ and $Y \cong Y'$, then $X \times Y \cong X' \times Y' \subseteq \mathbb{A}_k^{p+q}$, showing that this definition is embedding-independent. The projective case is not so simple. There is no N with $\mathbb{P}_k^n \times \mathbb{P}_k^m \cong \mathbb{P}_k^N$; what there is instead is a *closed embedding* into $\mathbb{P}_k^N$ for $N = (n+1)(m+1)-1$, whose image is a proper subvariety cut out by explicit quadratic equations. Finding that embedding is the business of this section. Defining the product also supplies the tool that section 10.2 needs.

To start, recall that $\mathbb{A}_k^n \times \mathbb{A}_k^m \cong \mathbb{A}_k^{n+m}$. A first attempt at understanding $\mathbb{P}_k^n \times \mathbb{P}_k^m$ would be to consider $\mathbb{P}_k^{n+m}$. However, this space is dependent on the representation of the homogeneous points in $\mathbb{P}_k^n$ and $\mathbb{P}_k^m$, for example if $[2:0] = [1:0] \in \mathbb{P}_k^1$, and $[3:0] = [1:0] \in \mathbb{P}_k^1$, then:

$$[1:1:0] \neq [2:3:0] \in \mathbb{P}_k^2$$

Furthermore, if we choose $[x_1 : x_2] \in \mathbb{P}_k^1$ and $[y_1 : y_2] \in \mathbb{P}_k^1$, which point in $\mathbb{P}_k^2$ ought to represent this point? To resolve this, we shall "twist-multiply"

$$[x_1y_1 : x_1y_2 : x_2y_1 : x_2y_2] \in \mathbb{P}_k^{(1+1)(1+1)-1}$$

Notice that for $[\lambda x_1 : \lambda x_2]$ and $[\mu y_1 : \mu y_2]$ then:

$$[\lambda\mu x_1y_1 : \lambda\mu x_1y_2 : \lambda\mu x_2y_1 : \lambda\mu x_2y_2] = [x_1y_1 : x_1y_2 : x_2y_1 : x_2y_2] \in \mathbb{P}_k^3$$

and hence is independent of representation. Generalizing this, we define for any

$$x = [x_0 : \cdots : x_n], y = [y_0 : \cdots : y_m] \qquad w = [w_{ij}] \in \mathbb{P}_k^{(n+1)(m+1)-1}, \ w_{ij} = x_iy_j$$

Consider now the embedding $\varphi : \mathbb{P}_k^n \times \mathbb{P}_k^m \to \mathbb{P}_k^{(n+1)(m+1)-1}$ given by the above. The image is indeed a closed subset, for it satisfies the equations:

$$w_{ij}w_{kl} - w_{il}w_{kj} = 0 \tag{10.1}$$

which hold because both sides equal $x_ix_ky_jy_l$ once $w_{ij} = x_iy_j$ is substituted. The construction is natural in the sense that it matches the standard affine charts on both sides. Write $\mathbb{A}^n$ for $\mathbb{A}_k^n$ and let $U_i \subseteq \mathbb{P}_k^n$ be the ith standard affine chart, so $U_i \cong \mathbb{A}^n$. Then:

$$\varphi(U_i \times U_j) = W_{ij} = \mathbb{A}_{ij}^N \cap \varphi(\mathbb{P}_k^n \times \mathbb{P}_k^m)$$

where $N = (n+1)(m+1) - 1$ and $\mathbb{A}_{ij}^N$ is the set of points of $\mathbb{P}_k^N$ where $w_{ij} \neq 0$. Consider now the set $\mathbb{A}_{00}^N$. Take a point in projective space where $w_{00} \neq 0$ so that it corresponds to a point in $\mathbb{A}_{00}^N$:

$$[w_{ij}] = \varphi(x, y) \in \mathbb{P}_k^N$$

Take $z_{ij} = w_{ij}/w_{00}$, $x_i = u_i/u_0$, $y_j = v_j/v_0$ as the inhomogeneous coordinates. Then:

$$z_{i0} = x_i \qquad z_{0j} = y_j \qquad z_{ij} = x_iy_j = z_{i0}z_{0j}$$

So $\mathbb{A}_{00}^N \cong \mathbb{A}^{n+m}$ with coordinates $(x_1, \ldots, x_n, y_1, \ldots, y_m)$, the isomorphism being induced by φ. Hence φ is indeed the natural choice for the embedding of $\mathbb{P}_k^n \times \mathbb{P}_k^m$. This may feel more natural if we recall that each point of projective space $[a_0 : \cdots a_n] \in \mathbb{P}_k^n$ is closed and given by a system of linear equations

$$a_ix_j - a_jx_i = 0$$

mirroring (10.1). Notice that we may interpret the point $[w_{ij}]$ as an $(n+1) \times (m+1)$ matrix given by multiplying the points x and y as a $(n+1, 1)$ matrix and a $(1, m+1)$ matrix respectively (note the resulting rank of the matrix will be 1). From this, we may define the product of two quasi-projective varieties

Definition 10.1.1: Segre Embedding: Product of Projective Space

Let $V \subseteq \mathbb{P}_k^n, W \subseteq \mathbb{P}_k^m$ be two quasi-projective varieties. Then:

$$V \times W \subseteq \varphi(\mathbb{P}_k^n \times \mathbb{P}_k^m) \subseteq \mathbb{P}_k^{(n+1)(m+1)-1}$$

is the product of the quasi-projective varieties V, W.

Example 10.1.2: Product of Projective Lines

Let $V = W = \mathbb{P}^1_k$ and consider $\varphi(\mathbb{P}^1_k \times \mathbb{P}^1_k) \subseteq \mathbb{P}^3_k$. This set satisfies the equation:

$$w_{11}w_{00} - w_{01}w_{10} = 0$$

which is a nondegenerate quadric Q in $\mathbb{P}^3_k$. Let us consider the set

$$\varphi(\{\alpha\} \times \mathbb{P}^1_k)$$

then given $\alpha = [\alpha_0 : \alpha_1]$, these are all points in $\mathbb{P}^3_k$ such that

$$\alpha_1 w_{00} - \alpha_0 w_{10} = 0 \qquad \alpha_1 w_{01} - \alpha_0 w_{11} = 0$$

These equations give lines in $\mathbb{P}^3_k$. Similarly for $\varphi(\mathbb{P}^1_k \times \{\beta\})$. The two rulings of the quadric are visible in the following picture:

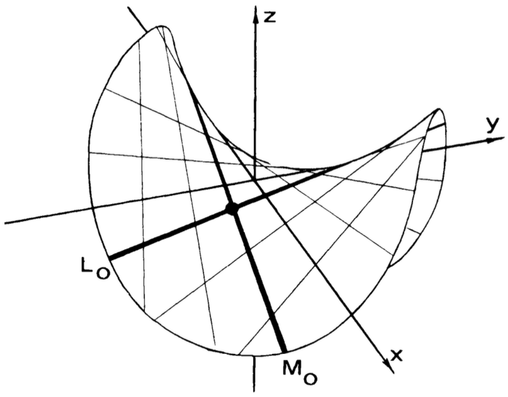


Figure 10.1: Segre Embedding of Product of Projective Lines

The closed subsets of im$\varphi \subseteq \mathbb{P}^{(n+1)(m+1)-1}_k$ correspond well with the closed subsets of $\mathbb{P}^n_k \times \mathbb{P}^m_k$. Let us start with seeing which subset of $\mathbb{P}^n_k \times \mathbb{P}^m_k$ are algebraic subsets under the Segre Embedding. Let $X \subseteq \mathbb{P}^{(n+1)(m+1)-1}_k$ be a sub-variety so that it is defined by homogeneous equations $f_i(w_{00}, ..., w_{nm})$. Then these induce the homogeneous polynomials $g_i(u_0, ..., u_n, v_0, ..., v_m)$ on $\mathbb{P}^n_k \times \mathbb{P}^m_k$ where the u_i, v_j are the same homogeneous degree. Conversely, if there are two systems of homogeneous equations taking variables of the same homogeneous degree in u_i, v_j, then there exists homogeneous polynomials taking variables in $u_i v_j$. If the degrees of the variables u_i and v_j differ, say the former are of degree r and the latter of degree s, then we can create the system of homogeneous equations:

$$v_j^{r-s} g_i(u_0, \ldots, u_n, v_0, \ldots, v_m) = 0$$

assuming $r > s$. One important consequence of this is that we can directly check for a set $Z \subseteq \mathbb{P}^n_k \times \mathbb{P}^m_k$ being closed by looking for a single polynomial in $\mathbb{P}^{(n+1)(m+1)-1}_k$ instead of looking for two polynomials in each separate projective space and verifying the set is the direct product of those two closed sets. Let us box this for future reference:

Proposition 10.1.3: Segre Embedding is Closed

Let $\varphi : \mathbb{P}^n_k \times \mathbb{P}^m_k \to \mathbb{P}^{(n+1)(m+1)-1}_k$ be the Segre embedding. Then $\operatorname{im}\varphi$ is closed. Furthermore, $X \subseteq \mathbb{P}^n_k \times \mathbb{P}^m_k$ is closed if and only if $\varphi(X) \subseteq \operatorname{im}\varphi$ is closed.

***Proof*:**
The image is closed. Substituting $w_{ij} = x_i y_j$ shows every point of $\operatorname{im}\varphi$ satisfies (10.1). Conversely let $w = [w_{ij}]$ satisfy all those relations, and pick (a, b) with $w_{ab} \neq 0$. Set $x_i = w_{ib}$ and $y_j = w_{aj}$. Then for all i, j,

$$x_i y_j = w_{ib} w_{aj} = w_{ij} w_{ab}$$

using (10.1), so $[x_i y_j] = [w_{ij}]$ after scaling by w_{ab}, and $w = \varphi([x], [y])$. Hence $\operatorname{im}\varphi = \mathcal{Z}(\{w_{ij}w_{kl} - w_{il}w_{kj}\})$ is closed. The same computation shows φ is injective, since $[x]$ and $[y]$ are recovered from w up to scalars.

Closed subsets correspond. If $\varphi(X)$ is closed in $\operatorname{im}\varphi$, then $X = \varphi^{-1}(\varphi(X))$ is closed because φ is continuous. Conversely let X be closed, cut out by bihomogeneous equations $g_i(u; v)$. By the degree-balancing displayed above we may assume each g_i has the same degree r in the u and in the v, and a bihomogeneous polynomial of bidegree (r, r) is a polynomial in the products $u_i v_j$, hence a form in the w_{ij}. Those forms cut out $\varphi(X)$ inside $\operatorname{im}\varphi$, as we sought to show.

Before leaving products, three facts about them are needed later, and none is formal. The first says the maps one expects to have are regular, the second is general topology in geometric clothing, and the third says that a product of irreducible varieties does not fall apart.

Proposition 10.1.4: Projections and Slices Are Regular

Let $X \subseteq \mathbb{P}^n_k$ and $Y \subseteq \mathbb{P}^m_k$ be quasi-projective varieties. Then both projections $\pi_1 : X \times Y \to X$ and $\pi_2 : X \times Y \to Y$ are regular. Moreover, for fixed $y \in Y$ the slice map $j_y : X \to X \times Y$, $x \mapsto (x, y)$, is an isomorphism onto $X \times \{y\}$, and symmetrically $i_x : Y \to X \times Y$ is an isomorphism onto $\{x\} \times Y$.

***Proof*:**
Work in the Segre coordinates $w_{ij} = x_i y_j$. Fix a and let $W_a \subseteq X \times Y$ be the open set where some $w_{aj} \neq 0$, which is where $x_a \neq 0$. On W_a,

$$[\, w_{a0} : \cdots : w_{am} \,] = [\, x_a y_0 : \cdots : x_a y_m \,] = [\, y_0 : \cdots : y_m \,]$$

so π_2 is given on W_a by $m + 1$ linear forms in the w_{ij}, not all zero there. The W_a cover $X \times Y$ and the formulas agree on overlaps, so π_2 is regular; the same argument with $[\, w_{0b} : \cdots : w_{nb} \,]$ handles π_1.

For the slices, fix $y = [b_0 : \cdots : b_m]$ and write $j_y(x) = [\, x_i b_j \,]$. The entries are linear forms in x, not all identically zero since some $b_j \neq 0$, so j_y is regular. Its image is $X \times \{y\}$, and $\pi_1 \circ j_y$ is the identity of X while $j_y \circ \pi_1$ is the identity of $X \times \{y\}$, so the two are mutually inverse isomorphisms. The map i_x is symmetric, as we sought to show.

Lemma 10.1.5: A Cover by Overlapping Irreducibles

Let Z be a topological space covered by open sets V_α, each irreducible, with $V_\alpha \cap V_\beta \neq \emptyset$ for every pair. Then Z is irreducible.

***Proof*:**
A space is irreducible exactly when each of its nonempty open subsets is dense. First, every V_α is dense in Z: given β, the set $V_\beta \cap V_\alpha$ is nonempty and open in V_β, hence dense in the irreducible V_β, so $V_\beta \subseteq \overline{V_\alpha}$; as the V_β cover Z this forces $\overline{V_\alpha} = Z$.

Now let $W \subseteq Z$ be nonempty and open. It meets some V_α, and $W \cap V_\alpha$ is a nonempty open subset of the irreducible V_α, hence dense in it, so $\overline{W} \supseteq V_\alpha$. Closures are closed, so $\overline{W} \supseteq \overline{V_\alpha} = Z$, as we sought to show.

Proposition 10.1.6: A Product of Irreducibles Is Irreducible

Let X and Y be irreducible quasi-projective varieties. Then $X \times Y$ is irreducible.

***Proof*:**
By proposition 10.1.4 each slice $\{x\} \times Y$ is isomorphic to Y, hence irreducible, and each slice map j_y is regular. Suppose $X \times Y = Z_1 \cup Z_2$ with both Z_i closed. For fixed x the slice $\{x\} \times Y$ is irreducible and is covered by the two relatively closed sets $Z_i \cap (\{x\} \times Y)$, so it lies wholly inside Z_1 or wholly inside Z_2. Set

$$X_i = \{x \in X \mid \{x\} \times Y \subseteq Z_i\}$$

so that $X = X_1 \cup X_2$. Each X_i is closed, being the intersection $X_i = \bigcap_{y \in Y} j_y^{-1}(Z_i)$ of closed sets. Since X is irreducible, $X = X_i$ for one of the two, and then $X \times Y = Z_i$, as we sought to show.

10.2 Completeness of Projective Varieties

The image of a regular map need not be closed: example 2.1.4 took $X = \mathcal{Z}(xy - 1)$, $Y = \mathbb{A}^1_k$ and $\pi(a, b) = a$, whose image is $\mathbb{A}^1_k \setminus \{0\}$. However, the image of a projective variety will always be projective! If the reader has done some analysis this should remind the reader of the fact that the image of compact sets are compact. Though all affine and projective algebraic sets are compact (they are closed subsets of the compact space $\mathbb{A}^n_k$ or $\mathbb{P}^n_k$), projective varieties behave more like compact sets. Scheme theory draws the distinction by calling projective varieties *proper*[1]. To see this, we require a few lemmas:

[1] In analysis a map is proper if the pre-image of every compact set is compact

Lemma 10.2.1: Graph of Regular Mapping

Let $\varphi : X \to Y$ be a regular mapping between quasi-projective varieties. Then the graph

$$\Gamma(\varphi) = \{(x, \varphi(x)) \mid x \in X\}$$

is closed in $X \times Y$

If the Zariski topology was Hausdorff, then we could have concluded using classical topological results, however some more care has to be done as the topology is not Hausdorff.

***Proof*:**
First, we may take Y to be a projective space; for an appropriate choice of m, $Y \subseteq \mathbb{P}_k^m$, so that $X \times Y \subseteq X \times \mathbb{P}_k^m$. Then by proposition 3.2.7, φ determines a mapping $\overline{\varphi} : X \to \mathbb{P}_k^m$ and $\Gamma_\varphi = \Gamma_{\overline{\varphi}} \cap (X \times Y)$. As $X \times Y$ is closed in $X \times \mathbb{P}_k^m$ by proposition 10.1.3, closed then Γ_f is closed.

Next define $(\overline{\varphi}, \mathrm{id})\colon X \times \mathbb{P}_k^m \to \mathbb{P}_k^m \times \mathbb{P}_k^m$. Then $\Gamma_{\overline{\varphi}}$ is the inverse image of the diagonal $\Delta = \Gamma_{\mathrm{id}}$ under this map. As this is a continuous map, the inverse image of a closed set is closed, and hence it suffices to show that Γ_i is closed.

Finally, the diagonal $\Delta \subseteq \mathbb{P}_k^m \times \mathbb{P}_k^m$ consists of the pairs $([u_0 : \cdots : u_m], [v_0 : \cdots : v_m])$ with $u_i v_j = u_j v_i$ for all i, j, which under the Segre embedding are exactly the points where the corresponding 2×2 minors vanish. By proposition 10.1.3 that is a closed condition, so Δ is closed, hence so is $\Gamma_{\overline{\varphi}}$ and therefore Γ_φ, completing the proof.

Proposition 10.2.2: Projection of Projective Variety

Let X be a projective variety and Y a quasi-projective variety. Then the projection map:

$$\pi_2 : X \times Y \to Y$$

is a closed map (the image of a closed set is closed)

Hence, the projection of a projective variety can always be written as a system of polynomial equations! This is not true for affine varieties: take $X = \mathcal{Z}(xy - 1)$ and $Y = \mathbb{A}_k^1$

***Proof*:**
Let us first reduce the problem: as $X \subseteq \mathbb{P}_k^n$ is closed, $X \times Y \subseteq \mathbb{P}_k^n \times Y$ is a closed subset, hence if $Z \subseteq X$ is closed, it is closed in $Z \times Y \subseteq \mathbb{P}_k^n \times Y$, and so it is enough to show it for the case of $X = \mathbb{P}_k^n$. Next, as being closed is affine local, it suffices to cover Y by affine open subsets U_i and prove the theorem on each of these. Thus, it suffices to assume that Y is an affine variety. Finally, as Y is closed in $\mathbb{A}_k^m$, $\mathbb{P}_k^n \times Y \subseteq \mathbb{P}_k^n \times \mathbb{A}_k^m$ is closed, and so we can reduce further to $Y = \mathbb{A}_k^m$.

Thus, let us characterize closed subsets $Z \subseteq \mathbb{P}_k^n \times \mathbb{A}_k^m$. By definition of the Zariski topology, and by a simple generalization of proposition 10.1.3, we see that Z is closed if and only if it is the zero set of the collection of polynomials

$$g_i([u_0 : \cdots : u_n], y_1, ..., y_m) \qquad 1 \leq i \leq t$$

In particular, we do not need to create two zero sets for each component and check that the their product produces Z: we can directly work with a set of polynomials. Let us write $g_i(u; y)$ to represent these polynomials with the two different inputs.

Now, take any $y_0 \in \mathbb{A}_k^m$. Then $\pi^{-1}(y_0)$ consists of all solutions (whether empty or non-empty) to the system of equations

$$g_i(u; y_0) = 0 \qquad 1 \leq i \leq t$$

Take the collection of y_0 such that the system of polynomials equations $g_i(u; y_0) = 0$ has a nonzero solution; call this collection $T \subseteq \mathbb{A}_k^m$. We want to show that T is closed, for then the projection π_2 is a closed map. By lemma 8.4.2, the system of polynomials equations $g_i(u; y_0) = 0$ has a nonzero solution if and only if:

$$I_s \not\subseteq (g_1(u, y_0), ..., g_t(u, y_0)) \qquad \forall s \in \mathbb{N}$$

where I_s is the irrelevant ideal of degree $s \geq 1$. Thus it suffices to show that the set T_s of points $y_0 \in \mathbb{A}_k^m$ for which $I_s \not\subseteq (g_1(u, y_0), ..., g_t(u, y_0))$ is closed, in particular if we find polynomial equations whose zero set is T_s, then as $T = \bigcap_s T_s$, T is closed which completing the proof.

To that end, let k_i be the homogeneous degree of $g_i(u; y)$ in the u_j variables. Let $(M^\alpha)_\alpha$ be the monomials of degree s in $u_0, ..., u_n$ with some total ordering (note this set is finite). Then $I_s \subseteq (g_1(u, y_0), ..., g_t(u, y_0))$ if and only if each monomial M^α can be expressed as:

$$M^\alpha = \sum_{i=1}^{t} g_i(u, y_0) F_{i,\alpha}(u) \tag{10.2}$$

As each g_i has homogeneous degree k_i, it must be that $F_{i,\alpha}$ has degree $s - k_i$ or $F_{i,\alpha} = 0$ if $k_i > s$. Then we may replace $F_{i,\alpha}$ with elements in $(N_i^\beta)_\beta$ which are monomial of degree $s - k_i$ written in some total ordering. Then (10.2) holds if and only if M^α is a linear combination of the polynomials $g_i(u, y_0)N_i^\beta$. As this is true for each M^α, this is equivalent to saying that the polynomials $g_i(u, y_0)N_i^\beta$ span the entire vector space S of homogeneous polynomials of degree s in $u_0, ..., u_n$. As $I_s \not\subseteq (g_1(u, y_0), ..., g_t(u, y_0))$, we have that the $g_i(u, y_0)N_i^\beta$ *do not* span S.

Now, take the matrix $(a_{\alpha\beta})$ that represents the coefficients of the M^α terms as a linear combination of the $g_i(u, y_0)N_i^\beta$. As not all M^α can be written as such terms, this matrix cannot be full rank. Hence all $\sigma \times \sigma$ minors (where $\sigma = \dim_k S$) are polynomials in the coefficients of the polynomials $g_i(u, y_0)$, in particular they are polynomials in the coordinates of the point y_0. So T_s is cut out by polynomial equations in y_0, hence closed.

Thus, since the arbitrary intersection of closed sets is closed $T = \bigcap_s T_s$ is closed, completing the proof.

For those interested in algebraic geometry, the above proof is in part in proving that projective varieties are *proper*.

Corollary 10.2.3: Image of Projective Variety is Closed

Let $f : X \to Y$ be a regular map where X is a projective variety. Then $\mathrm{im} f$ is closed

***Proof*:**
Let $\pi : X \times Y \to Y$ be the natural projection. Consider the graph $\Gamma_f \subseteq X \times Y$ of lemma 10.2.1. Then:
$$\mathrm{im} f = \pi(\Gamma_f)$$
and hence by proposition 10.2.2 $\mathrm{im} f$ is closed, completing the proof.

We can finally rigorously show that the collection of regular functions on projective space, and more generally projective varieties, are only constant functions:

Corollary 10.2.4: Regular Function on Projective Variety

Let φ be a regular function on an irreducible projective variety. Then $\varphi \in k$, i.e. φ is constant.

***Proof*:**
Take $\varphi : X \to \mathbb{A}^1_k$. As $\mathbb{A}^1_k$ is dense in $\mathbb{P}^1_k$, there is a unique extension to a regular map $\overline{\varphi} : X \to \mathbb{P}^1_k$. As X is a projective variety, by corollary 10.2.3 $\overline{\varphi}(X) \subseteq \mathbb{P}^1_k$ is closed. Since $\overline{\varphi}(X) \subseteq \mathbb{A}^1_k$, the point at infinity is not in the image, so $\overline{\varphi}(X)$ is a *proper* closed subset of $\mathbb{P}^1_k$. The proper closed subsets of $\mathbb{P}^1_k$ are exactly the finite sets, so $\overline{\varphi}(X)$ is finite. (It cannot be all of $\mathbb{A}^1_k$, which is not closed in $\mathbb{P}^1_k$.) Hence $\varphi(X)$ is a finite set of points. closed, their pre-image would make X the non-trivial union of closed sets, contradicting its irreducibility, and hence it must be that the image is a single point, in other words a constant, as we sought to show.

Corollary 10.2.5: Regular Map on Projective Variety

Let $f : X \to Y$ be a regular map from a projective variety to an affine variety. Then $\mathrm{im} f$ is a point

***Proof*:**
Take $f(x) = (\varphi_1(x), ..., \varphi_m(x))$ where each φ_i is a regular function. Then by corollary 10.2.4 these are all constant, but then
$$f(X) = (\alpha_1, ..., \alpha_m)$$
completing the proof.

10.2.6 Note: Completeness Is Elimination Theory The classical name for proposition 10.2.2 is *elimination theory*. Given equations in $x_1, \dots, x_n$ and $y_1, \dots, y_m$, one asks for the conditions on the y alone under which a solution in x exists; that is exactly asking for the image of a projection, and the theorem says the answer is again given by equations, provided the x range over projective space.
The classical tool is the resultant. For two one-variable polynomials

$$f = a_0 x^d + \cdots + a_d, \qquad g = b_0 x^e + \cdots + b_e$$

the resultant $\operatorname{Res}(f, g)$ is the determinant of the $(d+e) \times (d+e)$ Sylvester matrix in the coefficients, and it vanishes exactly when f and g have a common root, given a_0, b_0 not both zero. Since Res is a polynomial in the a_i and b_j, that is precisely a closed condition on the coefficients, and it eliminates x. Iterating in several variables is the classical proof of completeness, and [11, Resultants] develops the algebra.
The proof given above takes a different route, through lemma 8.4.2 and the irrelevant ideal, which is shorter and generalizes better. The resultant remains the computational tool: it is what a computer algebra system runs when asked to eliminate a variable, and example 12.4.1 could equally have been done by resultants.

10.3 Finite Maps

These maps shall be the mirror of integral extensions, and shall be one of the easiest examples of *proper morphisms* which is a somewhat complicated but fundamental type of morphism in scheme theory.

Definition 10.3.1: Finite Map [on Affine Varieties]

Let $f\colon X \to Y$ be a regular map of affine varieties with $f(X)$ dense in Y, so that the pullback $f^{\#}\colon k[Y] \hookrightarrow k[X]$ is injective. Then f is a *finite map* if $k[X]$ is integral over $f^{\#}(k[Y])$.

Example 10.3.2: Finite and Non-Finite Maps

1. *A finite map.* $\varphi\colon \mathbb{A}^1_k \to \mathbb{A}^1_k$, $\varphi(x) = x^2$. On coordinate rings it is $k[y] \to k[x]$, $y \mapsto x^2$, and $k[x]$ is generated as a $k[y]$-module by 1 and x, since $x^2 = y$. So the map is finite, and its fibres have one or two points: two over $a \neq 0$, one over 0.

2. *Finite maps are closed and surjective.* The image of φ is all of $\mathbb{A}^1_k$, as proposition 10.3.3 requires, because every a has a square root in the algebraically closed k.

3. *A non-example with finite fibres.* The projection $\mathcal{Z}(xy-1) \to \mathbb{A}^1_k$, $(x, y) \mapsto x$, has fibres of size at most one, yet it is not finite: $k[x, x^{-1}]$ is not a finitely generated $k[x]$-module, since no finite set of Laurent polynomials spans all negative powers. Its image misses the origin, so it is not even surjective. Finiteness is strictly stronger than having finite fibres.

4. *An open immersion is almost never finite.* The inclusion $D(x) \hookrightarrow \mathbb{A}^1_k$ is the previous example in disguise, by example 1.3.6(1).

The composition of finite maps is finite. If $f\colon X \to Y$ is finite then its fibres are finite. A reasonable assumption would be that a map is finite if and only if its fibers are finite, but that is not the case: an example of a non-finite map would be $\pi_1 : V(xy-1) \to \mathbb{A}^1_k$ as $k[x] \hookrightarrow k[x, 1/x]$ is not an integral extension. Instead, one way of thinking of a finite map $f : X \to Y$ is that given an element $y \in Y$, as y is moved around in Y and we look at $f^{-1}(y)$, none of the roots could tend to infinity, that is none can disappear. This intuition is captured by the following proposition:

Proposition 10.3.3: Finite Maps Are Surjective

Let $f : X \to Y$ be a finite map between affine varieties. Then f is surjective.

The tool is the lying-over property of integral extensions ([5, Integral Extension Prime Correspondence]), in the following form: if B is integral over a subring A and $\mathfrak{a} \subsetneq A$ is a *proper* ideal, then $\mathfrak{a}B \neq B$. Equivalently, a proper ideal of A cannot generate the unit ideal of B. This is what stops a fibre from being empty.

***Proof*:**
Let $f : X \to Y$ be a finite map between affine varieties. Take any $y \in Y$ and let $\mathfrak{m}_y \subseteq k[Y]$ be the functions that take the value 0 at y: if $f \in \mathfrak{m}_y$, then $f(y) = 0$. Let $x_1, ..., x_n$ be coordinate functions on Y and $y = (\alpha_1, ..., \alpha_n)$ so that

$$\mathfrak{m}_y = (x_1 - \alpha_1, \ldots, x_n - \alpha_n)$$

Then the equations of the variety $f^{-1}(y)$ have the form:

$$f^{\#}(x_1) = \alpha_1, \quad \ldots, \quad f^{\#}(x_n) = \alpha_n$$

By the Nullstellensatz, $f^{-1}(y) = \emptyset$ if and only if the elements $f^{\#}(x_i) - \alpha_i$ generate the unit ideal, that is:

$$(f^*(x_1) - \alpha_1, ..., f^*(x_n) - \alpha_n) = k[X]$$

Viewing $k[Y]$ as a subring of $k[X]$, we see that the above is equivalent to:

$$(x_1 - \alpha_1, \ldots, x_n - \alpha_n)\, k[X] = k[X]$$

In other words, $f^{-1}(y) = \emptyset$ would force $\mathfrak{m}_y k[X] = k[X]$. But $\mathfrak{m}_y$ is maximal, hence a proper ideal of $k[Y]$, and $k[X]$ is integral over it, so lying over gives $\mathfrak{m}_y k[X] \neq k[X]$. The two are contradictory, so $f^{-1}(y) \neq \emptyset$. As y was arbitrary, f is surjective, completing the proof.

Corollary 10.3.4: Finite Maps Are Closed Maps

Let $f : X \to Y$ be a finite map between affine varieties. Then f is a closed map.

***Proof*:**
It suffices to check on irreducible closed subsets $Z \subseteq X$. By the above theorem the restriction of f to Z, $f|_Z : Z \to \overline{f(Z)}$, is a finite map between affine varieties, and so by surjectivity we get $f(Z) = \overline{f(Z)}$, but then $f(Z)$ is closed, completing the proof.

Let us now build-up to defining finite maps for quasi-projective varieties. For that, we start by showing that finiteness is a local property:

Lemma 10.3.5: Finiteness is Local

Let $f : X \to Y$ be a regular map of affine varieties. Then if for every point $y \in Y$, there is an affine neighborhood U of y such that $V = f^{-1}(U)$ is affine and $f : V \to U$ is finite, then f is finite

***Proof*:**
Let $k[Y] = A$ and $k[X] = B$. Take the principal open set around each point of Y, and as Y is compact take a finite subcover, label these $D(g_\alpha)$ so that $Y = \bigcup_\alpha D(g_\alpha)$. Then:

$$V_\alpha = f^{-1}(D(g_\alpha)) = D(f^\#(g_\alpha))$$

and

$$k[D(g_\alpha)] = A[1/g_\alpha] \qquad k[V_\alpha] = B[1/f^\#(g_\alpha)]$$

As f is finite on this chart, $B[1/g_\alpha]$ is a finite module over $A[1/g_\alpha]$; fix a finite generating set $w_{i,\alpha}$. (Generating set, not basis: a finite module over a ring need not be free, and freeness is never used below.) We may take $w_{i,\alpha} \in B$, since clearing the denominator of $w_{i,\alpha}/g_\alpha^{m_i}$ multiplies by a unit of $A[1/g_\alpha]$ and so preserves generation.

Now take the union of all these finite sets. We claim they generate B as an A-module. To see this, take any $b \in B$. Then for each α we can write it as :

$$b = \sum_i \frac{a_{i,\alpha}}{g_\alpha^{n_\alpha}} w_{i,\alpha}$$

As the $g_\alpha^{n_\alpha}$ generate the unit ideal of A (the sets $D(g_\alpha)$ cover Y), there exists $h_\alpha \in A$ such that

$$\sum_\alpha g_\alpha^{n_\alpha} h_\alpha = 1$$

Thus:

$$b = b \sum_\alpha g_\alpha^{n_\alpha} h_\alpha = \sum_i \sum_\alpha a_{i,\alpha} h_\alpha w_{i,\alpha}$$

which exhibits an arbitrary $b \in B$ as an A-combination of the $w_{i,\alpha}$. Hence B is a finite A-module and f is finite, as we sought to show.

Thus, we can define finite maps for quasi-projective varieties:

Definition 10.3.6: Finite Map [on Quasi-Projective Varieties]

Let $f : X \to Y$ be a regular map between quasi-projective varieties. Then f is said to be *finite* if any point $y \in Y$ has an affine local neighborhood V such that $U = f^{-1}(V)$ is affine and $f : U \to V$ is a finite map between affine varieties.

Naturally, $f^{-1}(y)$ is finite for every $y \in Y$, and it follows easily that any finite map is surjective. Note that the density condition is now a local condition. One consequence of these results that we shall not prove is that if $f : X \to Y$ between quasi-projective is a regular map (not necessarily finite) and $f(X)$ dense in Y, then $f(X)$ contains an open subset of Y. This is *not* true in analysis, the canonical example being the wrapping of the real line around the torus with an irrational period which will be a dense map with empty interior.

10.4 Noether Normalization, Geometrically

An important culminating result is that every irreducible projective variety is in some way only a "finite extension" from projective space. These results reduce many arguments about irreducible varieties to arguments about affine or projective space. These ideas return in scheme-theoretic form in *Algebraic Geometry* [3]. We first need a technical lemma:

10.4.1 Normalization and Singularities Note that this is *not* to be confused with normalization of singularities! Rather confusingly, normalization of singularities shall be the map in the *opposite direction* to the normalization map that will be defined here. The two uses of the word are unrelated, and it is not the case that a normal resolution of singularities results in a map $\mathbb{A}^n_k \to X$ or $\mathbb{P}^n_k \to X$ for a variety X (where the normalization map we shall do will be map of the form $X \twoheadrightarrow \mathbb{A}^n_k$ or $X \twoheadrightarrow \mathbb{P}^n_k$).

Lemma 10.4.2: Projection is Finite Mapping

Let X be closed in $\mathbb{P}^n$ with $X \subset \mathbb{P}^n \setminus E$, where E is a d-dimensional linear subspace. Then the projection $\pi : X \to \mathbb{P}^{n-d-1}$ with center at E determines a finite mapping $X \to \pi(X)$.

***Proof*:**
Let $y_0, \dots, y_{n-d-1}$ be homogeneous coordinates in $\mathbb{P}^{n-d-1}$ and let π be given by the formulae $y_j = L_j(x)$, $x \in X$ $(j = 0, \dots, n-d-1)$. Naturally, $U_i = \pi^{-1}(\mathbb{A}^{n-d-1}) \cap X$ is given by the condition $L_i(x) \neq 0$ and is an affine open subset of X. We want to show that $\pi : U_i \to \mathbb{A}^{n-d-1} \cap \pi(X)$ is a finite mapping. To that end, first note that every function $g \in k[U_i]$ is of the form $g = G_i(x_0, \dots, x_n)/L_i^m$, where G_i is a form of degree m. Consider the mapping $\pi_1 : X \to \mathbb{P}^{n-d} : z_j = L_j^m(x)$ $(j = 0, \dots, n-d-1)$, $z_{n-d} = G_i(x)$, where $z_0, \dots, z_{n-d}$ are homogeneous coordinates in $\mathbb{P}^{n-d}$. It is a regular mapping and its image $\pi_1(X)$ is closed in $\mathbb{P}^{n-d}$. Let $F_1 = \cdots = F_s = 0$ be its equations. Since $X \subset \mathbb{P}^n - E$, the forms $L_i (i = 0, \dots, n-d-1)$ do not vanish simultaneously on X. Hence the point $[0 : \cdots : 0 : 1]$ is not contained in $\pi_1(X)$, in other words, that the equations $z_0 = \cdots = z_{n-d-1} = F_1 = \cdots = F_s = 0$ have no solution in $\mathbb{P}^{n-d}$. Hence by lemma 8.4.2, for some $l > 0$:

$$(z_0, \dots, z_{n-d-1}, F_1, \dots, F_s) \supseteq \mathfrak{m}^l$$

in particular, $z_{n-d}^l \in (z_0, \dots, z_{n-d-1}, F_1, \dots, F_s)$. This means that

$$z_{n-d}^l = \sum_{j=0}^{n-d-1} z_j H_j + \sum_{j=1}^{s} F_j P_j,$$

where H_j and P_j are polynomials. Denote by $H^{(q)}$ the homogeneous component of H of degree q. Then:

$$\Phi(z_0, \dots, z_{n-d}) = z_{n-d}^l - \sum z_j H_j^{(l-1)} = 0 \quad \text{on } \pi_1(X) \tag{10.3}$$

The homogeneous polynomial Φ is of degree l and as a polynomial in z_{n-d} it has the leading coefficient 1:

$$\Phi = z_{n-d}^l - \sum_{j=0}^{l-1} A_{l-j}(z_0, \dots, z_{n-d-1}) z_{n-d}^j \tag{10.4}$$

Substituting the transformation formulae for π_1 in (10.3) we find that $\Phi(L_0^m, \dots, L_{n-d-1}^m, G_i) = 0$ on X, with Φ of the form (10.4). Dividing this relation by L_i^{ml} we obtain the required relation

$$g^l + \sum_{j=0}^{l-1} A_{l-j}(x_0, \dots, 1, \dots, x_{n-d-1}) g^j = 0,$$

where $x_r = y_r/y_i$ are coordinates in $\mathbb{A}^{n-d-1}$. But then we have that the mapping is finite, completing the proof.

Theorem 10.4.3: Normalization of Projective Variety

Let X be an irreducible projective variety. Then there exists a finite mapping $\varphi : X \twoheadrightarrow \mathbb{P}_k^m$ for appropriate m.

***Proof*:**
Take $X \subseteq \mathbb{P}_k^n$. If $X = \mathbb{P}_k^n$ we're done so take $X \neq \mathbb{P}_k^n$. Take $x \in \mathbb{P}_k^n \setminus X$ and take the projection from x, $\varphi_x : X \to \mathbb{P}_k^{n-1}$. This map is regular by example 9.4.4, and the image is closed, hence projective. It is furthermore finite by lemma 10.4.2. If $\varphi(X) \neq \mathbb{P}_k^{n-1}$ we repeat the argument. is closed/finite. Each projection lowers the ambient dimension by one, so after finitely many steps the image fills its ambient space: the dimension cannot fall below $\dim X$, and a proper closed subvariety of the same dimension as its ambient is impossible by proposition 4.3.5. Composing the projections gives a finite surjection

$$\varphi \colon X \twoheadrightarrow \mathbb{P}_k^m \qquad m = \dim X$$

completing the proof.

Theorem 10.4.4: Normalization of Affine Variety

Let X be an irreducible affine variety. Then there exists a finite mapping $\varphi : X \twoheadrightarrow \mathbb{A}_k^m$ for appropriate m.

Note that this is the analog of Noether's Normalization theorem for rings that are finitely generated k-algebras without zero-divisors, and in fact can be proven using it: see the fifth exercise of exercise set 10.6.1.

***Proof*:**
Take $X \subseteq \mathbb{A}^n_k$ and suppose $X \neq \mathbb{A}^n_k$, the other case being trivial. Identify $\mathbb{A}^n_k$ with the chart $U_{n+1} \subseteq \mathbb{P}^n_k$. Then X is *closed in* $\mathbb{A}^n_k$, hence locally closed in $\mathbb{P}^n_k$, and we may take its projective closure $\overline{X}$.

Since X is a proper closed subvariety of $\mathbb{A}^n_k$ we have $\dim X < n$ by proposition 4.3.5, so $\overline{X}$ cannot contain the whole hyperplane at infinity; choose $x \in H_\infty$ with $x \notin \overline{X}$. Projecting from x gives a finite map $\overline{X} \to \mathbb{P}^{n-1}_k$ by lemma 10.4.2, and since the centre lies at infinity the projection carries X into the affine chart, giving a finite map $X \to \mathbb{A}^{n-1}_k$.

Repeating as in theorem 10.4.3 yields a finite surjection

$$\varphi \colon X \twoheadrightarrow \mathbb{A}^m_k \qquad m = \dim X$$

completing the proof.

Thus, though not all varieties are rational, all varieties are a finite cover of affine or projective space, and hence the study of which varieties are not rational will naturally lead to the question of how far away are they from being rational[2]. More on this in the exercises.

Example 10.4.5: Finite Maps and Normalization
Consider the curve $f : \mathbb{A}^1_k \to \mathbb{A}^2_k$ given by $f(t) = (t^2, t^3)$. This is associated to the pullback on coordinate rings $f^\# \colon k[x,y] \to k[t]$, $x \mapsto t^2$, $y \mapsto t^3$. Taking the quotient of the domain gives $k[x,y]/(x^3 - y^2) \cong k[t^2, t^3] \hookrightarrow k[t]$, which is injective but *not surjective*: nothing maps to t. As $(x^3 - y^2)$ is prime, we take the field of fraction and extend the map, which has $t^3/t^2 = t$ we have that $\mathrm{Frac}(k[t^2, t^3]) = k(t)$ giving a field isomorphism.

This is a 1-to-1 map. Define an n-to-1[a] map $f : \mathbb{A}^1_k \to \mathbb{A}^2_k$ given by $t \mapsto (t^{2n}, t^{3n})$. Notice the image is the same. The global section's are also isomorphic as $k[t^2, t^3] \cong k[t^{2n}, t^{3n}]$. Thus the field of fractions are isomorphic, $k(t) \cong k(t^n)$. Working with a specific n-to-1 map, the field of fraction of the global section of the image is $k(t^n)$ which has index n in the field of fractions of the domain, namely $k(t)$. This reflects the geometric nature of finite maps in algebraic properties.

Notice that $x^2 - y^3$ forms a curve with a cusp; it has a singularity at $(0,0)$. This curve maps onto

[a] note it is important to assume that either k is algebraically closed, or at minimum that k contains all the roots of unity

10.5 Dimension of Fibres

A map between varieties should decrease dimension by the dimension of its fibres, and this section makes that precise. The statement has three parts, and only the middle one is delicate: fibres can jump *up* in dimension over special points, but never down.

[2] Chevalley's theorem (chapter 11) is the statement in this direction that survives without the finiteness hypothesis: the image of a regular map is constructible even when it is neither open nor closed.

Theorem 10.5.1: Dimension of Fibres

Let $f\colon X \to Y$ be a dominating regular map of irreducible varieties, with $\dim X = n$ and $\dim Y = m$. Then:

1. $m \leq n$;
2. for every $y \in f(X)$, every irreducible component of the fibre $f^{-1}(y)$ has dimension at least $n - m$;
3. there is a nonempty open $U \subseteq Y$ such that $\dim f^{-1}(y) = n - m$ for every $y \in U$.

***Proof*:**
(1). Since f is dominating, $f^{\#}$ embeds $k(Y)$ into $k(X)$ by proposition 3.3.4(2), and transcendence degree cannot decrease under a field extension, so $m = \operatorname{trdeg}_k k(Y) \leq \operatorname{trdeg}_k k(X) = n$.

(2). The question is local on Y, so replace Y by an affine neighbourhood of y and X by its preimage. The local ring $\mathcal{O}_y(Y)$ has dimension m, so it admits a system of parameters ([5, System of Parameters]): m functions $g_1, \ldots, g_m$ generating an ideal whose zero set is $\{y\}$ near y. Then, near any point of the fibre,

$$f^{-1}(y) = \mathcal{Z}\big(f^{\#}(g_1), \ldots, f^{\#}(g_m)\big)$$

is cut out of X by m equations. Applying theorem 4.4.1 once for each equation drops the dimension by at most one each time, so every component of the result has dimension at least $n - m$.

(3). Choose affine open sets so that f corresponds to an injection $k[Y] \hookrightarrow k[X]$ of finitely generated domains. Over the fraction field $K = k(Y)$, the ring $k[X] \otimes_{k[Y]} K$ is a finitely generated K-algebra of dimension $n - m$, since its fraction field is $k(X)$ and $\operatorname{trdeg}_K k(X) = n - m$. Apply Noether normalization over K: there are $t_1, \ldots, t_{n-m}$ with $k[X] \otimes K$ finite over $K[t_1, \ldots, t_{n-m}]$. Each of the finitely many integral equations involved has coefficients in $k(Y)$; let $g \in k[Y]$ be a common denominator. Then over the open set $U = D(g)$,

$$k[X]_g \text{ is finite over } k[Y]_g[t_1, \ldots, t_{n-m}]$$

Hence for $y \in U$ the fibre $f^{-1}(y)$ is finite over $\mathbb{A}_k^{n-m}$, and a finite surjection preserves dimension by proposition 4.2.5, so $\dim f^{-1}(y) = n - m$, as we sought to show.

Part (2) is the useful half in practice, and it is worth saying what it does not claim. It gives a *lower* bound on fibre dimension, valid at every point; the matching upper bound holds only generically, which is part (3). Fibres genuinely do jump.

Example 10.5.2: Fibres That Jump
Let $X = \mathcal{Z}(xz - y) \subseteq \mathbb{A}_k^3$ and $f\colon X \to \mathbb{A}_k^1$ the projection $(x, y, z) \mapsto x$. Then X is isomorphic to $\mathbb{A}_k^2$ with coordinates (x, z), since y is determined, so $n = 2$ and $m = 1$, giving an expected fibre dimension of 1. Over $x = a \neq 0$ the fibre is the line $\{(a, az, z) \mid z \in k\}$, of dimension 1 as predicted. Over $x = 0$ it is $\{(0, 0, z) \mid z \in k\}$, again dimension 1.

Now blow the picture up: let $g\colon \mathbb{A}^2_k \to \mathbb{A}^2_k$ be $(u,v) \mapsto (u, uv)$. Here $n = m = 2$, so the expected fibre dimension is 0. Over (a,b) with $a \neq 0$ the fibre is the single point $(a, b/a)$; but over $(0,0)$ the fibre is the whole line $\{(0,v) \mid v \in k\}$, of dimension 1. The fibre has jumped, exactly as part (2) permits and part (3) confines to a proper closed subset. This map is the restriction to one chart of the blow-up of chapter 17, computed there as proposition 17.1.2(1).

A count that will be used in chapter 11 follows immediately by adding dimensions along a map whose fibres are all of the same dimension.

Corollary 10.5.3: Dimension of a Total Space

Let $f\colon X \to Y$ be a surjective regular map of irreducible varieties whose fibres are all irreducible of the same dimension d. Then $\dim X = \dim Y + d$.

***Proof*:**
By theorem 10.5.1(3) the generic fibre has dimension $\dim X - \dim Y$, and by hypothesis every fibre has dimension d, so $\dim X - \dim Y = d$, completing the proof.

Applied to a projection, this settles the dimension of a product, which chapter 4 could only gesture at because products of projective varieties were not yet available.

Corollary 10.5.4: Dimension of a Product

Let X and Y be irreducible quasi-projective varieties. Then $\dim(X \times Y) = \dim X + \dim Y$.

***Proof*:**
$X \times Y$ is irreducible by proposition 10.1.6, and the projection $\pi\colon X \times Y \to X$ is a surjective regular map whose fibre over x is the slice $\{x\} \times Y \cong Y$, irreducible of dimension $\dim Y$ at every point. Corollary 10.5.3 applies and gives the claim.

10.6 The Dimension of an Intersection

Theorem 4.4.1 says that cutting an affine variety by one equation costs at most one dimension. Iterating it bounds the dimension of an intersection, but only when one of the two varieties is cut out by as many equations as its codimension, and that fails in general: the twisted cubic of example 8.3.7 has codimension two in $\mathbb{P}^3_k$ and needs three equations. The repair is to intersect with the diagonal instead, and it costs nothing now that products are available.

Proposition 10.6.1: The Affine Dimension Theorem

Let $X, Y \subseteq \mathbb{A}^n_k$ be irreducible affine varieties with $\dim X = r$ and $\dim Y = s$. Then every irreducible component of $X \cap Y$ has dimension at least $r + s - n$.

***Proof*:**
Let $\Delta = \{(a,a) \mid a \in \mathbb{A}^n_k\} \subseteq \mathbb{A}^{2n}_k$ be the diagonal, which is $\mathcal{Z}(x_1 - y_1, \dots, x_n - y_n)$, cut out by exactly n equations. The map $a \mapsto (a,a)$ is an isomorphism

$$X \cap Y \xrightarrow{\sim} (X \times Y) \cap \Delta$$

with inverse the first projection, both regular.

Now $X \times Y$ is irreducible by proposition 10.1.6 and has dimension $r + s$ by corollary 10.5.4. Cutting it by the n equations one at a time and applying theorem 4.4.1 at each step, every component of the result has dimension at least $r + s - n$, since each equation costs at most one dimension: an equation either vanishes identically on a component, leaving it untouched, or cuts it by exactly one. Transporting through the isomorphism gives the claim, as we sought to show.

Passing to affine cones turns this into the projective statement, and the projective statement is strictly stronger: it also asserts that the intersection is nonempty.

Proposition 10.6.2: Dimension of the Affine Cone

Let $\emptyset \neq V \subseteq \mathbb{P}^n_k$ be a projective variety. Then $C(V)$ is an irreducible affine variety and

$$\dim C(V) = \dim V + 1$$

***Proof*:**
Irreducibility: $\mathcal{I}_{\mathbb{A}}(C(V)) = \mathcal{I}_{\mathbb{P}}(V)$ by proposition 8.3.2, and the latter is prime because V is a variety, so $k[C(V)] = S(V)$ is a domain.

For the dimension, choose a linear form t not vanishing identically on V, possible since $V \neq \emptyset$. Every element of $\mathrm{Frac}(S(V))$ is a ratio f/g of homogeneous elements, and

$$\frac{f}{g} = \frac{f/t^{\deg f}}{g/t^{\deg g}} \cdot t^{\deg f - \deg g}$$

exhibits it as a rational function of t with coefficients in the degree-zero part, which is $k(V)$ by definition 9.2.4. So $\mathrm{Frac}(S(V)) = k(V)(t)$, and t is transcendental over $k(V)$ because it has degree 1 while $k(V)$ sits in degree 0. Hence

$$\dim C(V) = \mathrm{kdim} S(V) = \mathrm{trdeg}_k \mathrm{Frac}(S(V)) = \mathrm{trdeg}_k k(V) + 1 = \dim V + 1$$

using proposition 4.2.5 twice, as we sought to show.

Theorem 10.6.3: The Projective Dimension Theorem

Let $X, Y \subseteq \mathbb{P}^n_k$ be projective varieties with $\dim X = r$ and $\dim Y = s$. Then every irreducible component of $X \cap Y$ has dimension at least $r + s - n$. If moreover $r + s - n \geq 0$, then $X \cap Y \neq \emptyset$.

***Proof*:**
Pass to the affine cones in $\mathbb{A}_k^{n+1}$. By proposition 10.6.2 they are irreducible of dimensions $r+1$ and $s+1$, so proposition 10.6.1 says every component of $C(X) \cap C(Y)$ has dimension at least

$$(r+1)+(s+1)-(n+1) = r+s-n+1$$

By proposition 8.3.2 the components of $C(X) \cap C(Y)$ other than the origin are the cones over the components of $X \cap Y$, whose dimensions are one less by proposition 10.6.2. That gives the bound $r+s-n$.

For nonemptiness, note that both cones contain the origin, so $C(X) \cap C(Y)$ is nonempty and the bound applies to a component through the origin: it has dimension at least $r+s-n+1 \geq 1$. A variety of dimension at least 1 is not a single point, so that component contains some $a \neq 0$, and a determines a point of $\mathbb{P}_k^n$ lying on both X and Y, as we sought to show.

The nonemptiness clause is what separates projective space from affine space. Both cones always meet, at the origin; the content is that when the dimensions are large enough they meet in more than the origin. No such statement is available affinely: in $\mathbb{A}_k^2$ the parallel lines $\mathcal{Z}(y)$ and $\mathcal{Z}(y-1)$ have $r=s=1$ and $n=2$, so $r+s-n=0$, and they are disjoint.

Example 10.6.4: Dimension Counts in Projective Space

1. *Two curves in* $\mathbb{P}_k^2$. Here $r=s=1$ and $n=2$, so $r+s-n=0$ and any two projective plane curves meet. Chapter 13 refines this into a count of the intersection points; the present theorem gives only that there is one, but it needs no hypothesis beyond dimension.

2. *A curve and a surface in* $\mathbb{P}_k^3$. With $r=1$, $s=2$, $n=3$ the bound is 0, so a curve and a surface in $\mathbb{P}_k^3$ always meet.

3. *Two curves in* $\mathbb{P}_k^3$. Now $r+s-n=-1$ and the theorem says nothing, correctly: two disjoint lines in $\mathbb{P}_k^3$ exist, as exercise set 8.5.1(4) exhibits.

4. *Hypersurfaces.* If $Y = \mathcal{Z}(F)$ is a hypersurface then $s = n-1$ and the bound reads $r-1$, with nonemptiness as soon as $\dim X \geq 1$. So a projective variety of positive dimension meets every hypersurface, which is another way of seeing why it carries no nonconstant regular function.

Exercise 10.6.1

1. Show that the image of the Segre embedding $\mathbb{P}_k^1 \times \mathbb{P}_k^1 \to \mathbb{P}_k^3$ is the quadric $\mathcal{Z}(w_{00}w_{11} - w_{01}w_{10})$, and that it contains two families of pairwise disjoint lines. Deduce, using exercise set 8.5.1(4), that this quadric is not isomorphic to $\mathbb{P}_k^2$.

2. Show that a variety that is both projective and affine is a single point.

3. Show that the map $\mathbb{A}_k^1 \to \mathbb{A}_k^1$, $x \mapsto x^2$, is finite, and that the inclusion $\mathbb{A}_k^2 \setminus \{0\} \hookrightarrow \mathbb{A}_k^2$ is not. For the second, which hypothesis of proposition 10.3.3 fails?

4. Exhibit a finite surjection $\mathcal{Z}(xy-1) \to \mathbb{A}_k^1$, and check directly that it is finite by writing the coordinate ring as a module over the image.

5. Prove the affine normalization theorem using Noether's normalization theorem (theorem 2.4.1): every affine variety admits a finite surjective map onto some $\mathbb{A}_k^m$, with m its dimension.
6. For $f\colon \mathbb{A}_k^2 \to \mathbb{A}_k^2$, $(u, v) \mapsto (u, uv)$, compute every fibre and verify all three parts of theorem 10.5.1 by hand. Which part fails if the word "component" is dropped from part (2)?
7. Using corollary 10.5.4, compute $\dim(\mathbb{P}_k^n \times \mathbb{P}_k^m)$ and check the answer against the Segre embedding for $n = m = 1$.
8. Show that the two projections $\mathbb{P}_k^1 \times \mathbb{P}_k^1 \to \mathbb{P}_k^1$ have all fibres isomorphic to $\mathbb{P}_k^1$, and use corollary 10.5.3 to recover the dimension of the product.

11

Projective Embeddings and Constructible Sets

Projective space carries more embeddings of itself than the linear ones, and the useful ones turn products, hypersurfaces, and linear subspaces into points of a larger projective space. This chapter collects them: the Veronese embedding, which linearizes degree, and the Plücker embedding, which realizes the Grassmannian of d-planes as a projective variety.

Then comes Chevalley's theorem. The image of a regular map is neither open nor closed in general, but it is always constructible, a finite union of locally closed sets, and that is the correct replacement for the closedness that chapter 10 supplies only in the projective case.

The chapter ends by putting the two halves together. Counting the lines that lie on a cubic surface needs the Grassmannian to parameterize the lines, projective space to parameterize the surfaces, and theorem 10.5.1 to conclude that the count is finite. It is the first argument in this book where the answer is a number nobody could have guessed.

11.1 The Veronese Embedding

A hypersurface of degree d in $\mathbb{P}^n$ is cut out by one equation, but that equation is not linear, and linear conditions are the ones that are easy to count. The Veronese embedding trades the degree for a dimension: it realizes $\mathbb{P}^n$ inside a much larger projective space in such a way that degree-d hypersurfaces downstairs become *hyperplane sections* upstairs.

Definition 11.1.1: Veronese Embedding

Fix $n, d \geq 1$ and let $M_0, \ldots, M_N$ be the monomials of degree d in $x_0, \ldots, x_n$, in some fixed order, where

$$N = \binom{n+d}{d} - 1$$

The *Veronese embedding* of degree d is

$$v_d \colon \mathbb{P}^n_k \to \mathbb{P}^N_k \qquad v_d([x_0 : \cdots : x_n]) = [M_0(x) : \cdots : M_N(x)]$$

The map is well defined: every M_i is homogeneous of the same degree d, so scaling x by λ scales every coordinate by λ^d, and the M_i never vanish simultaneously, since x_j^d is among them and some $x_j \neq 0$.

Theorem 11.1.2: The Veronese Is a Closed Embedding

v_d is injective, its image $V_{n,d} = v_d(\mathbb{P}^n_k)$ is closed in $\mathbb{P}^N_k$, and v_d is an isomorphism onto its image.

***Proof*:**
Injective. Write z_α for the coordinate of $\mathbb{P}^N$ attached to the monomial x^α. Suppose $v_d(x) = v_d(y)$. Some $x_j \neq 0$, so $z_{de_j} \neq 0$ at that point, hence $y_j \neq 0$ as well. Rescaling, take $x_j = y_j = 1$. Then the coordinate attached to $x_j^{d-1}x_i$ reads off x_i and y_i directly, so $x = y$.

Closed. The image satisfies the relations

$$z_\alpha z_\beta = z_\gamma z_\delta \qquad \text{whenever } \alpha + \beta = \gamma + \delta$$

since both sides equal $x^{\alpha+\beta}$. Conversely, suppose z satisfies them all and $z_{de_j} \neq 0$ for some j; normalize $z_{de_j} = 1$ and set $x_i = z_{(d-1)e_j+e_i}$. The relations then force $z_\alpha = x^\alpha$ for every α, by induction on how far α is from de_j. Hence the image is exactly the closed set cut out by those quadrics.

Isomorphism onto the image. The inverse is given on the open set $z_{de_j} \neq 0$ by $z \mapsto [z_{(d-1)e_j+e_0} : \cdots : z_{(d-1)e_j+e_n}]$, a tuple of forms of the same degree with no common zero there, hence regular by proposition 9.4.3. These agree on overlaps because both compute the same point, so they glue to a regular inverse, as we sought to show.

The payoff is the promised trade.

Corollary 11.1.3: Hypersurfaces Become Hyperplane Sections

Let F be a form of degree d in $x_0, \ldots, x_n$. Then there is a hyperplane $H \subseteq \mathbb{P}^N_k$ with

$$v_d\big(\mathcal{Z}(F)\big) = V_{n,d} \cap H$$

Consequently every degree-d hypersurface in $\mathbb{P}^n_k$ is the pullback of a hyperplane under v_d.

***Proof*:**
Write $F = \sum_\alpha c_\alpha x^\alpha$, the sum over monomials of degree d. Let $H = \mathcal{Z}(L)$ where $L = \sum_\alpha c_\alpha z_\alpha$, a linear form. Then $L(v_d(x)) = \sum_\alpha c_\alpha x^\alpha = F(x)$, so $v_d(x) \in H$ if and only if $F(x) = 0$, as we sought to show.

Example 11.1.4: Small Veronese Embeddings

1. $v_2 \colon \mathbb{P}^1_k \to \mathbb{P}^2_k$, $[s : t] \mapsto [s^2 : st : t^2]$. Here $N = \binom{3}{2} - 1 = 2$ and the image is the conic $\mathcal{Z}(z_0 z_2 - z_1^2)$. This is the isomorphism $\mathbb{P}^1 \cong$ smooth conic met in example 9.4.4(3), in coordinates.

2. $v_3 \colon \mathbb{P}^1_k \to \mathbb{P}^3_k$, $[s : t] \mapsto [s^3 : s^2 t : st^2 : t^3]$, whose image is the twisted cubic of example 8.3.7(4). So the twisted cubic is a Veronese image, which is another way of seeing that it is rational.

3. $v_2 \colon \mathbb{P}^2_k \to \mathbb{P}^5_k$ is the *Veronese surface*, the classical example of a smooth surface in $\mathbb{P}^5$ that is not contained in any hyperplane.

The construction also gives a clean way to think about the parameter space of section 9.2: the degree-d forms on $\mathbb{P}^n$, taken up to scalar, are the points of $\mathbb{P}^N_k$, and corollary 11.1.3 says the hypersurface a point cuts out is the corresponding hyperplane section of the Veronese. The following is a first payoff.

Proposition 11.1.5: Parameterizing Reducible Hypersurfaces

Fix n and d, and let $\mathbb{P}^N_k$, $N = \binom{n+d}{d} - 1$, be the projective space parameterizing degree-d forms in $n+1$ variables up to scalar, as in section 9.2. Then the set of points corresponding to *reducible* forms is closed in $\mathbb{P}^N_k$.

***Proof*:**
A reducible form of degree d factors as a product of forms of degrees j and $d-j$ for some $1 \leq j \leq d-1$. Writing Z_j for the set of points of $\mathbb{P}^N_k$ arising this way, the set of reducible forms is $\bigcup_{j=1}^{d-1} Z_j$, a finite union, so it suffices to show each Z_j is closed.

Let $N_j = \binom{n+j}{j} - 1$ and $N_{d-j} = \binom{n+d-j}{d-j} - 1$ be the corresponding parameter spaces. Multiplication of forms gives a map

$$\mu \colon \mathbb{P}^{N_j}_k \times \mathbb{P}^{N_{d-j}}_k \to \mathbb{P}^N_k$$

which is well defined, since scaling either factor scales the product, and regular, since the coefficients of a product are bilinear in the coefficients of the factors. Its image is exactly Z_j.

The source is a product of projective spaces, hence a projective variety by proposition 10.1.3, so $Z_j = \mu(\mathbb{P}^{N_j}_k \times \mathbb{P}^{N_{d-j}}_k)$ is closed by corollary 10.2.3. That is the whole point of completeness: the image of a projective variety is closed, which for an affine source would fail. Hence the reducible locus is closed, as we sought to show.

For $n = 1$ and $d = 2$ this recovers a familiar fact: a binary quadratic form $ax^2 + bxy + cy^2$ is reducible exactly when its discriminant $b^2 - 4ac$ vanishes, a single equation in the coefficients.

11.2 Grassmannians and the Plücker Embedding

The Veronese parameterizes hypersurfaces. The Grassmannian parameterizes linear subspaces, and it is the space one needs in order to ask questions like “how many lines lie on a surface”. As with the Veronese, the construction realizes a set of geometric objects as the points of a projective variety.

Definition 11.2.1: Grassmannian

Let $0 < r < n$. The *Grassmannian* $G(r, n)$ is the set of r-dimensional linear subspaces of k^n. Equivalently, since an r-dimensional subspace of k^n is an $(r-1)$-dimensional linear subspace of $\mathbb{P}_k^{n-1}$, it is the set of $(r-1)$-planes in $\mathbb{P}_k^{n-1}$.

The two readings differ only by a shift, and both are used: $G(2, 4)$ is the space of 2-dimensional subspaces of k^4, which is the space of *lines* in $\mathbb{P}_k^3$. That is the case section 11.4 needs, and there the columns of k^4 are indexed by $0, 1, 2, 3$ to match the coordinates $x_0, \ldots, x_3$ on $\mathbb{P}_k^3$, writing the Plücker coordinates as p_{ij} with $0 \le i < j \le 3$.

To make $G(r, n)$ a variety, represent a subspace by a basis and kill the ambiguity in that choice.

Definition 11.2.2: Plücker Embedding

Let $W \subseteq k^n$ be an r-dimensional subspace with basis $w_1, \ldots, w_r$. Writing the w_i as the rows of an $r \times n$ matrix, let $p_{i_1 \cdots i_r}$ be the $r \times r$ minor on columns $i_1 < \cdots < i_r$. The *Plücker embedding* is

$$\rho\colon G(r, n) \to \mathbb{P}_k^{\binom{n}{r}-1} \qquad \rho(W) = [p_{i_1 \cdots i_r}]_{i_1 < \cdots < i_r}$$

Proposition 11.2.3: The Plücker Embedding Is Well Defined and Injective

ρ does not depend on the choice of basis, and it is injective.

***Proof*:**
Changing basis replaces the matrix A by gA for some $g \in \mathrm{GL}_r(k)$, and the Cauchy-Binet formula multiplies every maximal minor by the single scalar $\det g \neq 0$. So the tuple of minors changes by an overall scalar and its class in projective space is unchanged. Not all minors vanish, since the rows are independent and so some r columns are independent.

For injectivity, recover W from $\rho(W)$: a vector $v \in k^n$ lies in W exactly when the $(r+1) \times n$ matrix obtained by appending v has all its maximal minors zero, and expanding those minors along the appended row writes each as a linear form in v with coefficients among the $p_{i_1 \cdots i_r}$. So the Plücker coordinates determine W as the common zero set of an explicit system of linear equations, as we sought to show.

Theorem 11.2.4: The Grassmannian Is a Projective Variety

The image of ρ is closed in $\mathbb{P}_k^{\binom{n}{r}-1}$, so $G(r,n)$ is a projective variety. Its dimension is

$$\dim G(r,n) = r(n-r)$$

***Proof*:**
Closed. The image is cut out by the *Plücker relations*, the quadratic equations expressing that a tuple of $\binom{n}{r}$ numbers arises as the maximal minors of an $r \times n$ matrix. These are quadrics in the $p_{i_1 \cdots i_r}$; example 11.2.5(2) writes them out in the first case where there is more than one. Their zero set is closed, and the argument of proposition 11.2.3 shows a point satisfying them determines an r-dimensional subspace, so the zero set is exactly imρ.

Dimension. Cover $G(r,n)$ by the open sets $U_{i_1 \cdots i_r} = \{p_{i_1 \cdots i_r} \neq 0\}$. On $U_{1 \cdots r}$, every subspace has a unique basis whose matrix is $[\, I_r | B \,]$ with B an arbitrary $r \times (n-r)$ matrix: row-reduce using the invertible left block. The entries of B are free coordinates, giving $U_{1 \cdots r} \cong \mathbb{A}_k^{r(n-r)}$. These charts are irreducible and pairwise meet, since any two nonempty open subsets of an irreducible space do, so $G(r,n)$ is irreducible by lemma 10.1.5, and corollary 4.3.3 reads its dimension off a single chart as $r(n-r)$, as we sought to show.

Example 11.2.5: Grassmannians

1. $G(1,n) = \mathbb{P}_k^{n-1}$: a line through the origin in k^n is a point of $\mathbb{P}^{n-1}$, and the Plücker map is the identity. Dimension $1 \cdot (n-1) = n-1$, as it should be.
2. $G(2,4)$, the lines in $\mathbb{P}_k^3$. Here $\binom{4}{2} = 6$, so ρ lands in $\mathbb{P}_k^5$ with coordinates $p_{12}, p_{13}, p_{14}, p_{23}, p_{24}, p_{34}$, and there is exactly one Plücker relation:

$$p_{12}p_{34} - p_{13}p_{24} + p_{14}p_{23} = 0$$

   So $G(2,4)$ is a quadric hypersurface in $\mathbb{P}_k^5$, of dimension $2 \cdot 2 = 4$, matching theorem 4.4.1, which gives $5 - 1 = 4$.
3. $G(r,n) \cong G(n-r,n)$, by sending a subspace to its annihilator in the dual space. So the lines in $\mathbb{P}^3$ and the planes in $\mathbb{P}^3$ form isomorphic families, both of dimension 4.

The count in item (2) is the one section 11.4 runs on: the lines in $\mathbb{P}^3$ form a 4-dimensional family, and asking how many of them lie on a given surface is a question about intersecting that family with a condition.

11.3 Constructible Sets and Chevalley's Theorem

Chapter 10 showed that the image of a *projective* variety under a regular map is closed. For a general regular map the image can be neither closed nor open: the projection of $\mathcal{Z}(xy-1)$ to the x-axis is $\mathbb{A}_k^1 \setminus \{0\}$, which is open, while the image of $(x,y) \mapsto (x, xy)$ is $(\mathbb{A}_k^1 \setminus \{0\}) \times \mathbb{A}_k^1$ together with the single point $(0,0)$, which is neither. Chevalley's theorem says the second example is as bad as it gets: images are always finite unions of pieces of that shape.

Definition 11.3.1: Locally Closed and Constructible Sets

Let X be a variety. A subset $S \subseteq X$ is *locally closed* if $S = U \cap Z$ with U open and Z closed, equivalently if S is open in its own closure. A subset is *constructible* if it is a finite union of locally closed sets.

Proposition 11.3.2: Constructible Sets Form a Boolean Algebra

The constructible subsets of X are closed under finite union, finite intersection, and complement. They are the smallest such family containing the open sets.

***Proof*:**
Closure under finite union is the definition. For intersection, $(U_1 \cap Z_1) \cap (U_2 \cap Z_2) = (U_1 \cap U_2) \cap (Z_1 \cap Z_2)$ is locally closed, and intersection distributes over the finite unions.

For complement, it suffices to complement one locally closed set, since the complement of a finite union is the finite intersection of the complements and intersections are handled. And

$$X \setminus (U \cap Z) = (X \setminus U) \cup (X \setminus Z)$$

a union of a closed set and an open set, both locally closed. Finally any family closed under these operations and containing the opens contains every $U \cap Z$, hence every constructible set, as we sought to show.

The theorem rests on the fact that a dominating map hits a dense open set. Section 10.3 stated this without proof; it follows from the generic finiteness of theorem 10.5.1(3).

Lemma 11.3.3: A Dominating Image Contains a Dense Open Set

Let $f \colon X \to Y$ be a dominating regular map of irreducible varieties. Then $f(X)$ contains a nonempty open subset of Y.

***Proof*:**
Passing to affine open subsets changes neither hypothesis nor conclusion, so assume X and Y affine. The proof of theorem 10.5.1(3) produced $g \in k[Y]$ and $t_1, \dots, t_{n-m}$ such that

$$k[X]_g \text{ is finite over } k[Y]_g[t_1, \dots, t_{n-m}]$$

Geometrically this says f restricted over $D(g)$ factors as a finite map followed by a projection $D(g) \times \mathbb{A}^{n-m}_k \to D(g)$. A finite map is surjective by proposition 10.3.3, and the projection is surjective, so the composite is surjective onto $D(g)$. Hence $f(X) \supseteq D(g)$, a nonempty open subset of Y, as we sought to show.

Theorem 11.3.4: Chevalley's Theorem

Let $f\colon X \to Y$ be a regular map of varieties. Then f carries constructible sets to constructible sets. In particular $f(X)$ is constructible.

***Proof*:**
Since a constructible set is a finite union of locally closed sets, and the image of a union is the union of the images, it suffices to treat $f(X)$ itself, with X replaced by a locally closed subset. Decomposing X into irreducible components (theorem 2.3.5) and Y into the closure of the image, we may assume X and Y irreducible and f dominating.

Argue by Noetherian induction on Y: suppose the result holds for every proper closed subvariety of Y. By lemma 11.3.3 there is a nonempty open $U \subseteq Y$ with $U \subseteq f(X)$. Then

$$f(X) = U \cup f\big(X \setminus f^{-1}(U)\big)$$

The set $X' = X \setminus f^{-1}(U)$ is closed in X, and $f(X') \subseteq Y \setminus U$, a proper closed subset of Y. Applying the inductive hypothesis to $f|_{X'}\colon X' \to Y \setminus U$, which is a regular map of varieties with strictly smaller target, shows $f(X')$ is constructible. As U is open, hence constructible, and constructible sets are closed under finite union (proposition 11.3.2), $f(X)$ is constructible.

The induction terminates because Y is a Noetherian topological space (proposition 1.3.7), so there is no infinite strictly descending chain of closed subsets, completing the proof.

Chevalley's theorem is the right general statement, and the closed-image theorem of corollary 10.2.3 is the special case where properness upgrades "constructible" to "closed". It is also the statement that survives into scheme theory unchanged, where it becomes the assertion that a finite-type morphism of Noetherian schemes has constructible image.

Example 11.3.5: The Image That Is Neither Open Nor Closed
Take $f\colon \mathbb{A}^2_k \to \mathbb{A}^2_k$, $f(x, y) = (x, xy)$. For $x = a \neq 0$ the fibre over (a, b) is the single point $(a, b/a)$, so every such (a, b) is hit; for $x = 0$ the image of the line $x = 0$ is the single point $(0, 0)$. Hence

$$\operatorname{im} f = \big((\mathbb{A}^1_k \setminus \{0\}) \times \mathbb{A}^1_k\big) \;\cup\; \{(0, 0)\}$$

The first piece is open, the second is closed, and the union is neither. It is constructible, being a union of two locally closed sets, exactly as theorem 11.3.4 requires. This is also the map of example 10.5.2, whose fibre over the origin jumped in dimension; the jump and the failure of openness are the same phenomenon.

11.4 Application: Lines on a Cubic Surface

The Grassmannian and the fibre-dimension theorem together answer a question that would otherwise be inaccessible: how many lines lie on a surface in $\mathbb{P}^3$? For a general cubic surface the answer is a finite number, and the argument that it is finite is a dimension count of exactly the kind this Part was built to support.

Fix the two parameter spaces. Lines in $\mathbb{P}^3_k$ form $G = G(2, 4)$, of dimension 4 by theorem 11.2.4.

Cubic surfaces in $\mathbb{P}^3_k$ are the nonzero degree-3 forms in four variables up to scalar, so they form $\mathbb{P}^{19}_k$: there are $\binom{3+3}{3} = 20$ monomials of degree 3 in four variables. Note that this parameter space contains reducible and singular surfaces too, which is harmless and in fact necessary, since a parameter space has to be complete to be useful.

The strategy is to put lines and surfaces into a single variety and project it two ways. Set

$$I = \left\{(\ell, S) \in G \times \mathbb{P}^{19}_k \;\middle|\; \ell \subseteq S\right\}$$

Projecting to G organizes I by the line; over a fixed line the cubics containing it form a linear system, and that side computes $\dim I$. Projecting to $\mathbb{P}^{19}$ organizes I by the surface, and its fibres are the sets of lines to be counted.

Lemma 11.4.1: The Incidence Variety

I is an irreducible projective variety of dimension 19, and $\pi_1 \colon I \to G$ is surjective with every fibre a linear subspace $\mathbb{P}^{15} \subseteq \mathbb{P}^{19}$.

***Proof*:**
Work chart by chart on G. By the proof of theorem 11.2.4 the open set $U \subseteq G$ where $p_{01} \neq 0$ is isomorphic to $\mathbb{A}^4_k$, with coordinates a, b, c, d naming the plane spanned by $(1, 0, a, b)$ and $(0, 1, c, d)$; the corresponding line $\ell_{abcd} \subseteq \mathbb{P}^3_k$ is the image of

$$[s : t] \longmapsto [\,s : t : as + ct : bs + dt\,]$$

Let $T = T_{abcd}$ be the linear automorphism of k^4 fixing e_2, e_3 and sending e_0, e_1 to $(1, 0, a, b)$ and $(0, 1, c, d)$. Its matrix is lower triangular with 1s on the diagonal, so $\det T = 1$, and both T and T^{-1} have entries that are polynomials in a, b, c, d. By construction T carries the line $\mathcal{Z}(x_2, x_3)$ onto ℓ_{abcd}, so for a cubic form F,

$$F \text{ vanishes on } \ell_{abcd} \iff F \circ T \text{ vanishes on } \mathcal{Z}(x_2, x_3) \iff F \circ T \text{ has no term in } x_0, x_1 \text{ alone}$$

the last step because the restriction of $F \circ T$ to $\mathcal{Z}(x_2, x_3)$ is obtained by setting $x_2 = x_3 = 0$, leaving exactly the four terms $x_0^3, x_0^2 x_1, x_0 x_1^2, x_1^3$.

Now consider

$$\Psi \colon U \times \mathbb{P}^{19}_k \to U \times \mathbb{P}^{19}_k \qquad \big((a, b, c, d), [F]\big) \mapsto \big((a, b, c, d), [F \circ T_{abcd}]\big)$$

The coefficients of $F \circ T$ are linear in those of F with coefficients polynomial in a, b, c, d, and $F \circ T \neq 0$ because T is invertible, so Ψ is a regular map; the same formula with T^{-1} gives its inverse, so Ψ is an isomorphism. The displayed equivalence says precisely that Ψ carries $\pi_1^{-1}(U)$ onto $U \times L$, where $L \subseteq \mathbb{P}^{19}$ is the *fixed* linear subspace of cubics whose four coefficients on $x_0^3, x_0^2 x_1, x_0 x_1^2, x_1^3$ vanish. Those four coordinates are independent, so $L \cong \mathbb{P}^{15}$, and

$$\pi_1^{-1}(U) \cong \mathbb{A}^4_k \times \mathbb{P}^{15}_k$$

Three consequences follow, all of them local statements verified on each of the six charts of G in turn, the computation being identical after permuting coordinates.

I is closed. $U \times L$ is closed in $U \times \mathbb{P}^{19}$, hence so is $\pi_1^{-1}(U)$, and being closed is a local condition on a cover. Since $G \times \mathbb{P}^{19}$ is projective by proposition 10.1.3, I is a projective variety.

π_1 is surjective with fibre $\mathbb{P}^{15}$. Under Ψ the fibre over (a, b, c, d) becomes $\{(a, b, c, d)\} \times L$, so it is a linear $\mathbb{P}^{15}$, in particular nonempty and irreducible of dimension 15.

I is irreducible of dimension 19*.* Each $\pi_1^{-1}(U)$ is irreducible by proposition 10.1.6, and any two of the six meet, since the charts of the irreducible G do and the fibres of π_1 are nonempty. Lemma 10.1.5 makes I irreducible, and then corollary 10.5.3 applied to π_1 gives $\dim I = 4 + 15 = 19$, as we sought to show.

The count now hinges on a single input from outside the dimension theory: one cubic surface whose lines can be counted by hand, and which has finitely many. The Fermat cubic is the traditional choice, but its 27 lines take work to enumerate; the following surface is singular, has only three lines, and settles the matter in half a page.

Lemma 11.4.2: A Cubic Surface With Exactly Three Lines

Let $S_0 = \mathcal{Z}(x_0x_1x_2 - x_3^3) \subseteq \mathbb{P}^3_k$. Then S_0 is an irreducible cubic surface, and the only lines contained in it are $\mathcal{Z}(x_3, x_0)$, $\mathcal{Z}(x_3, x_1)$ and $\mathcal{Z}(x_3, x_2)$.

***Proof*:**
Irreducibility. Read $x_0x_1x_2 - x_3^3$ as a polynomial in x_3 over the unique factorization domain $k[x_0, x_1, x_2]$. Its leading coefficient is -1, its middle coefficients are 0, and its constant term $x_0x_1x_2$ lies in (x_0) but not in (x_0^2). Eisenstein's criterion at the prime x_0 applies, so the polynomial is irreducible and S_0 is a variety.

The three lines lie on it. On $\mathcal{Z}(x_3, x_i)$ both $x_0x_1x_2$ and x_3^3 vanish identically.

There are no others. Let $\ell \subseteq S_0$ be a line, parameterized by $[s : t] \mapsto [g_0 : g_1 : g_2 : g_3]$, where the g_i are linear forms in s, t spanning a two-dimensional space, that being what makes the image a line rather than a point. The defining equation restricts to $g_0g_1g_2 = g_3^3$ in $k[s, t]$.

If $g_3 \equiv 0$ then $g_0g_1g_2 \equiv 0$, and $k[s, t]$ is a domain, so $g_i \equiv 0$ for some $i \leq 2$; then $\ell \subseteq \mathcal{Z}(x_3, x_i)$, and as both are lines they are equal.

If $g_3 \not\equiv 0$, change coordinates on the source $\mathbb{P}^1$ so that $g_3 = t$, which is possible because g_3 is a nonzero linear form. Then $g_0g_1g_2 = t^3$. None of g_0, g_1, g_2 is zero, and each divides t^3 in the unique factorization domain $k[s, t]$, in which t is prime; the only linear divisors of t^3 are the scalar multiples of t. So $g_i = c_it$ for every $i \leq 2$ as well, making all four g_i proportional, which contradicts the requirement that they span a two-dimensional space of linear forms. Hence this case does not occur, as we sought to show.

Theorem 11.4.3: Lines on a Cubic Surface

Every cubic surface in $\mathbb{P}^3_k$ contains a line, and there is a nonempty open subset $U \subseteq \mathbb{P}^{19}_k$ such that a cubic surface parameterized by a point of U contains only finitely many.

***Proof*:**
Write $\pi_2 \colon I \to \mathbb{P}^{19}_k$ for the second projection. By lemma 11.4.1 the source is an irreducible projective variety with $\dim I = 19 = \dim \mathbb{P}^{19}_k$, so $\pi_2(I)$ is closed by corollary 10.2.3 and irreducible, being the continuous image of an irreducible space.

π_2 is surjective. The fibre $\pi_2^{-1}(S_0)$ over the surface of lemma 11.4.2 is a set of three points, so it is nonempty of dimension 0; in particular $S_0 \in \pi_2(I)$. Suppose $\pi_2(I) \neq \mathbb{P}^{19}$. Then $\pi_2(I)$ is a proper closed subvariety, so $\dim \pi_2(I) \leq 18$ by proposition 4.3.5. Applying theorem 10.5.1(2) to the dominating map $\pi_2 : I \to \pi_2(I)$ forces every component of every fibre to have dimension at least $19 - 18 = 1$, contradicting the fibre over S_0. Hence $\pi_2(I) = \mathbb{P}^{19}$, which says exactly that every cubic surface contains a line.

The general fibre is finite. Now π_2 is a dominating map of irreducible varieties with $\dim I = \dim \mathbb{P}^{19}$, so theorem 10.5.1(3) supplies a nonempty open $U \subseteq \mathbb{P}^{19}$ with $\dim \pi_2^{-1}(S) = 19 - 19 = 0$ for every $S \in U$. Such a fibre is closed in the projective I, hence an algebraic set, and by theorem 2.3.5 it has finitely many irreducible components. Each component is closed, irreducible and of dimension 0, hence a single point: a closed irreducible set with two distinct points $p \neq q$ contains the proper closed subset $\{p\}$, giving a chain of length 1 and so dimension at least 1. The fibre is therefore finite, as we sought to show.

The dimension count says "finitely many". The classical refinement, that a *smooth* cubic surface contains exactly 27 lines, needs more than dimension theory: one exhibits the 27 on a single smooth cubic and then shows the number is constant across the smooth locus of the family. On the Fermat cubic $\mathcal{Z}(x_0^3 + x_1^3 + x_2^3 + x_3^3)$, in characteristic zero, they are the lines

$$x_0 = \alpha x_1, \quad x_2 = \beta x_3 \qquad \alpha^3 = \beta^3 = -1$$

together with the images of these nine under the two other ways of splitting $\{x_0, x_1, x_2, x_3\}$ into pairs, giving $3 \times 9 = 27$. The count is due to Cayley and Salmon in 1849; a full treatment is in Shafarevich or in Hartshorne's chapter on surfaces.

11.4.4 Note: Where the Numbers Come From The two numbers in the proof are worth keeping straight. The 19 is $\binom{6}{3} - 1$: cubic forms in four variables, up to scalar. The 15 is $19 - 4$, where the 4 counts the coefficients of a binary cubic, that is the number of conditions imposed by requiring a cubic to vanish on a fixed line. The coincidence $\dim I = \dim \mathbb{P}^{19}$ is exactly what makes the answer finite and nonzero. Had the incidence variety been larger, every cubic would carry infinitely many lines; had it been smaller, the general cubic would carry none, which is what happens for surfaces of degree 4 and above.

11.5 Degree, Linear Sections, and Bézout in $\mathbb{P}^n$

Definition 8.5.3 defined the degree of a projective variety as $d!$ times the leading coefficient of its Hilbert polynomial. That is a definition one can compute with and not one that says anything geometric. This section supplies the geometry: the degree counts points, and degrees multiply under intersection. Both statements need theorem 10.6.3 to know that the intersections in question are nonempty and of the right size.

Theorem 11.5.1: Degree Counts Points of a Linear Section

Let $X \subseteq \mathbb{P}^n_k$ be a projective variety of dimension r and degree d. Then for a general linear subspace $L \subseteq \mathbb{P}^n_k$ of dimension $n - r$, the intersection $X \cap L$ is a finite set of exactly d points.

***Proof*:**
The intersection is finite and nonempty. By theorem 10.6.3 every component of $X \cap L$ has dimension at least $r + (n - r) - n = 0$, and the intersection is nonempty. For a general L the dimension is exactly 0: choosing L as an intersection of r hyperplanes, each successive hyperplane may be chosen to contain no component of what remains, since a component is a proper closed subset of $\mathbb{P}^n_k$ and the hyperplanes containing a fixed subvariety form a proper linear subspace of the dual space. Each such cut drops the dimension by exactly one by theorem 4.4.1, so after r cuts the dimension is 0 and the set is finite.

The count. Write $S = S(X)$ for the homogeneous coordinate ring and let $\ell_1, \dots, \ell_r$ be the linear forms cutting out L. Since no ℓ_i vanishes on a component of the successive intersections, each ℓ_i is a nonzerodivisor on the successive quotients, so

$$H_{X \cap L}(m) = \dim_k \left(S/(\ell_1, \dots, \ell_r)\right)_m$$

and each quotient by a nonzerodivisor of degree 1 takes the mth graded piece to the difference of the mth and $(m-1)$st. Iterating r times, the Hilbert polynomial of $X \cap L$ is the rth finite difference of h_X. As h_X has degree r with leading coefficient $d/r!$, its rth finite difference is the constant d.

Finally $X \cap L$ is a finite set of points, and the Hilbert polynomial of a finite set of N distinct points in $\mathbb{P}^n_k$ is the constant N: the restriction map from forms of degree m to functions on the points is surjective for m large, by choosing forms separating the points. Hence $N = d$, as we sought to show.

So the degree of a plane curve is the number of points in which a general line meets it, the degree of the twisted cubic is the number of points in which a general plane meets it, which example 8.5.4(4) computed as 3, and the degree of $\mathbb{P}^r$ itself is 1, a general linear space of complementary dimension meeting it once.

The same bookkeeping gives the general form of Bézout's theorem. Stated for hypersurfaces it is a count of degrees; the plane-curve case of chapter 13 is the case $n = 2$, refined there to include local multiplicities.

Theorem 11.5.2: Bézout's Theorem in $\mathbb{P}^n$

Let $F_1, \dots, F_r$ be forms in $k[x_0, \dots, x_n]$ of degrees $d_1, \dots, d_r$ with $r \leq n$, and suppose $X = \mathcal{Z}(F_1, \dots, F_r)$ has dimension exactly $n - r$, the smallest its dimension can be. Then X has degree

$$\deg X = d_1 d_2 \cdots d_r$$

counting components with multiplicity.

***Proof*:**
Induct on r. For $r = 1$, the computation of example 8.5.4(2) gives $H_X(m) = \binom{m+n}{n} - \binom{m-d_1+n}{n}$, whose leading term is $d_1 m^{n-1}/(n-1)!$, so $\deg X = d_1$: the degree of a hypersurface is the degree of its equation.

For the inductive step, let $X' = \mathcal{Z}(F_1, \dots, F_{r-1})$, of dimension $n - r + 1$ by the hypothesis that each cut is proper, and of degree $d_1 \cdots d_{r-1}$ by induction. Cutting X' by F_r drops the dimension by exactly one, so F_r vanishes on no component of X' and is therefore a nonzerodivisor on

$S(X')$. Multiplication by F_r is then injective and raises degree by d_r, giving the exact sequence of graded pieces

$$0 \to S(X')_{m-d_r} \to S(X')_m \to S(X)_m \to 0$$

so $h_X(m) = h_{X'}(m) - h_{X'}(m - d_r)$. Write $N = n - r + 1$ and $h_{X'}(m) = \frac{e}{N!}m^N + \cdots$ with $e = \deg X'$. Since $m^N - (m - d_r)^N = N d_r m^{N-1} + \cdots$, the difference has degree $N - 1 = n - r$ with leading coefficient

$$\frac{e}{N!} \cdot N d_r = \frac{e\, d_r}{(N-1)!} = \frac{e\, d_r}{(n-r)!}$$

By definition 8.5.3 the degree of X is $(n-r)!$ times that leading coefficient, namely $e\, d_r = d_1 \cdots d_r$, as we sought to show.

Example 11.5.3: Degrees of Intersections

1. *Two quadric surfaces in* $\mathbb{P}^3_k$. Each has degree 2, and if their intersection is a curve then theorem 11.5.2 gives it degree 4. A general such curve is the elliptic quartic; it is not a plane curve, since a plane curve of degree 4 has genus 3 by definition 14.4.1 while this one has genus 1.

2. *The twisted cubic is not a complete intersection.* It has degree 3 and codimension 2 in $\mathbb{P}^3_k$. Were it $\mathcal{Z}(F_1, F_2)$ with $\deg F_i = d_i$, theorem 11.5.2 would force $d_1 d_2 = 3$, so $\{d_1, d_2\} = \{1, 3\}$ and the curve would lie in the plane $\mathcal{Z}(F_1)$. It does not: its parameterization $[s^3 : s^2 t : st^2 : t^3]$ spans $\mathbb{P}^3_k$, since the four coordinate functions are linearly independent. This is the promise made in example 8.3.7.

3. *A line and a plane in* $\mathbb{P}^3_k$. Degrees 1 and 1, dimensions 1 and 2, so the intersection has dimension 0 and degree 1: exactly one point, which is the familiar statement that a line not contained in a plane meets it once.

11.6 Cones, Projections, and Secant Varieties

The embeddings of the previous sections all enlarge the ambient space. This one goes the other way: given a variety in $\mathbb{P}^N_k$, project it into a smaller projective space and ask when nothing is lost. The construction has already been used twice without being set up — to identify a conic with $\mathbb{P}^1_k$ in theorem 14.4.2, and to exhibit a finite map in section 10.3 — and the obstruction to it is a variety worth naming in its own right.

Definition 11.6.1: Cone and Join

Let $X, Y \subseteq \mathbb{P}^N_k$ be disjoint projective varieties. The *join* $J(X, Y)$ is the closure of the union of the lines $\overline{xy}$ for $x \in X$ and $y \in Y$. When $Y = \{p\}$ is a single point, $J(X, p)$ is the *cone* over X with vertex p.

Definition 11.6.2: Projection from a Point

Let $p \in \mathbb{P}^N_k$ and let $H \cong \mathbb{P}^{N-1}_k$ be a hyperplane not containing p. The *projection from p* is

$$\pi_p \colon \mathbb{P}^N_k \setminus \{p\} \to H \qquad q \mapsto \overline{pq} \cap H$$

In coordinates with $p = [0 : \cdots : 0 : 1]$ and $H = \mathcal{Z}(x_N)$, it is $[x_0 : \cdots : x_N] \mapsto [x_0 : \cdots : x_{N-1}]$, visibly regular away from p.

Proposition 11.6.3: Projection Is a Finite Morphism

Let $X \subseteq \mathbb{P}^N_k$ be a projective variety and $p \notin X$. Then π_p restricts to a morphism $X \to \mathbb{P}^{N-1}_k$ whose image is closed of the same dimension as X, and whose fibres are finite.

***Proof*:**
Since $p \notin X$, the map π_p is defined at every point of X, and it is regular there by definition 11.6.2. Its image is closed by corollary 10.2.3, X being projective.

The fibre over a point h is $X \cap \overline{ph}$, the intersection of X with a line through p. That line is not contained in X, since $p \notin X$, so the intersection is a proper closed subset of the line and hence finite. Finiteness of the fibres forces $\dim \pi_p(X) = \dim X$ by theorem 10.5.1(3), whose generic fibre dimension is then 0, as we sought to show.

Projection is injective on X exactly when no line through p meets X twice, and it is an immersion when in addition no tangent line to X passes through p. Both conditions say that p avoids a certain subvariety, and naming those subvarieties is what makes the argument work.

Definition 11.6.4: Secant and Tangent Varieties

Let $X \subseteq \mathbb{P}^N_k$ be a projective variety. The *secant variety* $\mathrm{Sec}(X)$ is the closure of the union of the lines $\overline{xy}$ over pairs of distinct points $x, y \in X$. The *tangent variety* $\mathrm{Tan}(X)$ is the closure of the union of the embedded tangent spaces $T_x X \cong \mathbb{P}^r_k$ over the nonsingular points x of X, where $r = \dim X$.

Proposition 11.6.5: Dimension of the Secant Variety

Let $X \subseteq \mathbb{P}^N_k$ be a projective variety of dimension r. Then

$$\dim \mathrm{Sec}(X) \le 2r + 1 \qquad \dim \mathrm{Tan}(X) \le 2r$$

***Proof*:**
Consider the incidence variety

$$I = \overline{\{(x, y, z) \mid x \neq y \in X,\ z \in \overline{xy}\}} \ \subseteq\ X \times X \times \mathbb{P}^N_k$$

Projecting to $X \times X$ has fibre the line $\overline{xy}$, of dimension 1, over the open set where $x \neq y$. So $\dim I \le \dim(X \times X) + 1 = 2r + 1$ by corollary 10.5.4 and theorem 10.5.1. Now $\mathrm{Sec}(X)$ is the image of I under the third projection, and the image of a projective variety is closed of dimension at most that of the source, by corollary 10.2.3 and theorem 10.5.1(1). Hence $\dim \mathrm{Sec}(X) \leq 2r+1$.

For $\mathrm{Tan}(X)$ run the same argument over X alone. The incidence $\{(x, z) \mid x \in X \text{ nonsingular}, \ z \in T_x X\}$ projects to the smooth locus of X, of dimension r, with fibre the embedded tangent space $T_x X$, of dimension r. So it has dimension at most $2r$, and $\mathrm{Tan}(X)$, being its image, has dimension at most $2r$, as we sought to show.

The bound is what makes projection useful: as soon as the ambient space is large enough, a point can be chosen outside the secant variety, and projecting from it loses nothing.

Theorem 11.6.6: Every Projective Variety Embeds in $\mathbb{P}^{2r+1}$

Let $X \subseteq \mathbb{P}^N_k$ be a projective variety of dimension r. If $N > 2r+1$, then there is a point $p \notin \mathrm{Sec}(X)$, and π_p maps X isomorphically onto its image in $\mathbb{P}^{N-1}_k$. Iterating, X is isomorphic to a projective variety in $\mathbb{P}^{2r+1}_k$.

***Proof*:**
By proposition 11.6.5, $\dim \mathrm{Sec}(X) \leq 2r+1 < N$, so $\mathrm{Sec}(X)$ is a proper closed subset of $\mathbb{P}^N_k$ and a point p outside it exists. Note $X \subseteq \mathrm{Sec}(X)$ whenever $\dim X \geq 1$, so $p \notin X$ and proposition 11.6.3 applies.

π_p is injective on X: if $\pi_p(x) = \pi_p(y)$ with $x \neq y$ then p lies on the line $\overline{xy}$, hence in $\mathrm{Sec}(X)$, contrary to choice. It is also an immersion: a tangent line to X is a limit of secant lines, so $\mathrm{Tan}(X) \subseteq \mathrm{Sec}(X)$ and no tangent line to X passes through p either. An injective morphism of projective varieties that is an immersion is an isomorphism onto its closed image.

Repeating drops the ambient dimension by one each time and stops at $N = 2r+1$, as we sought to show.

Example 11.6.7: Projections in Practice

1. *A curve in* $\mathbb{P}^3_k$. Here $r = 1$, so $2r+1 = 3$ and theorem 11.6.6 says every projective curve embeds in $\mathbb{P}^3_k$. It does not say $\mathbb{P}^2_k$: projecting once more from a point of $\mathrm{Sec}(X)$ is unavoidable when $N = 3$, and the image acquires singularities. That is why plane curves are special and why chapter 12 must handle singular ones.

2. *The twisted cubic.* Its secant variety is all of $\mathbb{P}^3_k$, since $\dim \mathrm{Sec} \leq 3$ and the secant lines already sweep out a three-dimensional set. So no projection to $\mathbb{P}^2_k$ is an isomorphism, consistent with the twisted cubic being a rational curve of degree 3 while every plane cubic has genus 1.

3. *A conic.* Projection from a point of the conic itself, rather than from outside it, is the map of theorem 14.4.2: a line through p meets the conic once more, so the projection is a bijection to $\mathbb{P}^1_k$. This is the degenerate case $p \in X$, excluded from proposition 11.6.3, and it lowers the dimension of the image rather than preserving it.

Exercise 11.6.1

1. Write out the Veronese embedding $\mathbb{P}^1_k \to \mathbb{P}^2_k$ for $d = 2$ and identify its image. Then do $d = 3$ and show the image is the twisted cubic of example 8.3.7.
2. Compute $N = \binom{n+d}{d} - 1$ for $(n, d) = (2, 2)$ and describe the Veronese surface in $\mathbb{P}^5_k$. Show that it contains no lines, although it is covered by conics.
3. Verify corollary 11.1.3 explicitly for $n = 1$, $d = 2$: exhibit the hyperplane of $\mathbb{P}^2_k$ cutting out a prescribed pair of points of $\mathbb{P}^1_k$.
4. Take the 2-plane in k^4 spanned by $(1, 0, a, b)$ and $(0, 1, c, d)$. Compute its six Plücker coordinates and verify the relation $p_{01}p_{23} - p_{02}p_{13} + p_{03}p_{12} = 0$ directly.
5. Show that $G(1, n) \cong \mathbb{P}^{n-1}_k$ and that $G(n-1, n) \cong \mathbb{P}^{n-1}_k$, and check both against the dimension formula of theorem 11.2.4.
6. Show that a subset of $\mathbb{A}^1_k$ is constructible if and only if it is finite or cofinite. Deduce that every constructible subset of $\mathbb{A}^1_k$ is either open or closed, so the phenomenon of example 11.3.5 needs at least two dimensions.
7. Give a dominating regular map whose image is not open, and one whose image is not closed. Explain how theorem 11.3.4 accommodates both.
8. Repeat the dimension count of section 11.4 for surfaces of degree e in $\mathbb{P}^3_k$: show that the incidence variety has dimension $\binom{e+3}{3} - 1 + 3 - e$ and conclude that a general surface of degree $e \geq 4$ contains no line, while every quadric contains a one-parameter family.

Part IV

Curves

The fundamental theorem of algebra counts the roots of a polynomial in one variable, and does so exactly because $\mathbb{C}$ is algebraically closed. This Part asks for the two-variable analogue: given two plane curves, how many points do they share? Over $\mathbb{A}^2$ the question has no clean answer, since curves can miss each other or meet at infinity. In $\mathbb{P}^2$ it does, and the answer, Bézout's theorem, is that two curves of degrees m and n meet in exactly mn points.

Making that exact requires deciding what a point of intersection is worth, which is the intersection number of chapter 12. The counting then organizes itself into a group, the divisor class group of chapter 14, whose appearance on a cubic curve is the classical group law. The Part closes by removing the singularities that made the counting delicate in the first place.

12

Intersection Multiplicity

Two curves meeting at a point may cross transversally, be tangent, or meet at a singularity of one or both. Any count of intersections that is to be well behaved must weigh these cases differently. This chapter constructs the weight.

The route runs through the local ring at the point. The multiplicity of a curve at a point turns out to be readable off that ring alone, and at a nonsingular point the ring is a discrete valuation ring, so it carries an order function that already sees tangency. The intersection number of two curves is then the dimension of a single quotient of the local ring of the plane, and it is pinned down by eight properties which double as an algorithm for computing it.

12.1 Plane Curves, Multiplicity, and Tangent Lines

The fundamental theorem of algebra counts the roots of a one-variable polynomial over an algebraically closed field, and it can be read geometrically: a curve of degree n meets the line $x = 0$ in n points, provided the points are weighted correctly. This Part generalizes that reading to two curves, and the weight is what this chapter constructs.

Affine space is the wrong home for the statement. Take the two concentric circles $x^2 + y^2 - 1$ and $x^2 + y^2 - 4$ in $\mathbb{A}^2_{\mathbb{C}}$: both have degree 2, so the count should be 4, and yet they share no point at all. Passing to $\mathbb{P}^2_{\mathbb{C}}$ repairs it. The homogenizations $x^2 + y^2 - z^2$ and $x^2 + y^2 - 4z^2$ meet where $z = 0$ and $x^2 + y^2 = 0$, which is the pair of *circular points* $[1 : \pm i : 0]$ already met in example 8.3.4. Two points, not four, so the weights must be 2 each, and indeed the two circles are tangent to each other at both. The rest of this chapter builds the notion of weight that makes such counts come out, and chapter 13 proves they always do.

The development follows Fulton [19]. Throughout, k is algebraically closed.

A plane curve in the sense of this Part is not quite a variety. The point of the intersection theory is to count with multiplicity, and multiplicity is information that $\mathcal{Z}(f)$ throws away: $\mathcal{Z}(f)$ and $\mathcal{Z}(f^2)$ are the same set. So the object of study is the polynomial, taken up to scalar.

Definition 12.1.1: Plane Curve

An *affine plane curve* is an equivalence class of nonconstant polynomials in $k[x,y]$, where $f \sim g$ if and only if $f = \alpha g$ for some $\alpha \in k^*$. A *projective plane curve* is the same with $k[x,y]$ replaced by the nonconstant homogeneous polynomials of $k[x,y,z]$. The *degree* of a curve is the degree of any of its representatives.

This is a deliberate departure from definition 4.3.4, where a curve was a one-dimensional variety. The two agree on the underlying point set, since $\mathcal{Z}(f)$ for $f \in k[x,y]$ nonconstant is a hypersurface in $\mathbb{A}^2_k$ and so has dimension 1 by theorem 4.4.2; they differ in that the present notion remembers repeated factors. Homogenizing an affine plane curve and taking the closure in $\mathbb{P}^2_k$ gives a projective plane curve, as in section 8.3, and the two constructions are inverse away from the line at infinity.

Write $f = \prod_{i=1}^n f_i^{e_i}$ for the factorization into distinct irreducible factors, which is available because $k[x,y]$ and $k[x,y,z]$ are unique factorization domains. The curve f is *irreducible* if $n = 1$ and $e_1 = 1$; the component f_i is *simple* if $e_i = 1$ and *multiple* otherwise[1].

The local ring at a point is the algebraic object that stores intersection data, and it was already constructed in chapter 2.

Example 12.1.2: Curves Versus Their Zero Sets

1. Let $f = y - x^2$ and $g = (y - x^2)^2$. Then $\mathcal{Z}(f) = \mathcal{Z}(g)$ as subsets of $\mathbb{A}^2_k$, both being the parabola, yet f and g are different plane curves in the sense of definition 12.1.1: one has degree 2 and the other degree 4. Everything this Part counts distinguishes them. A general line meets $\mathcal{Z}(f)$ in 2 points and $\mathcal{Z}(g)$ in 4, once multiplicity is accounted for, which is what theorem 13.1.1 will record.

2. The multiplicity at a point behaves the same way. At the origin, $f = y - x^2$ has lowest form y, so $m_0(f) = 1$ and the origin is a nonsingular point; while $g = (y - x^2)^2$ has lowest form y^2, so $m_0(g) = 2$. The zero set is nonsingular as a set and the curve g is not.

3. *Irreducible components with multiplicity.* The curve $h = x^2 y = x \cdot x \cdot y$ has $\mathcal{Z}(h) = \mathcal{Z}(xy)$, the union of the two axes, but as a curve h has the y-axis as a *double* component and the x-axis as a simple one. The factorization $h = \prod f_i^{e_i}$ records $e = (2,1)$, and proposition 12.1.7 will read $m_0(h) = 2 \cdot 1 + 1 \cdot 1 = 3$ off it.

Keeping the polynomial rather than its zero set is exactly what lets these distinctions be made, and it is the reason chapter 19 points at schemes, where the same information is carried intrinsically.

[1] In [3] schemes recover this information intrinsically: the scheme $\operatorname{Spec} k[x,y]/(f^2)$ is not the scheme $\operatorname{Spec} k[x,y]/(f)$, even though the two have the same points. Working with polynomials up to scalar is the classical workaround.

Definition 12.1.3: Recall: Stalk At A Point

Let V be a variety and $p \in V$ a point. The *stalk* at p is

$$\mathcal{O}_p(V) = \{f/g \in k(V) \mid g(p) \neq 0\}$$

By proposition 3.2.10 it is a Noetherian local domain, with maximal ideal $\mathfrak{m}_p(V) = \{f/g \in \mathcal{O}_p(V) \mid f(p) = 0\}$.

The intuition is that $\mathcal{O}_p(V)$ holds exactly the rational functions defined at p, which is why a denominator is allowed as long as it does not vanish there. For a curve f, write $\mathcal{O}_p(f)$ for the stalk of $\mathcal{Z}(f)$, and $\mathfrak{m}_p(f)$ for its maximal ideal.

Almost every computation below is easier at the origin, and moving a point there costs nothing.

Definition 12.1.4: Linear Coordinate Change

Let $\mathbb{A}^n_k$ be affine space with coordinate ring $k[x_1, \ldots, x_n]$. A bijective affine transformation $T\colon \mathbb{A}^n_k \to \mathbb{A}^n_k$ is called a *linear change of coordinates.*

If $T = (T_1, \ldots, T_n)$ then T acts on functions by $f(x_1, \ldots, x_n) \mapsto f(T_1, \ldots, T_n)$, and any such T decomposes into a translation followed by a bijective linear transformation. The first invariant to define this way is the multiplicity, obtained by translating the point of interest to the origin and reading off the lowest-degree term.

Definition 12.1.5: Multiplicity

Let $f \in k[x, y]$ be an affine plane curve and $p \in \mathcal{Z}(f)$. After a translation taking p to the origin, write

$$f = f_m + f_{m+1} + \cdots + f_d$$

for the decomposition into homogeneous components, where $f_m \neq 0$. The *multiplicity* of f at p, written $m_p(f)$, is m.

The definition does not depend on the translation chosen, since a translation carrying the origin to itself is the identity, and $m_p(f) \geq 1$ exactly because $f(p) = 0$. The multiplicity measures how badly the curve degenerates at p: it is 1 precisely when p is a nonsingular point of f in the sense of chapter 5, because f_1 is the differential of f at the origin and $f_1 \neq 0$ is the condition that some partial derivative survive.

The lowest form f_m carries more than its degree. Over an algebraically closed field a nonzero binary form factors completely into linear factors, so

$$f_m = \prod_{i=1}^{s} L_i^{r_i} \qquad \sum_i r_i = m$$

with the L_i pairwise non-proportional linear forms.

Definition 12.1.6: Tangent Lines of a Plane Curve

With notation as above, the lines $\mathcal{Z}(L_i)$ are the *tangent lines* to f at p, and r_i is the multiplicity of $\mathcal{Z}(L_i)$ as a tangent. A line through p that is not one of the $\mathcal{Z}(L_i)$ is assigned tangent multiplicity 0. The form f_m itself is the *tangent cone*.
The point p is an *ordinary multiple point* if f has $m = m_p(f)$ distinct tangent lines there, that is if every $r_i = 1$. An ordinary double point is a *node*.

This is the plane-curve case of definition 6.1.2: the initial form f_m generates the initial ideal of (f), so the tangent cone algebra is $k[x,y]/(f_m)$ and its zero set is the union of the tangent lines, while $m_p(f)$ agrees with the multiplicity of definition 6.1.4 by proposition 6.1.5. What the plane case adds is the factorization: a binary form splits into linear factors over the algebraically closed k, so the cone is literally a union of lines counted with multiplicity, which is false for a cone in three or more variables.

At a nonsingular point $m = 1$, so there is a single tangent line $\mathcal{Z}(f_1)$, and f_1 is the linear part of f. That recovers the familiar formula: for $p = (a, b)$ a nonsingular point of f, the tangent line is

$$\frac{\partial f}{\partial x}(p)\,(x-a) + \frac{\partial f}{\partial y}(p)\,(y-b) = 0$$

which is the definition of chapter 5 specialized to a plane curve. For a projective plane curve $F \in k[x,y,z]$ the point p is nonsingular when the three partials $\partial_x F, \partial_y F, \partial_z F$ do not all vanish at p, and the tangent line is then $\mathcal{Z}\big(\sum_i \partial_i F(p)\,x_i\big)$; passing to an affine chart recovers the two-variable statement.

Multiplicities and tangents both behave multiplicatively across the factorization of f.

Proposition 12.1.7: Multiplicity Across Components

Let $f = \prod_i f_i^{e_i}$ be an affine plane curve, factored into distinct irreducibles, and let $p \in \mathcal{Z}(f)$. Then

$$m_p(f) = \sum_i e_i\, m_p(f_i)$$

and a line L tangent to f_i with multiplicity r_i is tangent to f with multiplicity $\sum_i e_i r_i$. Consequently p is a nonsingular point of f if and only if it lies on exactly one f_i, that component is simple, and p is a nonsingular point of it.

***Proof*:**
Translate p to the origin. In the graded domain $k[x,y]$ the lowest form of a product is the product of the lowest forms, so the lowest form of f is $\prod_i (f_i)_{m_i}^{e_i}$ where $m_i = m_p(f_i)$; taking degrees gives the first claim, and reading off the linear factors gives the second.

For the last claim, $m_p(f) = 1$ forces exactly one term of $\sum_i e_i m_p(f_i)$ to be nonzero and equal to 1, so exactly one f_i passes through p, its exponent e_i is 1, and $m_p(f_i) = 1$. The converse is the same computation read backwards, as we sought to show.

12.2 Local Rings of Plane Curves

The multiplicity was defined by writing down a polynomial and looking at its lowest term, which is a coordinate-dependent recipe even if the answer is not. This section shows that $m_p(f)$ is visible inside the local ring $\mathcal{O}_p(f)$ alone, and that the nonsingular points are exactly those where that ring is as simple as a one-dimensional local ring can be.

Theorem 12.2.1: Multiplicity and Local Ring

Let f be an irreducible affine plane curve and $p \in \mathcal{Z}(f)$. Write $\mathcal{O} = \mathcal{O}_p(f)$ and $\mathfrak{m}$ for its maximal ideal. Then for all sufficiently large n,

$$m_p(f) = \dim_k \left(\frac{\mathfrak{m}^n}{\mathfrak{m}^{n+1}} \right)$$

so the multiplicity of f at p is determined by the local ring $\mathcal{O}_p(f)$.

***Proof*:**
The quotient maps $\mathcal{O}/\mathfrak{m}^{n+1} \to \mathcal{O}/\mathfrak{m}^n$ have kernel $\mathfrak{m}^n/\mathfrak{m}^{n+1}$, giving a short exact sequence of k-vector spaces

$$0 \to \mathfrak{m}^n/\mathfrak{m}^{n+1} \to \mathcal{O}/\mathfrak{m}^{n+1} \to \mathcal{O}/\mathfrak{m}^n \to 0$$

Dimension is additive along a short exact sequence of finite-dimensional vector spaces, so it suffices to prove

$$\dim_k (\mathcal{O}/\mathfrak{m}^n) = n\, m_p(f) + s$$

for a constant s independent of n and all $n \geq m_p(f)$; subtracting consecutive values then leaves $m_p(f)$.

Translate so that $p = (0,0)$ and write $I = (x, y) \subseteq k[x, y]$, so that $\mathcal{I}(\{p\}) = I$. Since $\mathcal{Z}(I^n) = \{p\}$, corollary 12.2.7 and the third exercise of exercise set 12.4.1 give

$$\frac{k[x,y]}{(I^n, f)} \cong \frac{\mathcal{O}_p(\mathbb{A}^2_k)}{(I^n, f)\mathcal{O}_p(\mathbb{A}^2_k)} \cong \frac{\mathcal{O}_p(f)}{I^n\mathcal{O}_p(f)} \cong \mathcal{O}/\mathfrak{m}^n$$

the last isomorphism because $\mathfrak{m}$ is generated by the images of x and y. So the problem is to compute $\dim_k k[x, y]/(I^n, f)$.

Let $m = m_p(f)$. Multiplication by f gives a map

$$\psi\colon \frac{k[x,y]}{I^{n-m}} \to \frac{k[x,y]}{I^n} \qquad \psi(\overline{g}) = \overline{fg}$$

which is well defined because $f \in I^m$, and injective: $k[x, y]$ is a graded domain, so the lowest form of fg is the product of those of f and g, whence $fg \in I^n$ forces $g \in I^{n-m}$. Its image is $(I^n, f)/I^n$, the kernel of the projection $\varphi\colon k[x, y]/I^n \to k[x, y]/(I^n, f)$, so

$$0 \to \frac{k[x,y]}{I^{n-m}} \xrightarrow{\psi} \frac{k[x,y]}{I^n} \xrightarrow{\varphi} \frac{k[x,y]}{(I^n, f)} \to 0$$

is exact. Now $k[x, y]/I^t$ has as a basis the monomials of degree less than t, of which there are $1 + 2 + \cdots + t = \frac{t(t+1)}{2}$, as the second exercise of exercise set 12.4.1 records. Additivity therefore gives

$$\dim_k(\mathcal{O}/\mathfrak{m}^n) = \frac{n(n+1)}{2} - \frac{(n-m)(n-m+1)}{2} = nm - \frac{m(m-1)}{2}$$

valid for all $n \geq m$. This has the required shape with $s = -\frac{m(m-1)}{2}$, a constant, as we sought to show.

Example 12.2.2: Multiplicity from the Local Ring

Take p the origin and compute $\dim_k \mathfrak{m}^n/\mathfrak{m}^{n+1}$ in each case, which theorem 12.2.1 says stabilizes at $m_p(f)$.

1. *A nonsingular point.* $f = y - x^2$. Here $\mathcal{O}_p(f) \cong k[x]_{(x)}$, since $y = x^2$ on the curve, and $\mathfrak{m} = (x)$. So $\mathfrak{m}^n/\mathfrak{m}^{n+1}$ is spanned by x^n alone and has dimension 1 for every n. That matches $m_p(f) = 1$.
2. *The node.* $f = y^2 - x^2 - x^3$, with $m_p(f) = 2$. The formula of the proof gives $\dim_k \mathcal{O}/\mathfrak{m}^n = 2n - \binom{2}{2} = 2n - 1$ for $n \geq 2$, so $\dim_k \mathfrak{m}^n/\mathfrak{m}^{n+1} = (2n+1) - (2n-1) = 2$ for $n \geq 2$, matching $m_p(f) = 2$.
3. *The cusp.* $f = y^2 - x^3$, again with $m_p(f) = 2$ since the lowest form is y^2. The same computation gives 2 again.

The node and the cusp have the same multiplicity and the same sequence of dimensions, so the local ring sees the multiplicity and not the tangent cone. What separates them is the number of tangent lines, which is definition 12.1.6, or equivalently the number of branches, which example 17.3.3 separates by blowing up.

The theorem already separates the nonsingular points from the rest, since they are the ones where the successive quotients have dimension 1. The next result says what that means ring-theoretically.

Corollary 12.2.3: Simple Points and Multiplicity

Let f be an irreducible affine plane curve and $p \in \mathcal{Z}(f)$. Then p is a nonsingular point of f if and only if $\mathcal{O}_p(f)$ is a discrete valuation ring. In that case, if L is a line through p that is not tangent to f at p, the image of its defining linear form generates $\mathfrak{m}_p(f)$.

***Proof*:**
Suppose p is nonsingular and L is a line through p that is not tangent there. Change coordinates so that $p = (0,0)$, the tangent line is $\mathcal{Z}(y)$, and $L = \mathcal{Z}(x)$. It suffices to show that $\mathfrak{m} = \mathfrak{m}_p(f)$ is generated by the image of x, since $\mathfrak{m}$ is generated by the images of x and y.

Because $\mathcal{Z}(y)$ is the tangent line, the lowest form of f is a nonzero multiple of y, and after scaling f we may take $f = y + (\text{terms of degree} \geq 2)$. Collect the terms containing y: what remains involves x alone and has degree at least 2, since the only degree-one term of f is y. Hence

$$f = yg - x^2h \qquad g = 1 + (\text{terms of positive degree}), \quad h \in k[x,y]$$

In $k[\mathcal{Z}(f)]$ this reads $yg = x^2h$, and $g(p) = 1 \neq 0$ makes g a unit in $\mathcal{O}_p(f)$, so

$$y = x^2hg^{-1} \in (x)$$

Therefore $\mathfrak{m} = (x)$ is principal. By proposition 3.2.10 the ring $\mathcal{O}_p(f)$ is a Noetherian local domain, and it has Krull dimension 1 by proposition 4.3.2, since $\mathcal{Z}(f)$ is a curve and $\{p\}$ has codimension 1 in it. So proposition 5.5.2 applies, and its implication (4) ⇒ (2) makes $\mathcal{O}_p(f)$ a discrete valuation ring.

Conversely, if $\mathcal{O}_p(f)$ is a discrete valuation ring then $\mathfrak{m}/\mathfrak{m}^2$ is one-dimensional, generated by the image of a uniformizer, and the same holds for every $\mathfrak{m}^n/\mathfrak{m}^{n+1}$. By theorem 12.2.1 this common dimension is $m_p(f)$, so $m_p(f) = 1$ and p is nonsingular, completing the proof.

Now that $\mathcal{O}_p(f)$ is known to be a discrete valuation ring at a nonsingular point, its valuation can be named. This is the invariant that chapter 13 uses to read off intersection numbers.

Definition 12.2.4: Order Function at a Simple Point

Let f be an irreducible plane curve and p a nonsingular point of f, so that $\mathcal{O}_p(f)$ is a discrete valuation ring by corollary 12.2.3. The *order function* at p

$$\operatorname{ord}_p^f \colon \mathcal{O}_p(f) \setminus \{0\} \to \mathbb{N}$$

is the discrete valuation of $\mathcal{O}_p(f)$: writing t for a uniformizing parameter, $\operatorname{ord}_p^f(g)$ is the unique m with $g = ut^m$ for a unit $u \in \mathcal{O}_p(f)$. By convention $\operatorname{ord}_p^f(0) = \infty$. The definition extends to $k(f)$ by $\operatorname{ord}_p^f(g/h) = \operatorname{ord}_p^f(g) - \operatorname{ord}_p^f(h)$, taking values in $\mathbb{Z}$.

Geometrically $\operatorname{ord}_p^f(g)$ counts how many times g vanishes at p along f, and the computation just made says that the order function sees tangency. A line L meeting f transversally at p has $\operatorname{ord}_p^f(L) = 1$, because its linear form generates $\mathfrak{m}$. The tangent line has order at least 2: in the coordinates of the proof it is $\mathcal{Z}(y)$, and

$$\operatorname{ord}_p^f(y) = \operatorname{ord}_p^f(x^2hg^{-1}) = 2\operatorname{ord}_p^f(x) + \operatorname{ord}_p^f(hg^{-1}) \geq 2$$

since g^{-1} is a unit and $h \in \mathcal{O}_p(f)$.

The last ingredient of the chapter is a decomposition for the coordinate ring of a finite set of points. It says that such a ring sees each of its points separately, which is what will let a global intersection count be assembled from local ones.

Lemma 12.2.5: Coordinate Ring For Finite Points

Let $I \subseteq k[x_1, \dots, x_n]$ be an ideal with $\mathcal{Z}(I) = \{p_1, \dots, p_m\}$ finite, and write $\mathcal{O}_i = \mathcal{O}_{p_i}(\mathbb{A}_k^n)$. Then

$$\frac{k[x_1, \dots, x_n]}{I} \cong \prod_{i=1}^{m} \frac{\mathcal{O}_i}{I\mathcal{O}_i}$$

and in particular the left side is a finite-dimensional k-vector space.

***Proof*:**
Put $R = k[x_1, \dots, x_n]/I$. Its maximal ideals correspond by the Nullstellensatz (theorem 2.4.4) to the points of $\mathcal{Z}(I)$, so there are m of them and every prime of R is maximal, that is $\operatorname{kdim} R = 0$. As R is a finitely generated k-algebra it is Noetherian, so [5, Artinian Ring Equivalent Condition I] makes R Artinian, and [5, Artinian Ring Equivalent Condition II] makes it finite-dimensional

over k.

Now [5, Structure Theorem for Artinian Rings] decomposes R as a finite product of local Artinian rings, one for each maximal ideal, the factors being the localizations $R_{\mathfrak{m}_i}$. It remains only to identify $R_{\mathfrak{m}_i}$ geometrically: localizing $k[x_1, \dots, x_n]/I$ at the maximal ideal of p_i is the same as localizing $k[x_1, \dots, x_n]$ there and then dividing by the extended ideal, since localization is exact, and that is $\mathcal{O}_i/I\mathcal{O}_i$, as we sought to show.

This is a case where the algebra was done once upstream and should not be done again. The classical proof constructs idempotents e_i by hand and verifies the Chinese-remainder decomposition directly; the Artinian structure theorem is that construction in general form, and the only genuinely geometric step is the Nullstellensatz translation at the start.

Corollary 12.2.6: Dimension of Coordinate Ring of Finite Points

With notation as above,

$$\dim_k \frac{k[x_1, \dots, x_n]}{I} = \sum_{i=1}^{m} \dim_k \left(\mathcal{O}_i/I\mathcal{O}_i\right)$$

***Proof*:**
Dimension is additive over a finite direct product of vector spaces, so this is lemma 12.2.5 read on dimensions, completing the proof.

Corollary 12.2.7: Coordinate Ring at a Single Point

Let $I \subseteq k[x_1, \dots, x_n]$ satisfy $\mathcal{Z}(I) = \{p\}$. Then

$$\frac{k[x_1, \dots, x_n]}{I} \cong \frac{\mathcal{O}_p(\mathbb{A}^n_k)}{I\,\mathcal{O}_p(\mathbb{A}^n_k)}$$

***Proof*:**
This is lemma 12.2.5 with $m = 1$, completing the proof.

One further consequence of the local rings being discrete valuation rings deserves recording here, since chapter 15 needs it: on a smooth curve, a rational map to a projective variety has no points of indeterminacy at all.

Lemma 12.2.8: Rational Maps on Smooth Curves

Let C be an irreducible curve, $V \subseteq \mathbb{P}^N_k$ a *projective* variety, and $\varphi\colon C \dashrightarrow V$ a rational map. If $p \in C$ is a nonsingular point, then φ is regular at p. If C is smooth, then φ is a regular morphism.

***Proof*:**
Since p is a nonsingular point of the curve C, the local ring $\mathcal{O}_p(C)$ is a Noetherian local domain of Krull dimension 1 which is regular, hence a discrete valuation ring by proposition 5.5.2; write ord_p for its valuation and t for a uniformizer. Represent φ near p as $[f_0 : \cdots : f_N]$ with $f_i \in k(C)$ not all zero, and set

$$e = \min_{0 \le i \le N} \operatorname{ord}_p(f_i)$$

Then $\operatorname{ord}_p(t^{-e} f_i) \ge 0$ for every i, so each $t^{-e} f_i$ lies in $\mathcal{O}_p(C)$ and is defined at p, while $\operatorname{ord}_p(t^{-e} f_j) = 0$ for the index j achieving the minimum, so $t^{-e} f_j(p) \neq 0$ and the tuple is not identically zero at p. Since

$$[t^{-e} f_0 : \cdots : t^{-e} f_N] = [f_0 : \cdots : f_N]$$

as maps to $\mathbb{P}^N_k$, this exhibits φ as regular at p. The image lands in V because V is closed and $\varphi(C)$ is dense in it. If every point of C is nonsingular the argument applies at each, so φ is regular everywhere, as we sought to show.

Projectivity of the target is essential: the rational map $\mathbb{A}^1_k \dashrightarrow \mathbb{A}^1_k$ given by $t \mapsto 1/t$ is undefined at the origin and stays undefined, whereas the same formula into $\mathbb{P}^1_k$ extends by sending 0 to the point at infinity.

12.3 The Intersection Number

Let $f, g \in k[x, y]$ be affine plane curves and $p \in \mathbb{A}^2_k$. The goal is a number $I(p, f \cap g)$ counting how many times the two curves meet at p. Several plausible definitions present themselves, and rather than choose one and hope, the strategy is to write down the properties such a number must have and show that they pin it down uniquely. The list below is Fulton's [19, section 3.3]; for a complementary treatment leaning on complex analysis and on the local picture introduced in [5, Completion], see [1].

The list is deliberately not minimal. Computations rarely need its full strength, but proving that the intersection number satisfies all of it is what makes the number usable.

Say that f and g *intersect properly* at p if they have no common component passing through p. The properties are:

1. $I(p, f \cap g)$ is a non-negative integer whenever f and g intersect properly at p, and $I(p, f \cap g) = \infty$ otherwise.
2. $I(p, f \cap g) = 0$ if and only if $p \notin \mathcal{Z}(f) \cap \mathcal{Z}(g)$. In particular $I(p, f \cap g) = 0$ whenever f or g is a nonzero constant.
3. $I(p, f \cap g)$ depends only on the components of f and of g that pass through p.
4. $I(p, f \cap g) = I(p, g \cap f)$.
5. $I(p, f \cap g)$ is invariant under a linear change of coordinates: if T is one and $q = T^{-1}(p)$, then $I(q, f^T \cap g^T) = I(p, f \cap g)$.
6. $I(p, f \cap g) \ge m_p(f)\, m_p(g)$, with equality if and only if f and g have no tangent line in common at p. In particular two curves meeting transversally at a point nonsingular on both have intersection number 1.

7. If $f = \prod_i f_i^{r_i}$ and $g = \prod_j g_j^{s_j}$, then
$$I(p, f \cap g) = \sum_{i,j} r_i s_j \, I(p, f_i \cap g_j)$$
8. $I(p, f \cap g) = I(p, f \cap (g + hf))$ for every $h \in k[x, y]$.

Property (8) is the one that does the work: it says the intersection number sees the ideal (f, g) and not the particular generators, which is exactly the licence needed to run a Euclidean algorithm on the pair.

Theorem 12.3.1: Intersection Number

There is exactly one function $I(p, f \cap g)$ of a point and a pair of affine plane curves satisfying properties (1) through (8), namely
$$I(p, f \cap g) = \dim_k \left(\frac{\mathcal{O}_p(\mathbb{A}^2_k)}{(f, g)} \right)$$

***Proof*:**
Uniqueness. Suppose $I(p, f \cap g)$ satisfies the eight properties. The properties then determine its value, by an algorithm. By (1) it is enough to treat curves intersecting properly at p, and by (5) we may translate so that $p = (0, 0)$. If p does not lie on both curves, (2) gives the value 0, so assume $0 < I(p, f \cap g) < \infty$ and argue by strong induction: suppose the value is known whenever it is smaller than n, and let $I(p, f \cap g) = n$.

Consider the one-variable restrictions $f(x, 0)$ and $g(x, 0)$, of degrees r and s respectively, with the convention that the zero polynomial has degree $-\infty$. By (4) we may assume $r \leq s$.

1. *Case* $f(x, 0) \equiv 0$. Then $y \big| f$, say $f = yh$, and (7) gives
$$I(p, f \cap g) = I(p, y \cap g) + I(p, h \cap g)$$
Write $g(x, 0) = x^\ell(a_0 + a_1 x + \cdots)$ with $a_0 \neq 0$; here $\ell > 0$ because $g(0, 0) = 0$. Since $g(x, y) - g(x, 0)$ vanishes on the line $y = 0$, it is divisible by y, so property (8) applied to the pair (y, g) replaces g by $g(x, 0)$ without changing the value. Then (7) splits off the factor $a_0 + a_1 x + \cdots$, which does not vanish at p and so contributes nothing by (2):
$$I(p, y \cap g) = I(p, y \cap g(x, 0)) = I(p, y \cap x^\ell) = \ell \, I(p, y \cap x) = \ell$$
the last step by (6), the two lines meeting transversally. Since $\ell > 0$, the remaining term $I(p, h \cap g)$ is strictly less than n and the induction applies.
2. *Case* $r \geq 0$. Scale so that $f(x, 0)$ and $g(x, 0)$ are monic and put $h = g - x^{s-r} f$. The leading terms cancel, so $\deg h(x, 0) = t < s$, while (8) gives
$$I(p, f \cap g) = I(p, f \cap h)$$
Repeat on the pair (f, h), interchanging the two by (4) whenever $t < r$. Each step replaces the degree pair (r, s) by (r, t) with $t < s$, so the sum of the two degrees strictly decreases. It cannot decrease forever, so after finitely many steps one of the two restrictions is identically zero and the previous case applies.

So the eight properties leave no freedom, and at most one such function exists.

Existence. It remains to check that $\dim_k \mathcal{O}/(f,g)$, where $\mathcal{O} = \mathcal{O}_p(\mathbb{A}^2_k)$, has the eight properties. Properties (4), (5) and (8) are immediate, since the quantity depends only on the ideal $(f,g)\mathcal{O}$, which is symmetric in f and g, unchanged by $g \mapsto g + hf$, and carried to the corresponding ideal by a linear change of coordinates. Property (3) holds because a component of f not passing through p is a unit in $\mathcal{O}$ and so contributes nothing to the ideal. Property (2) holds because $\mathcal{O}/(f,g)$ is the zero ring exactly when $(f,g)\mathcal{O} = \mathcal{O}$, that is when some f or g fails to vanish at p. Translating by (5), assume from now on that $p = (0,0)$ and every component of f and g passes through p. What is left is (1), (6) and (7).

Property (1). If f and g have no common component then $\mathcal{Z}(f,g)$ is finite, so corollary 12.2.6 makes $\dim_k \mathcal{O}/(f,g)$ finite. If instead h is a common component, then $(f,g) \subseteq (h)$ and there is a surjection

$$\frac{\mathcal{O}}{(f,g)} \twoheadrightarrow \frac{\mathcal{O}}{(h)}$$

Now $\mathcal{O}/(h) \cong \mathcal{O}_p(\mathcal{Z}(h))$ contains $k[\mathcal{Z}(h)]$, which is infinite-dimensional over k because $\mathcal{Z}(h)$ is a curve. So the left side is infinite-dimensional too.

Property (7). By induction and symmetry it is enough to prove

$$I(p, f \cap gh) = I(p, f \cap g) + I(p, f \cap h)$$

By (1) we may assume f and gh have no common component. Let $\varphi\colon \mathcal{O}/(f,gh) \to \mathcal{O}/(f,g)$ be the natural surjection and let $\psi\colon \mathcal{O}/(f,h) \to \mathcal{O}/(f,gh)$ be the k-linear map $\psi(\overline{z}) = \overline{gz}$. Since dimension is additive along a short exact sequence, it suffices to show that

$$0 \to \frac{\mathcal{O}}{(f,h)} \xrightarrow{\psi} \frac{\mathcal{O}}{(f,gh)} \xrightarrow{\varphi} \frac{\mathcal{O}}{(f,g)} \to 0$$

is exact. Surjectivity of φ is clear, and $\operatorname{im}\psi = \ker\varphi$ because $\ker\varphi = (f,g)/(f,gh)$ is generated by the images of f and g, the first being zero. For injectivity of ψ, suppose $\overline{gz} = \overline{0}$, so that $gz = uf + vgh$ with $u, v \in \mathcal{O}$. Choose $s \in k[x,y]$ with $s(p) \neq 0$ clearing all denominators, and set $A = su$, $B = sv$, $C = sz$, all in $k[x,y]$. Then $gC = Af + Bgh$, that is

$$g(C - Bh) = Af$$

Since f and g have no common factor and $k[x,y]$ is a unique factorization domain, f divides $C - Bh$, say $C - Bh = Df$. Dividing by s, which is a unit in $\mathcal{O}$,

$$z = \frac{B}{s}h + \frac{D}{s}f \in (f,h)\mathcal{O}$$

so $\overline{z} = \overline{0}$ and ψ is injective.

Property (6). Let $m = m_p(f)$, $n = m_p(g)$ and $I = (x,y) \subseteq k[x,y]$. Consider

$$\begin{tikzcd}
\frac{k[x,y]}{I^n} \times \frac{k[x,y]}{I^m} \arrow[r, "\psi"] & \frac{k[x,y]}{I^{n+m}} \arrow[r, "\varphi"] & \frac{k[x,y]}{(I^{n+m},f,g)} \arrow[r] \arrow[d, "\alpha"] & 0 \\
& \frac{\mathcal{O}}{(f,g)} \arrow[r, "\pi"] & \frac{\mathcal{O}}{(I^{n+m},f,g)} \arrow[r] & 0
\end{tikzcd}$$

where φ, π, α are the natural maps and $\psi(\bar{a}, \bar{b}) = \overline{af + bg}$, which is well defined because $f \in I^m$ and $g \in I^n$. The top row is exact at the middle and right, though ψ need not be injective, and α is an isomorphism by corollary 12.2.7, since $\mathcal{Z}(I^{n+m}, f, g) \subseteq \{p\}$. Exactness gives

$$\dim_k \frac{k[x,y]}{(I^{n+m}, f, g)} = \dim_k \frac{k[x,y]}{I^{n+m}} - \dim_k \ker\varphi \qquad \dim_k \ker\varphi \leq \dim_k \frac{k[x,y]}{I^n} + \dim_k \frac{k[x,y]}{I^m}$$

with equality in the second exactly when ψ is injective. Since π is surjective, chaining these and using $\dim_k k[x,y]/I^t = \frac{t(t+1)}{2}$ gives

$$\begin{aligned}
I(p, f \cap g) &= \dim_k \frac{\mathcal{O}}{(f,g)} \\
&\geq \dim_k \frac{\mathcal{O}}{(I^{n+m}, f, g)} \\
&= \dim_k \frac{k[x,y]}{(I^{n+m}, f, g)} \\
&\geq \dim_k \frac{k[x,y]}{I^{n+m}} - \dim_k \frac{k[x,y]}{I^n} - \dim_k \frac{k[x,y]}{I^m} \\
&= \frac{(n+m)(n+m+1)}{2} - \frac{n(n+1)}{2} - \frac{m(m+1)}{2} \;=\; mn
\end{aligned}$$

which is the inequality. Equality holds exactly when both steps are equalities, that is when π is an isomorphism, equivalently $I^{n+m} \subseteq (f,g)\mathcal{O}$, and when ψ is injective. That both hold precisely when f and g have no common tangent is lemma 12.3.2, which completes the proof.

The lemma the last step needs is where the tangent lines finally enter the argument, and it is worth isolating because both of its parts are used again in chapter 13.

Lemma 12.3.2: Technical Lemma - Intersection and Tangents

Let f, g be affine plane curves meeting properly at $p = (0,0)$, with $m = m_p(f)$, $n = m_p(g)$, and let $I = (x,y)$. Write ψ for the map of theorem 12.3.1. Then

1. If f and g have no common tangent line at p, then $I^t \subseteq (f,g)\mathcal{O}$ for every $t \geq m+n-1$.

2. ψ is injective if and only if f and g have no common tangent line at p.

***Proof*:**
Let $L_1, \ldots, L_m$ be the tangent lines of f at p, listed with multiplicity, and $M_1, \ldots, M_n$ those of g. Extend the lists by $L_i = L_m$ for $i > m$ and $M_j = M_n$ for $j > n$, and set

$$A_{ij} = L_1 L_2 \cdots L_i \, M_1 M_2 \cdots M_j$$

with $A_{00} = 1$, so A_{ij} is a form of degree $i+j$. The whole argument rests on the following claim.

Claim: if f and g have no common tangent, then $\{A_{ij} \mid i+j = t\}$ is a k-basis for the space of forms of degree t. There are $t+1$ of them and the space has dimension $t+1$, so it suffices to prove independence. Write $P_i = L_1 \cdots L_i$ and $Q_j = M_1 \cdots M_j$ and suppose $\sum_{i=0}^{t} c_i P_i Q_{t-i} = 0$ with not all c_i zero. Let a be the least and b the greatest index with $c_i \neq 0$. If $a = b$ the relation reads $c_a P_a Q_{t-a} = 0$, impossible in the domain $k[x,y]$. So $a < b$. Every P_i with $i \geq a$ is divisible by P_a, and every Q_{t-i} with $i \le b$ is divisible by Q_{t-b}, so dividing the relation by $P_a Q_{t-b}$ leaves

$$\sum_{i=a}^{b} c_i \frac{P_i}{P_a} \cdot \frac{Q_{t-i}}{Q_{t-b}} = 0$$

Every term with $i > a$ is divisible by L_{a+1}. Hence L_{a+1} divides the $i = a$ term, which is $c_a\, Q_{t-a}/Q_{t-b}$, a nonzero scalar times a nonempty product of the M_j. But L_{a+1} is a tangent of f and no M_j is proportional to it, contradicting unique factorization. This proves the claim, and shows exactly where the hypothesis is used.

Part (1). First, $I^t \subseteq (f,g)\mathcal{O}$ for all sufficiently large t. Write $\mathcal{Z}(f,g) = \{p, q_1, \dots, q_s\}$, which is finite since f and g meet properly, and choose a polynomial h with $h(q_i) = 0$ for every i and $h(p) \neq 0$. Then xh and yh vanish on $\mathcal{Z}(f,g)$, so by the Nullstellensatz (theorem 2.4.4) there is N with $(xh)^N, (yh)^N \in (f,g)$. As h^N is a unit in $\mathcal{O}$, this gives $x^N, y^N \in (f,g)\mathcal{O}$, and every monomial of degree at least $2N$ is divisible by x^N or by y^N, so $I^{2N} \subseteq (f,g)\mathcal{O}$.

Now argue by descending induction on $i + j$ that $A_{ij} \in (f,g)\mathcal{O}$ whenever $i + j \ge m + n - 1$, the case $i + j \ge 2N$ being covered by the previous paragraph together with the claim. Given $i + j \ge m + n - 1$, one of $i \ge m$ or $j \ge n$ holds; say $i \ge m$, the other case being symmetric with g in place of f. Then $A_{ij} = A_{m0} B$ with $\deg B = i + j - m$. After scaling f so that its lowest form is exactly A_{m0}, write $f = A_{m0} + f'$ with every term of f' of degree at least $m + 1$. Multiplying by B and rearranging,

$$A_{ij} = Bf - Bf'$$

Here $Bf \in (f,g)\mathcal{O}$, and every term of Bf' is a form of degree at least $i + j + 1$, hence by the claim a combination of the $A_{i'j'}$ with $i' + j' \ge i + j + 1$, which lie in $(f,g)\mathcal{O}$ by the inductive hypothesis. So $A_{ij} \in (f,g)\mathcal{O}$. Since the A_{ij} with $i + j = t$ span the forms of degree t, and I^t is spanned by forms of degree at least t, this gives $I^t \subseteq (f,g)\mathcal{O}$ for $t \ge m + n - 1$.

Part (2). Suppose first that f and g have no common tangent, and let $\psi(\overline{A}, \overline{B}) = \overline{Af + Bg} = 0$, so that $Af + Bg \in I^{n+m}$. The domain of ψ is $\frac{k[x,y]}{I^n} \times \frac{k[x,y]}{I^m}$, so $\overline{A} \neq 0$ means $r < n$ and $\overline{B} \neq 0$ means $s < m$, where r and s are the orders of A and B, with lowest forms A_r and B_s. Assume for contradiction that $(\overline{A}, \overline{B}) \neq 0$ and consider the possibilities.

If $\overline{B} = 0$ then $B \in I^m$, so $Bg \in I^{m+n}$ and hence $Af \in I^{m+n}$. As $k[x,y]$ is a graded domain the order of Af is $r + m$, so $r + m \ge n + m$, giving $r \ge n$ and $\overline{A} = 0$, contrary to assumption. The case $\overline{A} = 0$ is symmetric.

So $\overline{A} \neq 0 \neq \overline{B}$, that is $r < n$ and $s < m$. The orders of Af and Bg are $r + m$ and $s + n$, both strictly less than $n + m$, so they must be equal, else $Af + Bg$ would have order less than $n + m$. Thus $r + m = s + n$ and the terms of that degree cancel:

$$A_r f_m = -B_s g_n$$

Now f_m and g_n have no common factor, since a common irreducible factor would be a linear form tangent to both curves. So $f_m \big| B_s$ and $g_n \big| A_r$, forcing $s \ge m$ and $r \ge n$ and contradicting $r < n$, $s < m$. Hence $(\overline{A}, \overline{B}) = 0$ and ψ is injective.

Conversely, suppose L is a common tangent and write $f_m = Lf'$, $g_n = Lg'$, so that $\deg f' = m - 1$ and $\deg g' = n - 1$. Then $\overline{g'} \neq 0$ in $k[x,y]/I^n$ and $\overline{f'} \neq 0$ in $k[x,y]/I^m$, and

$$\psi(\overline{g'}, \overline{-f'}) = \overline{g'f - f'g}$$

The two products $g'f$ and $f'g$ both have order $n+m-1$, and their forms of that degree are $g'f_m = g'Lf'$ and $f'g_n = f'Lg'$, which are equal. So they cancel, $g'f - f'g \in I^{n+m}$, and the image is zero on a nonzero element. Hence ψ is not injective, as we sought to show.

12.4 Computing Intersection Numbers

The proof of theorem 12.3.1 contains an algorithm, and in practice one rarely runs it to the end. Properties (7) and (8) do most of the work: (8) replaces a curve by any convenient representative of the same ideal, and (7) splits off factors. Property (6) then closes out any pair with no common tangent without further computation. The following example is worth working through slowly, because every step is one of those three.

Example 12.4.1: Intersection Number

Let

$$E = (x^2+y^2)^2 + 3x^2y - y^3 = x^4 + 2x^2y^2 + 3x^2y + y^4 - y^3$$
$$F = (x^2+y^2)^3 - 4x^2y^2 = x^6 + 3x^4y^2 + 3x^2y^4 - 4x^2y^2 + y^6$$

and $p = (0,0)$. Over the reals these are the three-leaf and four-leaf roses, $r = -\sin 3\theta$ and $r = |\sin 2\theta|$ in polar coordinates:

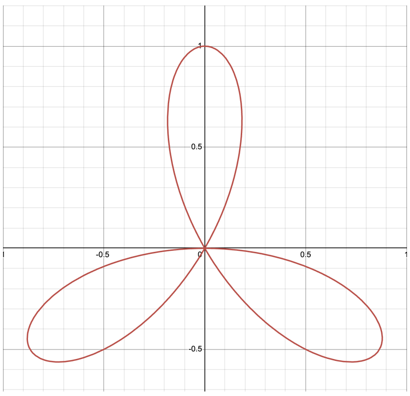


Figure 12.1: Three-Leaf Curve

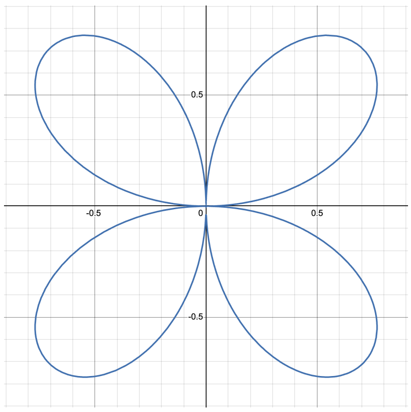


Figure 12.2: Four-Leaf Curve

Both pass through the origin, one with three branches and one with four, so the intersection number there should be substantial.

By (8) replace F with $F - (x^2+y^2)E$, which is

$$-4x^2y^2 - (x^2+y^2)(3x^2y - y^3) = y\big((x^2+y^2)(y^2-3x^2) - 4x^2y\big) = yG$$

so that by (7)

$$I(p, E\cap F) = I(p, E\cap yG) = I(p, E\cap y) + I(p, E\cap G)$$

For the second term, eliminate the quartic part of $G = -3x^4 - 2x^2y^2 - 4x^2y + y^4$ by replacing G with $G + 3E$, again legitimate by (8):

$$G + 3E = 4x^2y^2 + 5x^2y + 4y^4 - 3y^3 = y\big(5x^2 - 3y^2 + 4x^2y + 4y^3\big) = yH$$

Applying (7) once more,

$$I(p, E \cap F) = 2\, I(p, E \cap y) + I(p, E \cap H)$$

The first term is a one-variable computation: $E(x, 0) = x^4$, so by (8), (7) and (6),

$$I(p, E \cap y) = I(p, x^4 \cap y) = 4\, I(p, x \cap y) = 4$$

For the second, read off the tangent cones. The lowest form of E is $3x^2y - y^3 = y(\sqrt{3}x - y)(\sqrt{3}x + y)$, so $m_p(E) = 3$ with three distinct tangents; the lowest form of H is $5x^2 - 3y^2 = (\sqrt{5}x - \sqrt{3}y)(\sqrt{5}x + \sqrt{3}y)$, so $m_p(H) = 2$ with two more. No tangent of E is proportional to one of H, so (6) applies with equality and no further computation is needed:

$$I(p, E \cap H) = m_p(E)\, m_p(H) = 3 \cdot 2 = 6$$

Altogether

$$I(p, E \cap F) = 2 \cdot 4 + 6 = 14$$

Two consequences of theorem 12.3.1 are recorded here as further properties, continuing the numbering of the list. The first identifies the intersection number with the order function of definition 12.2.4 whenever one of the curves is nonsingular at p, which is the form chapter 13 uses. The second globalizes: the intersection numbers at all points assemble into a single dimension count.

Corollary 12.4.2: Intersection Number, Order, and the Global Count

Let f, g be affine plane curves. Then

9. If p is a nonsingular point of f, then $I(p, f \cap g) = \operatorname{ord}_p^f(g)$.
10. If f and g have no common component, so that $\mathcal{Z}(f) \cap \mathcal{Z}(g)$ is finite, then

$$\sum_p I(p, f \cap g) = \dim_k \frac{k[x, y]}{(f, g)}$$

***Proof*:**
(9). Since p is a nonsingular point of f, it lies on exactly one component of f and that component is simple by proposition 12.1.7; by property (3) we may replace f by that component and assume f irreducible. Then $\mathcal{O}_p(f)$ is a discrete valuation ring by corollary 12.2.3; let t be a uniformizer and $e = \operatorname{ord}_p^f(\overline{g})$, so that $\overline{g} = ut^e$ for a unit u. Hence $(\overline{g}) = (t^e)$ and

$$\frac{\mathcal{O}_p(f)}{(\overline{g})} = \frac{\mathcal{O}_p(f)}{(t^e)}$$

has $1, t, \ldots, t^{e-1}$ as a k-basis, the residue field being k because k is algebraically closed, so its dimension is e. Finally $\mathcal{O}_p(f) \cong \mathcal{O}_p(\mathbb{A}_k^2)/(f)$ by the third exercise of exercise set 12.4.1, so

$$\frac{\mathcal{O}_p(f)}{(\overline{g})} \cong \frac{\mathcal{O}_p(\mathbb{A}_k^2)}{(f,g)}$$

whose dimension is $I(p, f \cap g)$ by theorem 12.3.1.

(10). With no common component, $\mathcal{Z}(f,g)$ is finite, so corollary 12.2.6 applied to $I = (f,g)$ gives

$$\dim_k \frac{k[x,y]}{(f,g)} = \sum_i \dim_k \frac{\mathcal{O}_{p_i}(\mathbb{A}_k^2)}{(f,g)\mathcal{O}_{p_i}(\mathbb{A}_k^2)} = \sum_i I(p_i, f \cap g)$$

and the points not in $\mathcal{Z}(f,g)$ contribute 0 by property (2), so the sum over all p is the same, completing the proof.

The second statement is the bridge to chapter 13: it converts a sum of local invariants into the dimension of a single ring, and Bézout's theorem is what happens when that ring is computed for two projective curves by a graded argument.

Exercise 12.4.1

1. Let $p = (0, \ldots, 0) \in \mathbb{A}_k^n$. Show that the maximal ideal $\mathfrak{m} \subseteq \mathcal{O}_p(\mathbb{A}_k^n)$ is generated by $x_1, \ldots, x_n$.

2. Let $I = (x,y) \subseteq k[x,y]$. Show that the monomials of degree less than n form a k-basis of $k[x,y]/I^n$, and deduce that
$$\dim_k \left(\frac{k[x,y]}{I^n}\right) = 1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

3. Let $V \subseteq \mathbb{A}_k^n$ be a variety with ideal $\mathcal{I}(V) \subseteq R = k[x_1, \ldots, x_n]$, let $p \in V$, and let $J \subseteq R$ be an ideal containing $\mathcal{I}(V)$, with image $\overline{J}$ in $k[V]$. Show that
$$\frac{\mathcal{O}_p(\mathbb{A}_k^n)}{J\,\mathcal{O}_p(\mathbb{A}_k^n)} \cong \frac{\mathcal{O}_p(V)}{\overline{J}\,\mathcal{O}_p(V)}$$
and deduce the case $J = \mathcal{I}(V)$, namely $\mathcal{O}_p(\mathbb{A}_k^n)/\mathcal{I}(V)\mathcal{O}_p(\mathbb{A}_k^n) \cong \mathcal{O}_p(V)$.

4. Compute $m_p(f)$ and the tangent lines at the origin for $f = y^2 - x^3$, $f = y^2 - x^2 - x^3$, $f = x^4 - x^2y^2 + y^4$ and $f = (x^2+y^2)^2 + 3x^2y - y^3$. Which of the four have an ordinary singular point there?

5. Compute $I(p, f \cap g)$ at the origin for $f = y - x^2,\ g = y$; then for $f = y^2 - x^3,\ g = y$; then for $f = y^2 - x^3,\ g = x$. Check each against property (6).

6. Show that $I(p, f \cap g) = 1$ if and only if p is a nonsingular point of both curves and their tangent lines there are distinct.

7. Let f be an irreducible plane curve and p a nonsingular point of it. Show directly from definition 12.2.4 that $\operatorname{ord}_p^f(gh) = \operatorname{ord}_p^f(g) + \operatorname{ord}_p^f(h)$ and that $\operatorname{ord}_p^f(g+h) \geq \min(\operatorname{ord}_p^f(g), \operatorname{ord}_p^f(h))$, with an example where the second is strict.

13

Bézout's Theorem and Max Noether's Theorem

With the intersection number in hand the counting theorem is within reach. The proof is a dimension count in a graded ring: two curves with no common component generate an ideal whose quotient has dimension mn in every large degree, and the sum of the intersection numbers is exactly that dimension. Everything the theorem says about individual points is a consequence of that one budget being finite.

Max Noether's theorem then answers the converse question. Given that a curve passes through the points where two others meet, when is it an algebraic combination of them? The answer is local: a condition at each intersection point suffices, and checking it is cheap wherever the intersection is transversal. The chapter closes with two theorems of classical projective geometry, Pascal's and Pappus's, which fall out of a single application of that criterion to a cubic that splits into three lines.

13.1 Bézout's Theorem

Projective space was introduced so that intersections would stop disappearing, and for plane curves it delivers completely: two curves with no common component always meet, and the number of points, weighted by the intersection numbers of chapter 12, is exactly the product of the degrees.

The result is special to curves in $\mathbb{P}^2_k$, not a general feature of projective varieties. In $\mathbb{P}^3_k$ the point $\mathcal{Z}(x, y, w)$ and the plane $\mathcal{Z}(z)$ do not meet at all, and there is no weighting that would make them. What makes the plane-curve case work is that two curves in $\mathbb{P}^2_k$ have complementary dimension, so theorem 10.6.3 already guarantees they meet. Bézout's theorem sharpens that from "at least one point" to an exact count.

Theorem 13.1.1: Bézout's Theorem

Let f and g be projective plane curves of degrees m and n with no common component. Then $\mathcal{Z}(f) \cap \mathcal{Z}(g)$ is finite and

$$\sum_{p \in \mathbb{P}^2_k} I(p, f \cap g) = mn$$

***Proof*:**
Setting up. Since f and g have no common component, $\mathcal{Z}(f) \cap \mathcal{Z}(g)$ is a proper closed subset of the curve $\mathcal{Z}(f)$, hence has dimension 0 and is finite. The field k is infinite, so a linear change of coordinates moves the line at infinity off that finite set; assume from now on that no point of $\mathcal{Z}(f) \cap \mathcal{Z}(g)$ lies on $\mathcal{Z}(z)$. Write $f_* = f(x, y, 1)$ and $g_* = g(x, y, 1)$ for the dehomogenizations. Every intersection point now lies in the standard affine chart, where the intersection numbers agree with the affine ones, so property (10) of corollary 12.4.2 gives

$$\sum_p I(p, f \cap g) = \sum_p I(p, f_* \cap g_*) = \dim_k \frac{k[x, y]}{(f_*, g_*)}$$

Abbreviate

$$\Gamma_* = \frac{k[x, y]}{(f_*, g_*)} \qquad \Gamma = \frac{k[x, y, z]}{(f, g)} \qquad R = k[x, y, z]$$

Here R and Γ are graded, and R_d, Γ_d denote their degree-d pieces. The theorem follows from the two claims

$$\dim_k \Gamma_* \overset{1}{=} \dim_k \Gamma_d \overset{2}{=} mn \qquad (d \geq m + n)$$

Claim 2: $\dim_k \Gamma_d = mn$ *for* $d \geq m + n$. Define

$$\begin{aligned} \psi\colon R &\to R \times R & \psi(c) &= (gc, -fc) \\ \varphi\colon R \times R &\to R & \varphi(a, b) &= af + bg \\ \pi\colon R &\to \Gamma & \pi(h) &= \overline{h} \end{aligned}$$

The sequence

$$0 \to R \xrightarrow{\psi} R \times R \xrightarrow{\varphi} R \xrightarrow{\pi} \Gamma \to 0$$

is exact. Indeed ψ is injective because R is a domain; $\varphi\psi = 0$ by inspection; if $\varphi(a, b) = 0$ then $af = -bg$, and as f, g have no common factor in the unique factorization domain R we get $g \big| a$, say $a = gc$, whence $b = -fc$ and $(a, b) = \psi(c)$; finally $\operatorname{im}\varphi = (f, g) = \ker \pi$.

Everything is homogeneous, so the sequence restricts in each degree. Since $\deg f = m$ and $\deg g = n$,

$$0 \to R_{d-m-n} \xrightarrow{\psi} R_{d-m} \times R_{d-n} \xrightarrow{\varphi} R_d \xrightarrow{\pi} \Gamma_d \to 0$$

is exact, and for $d \geq m + n$ every term is a finite-dimensional space of forms, with $\dim_k R_j = \binom{j+2}{2} = \frac{(j+1)(j+2)}{2}$ for $j \geq 0$. Alternating dimensions along an exact sequence sum to zero, so

$$\dim_k \Gamma_d = \dim_k R_d - \dim_k R_{d-m} - \dim_k R_{d-n} + \dim_k R_{d-m-n}$$

Writing $P(j) = (j + 1)(j + 2)$, this is $\frac{1}{2}\big[P(d) - P(d - m) - P(d - n) + P(d - m - n)\big]$. The linear and constant parts of P cancel in that combination, leaving only the quadratic part:

$$d^2 - (d - m)^2 - (d - n)^2 + (d - m - n)^2 = 2mn$$

so $\dim_k \Gamma_d = mn$, which is Claim 2.

Multiplication by z is injective on Γ. This is the technical heart of Claim 1, and it is also used again in theorem 13.3.2. Let $\alpha\colon \Gamma \to \Gamma$ be $\alpha(\overline{h}) = \overline{zh}$ and suppose $\overline{zh} = \overline{0}$, so $zh = af + bg$ for some $a, b \in R$. For a form $u \in R$ write $u_0 = u(x, y, 0)$ for its restriction to the line at infinity. Because $\mathcal{Z}(f) \cap \mathcal{Z}(g) \cap \mathcal{Z}(z) = \emptyset$, the forms f_0 and g_0 have no common zero in $\mathbb{P}^1_k$ and so are relatively prime in $k[x, y]$. Setting $z = 0$ in $zh = af + bg$ gives

$$0 = a_0 f_0 + b_0 g_0$$

so $a_0 f_0 = -b_0 g_0$, and coprimality yields $c \in k[x, y]$ with

$$b_0 = f_0 c \qquad a_0 = -g_0 c$$

Now put $a_1 = a + cg$ and $b_1 = b - cf$. Then $a_1 f + b_1 g = af + bg + cgf - cfg = zh$, while

$$(a_1)_0 = -g_0 c + c g_0 = 0 \qquad (b_1)_0 = f_0 c - c f_0 = 0$$

so both are divisible by z, say $a_1 = za'$ and $b_1 = zb'$. Then $z(a'f + b'g) = zh$, and cancelling z in the domain R gives $h = a'f + b'g$, that is $\overline{h} = \overline{0}$. So α is injective.

Claim 1: $\dim_k \Gamma_* = \dim_k \Gamma_d$ *for* $d \geq m+n$. By the previous paragraph α restricts to an injection $\Gamma_d \hookrightarrow \Gamma_{d+1}$, and both have dimension mn by Claim 2, so it is an isomorphism. Consequently, if $\beta_1, \ldots, \beta_{mn} \in R_d$ have residues forming a basis of Γ_d, then for every $r \geq 0$ the residues of $z^r \beta_1, \ldots, z^r \beta_{mn}$ form a basis of Γ_{d+r}.

Let $b_i \in \Gamma_*$ be the residue of $(\beta_i)_* = \beta_i(x, y, 1)$. These span: given $\overline{h} \in \Gamma_*$ with $h \in k[x, y]$, let h^* be its homogenization, of some degree e, and choose r with $d + r \geq e$. Then $z^{d+r-e} h^*$ is a form of degree $d + r$, so

$$z^{d+r-e} h^* = \sum_{i=1}^{mn} \lambda_i z^r \beta_i + Af + Bg$$

for scalars λ_i and forms A, B. Dehomogenizing, that is setting $z = 1$, gives

$$h = \sum_i \lambda_i (\beta_i)_* + A_* f_* + B_* g_*$$

so $\overline{h} = \sum_i \lambda_i b_i$.

They are independent: suppose $\sum_i \lambda_i b_i = 0$, so that $\sum_i \lambda_i (\beta_i)_* = Af_* + Bg_*$ for some $A, B \in k[x, y]$. Homogenizing and clearing the powers of z that appear gives $r, s, t \geq 0$ with

$$z^r \sum_i \lambda_i \beta_i = z^s A^* f + z^t B^* g$$

so $\sum_i \lambda_i \overline{z^r \beta_i} = \overline{0}$ in Γ_{d+r}. As those residues form a basis, every $\lambda_i = 0$.

So $b_1, \ldots, b_{mn}$ is a basis of Γ_* and $\dim_k \Gamma_* = mn$, as we sought to show.

Example 13.1.2: Bézout in Practice

Every count below is over an algebraically closed field and in $\mathbb{P}^2_k$, where alone the statement holds.

1. *Two conics.* Degrees 2 and 2, so the intersection has total weight 4. For $\mathcal{Z}(x^2+y^2-z^2)$ and $\mathcal{Z}(x^2-yz)$ a direct substitution gives four distinct points, each counted once. Two general conics meet in four points.

2. *Tangency absorbs the count.* For the circle $\mathcal{Z}(x^2+y^2-z^2)$ and the line $\mathcal{Z}(y-z)$, substituting $y=z$ gives $x^2=0$, a double root: the line is tangent at $[0:1:1]$, and $I(p, C\cap L)=2$. The total is still $2\cdot 1=2$, all concentrated at one point.

3. *Where affine intuition fails.* The two parabolas $\mathcal{Z}(y-x^2)$ and $\mathcal{Z}(y-x^2-1)$ do not meet in $\mathbb{A}^2_k$ at all. Homogenizing gives $\mathcal{Z}(yz-x^2)$ and $\mathcal{Z}(yz-x^2-z^2)$, and subtracting, $z^2=0$: the two meet only where $z=0$, at $[0:1:0]$, with $I=4$. The four intersections predicted by $2\cdot 2$ are all at infinity, which is what projective space was introduced to see.

4. *A line and a cubic.* Degrees 1 and 3 give three points. For the cubic $\mathcal{Z}(y^2z-x^3+xz^2)$ and the line $\mathcal{Z}(z)$, substituting $z=0$ gives $x^3=0$: the line at infinity meets the cubic only at $[0:1:0]$, with multiplicity 3. That point is an inflection, which is why theorem 18.4.4 can put a genus-one curve in Weierstrass form with its base point there.

13.2 Consequences and Bounds on Multiplicity

Bézout's theorem is a budget: mn is all the intersection there is, and every local contribution is drawn against it. Most of its corollaries are that observation combined with the lower bound $I(p, f\cap g)\geq m_p(f)m_p(g)$ from property (6).

Corollary 13.2.1: Degree and Multiplicity

Let f, g be projective plane curves with no common component. Then

$$\sum_p m_p(f)\, m_p(g) \leq \deg(f)\deg(g)$$

***Proof*:**
Property (6) of the intersection number gives $I(p, f\cap g)\geq m_p(f)m_p(g)$ at every p, and summing over p against theorem 13.1.1 bounds the total by $\deg(f)\deg(g)$, completing the proof.

Corollary 13.2.2: Intersection and Simple Points

Let f, g be projective plane curves of degrees m and n with no common component. If $\mathcal{Z}(f)$ and $\mathcal{Z}(g)$ meet in mn distinct points, then every one of those points is a nonsingular point of both curves, and the two curves are transversal there.

***Proof*:**
The mn points each contribute at least 1 to a total of mn by theorem 13.1.1, so each contributes exactly 1. Property (6) then forces $m_p(f)m_p(g) \leq 1$, so both multiplicities are 1 and p is nonsingular on both by definition 12.1.5. Equality in (6) additionally says the curves share no tangent at p, which for two nonsingular points is transversality, completing the proof.

Corollary 13.2.3: Intersection and Common Component

If two projective plane curves of degrees m and n have more than mn points in common, they have a common component.

***Proof*:**
Were there no common component, theorem 13.1.1 would bound the number of common points by $\sum_p I(p, f \cap g) = mn$, since each contributes at least 1 by property (2), completing the proof.

The next bound goes the other way: it limits how singular a single irreducible curve can be. Both the crude and the sharp form are worth having, and the crude one is what licenses the argument for the sharp one.

Corollary 13.2.4: Bounding Multiplicity

Let f be an irreducible projective plane curve of degree n. Then f has finitely many singular points and

$$\sum_p \frac{m_p(f)\big(m_p(f)-1\big)}{2} \leq \frac{n(n-1)}{2}$$

***Proof*:**
Not every partial derivative of f vanishes identically. In characteristic zero this is because f is nonconstant. In characteristic $\rho > 0$, if all three vanished then every exponent appearing in f would be divisible by ρ, so $f \in k[x^\rho, y^\rho, z^\rho]$; as k is algebraically closed its elements are ρ-th powers, and the Frobenius is a ring homomorphism, so $f = h^\rho$ for some h, contradicting irreducibility. Let g be a nonzero partial derivative of f, a form of degree $n-1$.

At every point p, $m_p(g) \geq m_p(f) - 1$. Change coordinates so that $p = [0:0:1]$ and work in the chart $z = 1$, writing $f_* = f(x, y, 1)$ and $m = m_p(f)$. A linear change of coordinates replaces each partial derivative by a linear combination of the three partials, and multiplicity of a sum is at least the minimum of the multiplicities, so it is enough to check the bound for all three partials of the transformed form. Dehomogenization commutes with ∂_x and ∂_y, and differentiating drops the degree of each homogeneous piece of f_* by one, so $m_p(\partial_x f_*)$ and $m_p(\partial_y f_*)$ are both at least $m-1$. For the third, Euler's relation (exercise set 8.5.1(5)) $nf = x\partial_x f + y\partial_y f + z\partial_z f$ dehomogenizes to

$$(\partial_z f)_* = nf_* - x\,\partial_x f_* - y\,\partial_y f_*$$

Each term on the right has multiplicity at least m at the origin, the last two because a factor of x or y adds one to a multiplicity of at least $m-1$. So $m_p((\partial_z f)_*) \geq m$, which is more than required.

Since f is irreducible of degree n and g is a nonzero form of degree $n - 1 < n$, they have no common component. Corollary 13.2.1 therefore gives

$$\sum_p m_p(f)\big(m_p(f)-1\big) \leq \sum_p m_p(f)\, m_p(g) \leq n(n-1)$$

which is the inequality after dividing by 2. In particular only finitely many p have $m_p(f) \geq 2$, so the singular points are finite in number, as we sought to show.

The sharp form replaces n by $n-1$ on the right, and its proof is the standard count of how many conditions singularities impose on curves of degree $n-1$.

Corollary 13.2.5: Sharp Bound on Multiplicity

Let f be an irreducible projective plane curve of degree n. Then

$$\sum_p \frac{m_p(f)\big(m_p(f)-1\big)}{2} \leq \frac{(n-1)(n-2)}{2}$$

***Proof*:**
Write $m_p = m_p(f)$ and $M = \sum_p \binom{m_p}{2}$, a finite sum by corollary 13.2.4. The forms of degree $n-1$ in three variables constitute a projective space of dimension $\binom{n+1}{2} - 1 = \frac{(n-1)(n+2)}{2}$. Requiring a curve g of degree $n-1$ to satisfy $m_p(g) \geq m_p - 1$ is the vanishing of all partial derivatives of order less than $m_p - 1$ at p, which is $\binom{m_p}{2}$ linear conditions, and requiring g to pass through a further point is one more. Set

$$r = \frac{(n-1)(n+2)}{2} - M$$

This is non-negative: corollary 13.2.4 gives $M \leq \frac{n(n-1)}{2}$, and $\frac{n(n-1)}{2} \leq \frac{(n-1)(n+2)}{2}$. So r further conditions may be imposed, and since f has infinitely many points while only finitely many are singular, nonsingular points $q_1, \ldots, q_r \in \mathcal{Z}(f)$ may be chosen. The $M + r$ conditions cannot exhaust a projective space of dimension $M + r$, so a curve g of degree $n-1$ exists with $m_p(g) \geq m_p - 1$ at every singular point of f and $m_{q_i}(g) \geq 1$ for each i.

As before f is irreducible of degree n and g is a nonzero form of degree $n-1$, so they share no component and corollary 13.2.1 applies:

$$n(n-1) \;\geq\; \sum_p m_p(m_p-1) \;+\; r \;=\; 2M + \frac{(n-1)(n+2)}{2} - M$$

the r appearing because each q_i contributes $m_{q_i}(f)m_{q_i}(g) \geq 1$ and is not among the singular points already counted. Rearranging,

$$M \leq n(n-1) - \frac{(n-1)(n+2)}{2} = (n-1) \cdot \frac{2n-n-2}{2} = \frac{(n-1)(n-2)}{2}$$

as we sought to show.

The sharp bound is attained: the curve $\mathcal{Z}(x^n + y^{n-1}z)$ is irreducible of degree n and has multiplicity $n-1$ at $[0:0:1]$, since dehomogenizing there leaves $x^n + y^{n-1}$ with lowest form y^{n-1}. That single point already contributes $\binom{n-1}{2}$, so no other singularity is available. The quantity $\frac{(n-1)(n-2)}{2}$ is the *arithmetic genus* of a plane curve of degree n, and chapter 14 will meet it again.

13.3 Max Noether's Fundamental Theorem

Bézout's theorem counts the intersection of two curves; the natural next question is what that intersection determines. The data of an intersection is a set of points together with a weight at each, and such data forms a group under addition.

Definition 13.3.1: Zero-cycle

A *zero-cycle* on $\mathbb{P}^2_k$ is a formal sum $\sum_p n_p\, p$ over points $p \in \mathbb{P}^2_k$ with $n_p \in \mathbb{Z}$ and all but finitely many $n_p = 0$. Its *degree* is $\sum_p n_p$.

Adding coefficientwise makes the zero-cycles an abelian group, and writing $\sum_p n_p p \le \sum_p m_p p$ when $n_p \le m_p$ for every p orders it. For projective plane curves f, g with no common component, the *intersection cycle* is

$$f \bullet g = \sum_{p \in \mathbb{P}^2_k} I(p, f \cap g)\, p$$

which is a zero-cycle of degree $\deg(f)\deg(g)$ by theorem 13.1.1. The properties of the intersection number transfer directly:

1. $f \bullet g = g \bullet f$;
2. $f \bullet (gh) = f \bullet g + f \bullet h$;
3. $f \bullet (g + af) = f \bullet g$ for every form a of degree $\deg(g) - \deg(f)$.

It is tempting to hope that every effective zero-cycle, meaning one with all $n_p \ge 0$, arises as an intersection cycle. It does not. Take three points of $\mathbb{P}^2_k$ in general position and the cycle $p_1 + p_2 + p_3$, of degree 3. Realizing it as $f \bullet g$ would need $\deg(f)\deg(g) = 3$, so one of the curves is a line, and that line would have to contain all three points, which they are not chosen to admit. The obstruction is exactly the phenomenon that chapter 14 organizes into the divisor class group: which formal sums of points are cut out by equations is a question with a non-trivial answer, and the answer is an invariant of the curve.

The question this section settles is the relative version. Suppose f, g, h are curves with $g \bullet f \le h \bullet f$, so that h meets f at least as heavily as g does at every point. Is the difference itself an intersection cycle, that is, is there a curve b with

$$b \bullet f = h \bullet f - g \bullet f$$

Comparing degrees, such a b must have degree $\deg(h) - \deg(g)$. And it is enough to find forms a, b with

$$h = af + bg$$

for then property (3) followed by property (2) gives $h \bullet f = (bg) \bullet f = b \bullet f + g \bullet f$, which rearranges to the display. So the geometric question reduces to an algebraic one: when does h lie in the ideal generated by f and g, with the degrees working out? Max Noether's theorem answers it, and the answer is local.

Theorem 13.3.2: Max Noether's Fundamental Theorem

Let f, g, h be projective plane curves with f and g having no common component. Then there exist forms a, b of degrees $\deg(h) - \deg(f)$ and $\deg(h) - \deg(g)$ with

$$h = af + bg$$

if and only if for every $p \in \mathcal{Z}(f) \cap \mathcal{Z}(g)$ the local condition

$$h_* \in (f_*, g_*)\,\mathcal{O}_p(\mathbb{P}^2_k)$$

holds, where the subscript denotes dehomogenization in an affine chart containing p.

The local condition is called *Noether's condition* at p. That a global identity is equivalent to a finite list of local ones makes this a *local-to-global* theorem, and it is the reason the corollaries below are as easy as they are: each is a way of certifying the local condition cheaply.

***Proof*:**
One direction is immediate: if $h = af + bg$, then dehomogenizing at any p gives $h_* = a_* f_* + b_* g_*$, so h_* lies in $(f_*, g_*)\mathcal{O}_p(\mathbb{P}^2_k)$.

For the converse, change coordinates so that $\mathcal{Z}(f) \cap \mathcal{Z}(g) \cap \mathcal{Z}(z) = \emptyset$, which is possible because the intersection is finite, and dehomogenize in the chart $z = 1$. Noether's condition says the residue of h_* in $\mathcal{O}_p(\mathbb{A}^2_k)/(f_*, g_*)$ vanishes for every $p \in \mathcal{Z}(f_*, g_*)$. By lemma 12.2.5 the ring $k[x, y]/(f_*, g_*)$ is the product of those local quotients, so the residue of h_* there vanishes too:

$$h_* = \alpha f_* + \beta g_* \qquad \alpha, \beta \in k[x, y]$$

Homogenizing and clearing the powers of z introduced gives $r \geq 0$ and forms A, B with

$$z^r h = Af + Bg$$

Multiplication by z on $\Gamma = k[x, y, z]/(f, g)$ is injective, as established inside the proof of theorem 13.1.1, so cancelling z one factor at a time from $\overline{z^r h} = \overline{0}$ leaves $\overline{h} = \overline{0}$, that is $h = a'f + b'g$ for some forms a', b', not necessarily homogeneous.

Finally, extract the right graded pieces. Write $a' = \sum_i a'_i$ and $b' = \sum_i b'_i$ with a'_i, b'_i homogeneous of degree i. Comparing homogeneous components of degree $\deg h$ in $h = a'f + b'g$, and using that h is itself homogeneous, gives

$$h = a'_s f + b'_t g \qquad s = \deg(h) - \deg(f), \quad t = \deg(h) - \deg(g)$$

since those are the only pieces contributing in that degree. Taking $a = a'_s$ and $b = b'_t$ completes the proof.

Verifying Noether's condition directly is unpleasant. The following criteria cover every case that arises in practice.

Proposition 13.3.3: Noether's Criterion

Let f, g, h be projective plane curves and $p \in \mathcal{Z}(f) \cap \mathcal{Z}(g)$. Each of the following implies Noether's condition at p:

1. f and g meet transversally at p, and $p \in \mathcal{Z}(h)$;
2. p is a nonsingular point of f, and $I(p, h \cap f) \geq I(p, g \cap f)$;
3. f and g have no tangent line in common at p, and $m_p(h) \geq m_p(f) + m_p(g) - 1$.

***Proof*:**
(2). Since p is a nonsingular point of f, property (9) of corollary 12.4.2 converts both intersection numbers into orders, so the hypothesis reads $\operatorname{ord}_p^f(h_*) \geq \operatorname{ord}_p^f(g_*)$. In the discrete valuation ring $\mathcal{O}_p(f)$ that says

$$\overline{h}_* \in (\overline{g}_*) \subseteq \mathcal{O}_p(f)$$

Now $\mathcal{O}_p(f) \cong \mathcal{O}_p(\mathbb{A}_k^2)/(f_*)$ by the third exercise of exercise set 12.4.1, so $\mathcal{O}_p(f)/(\overline{g}_*) \cong \mathcal{O}_p(\mathbb{A}_k^2)/(f_*, g_*)$ and the residue of h_* there is zero, which is Noether's condition.

(3). Take $p = [0 : 0 : 1]$ and write $I = (x, y)$, $m = m_p(f)$, $n = m_p(g)$. The hypothesis $m_p(h) \geq m + n - 1$ says $h_* \in I^{m+n-1}$, and lemma 12.3.2(1) says $I^t \subseteq (f_*, g_*)\mathcal{O}_p(\mathbb{A}_k^2)$ for every $t \geq m + n - 1$ precisely because f and g have no common tangent. Hence $h_* \in (f_*, g_*)\mathcal{O}_p(\mathbb{A}_k^2)$.

(1). Transversality makes $m_p(f) = m_p(g) = 1$ with distinct tangents, so (3) applies with the requirement $m_p(h) \geq 1 + 1 - 1 = 1$, which is exactly $p \in \mathcal{Z}(h)$, completing the proof.

Corollary 13.3.4: Sufficient Condition For Max Noether's Theorem

Let f, g, h be projective plane curves with f and g having no common component, and suppose either

1. $\mathcal{Z}(f)$ and $\mathcal{Z}(g)$ meet in $\deg(f)\deg(g)$ distinct points and h passes through all of them, or
2. every point of $\mathcal{Z}(f) \cap \mathcal{Z}(g)$ is a nonsingular point of f, and $h \bullet f \geq g \bullet f$.

Then there is a curve b of degree $\deg(h) - \deg(g)$ with

$$b \bullet f = h \bullet f - g \bullet f$$

***Proof*:**
In case (1), corollary 13.2.2 says the $\deg(f)\deg(g)$ points are transversal intersections, so proposition 13.3.3(1) supplies Noether's condition at each. In case (2), the hypothesis $h \bullet f \geq g \bullet f$ is exactly $I(p, h \cap f) \geq I(p, g \cap f)$ at every p, so proposition 13.3.3(2) applies. Either way theorem 13.3.2 produces $h = af + bg$ with b of degree $\deg(h) - \deg(g)$, and the computation preceding that theorem turns the identity into the stated equation of cycles, completing the proof.

13.4 Application: Pascal's and Pappus's Theorems

Max Noether's theorem is a statement about ideals, and its geometric consequences are about incidence. Two of them are theorems of classical projective geometry, one from the fourth century and one from the seventeenth, and both come out of the same three-line computation: a cubic that splits into three lines, a conic, and the observation that Noether's condition costs nothing at a transversal intersection.

The shared setup is this. Let Q be a conic and let $p_1, \dots, p_6$ be six distinct points on $\mathcal{Z}(Q)$, thought of as the vertices of a hexagon in that cyclic order. Its six sides split into two alternating triples,

$$L_1 = \overline{p_1p_2}, \quad L_2 = \overline{p_3p_4}, \quad L_3 = \overline{p_5p_6} \qquad M_1 = \overline{p_2p_3}, \quad M_2 = \overline{p_4p_5}, \quad M_3 = \overline{p_6p_1}$$

and the three pairs of *opposite* sides are (L_1, M_2), (M_1, L_3) and (L_2, M_3). Write $C = L_1L_2L_3$ and $C' = M_1M_2M_3$, both cubics.

Lemma 13.4.1: The Hexagon Identity

With the notation above, suppose that no side of the hexagon meets $\mathcal{Z}(Q)$ outside the two vertices it joins. Then there are a constant $\lambda \in k$ and a linear form ℓ, not identically zero, with

$$M_1M_2M_3 = \lambda\, L_1L_2L_3 + \ell\, Q$$

***Proof*:**
Apply theorem 13.3.2 to $f = C$, $g = Q$ and $h = C'$. The curves C and Q have no common component, since each L_i meets $\mathcal{Z}(Q)$ in the two points p_{2i-1}, p_{2i} and so is not contained in it. By hypothesis $\mathcal{Z}(C) \cap \mathcal{Z}(Q)$ is exactly $\{p_1, \dots, p_6\}$, which is $\deg(C)\deg(Q) = 3 \cdot 2 = 6$ distinct points, so corollary 13.2.2 makes every one of them a transversal intersection. Each p_i also lies on C': the vertex p_j lies on M_1, M_2 or M_3 according to its index. So proposition 13.3.3(1) supplies Noether's condition at all six points, and theorem 13.3.2 gives

$$C' = aC + bQ$$

with $\deg a = 3 - 3 = 0$ and $\deg b = 3 - 2 = 1$; write $\lambda = a$ and $\ell = b$.

It remains to rule out $\ell = 0$, which would force $M_1M_2M_3 = \lambda L_1L_2L_3$ and hence, by unique factorization, make each M_i proportional to some L_j. That cannot happen: L_j meets $\mathcal{Z}(Q)$ in $\{p_{2j-1}, p_{2j}\}$ and M_i meets it in the corresponding pair for M_i, and the three pairs $\{p_1, p_2\}, \{p_3, p_4\}, \{p_5, p_6\}$ are disjoint from the three pairs $\{p_2, p_3\}, \{p_4, p_5\}, \{p_6, p_1\}$ as unordered pairs, the six vertices being distinct, as we sought to show.

Theorem 13.4.2: Pascal's Theorem

Let a hexagon with distinct vertices be inscribed in an irreducible conic. Then the three points in which the pairs of opposite sides meet are collinear.

***Proof*:**
Let Q be the conic and adopt the notation above. The hypothesis of lemma 13.4.1 holds: a line meets the irreducible conic $\mathcal{Z}(Q)$ in at most two points by theorem 13.1.1, since it is not a component of it, so each side meets $\mathcal{Z}(Q)$ exactly in its two vertices. Let λ and ℓ be as in that lemma.

Let r_1 be the point $L_1 \cap M_2$, the meeting of a pair of opposite sides, and define r_2 and r_3 from the pairs (M_1, L_3) and (L_2, M_3). Evaluating the identity of lemma 13.4.1 at r_1: the left side vanishes because $M_2(r_1) = 0$, and the term $\lambda L_1 L_2 L_3$ vanishes because $L_1(r_1) = 0$. Hence

$$\ell(r_1)\, Q(r_1) = 0$$

Now $Q(r_1) \neq 0$. For r_1 lies on L_1, whose only points on $\mathcal{Z}(Q)$ are p_1 and p_2, and on M_2, whose only points on $\mathcal{Z}(Q)$ are p_4 and p_5; were r_1 on $\mathcal{Z}(Q)$ it would belong to both pairs, which are disjoint. Therefore $\ell(r_1) = 0$, and the same computation gives $\ell(r_2) = \ell(r_3) = 0$. As ℓ is a nonzero linear form, $\mathcal{Z}(\ell)$ is a line containing all three points, as we sought to show.

Pappus's theorem is the case in which the conic degenerates into a pair of lines. The hexagon then has its vertices alternating between the two lines, and the argument runs unchanged.

Theorem 13.4.3: Pappus's Theorem

Let L and L' be distinct lines, and let p_1, p_2, p_3 be distinct points of $\mathcal{Z}(L)$ and q_1, q_2, q_3 distinct points of $\mathcal{Z}(L')$, none of the six equal to the point $\mathcal{Z}(L) \cap \mathcal{Z}(L')$. Put

$$r_1 = \overline{p_1 q_2} \cap \overline{q_1 p_2}, \qquad r_2 = \overline{q_2 p_3} \cap \overline{p_2 q_3}, \qquad r_3 = \overline{p_3 q_1} \cap \overline{q_3 p_1}$$

Then r_1, r_2, r_3 are collinear.

***Proof*:**
Take $Q = LL'$, a conic, and the hexagon with vertices $p_1, q_2, p_3, q_1, p_2, q_3$ in that cyclic order, all six lying on $\mathcal{Z}(Q)$ and all distinct. Its alternating side-triples are

$$\overline{p_1 q_2},\ \overline{p_3 q_1},\ \overline{p_2 q_3} \qquad \text{and} \qquad \overline{q_2 p_3},\ \overline{q_1 p_2},\ \overline{q_3 p_1}$$

and the three r_i are the intersections of the opposite pairs.

The hypothesis of lemma 13.4.1 holds. A side such as $\overline{p_1 q_2}$ joins a point of $\mathcal{Z}(L)$ to a point of $\mathcal{Z}(L')$, and it is equal to neither line, since $q_2 \notin \mathcal{Z}(L)$ and $p_1 \notin \mathcal{Z}(L')$ by the hypothesis on the six points. So it meets $\mathcal{Z}(L)$ only at p_1 and $\mathcal{Z}(L')$ only at q_2, and therefore meets $\mathcal{Z}(Q)$ exactly in its two vertices. Let λ and ℓ be as in the lemma.

Evaluating at r_1 exactly as in theorem 13.4.2 gives $\ell(r_1)Q(r_1) = 0$, so it remains to check $Q(r_1) \neq 0$, that is $r_1 \notin \mathcal{Z}(L) \cup \mathcal{Z}(L')$. Suppose $r_1 \in \mathcal{Z}(L)$. Then r_1 lies on $\overline{p_1 q_2}$, which meets $\mathcal{Z}(L)$ only at p_1, so $r_1 = p_1$; and r_1 lies on $\overline{q_1 p_2}$, which meets $\mathcal{Z}(L)$ only at p_2, so $r_1 = p_2$. These contradict $p_1 \neq p_2$. The case $r_1 \in \mathcal{Z}(L')$ is symmetric, using $q_2 \neq q_1$. Hence $\ell(r_1) = 0$, likewise for r_2 and r_3, and $\mathcal{Z}(\ell)$ is the required line, as we sought to show.

13.5 The Dual Curve and Plücker's Formulas

A nonsingular point of a plane curve has a tangent line, and a line in $\mathbb{P}^2_k$ is a point of the dual projective space. So a curve determines a second curve, and the numerical relations between the two are the oldest quantitative statements in the subject.

Definition 13.5.1: Dual Space and Dual Curve

The *dual projective space* $(\mathbb{P}^2_k)^\vee$ is the set of lines in $\mathbb{P}^2_k$, itself a $\mathbb{P}^2_k$ by sending $\mathcal{Z}(a_0x_0 + a_1x_1 + a_2x_2)$ to $[a_0 : a_1 : a_2]$. For an irreducible plane curve C, the *dual curve* $C^\vee \subseteq (\mathbb{P}^2_k)^\vee$ is the closure of the set of tangent lines to C at its nonsingular points. The *class* of C is $\deg C^\vee$.

By corollary 13.2.4 the nonsingular points of C are all but finitely many, so $C^\vee$ is the closure of the image of a map defined on a dense open set: the *Gauss map*

$$\gamma\colon C \dashrightarrow (\mathbb{P}^2_k)^\vee \qquad p \mapsto \big[\partial_0 F(p) : \partial_1 F(p) : \partial_2 F(p)\big]$$

whose coordinates are the partials of the defining form, by the tangent-line formula of definition 12.1.6. It is a rational map of curves, so $C^\vee$ is an irreducible curve by lemma 12.2.8 applied to a nonsingular model.

Proposition 13.5.2: The Class of a Nonsingular Curve

Let C be a nonsingular plane curve of degree $d \geq 2$ with $\operatorname{char} k = 0$. Then

$$\deg C^\vee = d(d-1)$$

***Proof*:**
A line of $(\mathbb{P}^2_k)^\vee$ is the set of lines of $\mathbb{P}^2_k$ through a fixed point q, so by theorem 11.5.1 the degree of $C^\vee$ is the number of points in which $C^\vee$ meets such a line for general q, counted through the Gauss map. In characteristic zero that map is birational onto its image for a nonsingular plane curve, so the count of points of $C^\vee$ equals the count of tangent lines to C through q; it is here, and only here, that the characteristic hypothesis is used.

A point $p \in C$ has its tangent line through $q = [q_0 : q_1 : q_2]$ exactly when

$$P_q(p) \;=\; q_0\,\partial_0 F(p) + q_1\,\partial_1 F(p) + q_2\,\partial_2 F(p) \;=\; 0$$

The form P_q has degree $d-1$ and is the *polar* of C with respect to q. It is not the zero form for general q: the partials $\partial_i F$ are not all zero, since $\operatorname{char} k = 0$ and F is nonconstant, and a general linear combination of forms that are not all zero is nonzero. And F does not divide P_q, because $\deg P_q = d-1 < d$ while F is irreducible. So C and $\mathcal{Z}(P_q)$ have no common component.

Theorem 13.1.1 therefore counts the intersection as $d(d-1)$ with multiplicity. For general q the intersections are transversal, so there are $d(d-1)$ distinct points and as many tangent lines through q, as we sought to show.

Example 13.5.3: Classes of Small Curves
A nonsingular conic has $d = 2$ and class 2: through a general point of the plane there pass exactly two tangent lines to a conic, and the dual of a conic is again a conic.

A nonsingular cubic has $d = 3$ and class 6. That number is the reason the dual of a smooth cubic is a sextic and not a cubic: dualizing does not preserve degree, and $(C^\vee)^\vee = C$ only because the two operations are inverse, not because the degrees match.

Singular points reduce the class, because a singular point has no well-defined tangent and the Gauss map is undefined there. The bookkeeping is Plücker's, and it is stated here rather than proved: the corrections require the genus and the behaviour of δ-invariants under duality.

13.5.4 Note: Plücker's Formulas Let C be an irreducible plane curve of degree d over a field of characteristic zero, with δ ordinary double points and κ ordinary cusps and no other singularities, and let $d^\vee$ be its class. Then

$$d^\vee = d(d-1) - 2\delta - 3\kappa$$

and dually, writing $\delta^\vee$ and $\kappa^\vee$ for the bitangents and inflections of C,

$$d = d^\vee(d^\vee - 1) - 2\delta^\vee - 3\kappa^\vee$$

The genus is common to both and is given by

$$g = \frac{(d-1)(d-2)}{2} - \delta - \kappa$$

consistently with the genus drop recorded in section 14.4, a node and a cusp each costing 1. For a nonsingular cubic, $\delta = \kappa = 0$ gives $d^\vee = 6$, and the second formula reads $3 = 30 - 2\delta^\vee - 3\kappa^\vee$; with $\delta^\vee = 0$, since a smooth cubic has no bitangent, this gives $\kappa^\vee = 9$. Those nine are the inflection points, recovering by a completely different route the count obtained in exercise set 15.3.1 from the Hessian and theorem 15.2.2.
A proof needs the resolution of chapter 17 together with the genus, and is in [19, chapter 9] and in [13] analytically.

Exercise 13.5.1

1. Let X be an irreducible curve. Show that $k[X] = \bigcap_{p \in X} \mathcal{O}_p(X)$, the intersection being taken inside $k(X)$.
2. Show that a line and an irreducible conic are nonsingular curves, either directly from definition 12.1.5 or from one of the corollaries of section 13.2.
3. Let C and C' be cubics with $C \cdot C' = \sum_{i=1}^{9} p_i$ for nine distinct points, and let Q be a conic with $Q \cdot C = \sum_{i=1}^{6} p_i$. Assuming $p_1, \ldots, p_6$ are nonsingular points of C, show that p_7, p_8, p_9 lie on a line. (Run the argument of lemma 13.4.1 with Q in place of the conic there.)
4. Show that the bound of corollary 13.2.5 fails for reducible curves by exhibiting a reducible curve of degree n violating it.

5. Verify that theorem 13.4.3 really is theorem 13.4.2 for a degenerate conic, by checking that the two proofs differ only in the verification that $r_i \notin \mathcal{Z}(Q)$.

6. Two conics meet in at most four points unless they share a component. Deduce that five points in general position determine a unique conic, and say what "general position" has to mean.

14

Divisors on Curves

The intersection cycles of section 13.3 are formal sums of points on a curve, and such sums form a group under addition. Not every formal sum is cut out by an equation, and the failure is measured by quotienting out the sums that arise from rational functions. What remains is the divisor class group, an invariant of the curve alone.

This chapter builds that group. Its degree-zero part is what chapter 15 identifies with the classical chord-tangent law on a cubic, and one book later it becomes the Jacobian. Its structure also detects rationality: a nonsingular plane curve is a copy of $\mathbb{P}^1$ exactly when distinct points on it are linearly equivalent, and that happens only in degrees 1 and 2. The invariant separating the cases is the genus, and the chapter closes by giving the genus-degree formula of section 8.5 that job.

14.1 Divisors and Their Degree

Throughout, C is an irreducible projective curve over the algebraically closed field k, and C is *nonsingular* unless said otherwise. Every local ring $\mathcal{O}_p(C)$ is then a discrete valuation ring by proposition 5.5.2, so every point carries an order function ord_p as in definition 12.2.4.

Section 7.5 already defined Weil divisors on any normal variety. A nonsingular projective curve is normal by corollary 5.5.3, and its prime divisors are its points, every point having codimension one. So the following is definition 7.5.1 specialized, restated here because the curve case is the one the rest of this Part uses.

Definition 14.1.1: Divisor on a Curve

A *divisor* on C is a formal sum

$$D = \sum_{p \in C} n_p\, p \qquad n_p \in \mathbb{Z}$$

with $n_p = 0$ for all but finitely many p. The divisors form an abelian group $\mathrm{Div}(C)$ under coefficientwise addition. The *degree* of D is $\deg D = \sum_p n_p$, and D is *effective*, written $D \geq 0$, if every $n_p \geq 0$. Write $D \geq D'$ for $D - D' \geq 0$.

A divisor is the same kind of object as a zero-cycle (definition 13.3.1), with the points confined to C. The intersection cycle of section 13.3 is the first source of divisors: if C is a nonsingular plane curve and G is a plane curve with no component in common, then $G \bullet C$ is an effective divisor on C, of degree $\deg(G)\deg(C)$ by theorem 13.1.1.

The second source is rational functions, and it is the one that matters.

Definition 14.1.2: Divisor of a Rational Function

Let $f \in k(C)^*$ be a nonzero rational function. Its *divisor* is

$$\mathrm{div}(f) = \sum_{p \in C} \mathrm{ord}_p(f)\, p$$

The divisor of zeros is $\mathrm{div}_0(f) = \sum_{\mathrm{ord}_p(f)>0} \mathrm{ord}_p(f)\, p$ and the divisor of poles is $\mathrm{div}_\infty(f) = -\sum_{\mathrm{ord}_p(f)<0} \mathrm{ord}_p(f)\, p$, so that $\mathrm{div}(f) = \mathrm{div}_0(f) - \mathrm{div}_\infty(f)$ with both parts effective.

The definition needs justification: the sum must be finite.

Proposition 14.1.3: A Rational Function Has Finitely Many Zeros and Poles

Let $f \in k(C)^*$. Then $\mathrm{ord}_p(f) = 0$ for all but finitely many $p \in C$, so $\mathrm{div}(f)$ is a divisor.

***Proof*:**
The set where $\mathrm{ord}_p(f) > 0$ is where f vanishes. Writing $f = g/h$ with g, h regular on some affine open $U \subseteq C$ and h not identically zero, the zeros of f in U lie in $\mathcal{Z}(g) \cap U$, a proper closed subset of the irreducible curve C, hence finite by proposition 4.3.5. The same argument applied to $1/f$ handles the poles, and $C \setminus U$ is finite for the same reason. So only finitely many points contribute, as we sought to show.

Example 14.1.4: Divisors on the Projective Line

Take $C = \mathbb{P}^1_k$ with coordinate x on the standard chart and ∞ the point at infinity.

1. *A function with a zero and a pole.* $\mathrm{div}(x) = 0 - \infty$, written $(0) - (\infty)$: the function x vanishes to order 1 at the origin and has a simple pole at infinity, since $1/x$ is a uniformizer there. Its degree is 0, as theorem 14.2.2 requires.

2. *Higher order.* $\operatorname{div}(x^n) = n(0) - n(\infty)$ for $n \in \mathbb{Z}$, negative n interchanging the roles. So every divisor supported on $\{0, \infty\}$ of degree 0 is principal.

3. *A general ratio.* For $f = (x-a)/(x-b)$ with $a \neq b$,

$$\operatorname{div}(f) = (a) - (b)$$

with no contribution at infinity, the numerator and denominator having equal degree. This is the computation behind proposition 14.2.4: every degree-zero divisor on the line is a combination of such differences.

4. *A divisor that is not principal.* The single point (0) has degree $1 \neq 0$, so by theorem 14.2.2 it is not $\operatorname{div}(f)$ for any f. On $\mathbb{P}^1_k$ the degree is the only obstruction, by proposition 14.2.4; on a curve of positive genus it is not, which is what theorem 14.2.5 shows.

The two sources are compatible in the case that matters, and the computation is worth doing once.

Proposition 14.1.5: Divisors from Ratios of Forms

Let $C \subseteq \mathbb{P}^2_k$ be a nonsingular plane curve and let G, H be forms of the same degree e, neither divisible by C. Then G/H defines a rational function on C and

$$\operatorname{div}\left(\frac{G}{H}\right) = G \bullet C - H \bullet C$$

***Proof*:**
The ratio G/H is a quotient of two forms of equal degree, so it is a well defined element of $k(C)$, nonzero because H does not divide G modulo C.

Fix $p \in C$. Property (9) of corollary 12.4.2 says that at a nonsingular point of C the intersection number is an order: $I(p, C \cap G) = \operatorname{ord}_p(G_*)$, where G_* is the dehomogenization of G in a chart containing p, and likewise for H. Choosing the same chart for both, $\operatorname{ord}_p(G_*/H_*) = \operatorname{ord}_p(G_*) - \operatorname{ord}_p(H_*)$ because ord_p is a valuation. The coefficient of p in $G \bullet C - H \bullet C$ is exactly that difference, and the chart cancels because G and H have the same degree, as we sought to show.

14.2 Principal Divisors and Linear Equivalence

Definition 14.2.1: Principal Divisor and Linear Equivalence

This is definition 7.5.3 for a curve. A divisor of the form $\operatorname{div}(f)$ for some $f \in k(C)^*$ is *principal*. The principal divisors form a subgroup $\operatorname{Prin}(C) \subseteq \operatorname{Div}(C)$, since $\operatorname{div}(fg) = \operatorname{div}(f) + \operatorname{div}(g)$. Two divisors D and D' are *linearly equivalent*, written $D \sim D'$, if $D - D'$ is principal.

The first structural fact is that the degree cannot see the difference between linearly equivalent divisors.

Theorem 14.2.2: Principal Divisors Have Degree Zero

Let C be a nonsingular projective plane curve and $f \in k(C)^*$. Then $\deg \operatorname{div}(f) = 0$. Consequently $D \sim D'$ implies $\deg D = \deg D'$.

***Proof*:**
Write $f = G/H$ with G, H forms of the same degree e, neither divisible by C; this is possible because $k(C)$ consists of exactly such ratios. By proposition 14.1.5, $\operatorname{div}(f) = G \bullet C - H \bullet C$. Both intersection cycles have degree $e \deg(C)$ by theorem 13.1.1, so the difference has degree 0. The consequence is immediate from additivity of the degree, completing the proof.

That Bézout's theorem is what makes this work is worth pausing on. The statement $\deg \operatorname{div}(f) = 0$ says a rational function on a projective curve has as many zeros as poles, and its analytic counterpart is the argument principle. Projectivity is essential: on $\mathbb{A}^1_k$ the function x has one zero and no poles.

The converse question is when a degree-zero divisor is principal, and the first thing to settle is which functions have no divisor at all.

Proposition 14.2.3: Functions with Trivial Divisor

Let C be a nonsingular projective curve and $f \in k(C)^*$ with $\operatorname{div}(f) = 0$. Then f is constant.

***Proof*:**
If $\operatorname{div}(f) = 0$ then $\operatorname{ord}_p(f) \geq 0$ at every p, so f lies in every local ring $\mathcal{O}_p(C)$ and is therefore a regular function on all of C. A regular function on a projective variety is constant by corollary 10.2.4, completing the proof.

On the simplest curve the converse of theorem 14.2.2 holds outright, and the computation is the model for everything later.

Proposition 14.2.4: Divisors on the Projective Line

Every divisor of degree 0 on $\mathbb{P}^1_k$ is principal. In particular any two points of $\mathbb{P}^1_k$ are linearly equivalent.

***Proof*:**
Let $D = \sum_i n_i p_i$ have degree 0, and write $p_i = [a_i : b_i]$ with $L_i = b_i x - a_i y$ the linear form vanishing exactly at p_i. Fix an index j with p_j distinct from the others. For $i \neq j$ the ratio L_i/L_j is a rational function on $\mathbb{P}^1_k$, and in the affine chart containing p_i the form L_i generates the maximal ideal at p_i while L_j is a unit there, so $\operatorname{ord}_{p_i}(L_i/L_j) = 1$; symmetrically $\operatorname{ord}_{p_j}(L_i/L_j) = -1$, and the order is 0 elsewhere. Hence $\operatorname{div}(L_i/L_j) = p_i - p_j$. Set

$$f = \prod_i \left(\frac{L_i}{L_j}\right)^{n_i}$$

a legitimate rational function since the exponents are integers. Then $\operatorname{div}(f) = \sum_i n_i(p_i - p_j) = D - (\deg D)\, p_j = D$, as we sought to show.

For curves of higher degree the answer is different, and the difference is the whole subject. The following theorem is the engine: on a plane curve of degree at least three, distinct points are never linearly equivalent. Its proof is a single application of Max Noether's theorem.

Theorem 14.2.5: Distinct Points Are Not Equivalent

Let C be a nonsingular projective plane curve of degree $d \geq 3$, and let $p, q \in C$ satisfy $p \sim q$. Then $p = q$.

***Proof*:**
Suppose $\operatorname{div}(f) = p - q$ and write $f = G/H$ with G, H forms of the same degree e, neither divisible by C. By proposition 14.1.5 this reads $G \bullet C - H \bullet C = p - q$, that is

$$G \bullet C + q = H \bullet C + p$$

an equality of effective divisors.

Choose a line L through p and write $L \bullet C = p + E$, so $E \geq 0$ has degree $d - 1 \geq 2$ by theorem 13.1.1. Then

$$(LH) \bullet C = L \bullet C + H \bullet C = p + E + H \bullet C = E + q + G \bullet C \;\geq\; G \bullet C$$

Every point of C is nonsingular, so corollary 13.3.4(2) applies to the curve C, the curve G and the curve LH, and produces a curve of degree $\deg(LH) - \deg(G) = (e + 1) - e = 1$, that is a line L', with

$$L' \bullet C = (LH) \bullet C - G \bullet C = E + q$$

So L and L' both cut E out of C, one with p left over and the other with q. Now E is effective of degree $d - 1 \geq 2$, and there are two cases. If its support contains two distinct points, then L and L' share those two points and hence coincide. Otherwise $E = (d - 1)r$ for a single point r, and since $d - 1 \geq 2$ both lines meet C at r with multiplicity at least 2, making each of them the tangent line to C at r, which is unique by definition 12.1.6; again they coincide.

With $L = L'$, comparing $L \bullet C = p + E$ against $L' \bullet C = q + E$ gives $p = q$, as we sought to show.

The hypothesis $d \geq 3$ is not an artefact. For $d = 1$ the curve is a line and proposition 14.2.4 says all its points are equivalent; for $d = 2$ the leftover divisor E has degree 1, a single point, which does not determine a line, and indeed a nonsingular conic is isomorphic to $\mathbb{P}^1_k$ by the computation in example 9.4.4. The theorem is the first place where a plane curve of degree three behaves differently from every curve seen so far.

14.3 The Picard Group

Definition 14.3.1: Picard Group

The *Picard group* of a nonsingular projective curve C, which is its class group $\mathrm{Cl}(C)$ in the sense of definition 7.5.3, is

$$\mathrm{Pic}(C) = \frac{\mathrm{Div}(C)}{\mathrm{Prin}(C)}$$

The class of a divisor D is written $[D]$. The subgroup of classes of degree zero is written $\mathrm{Pic}^0(C)$.

By theorem 14.2.2 the degree descends to a homomorphism $\deg\colon \mathrm{Pic}(C) \to \mathbb{Z}$, and it is surjective for a plane curve, since a hyperplane section has degree $d \neq 0$ and any class of degree n can be built from points. Its kernel is $\mathrm{Pic}^0(C)$, so there is a short exact sequence

$$0 \to \mathrm{Pic}^0(C) \to \mathrm{Pic}(C) \xrightarrow{\deg} \mathbb{Z} \to 0$$

All the geometry is in Pic^0.

Example 14.3.2: The Picard Group of the Line

For $C = \mathbb{P}^1_k$, proposition 14.2.4 says every degree-zero divisor is principal, so $\mathrm{Pic}^0(\mathbb{P}^1_k) = 0$ and the degree map is an isomorphism $\mathrm{Pic}(\mathbb{P}^1_k) \cong \mathbb{Z}$. The Picard group is as small as it can be, and this is exactly the statement that no formal sum of points on the line carries information beyond how many points it has.

The contrast with a curve of higher degree is total. On a nonsingular plane cubic, Pic^0 turns out to be in bijection with the points of the curve itself, and the bijection carries the chord-tangent operation of section 15.1 to addition of divisor classes. That is section 15.2, and the one general fact it needs is recorded here.

Lemma 14.3.3: Hyperplane Sections Are All Equivalent

Let C be a nonsingular plane curve and L, M lines, neither a component of C. Then $L \bullet C \sim M \bullet C$.

***Proof*:**
The ratio L/M is a quotient of forms of degree 1, so proposition 14.1.5 gives $\mathrm{div}(L/M) = L \bullet C - M \bullet C$, which is therefore principal, completing the proof.

More generally the same argument shows $G \bullet C \sim G' \bullet C$ for any two curves G, G' of equal degree meeting C properly, so the class of a plane section depends only on the degree. This is the statement that a plane curve carries a distinguished element of $\mathrm{Pic}(C)$ in each degree, and everything computed below is measured against it.

14.4 The Genus of a Plane Curve

Section 8.5 computed the arithmetic genus of a nonsingular plane curve of degree d from its Hilbert polynomial and found

$$p_a = \frac{(d-1)(d-2)}{2}$$

That computation was about the embedding. This section gives the number a job.

Definition 14.4.1: Genus of a Nonsingular Plane Curve

The *genus* of a nonsingular projective plane curve C of degree d is

$$g(C) = \frac{(d-1)(d-2)}{2}$$

which is its arithmetic genus in the sense of definition 8.5.3.

So a line and a conic have genus 0, a cubic has genus 1, a quartic genus 3, and no plane curve has genus 2: the sequence $0, 0, 1, 3, 6, 10, \dots$ skips it. Curves of genus 2 exist, but none of them embeds in $\mathbb{P}^2_k$ as a nonsingular curve.

The genus was introduced as a number attached to an embedding, and the following theorem is the first evidence that it is measuring the curve itself.

Theorem 14.4.2: Genus Zero Means Rational

Let C be a nonsingular projective plane curve of degree d. The following are equivalent:

1. $g(C) = 0$;
2. $d \leq 2$;
3. $C \cong \mathbb{P}^1_k$;
4. some two distinct points of C are linearly equivalent.

***Proof*:**
(1) $\Leftrightarrow$ (2) is arithmetic: $(d-1)(d-2)/2 = 0$ exactly when $d = 1$ or $d = 2$.

(2) $\Rightarrow$ (3): a line is $\mathbb{P}^1_k$. For a nonsingular conic C, fix $p \in C$ and project from p onto any line not through it. A line through p meets C in one further point by theorem 13.1.1, the tangent at p being the case where that point is p itself, so the projection is a bijection $C \to \mathbb{P}^1_k$; it is regular away from p, and regular at p by lemma 12.2.8, with a regular inverse for the same reason. Example 9.4.4(3) carries the computation out explicitly for $\mathcal{Z}(x^2 + y^2 - z^2)$.

(3) $\Rightarrow$ (4): an isomorphism carries rational functions to rational functions and preserves orders, hence carries principal divisors to principal divisors and linear equivalence to linear equivalence. Any two points of $\mathbb{P}^1_k$ are linearly equivalent by proposition 14.2.4, and a curve has more than one point, so the same holds on C.

(4) $\Rightarrow$ (2): this is theorem 14.2.5 in contrapositive form, which says $d \geq 3$ forces distinct points to be inequivalent, completing the proof.

Corollary 14.4.3: Nonsingular Plane Cubics Are Not Rational

A nonsingular projective plane cubic has genus 1 and is not isomorphic to $\mathbb{P}^1_k$. In particular it is not a rational curve.

***Proof*:**
Its genus is $(3-1)(3-2)/2 = 1 \neq 0$, so theorem 14.4.2 rules out $C \cong \mathbb{P}^1_k$. A rational curve is birational to $\mathbb{P}^1_k$, and a birational map between nonsingular projective curves is an isomorphism by lemma 12.2.8 applied in both directions, completing the proof.

This settles a claim made in passing in chapter 3, that the classification of elliptic curves up to birational equivalence needs the genus: the cubics are exactly the plane curves the genus first distinguishes from $\mathbb{P}^1_k$, and theorem 15.2.2 says what replaces the vanishing Picard group when it does.

Singular curves are handled by passing to a nonsingular model, which is what chapter 7 was built for, and the genus drops in a way controlled by the singularities.

14.4.4 Note: The Genus of a Singular Plane Curve Let C be an irreducible plane curve of degree d whose singular points are all *ordinary*, meaning m_p distinct tangent lines at a point of multiplicity m_p (definition 12.1.6), and let $\tilde{C}$ be its normalization (chapter 7), a nonsingular curve birational to C. Then

$$g(\tilde{C}) = \frac{(d-1)(d-2)}{2} - \sum_p \frac{m_p(m_p-1)}{2}$$

each singular point costing exactly $\binom{m_p}{2}$. The formula is proved in theorem 17.4.1, by counting adjoint curves of degree $d-3$ rather than by any computation on a blow-up.
That the right side is non-negative is corollary 13.2.5, proved in chapter 13 by a dimension count on curves of degree $d-1$, which now reads as the statement that a genus cannot be negative. The extremal curve $\mathcal{Z}(x^d + y^{d-1}z)$ of section 13.2 attains the bound and is rational, as section 17.4 exhibits explicitly.

Exercise 14.4.1

1. Show that $\operatorname{div}(fg) = \operatorname{div}(f) + \operatorname{div}(g)$ and $\operatorname{div}(1/f) = -\operatorname{div}(f)$, so that $\operatorname{Prin}(C)$ is a subgroup of $\operatorname{Div}(C)$ and $f \mapsto \operatorname{div}(f)$ is a homomorphism $k(C)^* \to \operatorname{Div}(C)$.

2. Deduce from proposition 14.2.3 that the kernel of that homomorphism is k^*, so that $\operatorname{Prin}(C) \cong k(C)^*/k^*$.

3. Let C be a nonsingular plane curve of degree d and H a line. Show that the class $[H \cdot C]$ generates the image of deg in $\mathbb{Z}$ if and only if $d = 1$, and describe the image in general.

4. Show that on a nonsingular plane cubic C with base point O, a point p satisfies $p \oplus p \oplus p = \varphi(O, O)$ if and only if p is an inflection point. Deduce, using theorem 15.2.2, that the inflection points form a subgroup of $E_k(C, O)$ when O is itself an inflection point.

5. Show that a divisor D on a nonsingular projective curve satisfies $D \sim 0$ and $D \geq 0$ only if $D = 0$.
6. Give an example of two divisors on a nonsingular plane cubic that have the same degree but are not linearly equivalent, and prove they are not.

15

The Group Law on Cubic Curves

A line meets a nonsingular cubic in three points, and that fact alone produces a group. The construction is elementary, its verification is not: associativity is a statement about the nine intersection points of two cubics, and the proof runs through Max Noether's theorem.

The chapter then explains where the group came from. Under the divisor theory of chapter 14 the chord-tangent law is addition in $\mathrm{Pic}^0(C)$, transported to the curve by $p \mapsto [p - O]$, and the hard-won associativity becomes the associativity of a quotient group. That description also settles the one loose end of the construction: it needed a base point, and nothing distinguishes one choice from another, but the target $\mathrm{Pic}^0(C)$ mentions no base point at all. The group obtained is the object that *Elliptic Curves* [6] takes as its subject.

15.1 The Chord-Tangent Construction

Let C be a nonsingular projective plane cubic. It is irreducible: were $C = C_1C_2$ a proper factorization, theorem 13.1.1 would give $\mathcal{Z}(C_1)$ and $\mathcal{Z}(C_2)$ a common point, and proposition 12.1.7 makes any such point singular on C. So no line is a component of C, and theorem 13.1.1 says every line meets it in exactly three points counted with multiplicity. Given $p, q \in \mathcal{Z}(C)$, not necessarily distinct, let L be the line through them, taken to be the tangent line at p when $p = q$. Then

$$L \bullet C = p + q + r$$

for a unique third point r, again with multiplicity. This defines a map

$$\varphi\colon \mathcal{Z}(C) \times \mathcal{Z}(C) \to \mathcal{Z}(C) \qquad \varphi(p, q) = r$$

which is visibly symmetric, and which satisfies $\varphi(p, \varphi(p, q)) = q$, since the line realizing the left side is the same L.

Figure 15.1 shows the two cases.

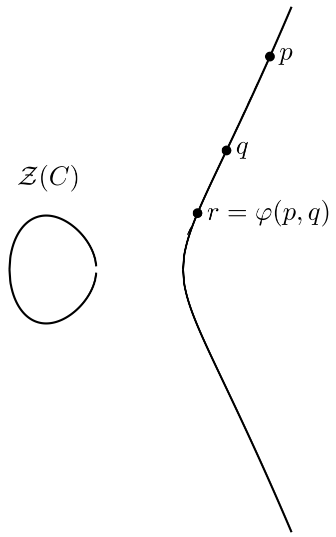


Figure 15.1: The chord through p and q meets the cubic in a third point. When $p = q$ the chord is replaced by the tangent line at p.

The map φ is a symmetric binary operation, but it is not a group law: there is no identity, since $\varphi(p, q) = q$ would force p to lie on the tangent at q for every q. The repair is to compose with φ once more against a chosen base point. Fix any $O \in \mathcal{Z}(C)$ and define

$$p \oplus q = \varphi\big(O, \varphi(p, q)\big)$$

Lemma 15.1.1: The Base Point Is an Identity

For every $p \in \mathcal{Z}(C)$, $O \oplus p = p \oplus O = p$.

***Proof*:**
Symmetry of φ gives $O \oplus p = p \oplus O$, so it is enough to compute one. Setting $q = \varphi(O, p)$, the involution property gives

$$O \oplus p = \varphi\big(O, \varphi(O, p)\big) = \varphi(O, q) = p$$

as we sought to show.

Before the group law can be proved, one point needs care. The natural guess for the inverse of p is $\varphi(p, O)$, and the natural guess for collinearity is that three collinear points sum to O. Both are false for a general base point, and the reason is that $\varphi(O, O)$, the third point on the tangent at O, need not be O itself. A point p with $\varphi(p, p) = p$ is one where the tangent line meets C triply, that is an *inflection point*, and a nonsingular cubic has only finitely many of them.

Lemma 15.1.2: Collinearity and the Sum

Let $p, q, r \in \mathcal{Z}(C)$. Then p, q, r are collinear, in the sense that $L \bullet C = p + q + r$ for some line L, if and only if

$$p \oplus q \oplus r = \varphi(O, O)$$

In particular, if the base point O is an inflection point then $\varphi(O, O) = O$ and three collinear points sum to the identity.

***Proof*:**
Suppose p, q, r are collinear, so $r = \varphi(p, q)$. Then $p \oplus q = \varphi(O, r)$, and

$$(p \oplus q) \oplus r = \varphi\Big(O, \varphi\big(\varphi(O, r), r\big)\Big)$$

The points O, r and $\varphi(O, r)$ are collinear by definition of φ, so $\varphi(\varphi(O, r), r) = O$, and the right side is $\varphi(O, O)$.

Conversely, suppose $p \oplus q \oplus r = \varphi(O, O)$, and let $r' = \varphi(p, q)$, so that p, q, r' are collinear. By the first part $p \oplus q \oplus r' = \varphi(O, O)$ as well, so r and r' have the same image under the map $x \mapsto (p \oplus q) \oplus x$. That map is injective once $\oplus$ is known to be a group law. Proposition 15.1.4 establishes this using only the involution property of φ and lemma 15.1.3, never the present converse, so there is no circularity. Hence $r = r'$ and p, q, r are collinear, as we sought to show.

The associativity of $\oplus$ is the hard point, and it rests on the following consequence of Max Noether's theorem: two cubics that agree with C in eight of their nine intersection points agree in the ninth as well.

Lemma 15.1.3: Cubic Point Counting

Let C be an irreducible cubic and C', C'' cubics with no component in common with C. Suppose

$$C' \bullet C = \sum_{i=1}^{9} p_i \qquad C'' \bullet C = \sum_{i=1}^{8} p_i + q$$

where the p_i are nonsingular points of C, not necessarily distinct. Then $q = p_9$.

***Proof*:**
Suppose $q \neq p_9$ and choose a line L through p_9 not passing through q, possible because the lines through p_9 sweep out the plane and only one of them contains q. Write $L \bullet C = p_9 + r + s$.

The quartic LC'' satisfies

$$(LC'') \bullet C = L \bullet C + C'' \bullet C = \sum_{i=1}^{9} p_i + q + r + s \ \geq \ C' \bullet C$$

Every point of $\mathcal{Z}(C') \cap \mathcal{Z}(C)$ is a nonsingular point of C by hypothesis, so corollary 13.3.4(2) applies to the triple (C, C', LC'') and produces a curve of degree $\deg(LC'') - \deg(C') = 4 - 3 = 1$, that is a line L', with

$$L' \bullet C = (LC'') \bullet C - C' \bullet C = q + r + s$$

Now L and L' coincide. If $r \neq s$, both contain the two distinct points r and s, and two distinct points determine a line. If $r = s$, then $L \cdot C \geq 2r$ and $L' \cdot C \geq 2r$, so both are tangent to C at the nonsingular point r, and the tangent line there is unique by definition 12.1.6.

But $L' = L$ forces q to lie on L, contradicting the choice of L. Hence $q = p_9$, as we sought to show.

Proposition 15.1.4: Group Law For Cubics

Let C be a nonsingular projective plane cubic and $O \in \mathcal{Z}(C)$ a base point. Then $\mathcal{Z}(C)$ with the operation $\oplus$ is an abelian group with identity O, in which the inverse of p is

$$\ominus p = \varphi\big(p, \varphi(O,O)\big)$$

Write $E_k(C,O)$ for this group.

***Proof*:**
Commutativity is the symmetry of φ, and O is an identity by lemma 15.1.1.

Inverses. Let $t = \varphi(O,O)$ and $\ominus p = \varphi(p,t)$. Only the involution property is needed: by definition of $\oplus$,

$$p \oplus \ominus p = \varphi\big(O, \varphi(p, \ominus p)\big) = \varphi\big(O, \varphi(p, \varphi(p,t))\big) = \varphi(O,t) = \varphi\big(O, \varphi(O,O)\big) = O$$

the third equality applying $\varphi(p,\varphi(p,-)) = \mathrm{id}$ and the last applying $\varphi(O,\varphi(O,-)) = \mathrm{id}$.

Associativity. Let $p,q,r \in \mathcal{Z}(C)$ and introduce six lines by

$$\begin{array}{lll} L_1 \cdot C = p + q + s' & M_1 \cdot C = O + s' + s & L_2 \cdot C = s + r + t' \\ M_2 \cdot C = q + r + u' & L_3 \cdot C = O + u' + u & M_3 \cdot C = p + u + t'' \end{array}$$

so that $s = \varphi(O,s') = p \oplus q$ and $u = \varphi(O,u') = q \oplus r$. Then

$$(p \oplus q) \oplus r = \varphi(O, \varphi(s,r)) = \varphi(O,t') \qquad p \oplus (q \oplus r) = \varphi(O, \varphi(p,u)) = \varphi(O,t'')$$

so it suffices to prove $t' = t''$. Put $C' = L_1L_2L_3$ and $C'' = M_1M_2M_3$, both cubics with no component in common with the irreducible C, since a line meets C in three points and so is not a component. Adding the three displayed cycles in each row,

$$C' \cdot C = p + q + s' + s + r + t' + O + u' + u \qquad C'' \cdot C = O + s' + s + q + r + u' + p + u + t''$$

The two agree in the eight points O, p, q, r, s, s', u, u', and every point of $\mathcal{Z}(C)$ is nonsingular, so lemma 15.1.3 gives $t' = t''$.

Thus $\oplus$ is an abelian group law, completing the proof.

The group depends on the base point only up to isomorphism, and the isomorphism is explicit. Given a second base point O', set $Q = \varphi(O,O')$ and

$$\alpha \colon E_k(C,O) \to E_k(C,O') \qquad \alpha(p) = \varphi(Q,p)$$

This sends O to O', since the line through Q and O is the line through O, O' and Q. That α is a group isomorphism is proved in section 15.3, once section 15.2 has identified $E_k(C,O)$ with a group of divisor classes and made the computation one line rather than a chain of collinearity arguments.

15.2 The Group Law via the Picard Group

The chord-tangent law was constructed by hand, and its associativity cost a nine-point argument through Max Noether. This section explains where the group comes from: it is $\operatorname{Pic}^0(C)$, transported to the curve.

Throughout, C is a nonsingular projective plane cubic with base point O, and φ, $\oplus$ are as in section 15.1. The bridge is a single linear equivalence.

Proposition 15.2.1: The Chord-Tangent Law Is Addition of Classes

For all $p, q \in C$,

$$p + q \;\sim\; (p \oplus q) + O$$

Consequently the map

$$\theta\colon E_k(C,O) \to \operatorname{Pic}^0(C) \qquad \theta(p) = [\,p - O\,]$$

is a group homomorphism.

***Proof*:**
Let $r = \varphi(p,q)$ and $s = \varphi(O,r) = p \oplus q$, and let L and M be the lines realizing them, so that

$$L \cdot C = p + q + r \qquad M \cdot C = O + r + s$$

By lemma 14.3.3 these are linearly equivalent, and subtracting the common r gives $p + q \sim O + s$, which is the display.

For the consequence, θ lands in Pic^0 because $\deg(p - O) = 0$, and

$$\theta(p) + \theta(q) = [\,p + q - 2O\,] = [\,(p \oplus q) + O - 2O\,] = [\,(p \oplus q) - O\,] = \theta(p \oplus q)$$

using the display in the middle step, as we sought to show.

Theorem 15.2.2: The Group Law Is the Picard Group

Let C be a nonsingular projective plane cubic with base point O. Then

$$\theta\colon E_k(C,O) \longrightarrow \operatorname{Pic}^0(C) \qquad \theta(p) = [\,p - O\,]$$

is an isomorphism of abelian groups.

***Proof*:**
It is a homomorphism by proposition 15.2.1.

Injective. If $\theta(p) = \theta(q)$ then $p - O \sim q - O$, so $p \sim q$, and theorem 14.2.5 gives $p = q$, the cubic having degree 3.

Surjective. Let D have degree 0 and write $D = \sum_{i=1}^{n} p_i - \sum_{j=1}^{n} q_j$, the two counts being equal because $\deg D = 0$; points may repeat, and $n = 0$ gives $D = 0 = \theta(O)$. First,

$$\sum_{i=1}^{n} p_i \;\sim\; (p_1 \oplus \cdots \oplus p_n) + (n-1)O$$

by induction on n. The case $n = 1$ is trivial. Given the statement for n, add p_{n+1} to both sides and apply proposition 15.2.1 to the pair $(p_1 \oplus \cdots \oplus p_n,\ p_{n+1})$, which replaces their sum by $(p_1 \oplus \cdots \oplus p_{n+1}) + O$; that is the statement for $n+1$.

Writing $P = p_1 \oplus \cdots \oplus p_n$ and $Q = q_1 \oplus \cdots \oplus q_n$ and subtracting the two instances, the $(n-1)O$ terms cancel and $D \sim P - Q$. Finally

$$[P - Q] = [P - O] - [Q - O] = \theta(P) - \theta(Q) = \theta(P \ominus Q)$$

so D lies in the image, as we sought to show.

Associativity of $\oplus$, which proposition 15.1.4 obtained from lemma 15.1.3 and so ultimately from Max Noether's theorem, is now visible as the associativity of addition in a quotient group. The two proofs are not independent: theorem 14.2.5 itself runs through Max Noether. What changes is where the difficulty sits. In section 15.1 it sits in a configuration of six lines; here it sits in the statement that a rational function on a cubic cannot have a single simple pole.

15.2.3 Note: Which Facts Are Doing the Work Two theorems carry theorem 15.2.2, and they pull in opposite directions. Lemma 14.3.3 says linear equivalence is coarse enough that all line sections agree, which is what makes θ additive. Theorem 14.2.5 says it is fine enough that points stay apart, which is what makes θ injective. On a conic the second fails and Pic^0 collapses to zero; on a line there is nothing to say at all. Degree three is the smallest case where both hold, which is the precise sense in which the group law is a phenomenon of cubics.

15.3 Independence of the Base Point

The construction of $\oplus$ required choosing O, and nothing recommends one point over another. Theorem 15.2.2 settles the matter at once, because its target does not mention O.

Theorem 15.3.1: Independence of the Base Point

Let C be a nonsingular projective plane cubic and $O, O' \in C$. Then

$$E_k(C,O) \;\cong\; E_k(C,O')$$

both being isomorphic to $\operatorname{Pic}^0(C)$, which is defined without reference to a base point.

***Proof*:**
Write θ_O and $\theta_{O'}$ for the two maps of theorem 15.2.2, each an isomorphism onto $\operatorname{Pic}^0(C)$. Then $\theta_{O'}^{-1} \circ \theta_O$ is an isomorphism $E_k(C,O) \to E_k(C,O')$, completing the proof.

The explicit map promised in section 15.1 is almost that one, differing by an inversion.

Proposition 15.3.2: The Explicit Isomorphism

Let $Q = \varphi(O, O')$ and $\alpha(p) = \varphi(Q, p)$. Then $\alpha\colon E_k(C,O) \to E_k(C,O')$ is a group isomorphism, and

$$\theta_{O'}\big(\alpha(p)\big) = -\,\theta_O(p)$$

***Proof*:**
The points Q, p and $\alpha(p)$ are collinear by definition of φ, and so are O, O' and Q. By lemma 14.3.3 the two line sections are linearly equivalent:

$$Q + p + \alpha(p) \;\sim\; O + O' + Q$$

Cancelling Q and rearranging gives $\alpha(p) - O' \sim O - p$, that is

$$\theta_{O'}\big(\alpha(p)\big) = [\,\alpha(p) - O'\,] = [\,O - p\,] = -[\,p - O\,] = -\theta_O(p)$$

So $\alpha = \theta_{O'}^{-1} \circ (-\mathrm{id}) \circ \theta_O$. Inversion is an automorphism of an abelian group, and θ_O, $\theta_{O'}$ are isomorphisms, so α is one too, as we sought to show.

That the natural explicit map inverts is not a defect of the choice; no map defined by a single application of φ can avoid it, since φ itself reverses the sign. Composing twice, $p \mapsto \varphi(O', \varphi(Q,p))$ is the base-point-free isomorphism $\theta_{O'}^{-1}\theta_O$ on the nose.

15.3.3 Note: What the Group Does and Does Not Depend On The *group* $E_k(C,O)$ depends on O only up to isomorphism, and the isomorphism class is that of $\operatorname{Pic}^0(C)$. What does depend on O is the identification of points of C with group elements: which point is the identity, and hence which triples sum to zero. Choosing O to be an inflection point makes lemma 15.1.2 read in its familiar form, three collinear points summing to the identity, and this is why the literature almost always makes that choice for a curve given in Weierstrass form, where the point at infinity is an inflection.

Exercise 15.3.1

1. Show that a nonsingular *projective* plane curve must be irreducible, and give a nonsingular *affine* plane curve that is not.
2. Let C be a nonsingular plane cubic with base point O. Show that $\varphi(O, O) = O$ if and only if O is an inflection point, and that in that case the inverse of proposition 15.1.4 takes the familiar form $\ominus p = \varphi(p, O)$.
3. With O an inflection point, show that the inflection points of C are exactly the elements of order dividing 3 in $E_k(C, O)$, and bound their number using theorem 13.1.1 and the Hessian.
4. Show that a nonsingular plane cubic has no point p with $\varphi(p, p) = p$ other than its inflection points.
5. Let C be a nonsingular plane cubic and $p, q \in C$ with $p \sim q$ as divisors. Show $p = q$, and explain precisely which step of the argument fails on a nonsingular conic.
6. Compare the labour of verifying associativity of $\oplus$ directly on $y^2 = x^3 + 1$ with $O = [0 : 1 : 0]$, for three points of your choosing, against the one-line argument of theorem 15.2.2.

16

Differential Forms and the Canonical Class

Every invariant of a curve met so far has been extrinsic: the degree depends on an embedding, the multiplicity on a choice of coordinates, the divisor class group on nothing at all but says little by itself. This chapter constructs the invariant that governs the rest, the canonical class, and it does so from the one piece of machinery not yet used, differentiation.

Derivations, the module $\Omega_{B/A}$ and its universal property, and the two fundamental exact sequences are commutative algebra, developed in [11, Derivation on Field]; this chapter takes them as given and does the geometry. What comes out is a divisor class attached to a curve with no choices at all, whose degree is $2g-2$, and which chapter 18⋆ shows to be the error term in Riemann's inequality.

16.1 Differential Forms on a Variety

Definition 16.1.1: Differentials on a Variety

Let X be an affine variety over k. The *module of differentials* of X is $\Omega_X = \Omega_{k[X]/k}$, the module of relative differential forms of [11, Module of Relative Differential Forms], together with the universal k-derivation $d\colon k[X] \to \Omega_X$. Its elements are the *differential forms* on X. For $p \in X$ the *forms at p* are $\Omega_{X,p} = \Omega_X \otimes_{k[X]} \mathcal{O}_p(X)$, and for an irreducible X the *rational differential forms* are $\Omega_{k(X)/k}$, a vector space over $k(X)$.

Example 16.1.2: Modules of Differentials

1. *Affine space.* $\Omega_{\mathbb{A}^n_k}$ is free of rank n over $k[x_1, \ldots, x_n]$ with basis $dx_1, \ldots, dx_n$, since a k-derivation is determined by arbitrary choices of the dx_i. This is the first exercise of exercise set 16.2.1.
2. *A nonsingular plane curve.* On $C = \mathcal{Z}(y - x^2)$ the relation $dy = 2x\,dx$ holds, so Ω_C is generated by dx alone and is free of rank $1 = \dim C$, as proposition 16.1.3 predicts.
3. *The cuspidal cubic.* On $C = \mathcal{Z}(y^2 - x^3)$ the relation is $2y\,dy = 3x^2\,dx$. At the origin both coefficients vanish, so the relation imposes nothing there and $\Omega_{C,0} \otimes k$ has dimension 2, while $\dim C = 1$. By proposition 16.1.3 the origin is singular, which is the fifth exercise of exercise set 16.2.1. Away from the origin the relation is nontrivial and Ω is free of rank 1.
4. *A form with a pole.* On $\mathbb{P}^1_k$ the form dx has $\operatorname{div}(dx) = -2(\infty)$, computed in the second exercise of exercise set 16.2.1. Its degree -2 is $2g - 2$ with $g = 0$, which is theorem 16.2.4 in the smallest case.

Two facts do all the work, and both are transported from the algebra.

Proposition 16.1.3: Differentials Detect Smoothness

Let X be a variety of dimension n and $p \in X$. Then

$$\dim_k \Omega_{X,p} \otimes_{\mathcal{O}_p(X)} k \;=\; \dim_k \mathfrak{m}_p/\mathfrak{m}_p^2$$

so p is a nonsingular point if and only if that dimension is n. If X is nonsingular then $\Omega_{X,p}$ is a free $\mathcal{O}_p(X)$-module of rank n.

***Proof*:**
Apply the second fundamental exact sequence ([11, Differentials and Exact Sequence II]) to $k[X] \to \mathcal{O}_p(X) \to k$ with $I = \mathfrak{m}_p$. Since $\Omega_{k/k} = 0$, it reads

$$\mathfrak{m}_p/\mathfrak{m}_p^2 \xrightarrow{\delta} \Omega_{X,p} \otimes k \to 0$$

and δ is injective because a k-linear functional on $\mathfrak{m}_p/\mathfrak{m}_p^2$ extends to a k-derivation of $\mathcal{O}_p(X)$, by sending f to the class of $f - f(p)$; so δ has a left inverse. That gives the displayed equality, and corollary 5.1.1 identifies the right side with n exactly at the nonsingular points.

For freeness, $\Omega_{X,p}$ is a finitely generated module over the Noetherian local ring $\mathcal{O}_p(X)$ by [11, Finite Generation of Differential Forms], whose fibre has dimension n; a finitely generated module over a regular local ring whose fibre dimension equals the generic rank is free, which is [11, Characterizing Free Modules using Residue and Quotient Field], completing the proof.

Corollary 16.1.4: Rational Differentials on a Curve

Let C be a nonsingular projective curve. Then $\Omega_{k(C)/k}$ is a one-dimensional vector space over $k(C)$. If t is a uniformizer at a point p, then dt is a basis, and every rational differential form is $f\,dt$ for a unique $f \in k(C)$.

***Proof*:**
By proposition 16.1.3 the module $\Omega_{C,p}$ is free of rank 1 over $\mathcal{O}_p(C)$, so tensoring up to the fraction field gives $\dim_{k(C)} \Omega_{k(C)/k} = 1$.

For the basis, $\mathcal{O}_p(C)$ is a discrete valuation ring with uniformizer t and $\mathfrak{m}_p/\mathfrak{m}_p^2$ is spanned by the class of t, so dt generates $\Omega_{C,p} \otimes k$ under the isomorphism of proposition 16.1.3. Nakayama's lemma then makes dt a generator of the free rank-one module $\Omega_{C,p}$, hence a basis after tensoring with $k(C)$, completing the proof.

16.2 The Canonical Class

A nonzero rational differential form on a curve has zeros and poles, exactly as a rational function does, and they assemble into a divisor. The point of the section is that the divisor class does not depend on the form.

Definition 16.2.1: The Divisor of a Differential Form

Let C be a nonsingular projective curve and $\omega \in \Omega_{k(C)/k}$ nonzero. For $p \in C$ with uniformizer t, write $\omega = f\,dt$ with $f \in k(C)$ as in corollary 16.1.4, and set $\operatorname{ord}_p(\omega) = \operatorname{ord}_p(f)$. The *divisor* of ω is

$$\operatorname{div}(\omega) = \sum_{p \in C} \operatorname{ord}_p(\omega)\, p$$

Proposition 16.2.2: The Divisor of a Form Is Well Defined

$\operatorname{ord}_p(\omega)$ does not depend on the choice of uniformizer, and $\operatorname{ord}_p(\omega) = 0$ for all but finitely many p, so $\operatorname{div}(\omega)$ is a divisor.

***Proof*:**
Let s be another uniformizer at p, so $s = ut$ with u a unit of $\mathcal{O}_p(C)$. Then $ds = u\,dt + t\,du$, and writing $du = u'\,dt$ with $u' \in \mathcal{O}_p(C)$ gives $ds = (u + tu')\,dt$. The coefficient $u + tu'$ is a unit, being u modulo $\mathfrak{m}_p$, so $\omega = f\,dt = f(u+tu')^{-1}ds$ and the order of the coefficient is unchanged.

For finiteness, cover C by finitely many affine opens; on each, $\omega = f\,dg$ for some $f, g \in k[U]$, and $\operatorname{ord}_p(\omega) \neq 0$ only where f or dg degenerates, a proper closed subset, hence finite by proposition 4.3.5, completing the proof.

Theorem 16.2.3: The Canonical Class

Let C be a nonsingular projective curve. Any two nonzero rational differential forms on C have linearly equivalent divisors, so their common class

$$K_C = [\operatorname{div}(\omega)] \;\in\; \operatorname{Pic}(C)$$

is an invariant of C, the *canonical class*. A divisor in that class is a *canonical divisor*.

***Proof*:**
Let ω, ω' be nonzero. By corollary 16.1.4 the space $\Omega_{k(C)/k}$ is one-dimensional over $k(C)$, so $\omega' = f\omega$ for a unique $f \in k(C)^*$. At each p, writing $\omega = g\,dt$, gives $\omega' = fg\,dt$ and hence $\operatorname{ord}_p(\omega') = \operatorname{ord}_p(f) + \operatorname{ord}_p(\omega)$. Summing,

$$\operatorname{div}(\omega') = \operatorname{div}(f) + \operatorname{div}(\omega)$$

so the two divisors differ by a principal one and are linearly equivalent by definition 14.2.1, completing the proof.

That one line is the whole point of introducing differentials: a curve carries a distinguished divisor class, obtained without choosing an embedding, a point, or a coordinate. Its degree is the genus in disguise.

Theorem 16.2.4: The Degree of the Canonical Class

Let C be a nonsingular projective plane curve of degree d, with genus $g = \binom{d-1}{2}$ as in definition 14.4.1. Then

$$K_C = (d-3)\,H \cdot C \qquad \text{and} \qquad \deg K_C = 2g - 2$$

where H is a line.

***Proof*:**
Work in the affine chart $z = 1$ with $C = \mathcal{Z}(f)$, $f = F(x, y, 1)$, and consider the rational form

$$\omega = \frac{dx}{\partial f/\partial y}$$

On C one has $f = 0$, so $0 = df = \partial_x f\,dx + \partial_y f\,dy$ and therefore $\omega = -\,dy/\partial_x f$ as well. At a point where $\partial_y f \neq 0$ the function x is a uniformizer, since $\mathcal{O}_p(C)$ is a discrete valuation ring and dx generates $\Omega_{C,p}$ by proposition 16.1.3; there ω has neither zero nor pole. Where $\partial_y f = 0$ the second expression applies and $\partial_x f \neq 0$, again giving neither zero nor pole, since C is nonsingular and the two partials cannot both vanish. So ω has no zeros and no poles in the affine chart.

It remains to compute the behaviour on the line at infinity. Homogenizing, $x = X/Z$ and $\partial f/\partial y$ has degree $d-1$ as a form, so ω is a ratio in which the numerator contributes -2 and the denominator $-(d-1)$ in the variable Z; collecting, ω acquires a zero of order $d-3$ along each point of C at infinity. Since the points at infinity are $H \cdot C$ for the line $H = \mathcal{Z}(Z)$, this reads

$$\operatorname{div}(\omega) = (d-3)\,H \cdot C$$

Taking degrees, $\deg K_C = (d-3) \cdot d$ by theorem 13.1.1, and

$$d(d-3) = (d-1)(d-2) - 2 = 2g - 2$$

as we sought to show.

This settles what chapter 18⋆ previously had to cite Fulton for: the identification $K = (d-3)H \bullet C$ and the degree $2g-2$ are now proved, and Riemann-Roch may quote them.

Example 16.2.5: Canonical Classes of Small Curves

1. *A line*, $d = 1$: $K_C = -2H \bullet C$ of degree $-2 = 2 \cdot 0 - 2$. On $\mathbb{P}^1_k$ the form dx has a double pole at infinity and no zeros, which is the whole story.
2. *A conic*, $d = 2$: $K_C = -H \bullet C$ of degree -2, consistent with a conic being isomorphic to $\mathbb{P}^1_k$ by theorem 14.4.2.
3. *A cubic*, $d = 3$: $K_C = 0$, so the canonical class is trivial and $\deg K_C = 0 = 2g - 2$ with $g = 1$. A smooth cubic carries a nowhere-zero, nowhere-polar differential form, which is the invariant differential of the group law of chapter 15 and the reason theorem 18.4.4 could take $K \sim 0$.
4. *A quartic*, $d = 4$: $K_C = H \bullet C$ of degree $4 = 2 \cdot 3 - 2$, so the canonical class is the hyperplane class and the canonical map is the original embedding, as exercise set 18.4.1 observes.

Exercise 16.2.1

1. Show that $\Omega_{\mathbb{A}^n_k}$ is free of rank n over $k[x_1, \ldots, x_n]$ with basis $dx_1, \ldots, dx_n$, directly from the universal property.
2. Compute $\operatorname{div}(dx)$ on $\mathbb{P}^1_k$ and verify $\deg K = -2$.
3. Let C be the nonsingular cubic $\mathcal{Z}(y^2z - x^3 - axz^2 - bz^3)$. Show that $\omega = dx/y$ is regular and nowhere zero on C, so that $K_C = 0$ directly.
4. Show that a nonsingular plane curve of degree $d \geq 4$ has $\deg K_C > 0$, and deduce that it admits no nonconstant map to $\mathbb{P}^1_k$ of degree 1.
5. Use proposition 16.1.3 to give a second proof that the cuspidal cubic $\mathcal{Z}(y^2 - x^3)$ is singular at the origin, by computing Ω there.
6. Show that $\operatorname{div}(\omega)$ for ω a nonzero rational form depends on ω only up to linear equivalence, and that every divisor in the class K_C arises as $\operatorname{div}(\omega)$ for some ω.

17

Blowing Up and Resolution of Curve Singularities

Chapter 7 already resolved the singularities of a curve: corollary 7.4.2 says the normalization of any curve is smooth. That proof is an existence statement obtained by taking an integral closure, and it says nothing about what the smooth model looks like or how singular the original was.

This chapter gives the geometric construction. Blowing up replaces a point by the set of directions through it, separating branches that were tangled together, and it is explicit enough to compute with: the multiplicity of a curve at a point is visible as the multiplicity with which the exceptional divisor appears in the total transform, and the tangent lines of definition 12.1.6 reappear as the points where the strict transform crosses it. The construction also survives into dimensions where normalization is not enough, which is why it, and not the integral closure, is what resolution of singularities means in general.

17.1 The Blow-Up of the Plane at a Point

A point of $\mathbb{P}^1_k$ is a direction through the origin of $\mathbb{A}^2_k$. The blow-up is the variety obtained by attaching all of those directions to the plane at once, in such a way that a curve through the origin remembers which direction it arrived from.

Definition 17.1.1: Blow-Up of the Plane at the Origin

Let $0 = (0,0) \in \mathbb{A}^2_k$ and give $\mathbb{P}^1_k$ homogeneous coordinates $[s:t]$. The *blow-up* of $\mathbb{A}^2_k$ at the origin is

$$\mathrm{Bl}_0\mathbb{A}^2_k = \big\{\big((x,y),[s:t]\big) \in \mathbb{A}^2_k \times \mathbb{P}^1_k \;\big|\; xt = ys\big\}$$

with $\pi\colon \mathrm{Bl}_0\mathbb{A}^2_k \to \mathbb{A}^2_k$ the first projection. The *exceptional divisor* is $E = \pi^{-1}(0)$.

The defining equation says that the vector (x,y), when nonzero, points in the direction $[s:t]$. So away from the origin the second coordinate is determined by the first, and over the origin it is unconstrained.

Proposition 17.1.2: Basic Properties of the Blow-Up

Let $B = \mathrm{Bl}_0\mathbb{A}^2_k$. Then:

1. B is closed in $\mathbb{A}^2_k \times \mathbb{P}^1_k$, and is covered by two open sets each isomorphic to $\mathbb{A}^2_k$;
2. B is an irreducible quasi-projective variety of dimension 2;
3. π restricts to an isomorphism $B \setminus E \to \mathbb{A}^2_k \setminus \{0\}$;
4. $E = \{0\} \times \mathbb{P}^1_k \cong \mathbb{P}^1_k$, and π is not an isomorphism at any point of E.

***Proof*:**
(1). The equation $xt = ys$ is homogeneous of degree 1 in $[s:t]$, so its vanishing is well defined on $\mathbb{A}^2_k \times \mathbb{P}^1_k$, and being closed is a local condition on the cover by $\{s \neq 0\}$ and $\{t \neq 0\}$. On $\{s \neq 0\} \cong \mathbb{A}^2_k \times \mathbb{A}^1_k$, with $v = t/s$ and coordinates (x,y,v), the equation reads $y = xv$, whose zero set is the graph of $(x,v) \mapsto xv$ and so is closed and isomorphic to $\mathbb{A}^2_k$ with coordinates (x,v). Call this chart U_s; then

$$\pi|_{U_s}(x,v) = (x,xv)$$

Symmetrically the chart U_t where $t \neq 0$ is $\mathbb{A}^2_k$ with coordinates (u,y), where $u = s/t$ and $x = uy$, and $\pi|_{U_t}(u,y) = (uy,y)$. The two charts cover B, which is therefore closed.

(2). Each U_s, U_t is irreducible of dimension 2, being isomorphic to $\mathbb{A}^2_k$, and they meet, since the point $((1,1),[1:1])$ lies in both. Lemma 10.1.5 makes B irreducible, and corollary 4.3.3 reads its dimension off either chart as 2.

(3). For $(x,y) \neq (0,0)$ the equation $xt = ys$ determines $[s:t]$ uniquely, namely $[x:y]$, so π is bijective there with inverse $(x,y) \mapsto ((x,y),[x:y])$, which is regular because x and y do not vanish simultaneously.

(4). Setting $x = y = 0$ leaves $[s:t]$ free, so $E = \{0\} \times \mathbb{P}^1_k$. Any nonempty open neighbourhood of a point of E meets E in an infinite set, since $E \cong \mathbb{P}^1_k$ is irreducible, and π sends all of it to the origin. So π is not injective on any such neighbourhood, completing the proof.

The chart map $\pi|_{U_s}(x,v) = (x,xv)$ has been seen before. It is the map of example 10.5.2, whose fibre over the origin jumped from a point to a line, and that jump is now identified: the fibre is the exceptional divisor, and the jump is the whole point of the construction rather than a pathology.

What E records is direction, and the following makes that precise.

Proposition 17.1.3: The Exceptional Divisor Parameterizes Directions

Let $L = \mathcal{Z}(bx - ay) \subseteq \mathbb{A}^2_k$ be a line through the origin, with $[a : b] \in \mathbb{P}^1_k$. Then the closure in B of $\pi^{-1}(L \setminus \{0\})$ is a curve meeting E in the single point $(0, [a : b])$, and distinct lines meet E in distinct points.

***Proof*:**
Parameterize $L \setminus \{0\}$ by $\lambda \mapsto (\lambda a, \lambda b)$ with $\lambda \neq 0$. Its preimage under the isomorphism of proposition 17.1.2(3) is $\big((\lambda a, \lambda b), [\lambda a : \lambda b]\big) = \big((\lambda a, \lambda b), [a : b]\big)$, so the preimage lies in $\mathbb{A}^2_k \times \{[a : b]\}$, a closed set. Its closure therefore also lies there, and adding the limit $\lambda \to 0$ gives the point $(0, [a : b])$. Distinct lines through the origin have distinct $[a : b]$, completing the proof.

So two lines through the origin, which meet in the plane, become disjoint after blowing up: they are pulled apart by exactly the amount needed to record that they arrived along different directions. That is the mechanism by which a node will be resolved in section 17.3.

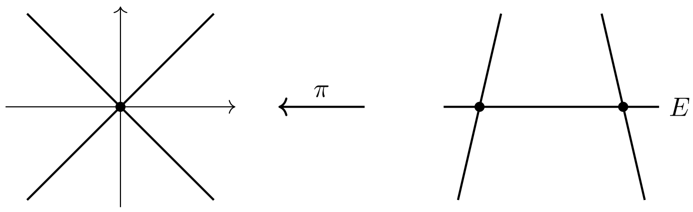


Figure 17.1: Blowing up separates the branches of a node. The exceptional divisor E records the direction along which each branch approaches the origin.

17.2 Blowing Up a Variety Along a Subvariety

Nothing in definition 17.1.1 was special to dimension two.

Definition 17.2.1: Blow-Up of Affine Space at the Origin

Give $\mathbb{P}^{n-1}_k$ homogeneous coordinates $[\ell_1 : \cdots : \ell_n]$. The blow-up of $\mathbb{A}^n_k$ at the origin is

$$\mathrm{Bl}_0\mathbb{A}^n_k = \big\{(x, [\ell]) \in \mathbb{A}^n_k \times \mathbb{P}^{n-1}_k \;\big|\; x_i\ell_j = x_j\ell_i \text{ for all } i, j\big\}$$

with π the first projection and $E = \pi^{-1}(0) \cong \mathbb{P}^{n-1}_k$.

The equations are exactly those saying the matrix with rows x and ℓ has rank at most one, which is the 2×2 minor condition of proposition 10.1.3. The proof of proposition 17.1.2 carries over verbatim with n charts in place of two: on the chart $\ell_i \neq 0$, setting $v_j = \ell_j/\ell_i$ gives $x_j = x_iv_j$ for $j \neq i$, so the chart is $\mathbb{A}^n_k$ with coordinates $(x_i, v_j)_{j\neq i}$ and

$$\pi(x_i, v) = (x_iv_1, \ldots, x_i, \ldots, x_iv_n)$$

Hence $\mathrm{Bl}_0\mathbb{A}_k^n$ is an irreducible quasi-projective variety of dimension n, isomorphic to $\mathbb{A}_k^n$ away from E, and E is a $\mathbb{P}_k^{n-1}$ parameterizing the lines through the origin.

Blowing up a subvariety rather than a point is the same construction with the point replaced by a linear subspace, at least locally.

Definition 17.2.2: Blow-Up Along a Coordinate Subspace

Let $Z = \mathcal{Z}(x_1, \dots, x_r) \subseteq \mathbb{A}_k^n$ be a coordinate subspace, $0 < r \leq n$. The blow-up of $\mathbb{A}_k^n$ along Z is

$$\mathrm{Bl}_Z\mathbb{A}_k^n = \left\{(x, [\ell]) \in \mathbb{A}_k^n \times \mathbb{P}_k^{r-1} \;\middle|\; x_i\ell_j = x_j\ell_i,\ 1 \leq i, j \leq r\right\}$$

with π the first projection. Its exceptional locus $\pi^{-1}(Z)$ is $Z \times \mathbb{P}_k^{r-1}$.

The last $n - r$ coordinates are carried along untouched, so $\mathrm{Bl}_Z\mathbb{A}_k^n \cong \mathrm{Bl}_0\mathbb{A}_k^r \times \mathbb{A}_k^{n-r}$; the construction is the point case in the transverse directions, applied uniformly along Z. The general blow-up along an arbitrary closed subvariety is not this concrete: it is defined as the Proj of the Rees algebra $\bigoplus_{d\geq 0} I^d$ of the ideal I of the subvariety, a construction that needs the language of [3]. Everything in this chapter happens at a point of a curve, where definition 17.2.1 suffices.

What is blown up is the ambient space; what one wants is the effect on a subvariety sitting inside it.

Definition 17.2.3: Strict and Total Transform

Let $X \subseteq \mathbb{A}_k^n$ be a closed subvariety and $\pi\colon \mathrm{Bl}_0\mathbb{A}_k^n \to \mathbb{A}_k^n$ the blow-up at a point $p \in X$. The *total transform* of X is $\pi^{-1}(X)$. The *strict transform* is

$$\tilde{X} = \overline{\pi^{-1}\big(X \setminus \{p\}\big)}$$

the closure being taken in $\mathrm{Bl}_0\mathbb{A}_k^n$.

The total transform always contains the whole exceptional divisor, which carries no information about X. The strict transform discards it and keeps only what is approached from X itself. By proposition 17.1.2(3) the map $\tilde{X} \to X$ is an isomorphism away from p, so $\tilde{X}$ is birational to X; and $\tilde{X}$ is irreducible whenever X is, being the closure of a set isomorphic to the irreducible $X \setminus \{p\}$.

17.3 Resolving Plane Curve Singularities

Now specialize to a plane curve. Everything reduces to one computation, carried out in the chart $\pi(x, v) = (x, xv)$.

Theorem 17.3.1: The Blow-Up of a Plane Curve at a Point

Let $f \in k[x, y]$ be an affine plane curve with $m = m_0(f) \geq 1$, and write $f = f_m + f_{m+1} + \cdots$ for its homogeneous decomposition, so that the tangent cone is $f_m = \prod_i L_i^{r_i}$ with $\sum_i r_i = m$. Then in the chart U_s with coordinates (x, v),

$$f(x, xv) = x^m \tilde{f}(x, v) \qquad \tilde{f}(x, v) = f_m(1, v) + x\, f_{m+1}(1, v) + x^2 f_{m+2}(1, v) + \cdots$$

and $\mathcal{Z}(\tilde{f})$ is the strict transform $\tilde{C}$. Moreover:

1. the exceptional divisor occurs in the total transform with multiplicity exactly m;
2. $\tilde{C}$ meets E exactly at the points corresponding, under proposition 17.1.3, to the tangent lines $\mathcal{Z}(L_i)$, and the intersection multiplicity there is r_i; in particular $\tilde{C} \cdot E$ has degree m;
3. at the point $q_i \in E$ corresponding to L_i, the multiplicity satisfies $m_{q_i}(\tilde{C}) \leq r_i \leq m$.

***Proof*:**
Each f_j is homogeneous of degree j, so $f_j(x, xv) = x^j f_j(1, v)$, and summing over $j \geq m$ gives the displayed factorization with $\tilde{f}$ as stated. Since E is cut out by $x = 0$ in this chart, the factor x^m is the exceptional divisor with multiplicity m, which is (1). The remaining factor $\tilde{f}$ does not vanish identically on E: writing $f_m = \sum_j c_j x^{m-j} y^j$ gives $\tilde{f}(0, v) = f_m(1, v) = \sum_j c_j v^j$, which is the zero polynomial only if every $c_j = 0$, that is only if $f_m = 0$. So $\mathcal{Z}(\tilde{f})$ contains no component of E, and away from E it agrees with $\pi^{-1}(C)$; hence $\mathcal{Z}(\tilde{f})$ is the closure of $\pi^{-1}(C \setminus \{0\})$, which is $\tilde{C}$.

(2). On E the equation of $\tilde{C}$ is $f_m(1, v) = 0$. Writing $L_i = b_i x - a_i y$, so that $\mathcal{Z}(L_i)$ is the line through the origin in direction $[a_i : b_i]$, gives $f_m(1, v) = \prod_i (b_i - a_i v)^{r_i}$ up to a scalar, whose roots are $v = b_i / a_i$ when $a_i \neq 0$. Under proposition 17.1.3 the point $v = b_i/a_i$ of E is the direction $[a_i : b_i]$, that is the tangent line $\mathcal{Z}(L_i)$; the roots with $a_i = 0$ are the tangent line $\mathcal{Z}(x)$ and appear in the chart U_t. The multiplicity of the root is r_i, and $\sum_i r_i = m$.

(3). Translate so that q_i is the origin of the chart, replacing v by $v_i + w$. Then

$$\tilde{f} = f_m(1, v_i + w) + x\, f_{m+1}(1, v_i + w) + \cdots$$

The first summand has order exactly r_i in w, by (2). Every later summand is divisible by x, hence has order at least 1. So the lowest-degree part of $\tilde{f}$ has degree at most r_i, which is (3), as we sought to show.

Part (3) is the statement that blowing up never makes things worse, and for the singularities named in definition 12.1.6 it makes them disappear outright.

Corollary 17.3.2: One Blow-Up Resolves an Ordinary Singularity

Let p be an ordinary singular point of multiplicity m on a plane curve C, so that C has m distinct tangent lines at p. Then $\tilde{C}$ meets E in m distinct points, each a nonsingular point of $\tilde{C}$, and each intersection is transversal.

***Proof*:**
Ordinariness says every $r_i = 1$ and there are m of them, so theorem 17.3.1(2) gives m distinct intersection points, each with intersection multiplicity 1, which is transversality. Theorem 17.3.1(3) bounds $m_{q_i}(\tilde{C}) \leq r_i = 1$, and a multiplicity is at least 1 at a point of the curve, so each q_i is nonsingular, completing the proof.

Example 17.3.3: Resolving a Node, a Cusp, and a Higher Cusp
All three computations use theorem 17.3.1 in the chart (x, v), where $y = xv$.

1. *The node* $f = y^2 - x^2 - x^3$. Here $f_2 = y^2 - x^2 = (y - x)(y + x)$, two distinct tangents, so the point is an ordinary double point. Substituting,
$$f(x, xv) = x^2v^2 - x^2 - x^3 = x^2\big(v^2 - 1 - x\big)$$
so $\tilde{f} = v^2 - 1 - x$, whose partial derivative in x is -1, nowhere zero: the strict transform is nonsingular. It meets $E = \mathcal{Z}(x)$ at $v = \pm 1$, the two tangent directions, as corollary 17.3.2 predicts.
2. *The cusp* $f = y^2 - x^3$. Here $f_2 = y^2$, a single tangent with $r_1 = 2$, so the point is *not* ordinary. Nonetheless
$$f(x, xv) = x^2v^2 - x^3 = x^2\big(v^2 - x\big)$$
and $\tilde{f} = v^2 - x$ is nonsingular. One blow-up suffices here too, though corollary 17.3.2 does not apply; the strict transform meets E at the single point $v = 0$, with intersection multiplicity 2.
3. *A higher cusp* $f = y^2 - x^5$. Again $f_2 = y^2$, and
$$f(x, xv) = x^2v^2 - x^5 = x^2\big(v^2 - x^3\big)$$
so $\tilde{f} = v^2 - x^3$: the blow-up has replaced the singularity by an ordinary cusp, not by a nonsingular point. Blowing up once more, by (2), resolves it. Two blow-ups are needed, and no fewer.

Item (3) is the general situation: one blow-up reduces a singularity but need not remove it, and the content of the resolution theorem is that the process stops.

Theorem 17.3.4: Resolution of Plane Curve Singularities

Let C be an irreducible plane curve. Then after finitely many blow-ups at points, the strict transform is a nonsingular curve birational to C.

***Proof*:**
Each blow-up is an isomorphism away from the point blown up by proposition 17.1.2(3), so the strict transform stays birational to C throughout, and C has only finitely many singular points by corollary 13.2.4, so they may be treated one at a time. Theorem 17.3.1(3) says the multiplicity at every point above the one blown up is at most the multiplicity below, so blowing up never makes a singularity worse, and corollary 17.3.2 removes an ordinary point in a single

step.

What those facts leave open is whether the multiplicity must eventually reach 1, rather than stabilizing forever at some $m \geq 2$. The input needed is that each singular point p carries a non-negative integer δ_p, the drop in arithmetic genus recorded in section 14.4, which strictly decreases at every blow-up above p; a strictly decreasing sequence of non-negative integers is finite, so the process stops. That δ_p behaves this way is a comparison of the local rings of C and of its strict transform, the same computation section 14.4 deferred, and it is carried out in [19, chapter 7]; the cohomological account is in [3]. Granting it, the process terminates and the final strict transform is nonsingular and birational to C, as we sought to show.

The theorem is not new information: corollary 7.4.2 already produced a smooth model, and section 7.3 showed it is unique in its birational class. So the smooth curve reached by blowing up is the normalization, arrived at by a different road. What blowing up adds is that the road is explicit and that each step is measured, which is the subject of the last section.

17.4 Multiplicity, Genus, and the Effect of Blowing Up

Blowing up trades a singular point for an intersection with the exceptional divisor, and theorem 17.3.1 says the trade is exact: a point of multiplicity m becomes m points of intersection with E, counted properly. That bookkeeping is what turns resolution into a computation of the genus.

Recall from section 14.4 that a nonsingular plane curve of degree d has genus $\binom{d-1}{2}$, and that for a singular curve the genus of the smooth model drops by $\binom{m_p}{2}$ at each ordinary singular point. Blowing up is where that number comes from.

Theorem 17.4.1: The Genus Drop at an Ordinary Singularity

Let C be an irreducible plane curve of degree d whose singular points $p_1, \ldots, p_s$ are all ordinary, of multiplicities $m_1, \ldots, m_s$, and let $\tilde{C}$ be its nonsingular model. Then

$$g(\tilde{C}) \;=\; \frac{(d-1)(d-2)}{2} \;-\; \sum_{i=1}^{s} \binom{m_i}{2}$$

***Proof*:**
The genus is computed by the space of regular differential forms, whose dimension is g by theorem 18.3.2, and on a plane curve those forms are described by *adjoints*. Say a curve G of degree e is an *adjoint* of C if $m_{p_i}(G) \geq m_i - 1$ at every singular point.

Adjoints of degree $d-3$ give the regular differentials. By theorem 16.2.4 a canonical divisor on a nonsingular plane curve of degree d is cut by a curve of degree $d-3$. For a curve with ordinary singularities the same computation applies on $\tilde{C}$, with the correction that a form $G/\partial_y f$ is regular at the points of $\tilde{C}$ over p_i exactly when G vanishes to order at least $m_i - 1$ at p_i: the denominator $\partial_y f$ vanishes to order $m_i - 1$ along each of the m_i branches, and $\tilde{C}$ separates the branches by corollary 17.3.2. So the regular differential forms on $\tilde{C}$ correspond to the adjoint curves of degree $d-3$, modulo those divisible by f.

Counting them. The curves of degree $d-3$ form a space of dimension $\binom{d-1}{2}$, since $\binom{d-3+2}{2} =$

$\binom{d-1}{2}$. Requiring $m_{p_i}(G) \geq m_i - 1$ imposes $\binom{m_i}{2}$ linear conditions, being the vanishing of all partial derivatives of order less than $m_i - 1$, and for ordinary singularities in distinct positions these conditions are independent, by the same argument that established independence in corollary 13.2.5. Nothing of degree $d - 3 < d$ is divisible by f. Hence

$$g(\tilde{C}) = \binom{d-1}{2} - \sum_{i=1}^{s} \binom{m_i}{2}$$

as we sought to show.

The formula was quoted in section 14.4 and used in corollaries 17.4.3 and 17.4.4; it is now proved, and those two corollaries are unconditional.

17.4.2 Note: Two Routes to the Same Number The proof above counts conditions on adjoint curves and never mentions the blow-up. There is a second route, standard in the literature, which computes the arithmetic genus of the strict transform on the blown-up surface: the total transform is $\tilde{C} + mE$, and the correction $\frac{1}{2}m(m-1)$ falls out of the self-intersection $E \cdot E = -1$ together with $\tilde{C} \cdot E = m$ from theorem 17.3.1(2).
That route was not taken here, deliberately. It requires an intersection pairing for divisors on a surface, which this book does not build and which [3] develops as its first genuine intersection product, with the general theory in [8]. The adjoint count needs only linear systems and the canonical class, both of which are now in hand, so the hub proves what it needs without borrowing from downstream.

Two consequences can be read off without that ingredient, because they only use the non-negativity of a genus.

Corollary 17.4.3: Blowing Up Terminates for Ordinary Singularities

Let C be an irreducible plane curve of degree d all of whose singular points are ordinary. Then one blow-up at each removes them all, the total drop in arithmetic genus is $\sum_p \binom{m_p}{2}$, and the number of blow-ups is at most

$$\frac{(d-1)(d-2)}{2}$$

***Proof*:**
A blow-up at an ordinary point of multiplicity m removes that point by corollary 17.3.2, and by theorem 17.4.1 it drops the arithmetic genus by $\binom{m}{2}$, which is at least 1 because a singular point has $m \geq 2$. Summing over the singular points, finite in number by corollary 13.2.4, gives the total drop.

The arithmetic genus starts at $\binom{d-1}{2}$ by definition 14.4.1 and ends at the genus of a smooth curve, which is non-negative, so the total drop is at most $\binom{d-1}{2}$. Since each blow-up contributes at least 1 to that drop, there are at most $\binom{d-1}{2}$ of them, completing the proof.

Corollary 17.4.4: The Multiplicity Bound, Read Geometrically

For an irreducible plane curve of degree d with ordinary singularities,

$$\sum_p \binom{m_p}{2} \le \frac{(d-1)(d-2)}{2}$$

with equality exactly when the smooth model has genus 0.

***Proof*:**
This is corollary 17.4.3 restated: the total drop is bounded by the starting genus, with equality exactly when the final genus is 0, completing the proof.

The equality case is a rationality statement, though not by way of theorem 14.4.2, which was about plane curves and the smooth model of a singular plane curve is generally not one. That a smooth projective curve of genus 0 is isomorphic to $\mathbb{P}^1_k$ is true and is one of the payoffs of chapter 18⋆. For the extremal curve it can be seen by hand: on $\mathcal{Z}(x^d + y^{d-1}z)$ one may solve $z = -x^d/y^{d-1}$, so

$$\mathbb{P}^1_k \to \mathcal{Z}(x^d + y^{d-1}z) \qquad [x : y] \mapsto [\, xy^{d-1} : y^d : -x^d \,]$$

parameterizes it.

That inequality was already proved in chapter 13, as corollary 13.2.5, by an entirely different argument: a dimension count on the curves of degree $d-1$ passing through the singular points with prescribed multiplicity. The two proofs agree, and the agreement is informative. The Bézout argument says the bound holds because there are not enough curves of degree $d-1$ to impose more conditions; the blow-up argument says it holds because a genus cannot be negative. The extremal curve $\mathcal{Z}(x^d+y^{d-1}z)$ of section 13.2 attains the bound, and corollary 17.4.4 now explains why it is rational.

17.5 Cubic Surfaces and the Twenty-Seven Lines

Theorem 11.4.3 showed that a general cubic surface carries finitely many lines, by a dimension count that never produced a number. The number is 27, and the route to it is the blow-up: a smooth cubic surface is $\mathbb{P}^2_k$ blown up at six points, and under that description the lines can simply be listed.

Throughout, $\operatorname{char} k \neq 2$ and the six points $p_1, \ldots, p_6 \in \mathbb{P}^2_k$ are in *general position*, meaning no three collinear and not all six on a conic.

Proposition 17.5.1: The Blow-Up of Six Points Is a Cubic Surface

Let $p_1, \ldots, p_6 \in \mathbb{P}^2_k$ be in general position and let S be the blow-up of $\mathbb{P}^2_k$ at the six points. The cubics through the six points form a linear system of projective dimension 3, and the map it defines embeds S as a cubic surface in $\mathbb{P}^3_k$.

***Proof*:**
Cubic forms in three variables constitute a space of dimension $\binom{5}{3} = 10$, so the cubics of $\mathbb{P}^2_k$ form a $\mathbb{P}^9_k$. Passing through a point is one linear condition, and for six points in general position the six conditions are independent: were they dependent, some p_i would lie on every cubic through the other five, which fails because the union of a conic through five of them and a line missing the sixth, or of three lines pairing them up, gives cubics through five and not the sixth. So the system has projective dimension $9 - 6 = 3$.

Let $\varphi\colon S \to \mathbb{P}^3_k$ be the map defined by that system, which is a morphism on S because blowing up the six points resolves the base locus. Its image is a surface, and the degree of the image is the number of points in which two general members of the system meet away from the p_i: two cubics meet in 9 points by theorem 13.1.1, of which six are the base points, leaving 3. So the image is a cubic surface, and φ is birational onto it since a general pair of members separates points of $\mathbb{P}^2_k$ away from the base locus.

That φ is an embedding, rather than merely birational, is the statement that the system separates points and tangent vectors on S; it does so exactly because no three of the p_i are collinear and the six do not lie on a conic, which are the configurations making two members of the system agree to first order somewhere. See [14, chapter V.4] for the verification, as we sought to show.

With the description in hand the lines can be counted, and the count is a matter of listing three families.

Theorem 17.5.2: The Twenty-Seven Lines

Let $S \subseteq \mathbb{P}^3_k$ be a smooth cubic surface, realized as the blow-up of $\mathbb{P}^2_k$ at six points $p_1, \dots, p_6$ in general position. Then S contains exactly 27 lines, namely

1. the six exceptional curves $E_1, \dots, E_6$ over the six points;
2. the $\binom{6}{2} = 15$ strict transforms of the lines $\overline{p_i p_j}$;
3. the six strict transforms of the conics through five of the six points.

In total $6 + 15 + 6 = 27$.

***Proof*:**
These are lines. A curve on S is a line exactly when the linear system of proposition 17.5.1 cuts it in a single point, that is when its image under φ has degree 1. The system consists of cubics through the six points, so its degree on a curve $D \subseteq \mathbb{P}^2_k$ is $3 \deg D$ less one for each base point on D, counted with multiplicity.

For an exceptional curve E_i the system restricts to the pencil of cubics through p_i with varying tangent direction, cutting E_i once, so $\varphi(E_i)$ is a line. For the line $\overline{p_i p_j}$ we get $3 \cdot 1 - 1 - 1 = 1$. For a conic Q through five of the six, $3 \cdot 2 - 5 = 1$. So all 27 are lines.

There are no others. Let $L \subseteq S$ be a line and let $D \subseteq \mathbb{P}^2_k$ be its image, of degree $e \geq 0$, passing through p_i with multiplicity m_i. The degree computation above reads

$$3e - \sum_{i=1}^{6} m_i = 1$$

and a second constraint comes from the genus: $L \cong \mathbb{P}^1_k$ has genus 0, while by section 14.4 the normalization of a plane curve of degree e with multiplicities m_i has genus

$$\frac{(e-1)(e-2)}{2} - \sum_{i=1}^{6} \binom{m_i}{2} = 0$$

So $\sum_i m_i = 3e - 1$ and $\sum_i \binom{m_i}{2} = \binom{e-1}{2}$. The case $e = 0$ is an exceptional curve. For $e \geq 1$ these two equations already bound e. The function $m \mapsto \binom{m}{2}$ is convex, so by Jensen's inequality applied to the six values m_i, whose mean is $(3e-1)/6$,

$$\binom{e-1}{2} = \sum_{i=1}^{6} \binom{m_i}{2} \geq 6\binom{(3e-1)/6}{2} = \frac{(3e-1)(3e-7)}{12}$$

Multiplying by 12 gives $6(e-1)(e-2) \geq (3e-1)(3e-7)$, that is $3e^2 - 6e - 5 \leq 0$, so $e \leq 2$.

It remains to check $e = 1$ and $e = 2$. For $e = 1$ the genus equation reads $\sum \binom{m_i}{2} = 0$, so every $m_i \leq 1$, and $\sum m_i = 2$ picks out two of the six points: a line through p_i and p_j. For $e = 2$ again every $m_i \leq 1$, and $\sum m_i = 5$ picks out five: a conic through five of the six. Counting the choices gives 6, $\binom{6}{2} = 15$ and $\binom{6}{5} = 6$, as we sought to show.

17.5.3 Note: What the Count Is Really About Three features of the argument deserve separating.

The number 27 does not depend on the six points. Every smooth cubic surface arises as such a blow-up, so the count is intrinsic, and the configuration of the 27 lines is the same for all of them: each line meets exactly 10 others, and the whole configuration has a symmetry group of order 51840, the Weyl group of the root system E_6. That is the beginning of a story this book does not tell; see [14] and, for the representation theory, [12].

The dimension count of section 11.4 and the present listing answer different questions. The first shows finiteness for a *general* cubic without exhibiting a single line; the second exhibits all of them for a *smooth* one. Neither subsumes the other, and the first is what guarantees the second is not vacuous.

The general-position hypothesis is doing real work. If three of the six points are collinear, the strict transform of that line has $3 \cdot 1 - 3 = 0$ and is contracted rather than embedded, and the image acquires a singular point. Cubic surfaces with singularities have fewer lines: the surface $\mathcal{Z}(x_0x_1x_2 - x_3^3)$ of lemma 11.4.2 has three.

Exercise 17.5.1

1. Show that $\mathrm{Bl}_0\mathbb{A}^2_k$ is not isomorphic to $\mathbb{A}^2_k$. Suggestion: $E \cong \mathbb{P}^1_k$ is a complete curve sitting inside it, whereas a curve inside $\mathbb{A}^2_k$ carries nonconstant regular functions, which corollary 10.2.4 forbids on a projective variety.

2. Compute the strict transform of the curve $y^2 = x^4 + x^5$ under one blow-up at the origin, identify its singularities, and say how many further blow-ups are needed.

3. Show that the blow-up of $\mathbb{A}^2_k$ at the origin, restricted to the chart U_t, is the map $(u, y) \mapsto (uy, y)$, and redo example 17.3.3(2) in that chart. Explain why the answer looks different and why

the two are consistent.

4. Let C be a plane curve with an ordinary point of multiplicity m at the origin. Show that the strict transform meets E transversally in m points, and deduce that $\tilde{C} \cdot E$ has degree m whether or not the point is ordinary.

5. Show that $\mathrm{Bl}_Z \mathbb{A}^n_k \cong \mathrm{Bl}_0 \mathbb{A}^r_k \times \mathbb{A}^{n-r}_k$ for Z a coordinate subspace of codimension r, as claimed after definition 17.2.2.

6. A curve of degree 4 can have at most three double points. Prove this from corollary 17.4.4, and exhibit a quartic attaining the bound.

18⋆

Riemann-Roch for Plane Curves

⋆ *Optional. This chapter is an aside; nothing later in the book depends on it.*

Every question this book has asked about a curve reduces, sooner or later, to the same one: given a divisor D, which rational functions have poles no worse than D allows? Chapter 14 met the question in the special case $D = 0$, where proposition 14.2.3 answered it, and again in theorem 15.2.2, where the answer for $D = p - q$ decided injectivity. The Riemann-Roch theorem answers it in general, up to a correction term that vanishes once D is large.

Nothing earlier in the book depends on this chapter, and the theorem is proved in full elsewhere in the series: analytically in *Complex Analysis* [17], cohomologically in *Algebraic Geometry* [3]. What is offered here is the plane-curve account, where the statement can be made concrete and the consequences are immediate. Section 18.1 builds the vector spaces and proves what can be proved directly; sections 18.2 and 18.3 state Riemann's theorem and Riemann-Roch and say how the proof goes; section 18.4 spends the theorem, including on two debts left open in chapters 14 and 17.

Throughout, C is a nonsingular projective curve over the algebraically closed field k, and divisors are as in chapter 14.

18.1 The Space of Functions with Prescribed Poles

Definition 18.1.1: The Space $L(D)$

Let D be a divisor on C. The *Riemann-Roch space* of D is

$$L(D) = \{f \in k(C)^* \mid \operatorname{div}(f) + D \geq 0\} \cup \{0\}$$

and $\ell(D) = \dim_k L(D)$.

Writing $D = \sum_p n_p\, p$, the condition $\operatorname{div}(f) + D \geq 0$ says $\operatorname{ord}_p(f) \geq -n_p$ at every p: where $n_p > 0$ the function may have a pole of order at most n_p, and where $n_p < 0$ it is required to vanish to order at least $-n_p$. So $L(D)$ is the space of functions whose poles are bounded by D and whose zeros are forced by it.

Proposition 18.1.2: Basic Properties of $L(D)$

Let D and D' be divisors on C. Then:

1. $L(D)$ is a k-vector space;
2. $L(0) = k$, so $\ell(0) = 1$;
3. if $\deg D < 0$ then $L(D) = 0$;
4. if $D \leq D'$ then $L(D) \subseteq L(D')$ and $\ell(D') - \ell(D) \leq \deg D' - \deg D$;
5. $\ell(D) \leq \deg D + 1$ whenever $\deg D \geq 0$;
6. if $D \sim D'$ then $\ell(D) = \ell(D')$, so ℓ is a function on $\operatorname{Pic}(C)$.

***Proof*:**
(1). For nonzero $f, g \in L(D)$ and $\lambda \in k^*$, the valuation inequality $\operatorname{ord}_p(f+g) \geq \min(\operatorname{ord}_p(f), \operatorname{ord}_p(g))$ holds at every p, and $\operatorname{ord}_p(\lambda f) = \operatorname{ord}_p(f)$, so $L(D)$ is closed under addition and scaling.

(2). $L(0)$ consists of the f with $\operatorname{div}(f) \geq 0$, that is with $\operatorname{ord}_p(f) \geq 0$ at every p. Such an f lies in every local ring $\mathcal{O}_p(C)$ and so is regular on all of C, hence constant by corollary 10.2.4. Conversely every constant lies in $L(0)$.

(3). If $f \in L(D)$ is nonzero then $\operatorname{div}(f) + D \geq 0$, so $\deg \operatorname{div}(f) + \deg D \geq 0$; but $\deg \operatorname{div}(f) = 0$ by theorem 14.2.2, so $\deg D \geq 0$.

(4). The inclusion is immediate from the definition. For the inequality it suffices, by induction on $\deg D' - \deg D$, to treat $D' = D + p$ for a single point p, and to show $\ell(D+p) \leq \ell(D) + 1$. Let n be the coefficient of p in D and let t be a uniformizer at p. The map

$$L(D+p) \to k \qquad f \mapsto \left(t^{n+1} f\right)(p)$$

is well defined, since $f \in L(D+p)$ has $\operatorname{ord}_p(f) \geq -n-1$ so $t^{n+1}f$ lies in $\mathcal{O}_p(C)$, and it is k-linear. Its kernel is exactly $L(D)$, because $\left(t^{n+1}f\right)(p) = 0$ says $\operatorname{ord}_p(f) \geq -n$. So $L(D+p)/L(D)$ embeds in k and the codimension is at most 1.

(6). Suppose $D' = D + \mathrm{div}(h)$ for some $h \in k(C)^*$, and consider $f \mapsto f/h$. For $f \in L(D)$,

$$\mathrm{div}(f/h) + D' = \mathrm{div}(f) - \mathrm{div}(h) + D + \mathrm{div}(h) = \mathrm{div}(f) + D \;\geq\; 0$$

so $f/h \in L(D')$. The map is k-linear and multiplication by h inverts it, so $L(D) \cong L(D')$ and $\ell(D) = \ell(D')$.

(5). If $L(D) = 0$ then $\ell(D) = 0 \leq \deg D + 1$. Otherwise choose a nonzero $f \in L(D)$ and put $D_0 = \mathrm{div}(f) + D$, which is effective by the definition of $L(D)$ and satisfies $D_0 \sim D$, hence $\deg D_0 = \deg D$ and $\ell(D_0) = \ell(D)$ by (6). Now $0 \leq D_0$, so (4) applied to the pair $(0, D_0)$ gives

$$\ell(D_0) - \ell(0) \;\leq\; \deg D_0 - 0 = \deg D$$

and $\ell(0) = 1$ by (2), so $\ell(D) = \ell(D_0) \leq \deg D + 1$, as we sought to show.

Item (5) already shows the size of $L(D)$ is controlled by $\deg D$ from above. Riemann's theorem is the statement that it is controlled from below as well, by the same quantity shifted by an invariant of the curve.

Example 18.1.3: The Spaces $L(D)$ on the Projective Line

Take $C = \mathbb{P}^1_k$ and $D = n\,\infty$ for $n \geq 0$, where $\infty = [1:0]$. A rational function with poles only at ∞, of order at most n, is a polynomial in the affine coordinate x of degree at most n. So

$$L(n\infty) = \{a_0 + a_1 x + \cdots + a_n x^n \mid a_i \in k\} \qquad \ell(n\infty) = n + 1$$

The bound of proposition 18.1.2(5) is attained for every $n \geq 0$. Since every divisor of degree n on $\mathbb{P}^1_k$ is linearly equivalent to $n\infty$ by proposition 14.2.4, item (6) gives $\ell(D) = \deg D + 1$ for every divisor of non-negative degree on the line. The line is the curve on which $L(D)$ is as large as it can possibly be.

18.2 Riemann's Theorem

On $\mathbb{P}^1_k$ the inequality $\ell(D) \leq \deg D + 1$ is an equality. On any other curve it is not, and the deficiency is bounded.

Theorem 18.2.1: Riemann's Theorem

Let C be a nonsingular projective curve. There is a smallest integer $g \geq 0$, the *genus* of C, such that

$$\ell(D) \;\geq\; \deg D + 1 - g$$

for every divisor D on C. Moreover equality holds for all D of sufficiently large degree.

Two things are worth extracting before saying where the proof comes from. First, taking $D = 0$ gives $\ell(0) = 1 \geq 1 - g$, so $g \geq 0$ automatically; and $g = 0$ forces $\ell(D) \geq \deg D + 1$, which combined with proposition 18.1.2(5) makes it an equality, the situation of example 18.1.3. Second, the genus so defined is an invariant of C as an abstract curve, since $L(D)$ and $\deg D$ are defined from $k(C)$ and the valuations alone. That is what section 14.4 could not say about $\binom{d-1}{2}$, which was read off an embedding.

18.2.2 Note: Where the Proof Comes From For a plane curve the proof is a counting argument built on machinery this book already has. Let C be a nonsingular plane curve of degree d and H a line, so $H \cdot C$ is a divisor of degree d. One computes $\ell(mH \cdot C)$ for large m by identifying $L(mH \cdot C)$ with the space of curves of degree m modulo those containing C, and the identification is exactly the local-to-global statement of theorem 13.3.2: a form of degree m cuts out a divisor $\geq$ a given one precisely when Noether's condition holds at each point. The dimension of the space of forms of degree m in three variables is $\binom{m+2}{2}$, and subtracting the forms divisible by C, of which there are $\binom{m-d+2}{2}$, leaves

$$\binom{m+2}{2} - \binom{m-d+2}{2} = md - \frac{d(d-3)}{2} = \deg(mH \cdot C) + 1 - \frac{(d-1)(d-2)}{2}$$

which is Riemann's bound with $g = \binom{d-1}{2}$, the number definition 14.4.1 already called the genus. Extending from the divisors $mH \cdot C$ to arbitrary D is proposition 18.1.2(4) together with the fact that every divisor is dominated by some $mH \cdot C$. The details, in particular the surjectivity of the map from forms to $L(mH \cdot C)$, are in [19, chapter 8].

The computation in that box is worth pausing on, because it is the same arithmetic as example 8.5.4(3): $md - d(d-3)/2$ is the Hilbert polynomial of a plane curve of degree d. The Hilbert polynomial counts forms of degree m modulo those vanishing on C, and Riemann's theorem counts functions with poles bounded by $mH \cdot C$; Max Noether's theorem is what identifies the two counts.

Corollary 18.2.3: Genus of a Nonsingular Plane Curve

The genus of theorem 18.2.1 agrees, for a nonsingular plane curve of degree d, with the genus of definition 14.4.1:

$$g(C) = \frac{(d-1)(d-2)}{2}$$

***Proof*:**
The computation of box 18.2.2 produces the bound of theorem 18.2.1 with $g = \binom{d-1}{2}$, so the genus is at most that. It is not smaller: were the bound to hold with some $g' < g$, then taking $D = mH \cdot C$ for large m would give $\ell(D) \geq \deg D + 1 - g' > \deg D + 1 - g$, contradicting that the same computation evaluates $\ell(D)$ exactly as $\deg D + 1 - g$ for large m, completing the proof.

This is the statement section 14.4 could only assert: the number computed from the embedding is the intrinsic invariant. It is also what makes theorem 14.4.2 more than a coincidence of small degrees.

18.3 The Canonical Divisor and Riemann-Roch

Riemann's theorem leaves an inequality, and the error term has a name. It is measured by a divisor class attached to the curve itself.

Definition 18.3.1: Canonical Divisor

A *canonical divisor* on C is a divisor K with the property that

$$\ell(D) - \ell(K - D) = \deg D + 1 - g$$

for every divisor D. Its class in $\mathrm{Pic}\,(C)$ is the *canonical class.*

Stated this way the definition presumes what has to be proved, namely that such a K exists. Theorem 16.2.3 has already constructed the class: it is the divisor class of any nonzero rational differential form, and theorem 16.2.4 computes it for a plane curve. What is still unproved at this point is that the class so constructed satisfies the displayed identity, which is the content of theorem 18.3.2.

Theorem 18.3.2: Riemann-Roch

Let C be a nonsingular projective curve of genus g. Then a canonical divisor K exists, and for every divisor D,

$$\ell(D) - \ell(K - D) = \deg D + 1 - g$$

Moreover $\deg K = 2g - 2$ and $\ell(K) = g$.

The last two assertions are consequences of the first. Taking $D = 0$ gives $1 - \ell(K) = 1 - g$, so $\ell(K) = g$; taking $D = K$ gives $\ell(K) - 1 = \deg K + 1 - g$, and substituting $\ell(K) = g$ leaves $\deg K = 2g - 2$. Riemann's theorem follows too: $\ell(K - D) \geq 0$ always, so $\ell(D) \geq \deg D + 1 - g$, and once $\deg D > 2g - 2$ the divisor $K - D$ has negative degree, so $\ell(K - D) = 0$ by proposition 18.1.2(3) and the inequality becomes an equality. That is the precise form of "sufficiently large degree" in theorem 18.2.1.

Proposition 18.3.3: The Canonical Divisor of a Plane Curve

Let C be a nonsingular plane curve of degree d and H a line. Then the canonical class of C is

$$K = (d - 3)\, H \bullet C$$

***Proof*:**
This is theorem 16.2.4, proved in chapter 16 by exhibiting the form $\omega = dx/\partial_y f$ and computing its divisor: it has neither zeros nor poles in the affine chart, because C is nonsingular and its two partials cannot vanish together, and it acquires a zero of order $d - 3$ at each point of C at infinity.

The degrees are consistent with theorem 18.3.2, as they must be: $H \bullet C$ has degree d by theorem 13.1.1, so the right side has degree $d(d - 3)$, while $\deg K = 2g - 2$ with $g = \binom{d-1}{2}$ by corollary 18.2.3 gives

$$2g - 2 = (d - 1)(d - 2) - 2 = d^2 - 3d = d(d - 3)$$

completing the proof.

The degree check alone would not have sufficed: a divisor class is not determined by its degree,

$\mathrm{Pic}^0(C)$ being generally nonzero, and on a cubic theorem 15.2.2 makes it as large as the curve. It is the explicit form ω that identifies the class.

The formula explains the numerology of low-degree plane curves at a glance. For $d = 1$ and $d = 2$ the canonical class has negative degree, for $d = 3$ it is zero, and for $d \geq 4$ it is positive. Those three regimes are the trichotomy that organizes the classification of curves, and the next section is the case analysis.

18.4 Application: Curves of Low Genus

Riemann-Roch is used the same way every time: choose D so that $\ell(K - D)$ vanishes for degree reasons, read off $\ell(D)$, and interpret a function in $L(D)$ geometrically. One input is needed from outside, and it is the only one.

18.4.1 Note: The Degree of a Map to the Line Let $f \in k(C)$ be nonconstant. By lemma 12.2.8 it defines a morphism $F\colon C \to \mathbb{P}^1_k$, and the fact used below is that

$$\deg \mathrm{div}_\infty(f) = \left[\, k(C) : k(f) \,\right]$$

so that a function with a single simple pole generates the whole function field. Both sides count the fibre of F, one geometrically and one algebraically; the proof is in [19, chapter 8].

Theorem 18.4.2: Genus Zero Curves Are Rational

Let C be a nonsingular projective curve of genus 0. Then $C \cong \mathbb{P}^1_k$.

***Proof*:**
Pick any $p \in C$ and apply theorem 18.3.2 with $D = p$. Then $\deg(K - p) = -2 < 0$, so $\ell(K - p) = 0$ by proposition 18.1.2(3), and

$$\ell(p) = \deg(p) + 1 - 0 = 2$$

Since $L(0) = k$ has dimension 1 by proposition 18.1.2(2), there is a nonconstant $f \in L(p)$. Its only possible pole is a simple one at p, and it must occur, since otherwise $\mathrm{div}(f) \geq 0$ would make f constant. So $\mathrm{div}_\infty(f) = p$ has degree 1, and box 18.4.1 gives $k(C) = k(f)$, that is C is birational to $\mathbb{P}^1_k$. A birational map between nonsingular projective curves is an isomorphism by lemma 12.2.8 applied in both directions, completing the proof.

Example 18.4.3: Genus Zero, Worked
Take $C = \mathcal{Z}(x^2 + y^2 - z^2) \subseteq \mathbb{P}^2_k$, a nonsingular conic, so $d = 2$ and $g = \binom{1}{2} = 0$ by definition 14.4.1.

Follow the proof of theorem 18.4.2 with $p = [1 : 0 : 1]$. Riemann-Roch gives $\ell(p) = 2$, so there is a nonconstant f with at most a simple pole at p and no other pole. Concretely, in the affine chart $z = 1$ the function

$$f = \frac{y}{1 - x}$$

does it. Substituting the parameterization $x = (t^2-1)/(t^2+1)$, $y = 2t/(t^2+1)$ gives $f = t$ identically, and that parameterization sends $t = 0$ to $[-1:0:1]$ and $t = \infty$ to p. So

$$\operatorname{div}(f) = \big([-1:0:1]\big) - \big([1:0:1]\big)$$

a single simple zero and a single simple pole. Hence $\deg \operatorname{div}_\infty(f) = 1$ and box 18.4.1 gives $k(C) = k(f)$.

So the abstract argument and the explicit parameterization of example 3.3.2(1) are the same map, and the genus is what guaranteed in advance that one exists.

This settles the debt left in section 17.4: the smooth model of a plane curve attaining the bound of corollary 13.2.5 has genus 0 and is therefore rational, so the extremal curves are exactly the rational ones.

Theorem 18.4.4: Genus One Curves Are Plane Cubics

Let C be a nonsingular projective curve of genus 1 and $O \in C$. Then C is isomorphic to a nonsingular plane cubic, embedded by $L(3O)$, and O maps to an inflection point.

***Proof*:**
Here $\deg K = 2g - 2 = 0$ and $\ell(K) = g = 1$, so $L(K)$ contains a nonzero f with $\operatorname{div}(f) + K \geq 0$ of degree 0, hence $\operatorname{div}(f) + K = 0$ and $K \sim 0$. So for any D of positive degree, $\deg(K - D) < 0$ and theorem 18.3.2 reads

$$\ell(D) = \deg D$$

Applying this to O, $2O$ and $3O$ gives dimensions $1, 2, 3$. Choose $x \in L(2O) \setminus L(O)$ and $y \in L(3O) \setminus L(2O)$, so that x has a double pole at O and y a triple one.

Now count in $L(6O)$, of dimension 6. The seven functions

$$1,\ x,\ y,\ x^2,\ xy,\ x^3,\ y^2$$

all lie in $L(6O)$, their pole orders at O being $0, 2, 3, 4, 5, 6, 6$. Seven vectors in a six-dimensional space are dependent, and any dependence must involve both x^3 and y^2, since they are the only two with a pole of order exactly 6 and every other listed function has a distinct pole order. Normalizing their coefficients gives a relation

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$$

which is a plane cubic W in Weierstrass form.

The map $p \mapsto [x(p) : y(p) : 1]$, defined on $C \setminus \{O\}$ and regular everywhere by lemma 12.2.8, has image in W. It is birational onto W: the subfield $k(x, y) \subseteq k(C)$ satisfies $[k(C) : k(x)] = 2$ and $[k(C) : k(y)] = 3$ by box 18.4.1, since $\deg \operatorname{div}_\infty(x) = 2$ and $\deg \operatorname{div}_\infty(y) = 3$, so $[k(C) : k(x, y)]$ divides both 2 and 3 and is therefore 1.

Hence W is birational to C, and W is not rational: a rational W would make C birational to $\mathbb{P}^1_k$ and so isomorphic to it by lemma 12.2.8, giving C genus 0 rather than 1.

But a singular irreducible plane cubic *is* rational. If q is a singular point of such a cubic W', then $m_q(W') \geq 2$, so every line L through q has $I(q, W' \cap L) \geq 2$ by property (6) of the intersection number, and $\deg W' \deg L = 3$, so theorem 13.1.1 leaves exactly one further intersection point, counted with multiplicity. Sending L to that point is a birational map $\mathbb{P}^1_k \dashrightarrow W'$, the lines through q forming a $\mathbb{P}^1_k$. So W is nonsingular, and a birational map between nonsingular projective curves is an isomorphism by lemma 12.2.8 applied in both directions.

Finally, homogenizing the Weierstrass equation as $y^2z + a_1xyz + a_3yz^2 = x^3 + a_2x^2z + a_4xz^2 + a_6z^3$ and intersecting with the line $z = 0$ leaves $x^3 = 0$, so that line meets W only at $[0 : 1 : 0]$ and with multiplicity 3. That point is the image of O, since x and y both blow up there with y of higher order. So O is an inflection point in the sense of definition 12.1.6, completing the proof.

That last sentence closes the loop with chapter 15. Lemma 15.1.2 took its simplest form when the base point was an inflection point, and the Weierstrass embedding produces one for free: any point of a genus-one curve can be made the inflection at infinity. So the classical picture, in which three collinear points on a cubic sum to zero, is available for every choice of base point after re-embedding.

18.4.5 Note: The Genus Trichotomy The three cases are governed by the sign of $\deg K = 2g - 2$.

- $g = 0$: $\deg K = -2$. The curve is $\mathbb{P}^1_k$ by theorem 18.4.2, there is exactly one such curve, and its group Pic^0 is trivial by example 14.3.2.
- $g = 1$: $\deg K = 0$. The curve is a plane cubic by theorem 18.4.4, there is a one-parameter family of them, and $\operatorname{Pic}^0(C) \cong C$ by theorem 15.2.2.
- $g \geq 2$: $\deg K = 2g - 2 > 0$. The canonical class itself embeds the curve for $g \geq 3$ non-hyperelliptic, the curves form a $(3g - 3)$-dimensional family, and Pic^0 is an abelian variety of dimension g, the Jacobian.

Only the first two rows are proved here. The third is the beginning of the theory of moduli and of abelian varieties, and it is where this book stops.

Exercise 18.4.1

1. Show that $\ell(D) = 0$ when $\deg D = 0$ and $D \not\sim 0$, and $\ell(D) = 1$ when $D \sim 0$.
2. Let C have genus g and let D be effective of degree $\deg D > 2g-2$. Show $\ell(D) = \deg D + 1 - g$ exactly, and deduce $\ell(K) = g$ without using theorem 18.3.2 twice.
3. Compute $\ell(nO)$ for all $n \geq 0$ on a nonsingular plane cubic, and check the answer against example 18.1.3 to see where the two curves first differ.
4. A curve of genus 2 has $\deg K = 2$ and $\ell(K) = 2$. Deduce that it admits a degree-two map to $\mathbb{P}^1_k$, so that every genus-two curve is hyperelliptic.
5. Show that a nonsingular plane quartic has genus 3 and that its canonical class is $H \cdot C$, so that the canonical map is the original embedding. Conclude that no nonsingular plane quartic is hyperelliptic.
6. Use theorem 18.4.4 to show that the group $E_k(C, O)$ of chapter 15 is defined for any nonsingular projective curve of genus 1, not just for those presented as plane cubics.

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What Next?

This book has worked over a fixed algebraically closed field, with varieties as sets of points and rings of functions on them. That framework is what makes the subject concrete, and every one of its limitations points at a successor theory.

Three Things This Book Could Not Say

Three separate difficulties have surfaced, and it is worth collecting them, because each is resolved by a different extension.

Multiplicity is not carried by the points. Chapter 12 had to define a plane curve as a polynomial up to scalar rather than as its zero set, precisely because $\mathcal{Z}(f)$ and $\mathcal{Z}(f^2)$ are the same set of points while the two curves ought to be different. Every intersection number in chapter 13 was then computed from the ideal, not from the set, and the book never had an object whose points and whose multiplicities were carried together.

The field is fixed and algebraically closed. The Nullstellensatz (theorem 2.4.4) is the foundation of the whole algebra-geometry dictionary, and it fails without algebraic closure. So nothing here applies to the geometry of equations over $\mathbb{Q}$ or over a finite field, and those are exactly the cases number theory treats.

Nilpotents are invisible. Chapter 2 matched varieties with *radical* ideals, discarding everything else. A family of varieties degenerating to a limit, say two points colliding or a smooth conic degenerating to a double line, has a limit whose natural equation is not radical, and the theory built here cannot see the difference between that limit and the reduced set underneath it.

Schemes

All three difficulties have one resolution: stop insisting that the geometric object be a set of points with a ring of functions on it, and let an arbitrary commutative ring define a space directly. The points of $\operatorname{Spec} R$ are the prime ideals of R, the structure sheaf is built from localizations, and the Nullstellensatz is no longer needed because the dictionary is definitional rather than a theorem.

That move recovers everything above as a special case. A variety over k becomes a scheme whose ring is a finitely generated reduced k-algebra, and dropping either adjective is what gives the three resolutions: dropping "reduced" makes $\operatorname{Spec} k[x, y]/(f^2)$ a genuinely different object from $\operatorname{Spec} k[x, y]/(f)$, and dropping "over an algebraically closed k" allows $\operatorname{Spec} \mathbb{Z}$ and its relatives. Chevalley's theorem (theorem 11.3.4) survives verbatim, the Picard group of section 14.3 becomes a functor, and Riemann-Roch becomes a statement about the cohomology of a sheaf.

This is *Algebraic Geometry* [3], which is the direct continuation of this book and the one to read next. Its prerequisite is exactly what has been developed here: the reason to endure the machinery of schemes is that one already knows what it is supposed to do.

Four Directions

Schemes are the trunk. Four branches lead off from the material of this book in particular.

The analytic theory. Over $\mathbb{C}$ a nonsingular projective curve is a compact Riemann surface, and the genus of chapter 18⋆ is its topological genus, the number of handles. The Riemann-Roch theorem is then a statement about holomorphic differentials, proved by analysis rather than by Max Noether's theorem, and the group law of chapter 15 becomes the quotient $\mathbb{C}/\Lambda$ by a lattice. *Complex Analysis* [17] develops this. The two theories give the same answers because the analytic and algebraic categories agree for projective varieties, which is a theorem and not merely a correspondence of vocabulary.

Intersection theory. Chapter 13 counted intersections of plane curves, and section 11.4 counted lines on a cubic surface, but both counts were run by hand. The general machinery replaces the intersection cycle of section 13.3 by the Chow ring, in which classes of subvarieties multiply and the answer to an enumerative question is a coefficient. The self-intersection self-intersection $E \cdot E = -1$, which box 17.4.2 explains this book does not need but which any surface theory must supply, is among its first computations. This is *Intersection Theory* [8].

Toric varieties. Sections 10.1 and 11.1 constructed embeddings by writing down monomials, and the varieties so obtained are governed entirely by the combinatorics of which monomials appear. Toric geometry makes that systematic: a fan of cones determines a variety, and its geometry (its singularities, its divisors, its cohomology) is read off the combinatorics. It is the best available source of examples that are complicated enough to be interesting and explicit enough to compute with. This is *Toric Geometry* [10].

Abelian varieties. Theorem 15.2.2 identified a cubic curve with its own Pic^0. For a curve of genus $g \geq 2$ that fails, and Pic^0 becomes a g-dimensional projective variety with a group law, the Jacobian. Abelian varieties are the projective group varieties, and they are where the group law of chapter 15 generalizes. This is *Abelian Varieties* [2]; the arithmetic of the genus-one case is *Elliptic Curves* [6].

Two Books to Read Alongside

Two external texts have been cited throughout and are worth naming as companions rather than as references.

Fulton's *Algebraic Curves* [19] is the source for chapters 12 and 13 and for the parts of chapter 18⋆ that were stated rather than proved. It is short, entirely classical, and available freely; a reader who wants the adjoint-curve proof of Riemann-Roch should go there first.

Shafarevich's *Basic Algebraic Geometry* [18] covers the varieties half of this book at greater length and with more examples, and its second volume treats schemes and complex manifolds side by side. Where this book has compressed, Shafarevich has room. Hartshorne [14] remains the standard reference for the scheme-theoretic account, but its first chapter is a compressed version of what has been done here, and it is easier read second.

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